https://dx.doi.org/10.1007/978-3-030-71098-9

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Explainable Fuzzy Systems

Paving the Way from Interpretable Fuzzy Systems to Explainable AI Systems

October 2, 2020

Springer

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For everyone, especially family and friends,
who contributed to make this book a reality.

Foreword

The explanability of decisions based on Artificial Intelligence systems goes beyond a scientific issue, as Artificial Intelligence is nowadays ubiquitous and many organizations have drawn our attention to the necessity to provide the decision-maker and the decision subject with an easy to understand explanation. After the DARPA's advocacy for Explainable AI[1], the European Commission published guidelines[2] to make AI systems and their decisions transparent and explained to all parties involved. Many efforts are currently made by AI researchers to provide solutions to this problem, generally approaching only one of the aspects of explanability, mainly interpretability, understandability, expressiveness or transparency. These aspects are clearly interrelated and lead to a better acceptability of decisions, as well as trustworthiness in the decision-making process.

Of all these components, interpretability and expressiveness are often considered to be of primary importance for the acceptability of systems based on artificial intelligence by users, considered from the point of view of natural language expression of decisions and their reasons. However, most knowledge in human brains is implicit and therefore not verbalizable, so that human beings themselves are used to decisions based on non-expressible reasoning. It should also be noted that words are not the only knowledge representation easily understandable by users, as graphs, charts, histograms or statistics can be well appreciated, depending on their clarity and the expertise of the user.

In addition to interpretability and expressiveness, Explainable Artificial Intelligence requires that the reasons for the decision based on artificial intelligence be presented to the user. The automatic generation of explanations has various forms and we can cite for instance the use of feature importance vectors that quantify the relative impact of each feature in the prediction[3], an example of which is the in-

[1] `https://www.darpa.mil/program/explainable-artificial-intelligence`

[2] `https://ec.europa.eu/digital-single-market/en/news/ethics-guidelines-trustworthy-AI`

[3] Baehrens, D., Schroeter, T., Harmeling, S., Motoaki, K., Hansen, K., Muller, K.R.: How to Explain Individual Classification Decisions. Journal of Machine Learning Research 11:1803-1831, 2010.

terpretability method LIME[4]. Visualization provides another form of explanation, in particular through partial dependence plots[5] which show the marginal effect of a feature over the model outcome. The comparison of decisions with particular ones is another solution to explain the decision, for instance using prototypes[6] or counterfactuals[7]. The detection of the most influential training instances for a given prediction gives another form of explanation[8]. A general overview of these approaches[9] provides more aspects of automatic explanations.

The most popular methods to address the problem of the explanability of models based on artificial intelligence are based on decision rules[10] and decision trees[11]. They are at the core of the very comprehensive book prepared by J. M. Alonso, C. Castiello, L. Magdalena and C. Mencar in the specific realm of a fuzzy knowledge representation.

A certain conception of interpretability and expressiveness can be dealt with immediately by fuzzy systems, since fuzzy sets were created by Lotfi Zadeh to treat classes or concepts in a manner similar to the human way of manipulating them. The paradigm of Computing With Words[12] has, in particular, been introduced to establish a bridge between computation and natural language. Lotfi Zadeh showed later that a fuzzy knowledge-based representation can be employed to represent the meaning by means of a constraint-based semantics of natural languages[13].

To approach the meaning of artificial intelligence-based decision supports is certainly one of the requests of Explainable Artificial Intelligence. In this sense, fuzzy classes or categories appear more acceptable to users than crisp classes, as they avoid the risk of an arbitrary boundary preventing the users to reach an expected

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[4] Ribeiro, M.T., Singh, S., Guestrin, C.: Why should I trust you?: Explaining the predictions of any classifier. In: ACM SIGKDD International Conference on Knowledge Discovery and Data Mining (KDD), pp. 1135–1144, 2016.

[5] Friedman, J.H.: Greedy function approximation: A gradient boosting machine. The Annals of Statistics, 29(5):1189–1232, 2001.

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[9] Laugel, Th.: Local Post-hoc Interpretability for Black-box Classifiers, PhD Thesis, Sorbonne Université, July 2020.

[10] Turner, R.: A model explanation system. In: NIPS Workshop on Black Box Learning and Inference, 2015.

[11] Guidotti, R., Monreale, A., Ruggieri, S., Pedreschi, D., Turini, F., Giannotti, F.: Local rule-based explanations of black box decision systems. arXiv preprint 1805.10820, 2018.

[12] Zadeh, L.A.: Fuzzy logic = computing with words, IEEE Transactions on Fuzzy Systems, 4(2):103-111, 1996.

[13] Zadeh, L.A.: From computing with numbers to computing with words. From manipulation of measurements to manipulation of perceptions. In: Annals of the New York Academy of Sciences, 929:221-252, 2001.

decision category of which they are very close. Expressed in natural language, such categories look familiar to the user and easily understandable. Moreover, fuzzy rule-based systems and fuzzy decision trees add to the general capabilities of decision rules and decision trees to be directly approached the flexibility and familiarity of classes with gradual boundaries. This remarkable book explains in details how fuzzy models can participate in the construction of decision systems that are interpretable, and moreover explainable. It goes far beyond the classic representation of fuzzy classes and fuzzy rule-based systems. It establishes a bridge with natural language processing and automatic text generation and it proposes solutions to deal with the well-known interpretability-accuracy trade-off. All aspects of interpretability of fuzzy rule-based systems are covered, from interpretability rating to directly exploitable software.

It paves the way for a greater involvement of fuzzy methods in Explainable Artificial Intelligence, participating in the general search for greater acceptability of artificial intelligence-based decision systems by users, more confidence in the resulting decisions and, hopefully, more fairness in decisions.

Paris, July 2020 *Bernadette Bouchon-Meunier*

Preface

This book has been written during a challenging journey lasted for more than ten years. In 2005, Jose M. Alonso, still a PhD student under supervision of Luis Magdalena, attended for the first time the conference of the EUropean Society for Fuzzy Logic And Technology (EUSFLAT) in Barcelona. He presented the papers "A simplification process of linguistic knowledge bases" and "Integrating induced knowledge in an expert fuzzy-based system for intelligent motion analysis on ground robots", which will be later an important part of his PhD thesis. In addition, he met Corrado Mencar who presented the work "Some fundamental interpretability issues in fuzzy modeling". This meeting was the beginning of a nice friendship and fruitful collaboration. In the following years, Jose M. Alonso happened to stay in Bari twice (2010 and 2017) as a visiting researcher at the Università degli Studi di Bari Aldo Moro. In those occasions, a collaboration was first established and then strengthened with Ciro Castiello. Indeed, the four authors of this book have been doing research, mostly together, on interpretability of fuzzy systems (and beyond) for more than fifteen years, and have published several conference and journal papers, as well as organized a number of conference events (special sessions, tutorials, panels, etc.). Moreover, the four authors are all members of the Task Force (TF) on EXplainable Fuzzy Systems (EXFS) which is under the umbrella of the Fuzzy Systems Technical Committee (FSTC) in the Computational Intelligence Society (CIS) of the Institute of Electrical and Electronics Engineers (IEEE). Actually, Corrado Mencar and Jose M. Alonso are the chairs (and founders in 2020) of the IEEE-CIS TF-EXFS which has the final goal of paving the way from interpretable fuzzy systems towards eXplainable Artificial Intelligence (eXplainable AI, or XAI for short).

The very first idea of writing this book dates back to 2010. The demand for interpretable models was receiving a growing echo even outside the academic buildings. Just a few years before had started a severe worldwide financial crisis which urged analysts to argue against decisions made by global credit rating agencies on the basis of black-box models which no one could understand. At the time, the authors' concern was mostly related to designing interpretable fuzzy models that might replace black-box models in transparent decision-making support systems. It was 2016 when the focus was attuned to creating a bridge between interpretable fuzzy systems and XAI. In that year, the USA Defense Advanced Research Projects Agency (DARPA) launched the first challenge for a new generation of XAI systems remarking that “even though current AI systems offer many benefits in many applications, their effectiveness is limited by a lack of explanation ability when interacting with humans”. In addition, the European General Data Protection Regulation (GDPR) was approved in 2016 and became effective since May 2018, remarking that European citizens have the “right to an explanation” of decisions affecting them, no matter who (or what AI system) makes such decision.

Since 2018, there is an increasing worldwide interest in XAI. It is worth noting that about 30% of publications in Scopus (before October 2017) concerning XAI came from authors well recognized in the community of researchers in Fuzzy Logic. Most of these publications pay attention either to characterizing interpretability issues of fuzzy systems, or designing interpretable fuzzy systems, or building fuzzy systems with a good balance between interpretability and accuracy. EXFS go a step ahead of interpretable fuzzy systems. They benefit of interpretable fuzzy-grounded knowledge representation and reasoning while enhancing also human-machine interaction through multi-modal (e.g., graphical or textual modalities) effective explanations. Given the multifaceted nature of explainability, there are explanations which are factual, counterfactual, contrastive, task-oriented, etc. Accordingly, designing EXFS is a matter of careful human-centered design which goes beyond the usual topics treated by the community of researchers in Fuzzy Logic. Thus, the multidisciplinary field of XAI demands researchers, from both academy and industry, with a holistic view of fundamentals and current research trends in the field of Computational Intelligence (with special attention to Fuzzy Logic but also addressing XAI challenges on Neural Networks, Evolutionary Computation, Bayesian Networks, Bio-inspired algorithms, etc.), as well as researchers with complementary background on Cognitive and Social Science, Neuroscience, Computational Linguistics, Human-machine Interaction, etc.

The book is structured so as to give a gentle introduction on the key concepts related to EXFS, with a special focus on Fuzzy Rule-Based Systems (FRBSs). Chap. 1 provides the reader with some general ideas related to XAI and motivates the adoption of Granular Computing in general, and Fuzzy Logic in particular, as key methodologies for XAI. Since the focus of the book is on FRBSs, Chap. 2 is devoted to introduce the fundamental concepts, definitions and notation related to Fuzzy Set Theory and Fuzzy Systems. A necessary condition for explainability in fuzzy systems is the interpretability of the knowledge base. Interpretability is multifaceted, subjective and blur in its nature. To give a computational machinery for designing interpretable fuzzy systems, a constraint-based approach is generally used, whereas interpretability is characterized by a number of criteria. For such reasons, Chap. 3 is entirely focused on defining such constraints and criteria, which are organized in accordance to the hierarchy that is usually established in an FRBS (fuzzy sets, linguistic variables, information granules, rules, model, model adaptation). Chap. 4 describes indexes for interpretability assessment, which can be categorized in two main classes: structural-based interpretability indexes, which pertain to the symbolic representation of the knowledge base (linguistic terms, rules, etc.), and semantic-based interpretability indexes related to the functional definition of the symbolic structures. Having a toolbox of interpretability constraints and criteria along with their assessment indexes is not enough to ensure interpretability in fuzzy systems. To this end, specific design approaches are required, which are described in Chap. 5. The design process requires several decision steps and tasks, which have been divided in design of the knowledge base and design of the processing structure. Noticeably, the role of the designer is crucial in making the appropriate choices for balancing the required level of interpretability with the required precision of the resulting system. Chap. 6 reports a case study of design and validation of an explainable FRBS, which combines expert knowledge and knowledge automatically extracted from data, on a real-world problem. Interpretability is achieved by using a specific methodology—HILK which stands for Highly Interpretable Linguistic Knowledge— and related tool—GUAJE which stands for Generating Understandable and Accurate fuzzy systems in a Java Environment—. They gather all the concepts described in the previous chapters to help the designer in producing interpretable fuzzy systems. Then, explainability is achieved through the Linguistic Description of Complex Phenomena (LDCP) methodology for generating explanations in natural language of the inference provided by such fuzzy systems. Finally, Chap. 7 gives an outlook on the evolution from interpretable fuzzy systems to EXFS and their key role in XAI.

The book is intended for researchers and practitioners who wish to explore the main ideas on EXFS, which are presented in a comprehensive and homogeneous way. The researchers in the theory of interpretable fuzzy systems will find information on the state-of-art on interpretability, as well as many hints on future developments and open challenges. The researchers interested in applying EXFS in real-world problems will find a thorough description of the methodological processes required to design interpretable fuzzy systems as well as the explanatory tools for generating visualizations and natural language descriptions of the outcome of such systems. The book also offers a guide for practitioners who want to build intelligent solutions based on EXFS by using the available software tools and libraries.

EXFS can play a central role in XAI but there is still a lot of work to do in the race for building fully self-explaining machines. We hope both theorists and practitioners find this book useful. All data, software and scripts needed to reproduce the examples provided along the book are freely available at:

`https://gitlab.citius.usc.es/jose.alonso/bookexfs/`

Moreover, we hope this book is especially inspiring for the youngest researchers who will be in charge of designing and developing future XAI systems. While AI systems are already integrated in many applications of our daily live; fair, trustful and comprehensive XAI systems will surround us soon.

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Acknowledgements

This book has been written while authors were working in different institutions and taking part in many different projects. It is impossible to name here all researchers who authors met in productive discussions about explainability and interpretability issues of fuzzy systems in any of the organized seminars, invited talks, workshops, panels, tutorials, journal special issues or conference special sessions. Anyway, authors want to recognize their implicit contribution to this book because they constitute an active community of researchers.

In particular, Jose M. Alonso wants to express his gratitute to the former colleagues in the European Center for Soft Computing as well as current collaborators in the Research Centre in Intelligent Technologies of the University of Santiago de Compostela. Dr. Alonso also recognizes the support of the Spanish Ministry of Science, Innovation and Universities [Current active projects and grants are RYC-2016-19802, RTI2018-099646-B-I00, TIN2017-84796-C2-1-R, TIN2017-90773-REDT, and RED2018-102641-T], the Galician Ministry of Education, University and Professional Training [Current active projects and grants are ED431F 2018/02, ED431C 2018/29, and ED431G 2019/04], and the IEEE-CIS TF-EXFS.

Ciro Castiello and Corrado Mencar are members of the INdAM Research group GNCS. They recognize the support of Ministero dell'Istruzione, dell'Università e della Ricerca (MIUR) [Current active projects are PON ARS01_00141 "CLOSE", PON ARS01_00820 "RPASInAir", and PON ARS01_01116 "TALIsMAn"] and Ministero dello Sviluppo Economico [Current active project F/190030/01-03/X44 "LIFT"].

Luis Magdalena is a member of GB2S (Universidad Politécnica de Madrid) and DISCUS (Universidad Complutense de Madrid) Research groups. He would like to recognize the support of the Government of Spain [Current active grant PGC2018-096509-B-I00].

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Acronyms

ACM	Association for Computing Machinery
ACO	Ant-Colony Optimization
AHP	Analytic Hierarchy Process
AI	Artificial Intelligence
ANFR	Average Number of Fired Rules
ARL	Average Rule Length
BADD	BAsic Defuzzification Distribution
CIS	Computational Intelligence Society
CF	Certainty Factor
CD	Completeness Degree
CG	Centre of Gravity
COF	CO-Firing
COFCI	COF Comprehensibility Index
COR	COoperative Rules
CP	Computational Perception
CV	Cross-Validation
DARPA	Defense Advanced Research Projects Agency
DB	Data Base
DC	Double Clustering
DNF	Disjunctive Normal Form
EUSFLAT	EUropean Society for Fuzzy Logic And Technology
EXFS	EXplainable Fuzzy Systems
FATI	First Aggregate Then Infer
FCM	Fuzzy C-Means
FD	Firing Degree
FDT	Fuzzy Decision Tree algorithm
FINGRAM	Fuzzy INference-GRAM
FIS	Fuzzy Inference System
FITA	First Infer Then Aggregate
FLC	Fuzzy Logic Controller
FM	Fuzzy Modeling

FoT	Frame of Cognition
FPA	Fast Prototyping Algorithm
FRB	Fuzzy Rule Base
FRBC	Fuzzy Rule-Based Classifier
FRBS	Fuzzy Rule-Based System
FS	Fuzzy System
FST	Fuzzy Set Theory
FSTC	Fuzzy Systems Technical Committee
FURIA	Fuzzy Unordered Rule Induction Algorithm
GA	Genetic Algorithm
GDPR	General Data Protection Regulation
GLMP	Granular Linguistic Model of a Phenomenon
GMP	Generalized Modus Ponens
GrC	Granular Computing
GUAJE	Generating Understandable and Accurate fuzzy systems in a Java Environment
HFP	Hierarchical Fuzzy Partitioning
HFS	Hierarchical Fuzzy System
HILK	Highly Interpretable Linguistic Knowledge
ID	Incompleteness Degree
IEEE	Institute of Electrical and Electronics Engineers
IFS	Interpretable Fuzzy System
JFML	Java Fuzzy Markup Language
KB	Knowledge Base
LDCP	Linguistic Description of Complex Phenomena
LFM	Linguistic Fuzzy Modeling
LV	Linguistic Variable
LVI	Logical View Index
LVQ	Linear Vector Quantization
MF	Membership Function
MIMO	Multiple Inputs Multiple Outputs
MISO	Multiple Inputs Single Output
MOEA	Multi-Objective Evolutionary Algorithm
MOM	Mean Of Maxima
NL	Natural Language
NLG	NL Generation
NLP	NL Processing
NSGA	Non-dominated Sorting Genetic Algorithm
OWA	Ordered Weighted Averaging
PAES	Pareto Archived Evolutionary Strategy
PFM	Precise Fuzzy Modeling
PM	Perception Mapping
RB	Rule Base
RBC	Rule Base Complexity
RGI	Readability Global Index

RI	Redundancy Index
SFP	Strong Fuzzy Partition
SFR	Simultaneously Fired Rules
SIGI	Semantic Interpretability Global Index
SPEA	Strength Pareto Evolutionary Algorithm
SRC	Similarity of Rule Consequents
SRP	Similarity of Rule Premises
TRL	Total Rule Length
TSK	Takagi Sugeno Kang
UoD	Universe of Discourse
WM	Wang and Mendel algorithm
XAI	eXplainable Artificial Intelligence
XML	eXtensible Markup Language

Symbols

$\otimes$	Triangular norm (t-norm)
$\oplus$	Triangular conorm (t-conorm)
$\rhd$	Implication operator
$A, B, \ldots$	Fuzzy set
$\mathscr{F}(U)$	Set of all fuzzy sets on U
$\mathbf{G}$	Information granule
$\mathrm{hgt}(A)$	Height of fuzzy set A
$[A]_{\underline{0}}$	Support of fuzzy set A
$[A]_1$	Core of fuzzy set A
$[A]_\alpha$	α-cut of fuzzy set A
$[A]_{\underline{\alpha}}$	Strict α-cut of fuzzy set A
μ	Interpretation function
$\mu_A, \mu_{\mathrm{BIG}}$	Membership function of a fuzzy set
m_U	Infimum of set U
M_U	Supremum of set U
n_I	Number of inputs
n_O	Number of outputs
n_R	Number of rules
$\mathscr{P}$	Fuzzy partition
$\mathbb{R}$	Set of real numbers
σ	Width of a Gaussian fuzzy set
s.t.	such that
T	Set of linguistic terms
τ	Linguistic term
U, U_i	Universe of Discourse
X	Name of a linguistic variable
ω	Centre of a Gaussian fuzzy set
Dm	Distance measure
Sm	Similarity measure

As a general rule, boldface symbols are used for multi-dimensional objects. E.g., x denotes a crisp/fuzzy scalar input in a one-dimensional domain; $\mathbf{x}$ denotes a crisp/fuzzy vector in a multi-dimensional domain.

Labels of fuzzy sets are denoted as MEDIUM, BIG.

Chapter 1
Toward Explainable Artificial Intelligence through Fuzzy Systems

Abstract Explainable Artificial Intelligence is a novel paradigm conjugating the effectiveness of machine learning with the new requirements coming from the integration of intelligent systems in the human society. Explainable Artificial Intelligence can find successful application in a plethora of contexts, endowing classical intelligent systems with a crucial added value: the possibility for users to interact with machines, validate their results and ultimately trust their behavior. Fuzzy Set Theory provides a mathematical framework which is especially suitable to model concepts and perceptions of physical reality, thus injecting a kind of common-sense reasoning into machine learning algorithms and realizing a human-centric information processing which is the core of the Granular Computing paradigm. In particular, the exploitation of natural language enables Fuzzy Set Theory to act as a key-element for designing explainable models which can be able to provide explanations useful for human beings. We argue on those topics in order to show how the development of explainable fuzzy systems is a promising direction for paving the way from interpretable fuzzy systems to explainable intelligent systems.

1.1 Introduction

In 2013, Eric Loomis was found driving a car that had been used in a crime. The judge sentenced him six years of prison, which was determined in part by his score on the COMPAS scale[1], an algorithmically determined assessment used to predict an individual's risk of recidivism. COMPAS is a proprietary algorithm, and its risk assessment procedure is opaque to the public. Mr. Loomis appealed against the sentence by objecting that the use of a predictive algorithm violated the principle of a due process. Nevertheless, the Wisconsin Supreme Court ruled against Mr. Loomis

[1] COMPAS classification software is property of Equivant, USA: `http://www.equivant.com/solutions/inmate-classification`

because he would have deserved the same sentence based solely on the usual factors, including his crime and his criminal history [56].[2]

Independently from the final specification from the court, Loomis' case is perhaps one of the first and most apparent examples of Artificial Intelligence (AI) used to determine the course of a person's life. More and more cases accumulated in recent years, in very disparate situations, including autonomous vehicles, robot-assisted surgery, health-care, warfare, and so on. Indeed, the European Commission has identified AI as the "most strategic technology of the 21st century"[3]. Moreover, the economic impact of AI is estimated in trillions of euros by 2025 [62]. Thus, AI is preponderantly entering our life and we must ask ourselves if we are willing to accept this new presence and under what conditions.

The scientific community already realized this new trend and began to react accordingly. In 2017, the Association for Computing Machinery (ACM) issued a statement on algorithmic transparency and accountability[4] which recognizes that computer algorithms have far-reaching impacts, and their use may consciously or unconsciously result in harmful discrimination.[5] As a consequence, ACM recommends to use in the context of AI systems the same standards followed by institutions where humans traditionally come to decisions: a set of principles have been outlined, including the ability of explanation (a.k.a. *explainability*) which encourages to produce explanations regarding both the procedures followed by an algorithm and the specific decisions that are made:

> Systems and institutions that use algorithmic decision-making are encouraged to produce explanations regarding both the procedures followed by the algorithm and the specific decisions that are made. This is particularly important in public policy contexts. [Source: ACM]

From a political standpoint, the importance of data and their processing has been recognized and regulated. The General Data Protection Regulation (GDPR) is an European regulation, issued in 2016 and implemented in 2018, for the protection of natural persons with regard to the processing of personal data and on the free movement of such data.[6] GDPR is motivated, among other things, by the right to obtain

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[2] The full history was reported by The New York Times, on May 2, 2017, p. A22: "Sent to Prison by a Software Program's Secret Algorithms" (see `https://nyti.ms/2qoe8FC`).

[3] European Commission, Artificial Intelligence for Europe, Brussels, Belgium, "Communication from the Commission to the European Parliament, the European Council, the Council, the European Economic and Social Committee and the Committee of the Regions", Tech. Rep., 2018, (SWD(2018) 137 final) `https://ec.europa.eu/digital-single-market/en/news/communication-artificial-intelligence-europe`

[4] `https://www.acm.org/binaries/content/assets/public-policy/2017_joint_statement_algorithms.pdf`

[5] As an example, journalists at propublica.org have shown that the model adopted in Mr. Loomis' case has a strong ethnic bias: blacks who did not reoffend were classified as high risk twice as much as whites who did not reoffend. See `https://www.propublica.org/article/machine-bias-risk-assessments-in-criminal-sentencing`

[6] `https://eur-lex.europa.eu/legal-content/EN/ALL/?uri=CELEX:32016R0679`

an explanation of the decision reached after any assessment provided by automatic procedures:[7]

> [...] decision-making based on such processing, including profiling, should be allowed [...]. In any case, such processing should be subject to suitable safeguards, which should include [...] the right to obtain human intervention, to express his or her point of view, to obtain an explanation of the decision reached after such assessment and to challenge the decision. [Source: EUR-Lex]

Actually, GDPR is quite timid in affirming the right of explanation [70], thence the need of more precise regulations on the subject in future.

1.2 Explainable AI in a Nutshell

Explainable AI (XAI) is a new approach to AI where the ability to explain the decisions provided by algorithms is the primary objective. The acronym XAI was made popular by the USA Defense Advanced Research Projects Agency (DARPA) when launching to the research community (including both academy and industry) the challenge of designing self-explanatory AI systems[8]. The XAI program was firstly defined by DARPA with the objective of creating machine learning techniques that produce more explainable models, while maintaining a high level of prediction accuracy and "enable human users to understand, appropriately trust, and effectively manage the emerging generation of artificially intelligent partners".

Figure 1.1 illustrates the differences between current AI (focusing on machine learning) and XAI according to DARPA: the model that is trained from data (pay attention to the box *Learned Function* in the upper part of the picture) is replaced by an explainable model and an explanation interface (see the bottom part of the picture) for helping users understanding the results of a machine learning process. Accordingly, non-expert users, i.e., users without a strong background on AI, require a new generation of XAI systems. Such systems are expected to naturally interact with humans by providing comprehensible explanations of decisions that are automatically made [18]. XAI systems can be also considered as an important step forward towards collaborative intelligence [27] which promises a fully accepted integration of AI in our society.

XAI brings AI to intersect its way with other major branches of science, such as Human-machine Interaction and Social Sciences [50]. The human factor is central in XAI systems because acquired knowledge should be used to provide explanations to people; also, users should be able to challenge and question the systems in order to understand the rationale of a decision according to their specific needs. Therefore, human-machine interaction plays a pivotal role in such systems. However, it is not easy to define and capture the concept of explanation in all of its facets. Social sciences may help to understand how people communicate and share knowledge.

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[7] See note (71) in the preamble of GDPR.

[8] `https://www.darpa.mil/program/explainable-artificial-intelligence`

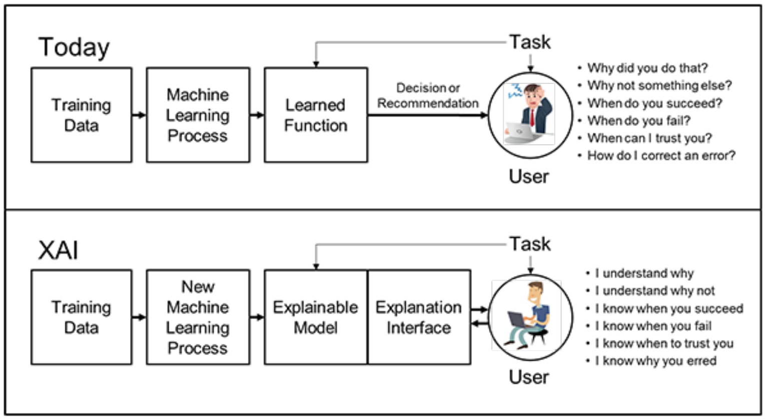

Fig. 1.1: XAI according to DARPA. [Source: DARPA]

Also, social sciences may give hints on how to effectively evaluate the quality of an explanation provided by a machine (i.e., a computer or any other electronic device like a smartphone).

There are several reasons that justify the adoption of XAI systems [58]. The most important among them is allowing people to decide whether to accept or to reject the systems' decisions. Such a choice can be possibly taken on non-technical basis, i.e., people involved in the decision process are not required to understand the machinery of the intelligent system, in the same way a person listening to another is not required to be a cognitive scientist in order to understand the reasons behind the decision taken by her fellow.

In critical contexts, an XAI system must be able to explain its choices in order to guarantee that some safety concerns are met. As an example, an autonomous car must be able to explain the reasons leading to the maneuver just before a car crash. Likewise, in a surgery room, an XAI system must be able to explain that the undertaken decisions meet lifesaver standards. In general terms, whenever safety is a critical issue, intelligent systems must be able to ensure people that all actions are taken in ways that minimize a risk or reduce a damage.

The capability to explain its own decisions enables AI systems to earn trust from their users. Trust leads to the use of such systems by relying on their decisions without asking for an explanation every time a decision is taken. However, whenever a mistake is suspected, systems must be able to explain each decision upon request. Trust is often established when users are not experienced, hence they are unable to verify if the system is right or not, simply for a lack of knowledge. In such cases, XAI systems can be adopted only if they have been previously checked by expert users who can validate them after assessing the provided explanations.

Whenever intelligent systems deal with personal data, their explanations must ensure fair, ethical, and legally bound decisions. It is important to notice that machine learning systems acquire knowledge from data that, more often than not, have been generated through previous processes, possibly biased by discrimination, fraudulence and other illegal acts. Such systems learn biased knowledge in "good faith" but it is unacceptable that they propagate the bias through future prediction. In sporadic (yet possible) cases, biases can be maliciously injected in the system knowledge base: checking for bias through an explanation may be the only way to detect malicious behaviors.

Intelligent systems are increasingly used also in scientific research, in order to find hidden relations in massive amounts of data. The so-called "fourth paradigm" of scientific discovery makes use of data mining processes to find patterns in data [38]. However, this approach to scientific discovery makes sense if the patterns extracted from data are meaningful, i.e., the involved intelligent systems must be able to explain the reasons underlying an observed phenomenon. Otherwise, far from solving a scientific problem, it would just be shifted to a black box that behaves as an "oracle" (accurate, but useless from a scientific point of view).

In addition, as the complexity of machine learning systems increases, explanations are helpful even to assist users in system debugging or in better understanding the system behavior, especially while dealing with previously unconsidered situations. The well-known case of adversarial image examples for deep neural networks—where images are modified in a way that is imperceptible to people but change the category of the image—exemplifies the situation that strange (and often undesirable) behavior is a consequence of a lack in understanding the inner machinery of the intelligent systems [31].

As a further notice, the adopted XAI system should also adjust (or personalize) its explanation according to different classes of users. Lay (or regular) users are those who do not have specific knowledge either in the application domain, or in the technicalities of the systems. Such people require a simplified language and short explanations, so that they can decide to accept (or not) the provided decisions. As an example, in loan application the regular user is the client asking the loan who should be convinced that the decision on the loan is not taken on discriminating bases. Expert users, on the other hand, have deep knowledge of the problem domain but little or no knowledge of the system's machinery. In the previous example, the clerk processing the loan application is an expert user. In this case, explanations can be more detailed and a more specific language can be adopted. Another category of users is represented by the system developers who need explanations demonstrating the soundness of the inference process underlying the ultimate decision. Finally, external entities (e.g., lawyers) represent a different category of users whenever they are involved in verifying that the system behaves according to legal, ethical and other non-technical constraints.

While XAI is based on the concept of explanation—i.e., a reason or justification for an outcome, an action or belief—a precise definition of explanation can be hardly achieved. In other words, a mathematical model of explanation does not exist. As a consequence of such vagueness, models are often judged explainable or not according to subjective interpretations and application needs [43]. Due to the centrality of human beings as final recipients of the explanation, any theory of explainability must take into account human-centric information processing.

1.3 What is an Explanation?

XAI is grounded on the concept of "explanation", meant as the ensemble of reasons underlying the decision taken by an intelligent system. Explainability is, therefore, the ability to provide an explanation upon user's request. But what is an explanation? There is not an easy way to answer this question. The literature abounds of many different terms, whose meanings are often overlapping. Also, different authors give the same term different (sometimes very different) meanings.

In the context of AI and Social Sciences, we can resort to some speculative discussions concerning the meaning of explanations [50]. Even though researchers have discussed about the nature of explanations for long [55, 63], there is not any definition admitted worldwide yet. There are many different approaches to deal with explanations regarding, among others, logical [35], causal [51], pragmatic [24, 69], unificationist [29], and cognitive [39] issues. For example, according to Lewis, "to explain an event is to provide some information about its causal history" [42], while Josephson & Josephson claim that "an explanation is an assignment of causal responsibility" [41]. In both cases, we find that the concept of *causality* is at the basis of an explanation.

Put in simpler terms, an explanation can be seen as an answer to a "why?" question. There are at least two senses of a "why?" question: (i) in the sense of "how come?", the question asks for the causes of a statement (e.g., the decision of an intelligent system); (ii) in the sense of "what for?", the question asks for the goals underlying the statement. For example, asking "why are you reading this book?" may have at least two answers: in the "how come?" sense, the answer could be "because my eyes are perceiving the signs on a piece of paper, which my brain interprets as characters and groups them into words which form sentences, paragraphs and so on"; in the "what-for?" sense, the answer could be "because I want to learn a bit more on XAI". According to the context, "how-come" and "what-for" senses are more or less useful (people are very clever in selecting the right sense). There is also a third sense, which may be called "contrastive", for example: "why are you reading this book? (and not watching TV)" where the contrastive statement is often implicit in the context. In this case the answer highlights the causative elements that justify one choice with respect to another (e.g. "because my favorite program starts later"). Contrastive questions are very frequent in people's communication and often are those asked to explainable models [50]. In fact, contrastive reasoning forms the basis of abductive inference [28] and contrastive explanations are claimed to be inherent to human cognition [20].

Whatever the sense, asking a "why?" question usually requires a causal explanation. In its turn, this would demand for the definition of a causal model. In [52, 53], Judea Pearl proposed the following three-level causal hierarchy:

Level 1 Association;
Level 2 Intervention;
Level 3 Counterfactual.

Briefly speaking, the first level associates different pieces of information simply observing the relationships defined by raw data. The second level goes a step further since it aims at deriving causal relationships over a modified environment, which would come up from the observer's intervention over reality. The third level is on the top of the hierarchy and investigates a scenario where hypothetical situations are imagined to take place assuming that some past actions would have gone differently. It is worth noting that counterfactuals (Level 3) refer to speculated alternatives to observed (factual) events, i.e., "contrary-to-fact" (counterfactual) events [57]. Thus, cognitive scientists talk about counterfactual thinking [20] and counterfactual reasoning [54]. On the one hand, this hierarchy reflects the superiority of the speculative/imaginative processes (related to levels 2, 3), that are typically carried on by humans to express creativity and to produce scientific advancements. On the other hand, the Pearl's view implies a limitation of AI systems which are restricted to the first level of the hierarchy, as long as they are conceived as data-centric models.

If intelligent systems cannot build causal models, how can they provide an explanation? This gap must be filled by the system developers who should take charge of making explicit the intrinsic relationships underlying any decision made, so that coherent explanations can be generated. Since causal relations must be investigated, the Aristotelian causal modes can be considered to assist the developer's task [34]. Particularly, the *material cause* is the substance composing things and beings; the *formal cause* is the shape or arrangement characterizing things and beings; the *efficient cause* is something which is apart from things and beings, but determines their states; the *final cause* is the ultimate end or goal of things and beings. In this way, when moving to the context of intelligent systems, a possible explanation scheme may be set as follows: the material cause is represented by the dataset or any other source of information about the problem at hand; the formal cause is given by the data representation (e.g., feature configuration); the efficient cause consists in the inference process that derives the decision; and the final cause is linked to the objective the intelligent system is aimed to. It is worth noting that all those kinds of explanations may be differently addressed to the various classes of users (lay, expert, etc.) mentioned in Sect. 1.2. Whenever every kind of explanation can be properly provided, the system can be regarded as *fully explainable*.

1.4 Interpretability and Explainability

Currently, any developed intelligent system could be hardly tagged as fully explainable because explainability is a highly complex property to be injected into an artificial machinery. How to achieve explainability is matter of current research. However, in literature a different term has been largely employed to express the property of a system of being *interpretable*. Interpretability is somewhat less demanding than explainability: far from possessing some additional capabilities (i.e., the formalization and the communication of explanations adhering to some causal model), an interpretable system should only lend itself to the inspection of a human user who should be able to readily comprehend its working engine. Although regarded as an *intermediate* quality, interpretability stands at the basis of an explainable system and should be exploited to generate effective explanations. Yet, the definition of interpretability is an elusive task too: the concept cannot be mathematically formalized and the term has been variously adopted with many nuances of meaning. The very term *interpretability* has been often employed in interchange with a number of synonyms, like *transparency*, *comprehensibility*, etc. An attempt to systematize this terminology has been made by Lipton [43] who distinguished between a couple of concepts which may both concur to the more general definition of interpretability:

Transparency is a quality connected to the understanding of the working mechanisms of a model. The quality of transparency can be further refined in: *simulatability* (i.e., a model is simple enough that a person could reproduce its inference process), *decomposability* (i.e., each part of the model has a clear meaning) and *algorithmic transparency* (i.e., the learning algorithm is transparent itself);

Post-hoc interpretability is a quality of methods that provide end users with useful information about the decisions carried out by the model. Methods for post-hoc interpretability include *textual explanations* to verbally justify decisions, and *non-textual explanations*. The latter include *visualizations* to qualitatively show what a model learned about data. More precisely, saliency maps show which parts of the input mostly concurred to the decision carried out by the model. In addition, the so-called *explanation by example* uses analogy with known examples that are taken as reference to justify the decision.

In the realm of Machine Learning, the most immediate approach to ensure interpretability is to use transparent models, such as linear models, decision trees, classification rules, Bayes networks, Fuzzy Rule-Based Systems (FRBSs), and so on. Transparent models have the clear advantage of offering a representation of knowledge that is in principle accessible to expert users. For example, linear models highlight the contributions of each feature in the determination of the output, while decision trees guide the user in the decision through a cascade of binary conditions. Classification rules represent knowledge in terms of sets of condition-decision pairs: whenever a condition is met, the corresponding decision is taken. Bayes networks represent the correlation of variables in terms of conditional probabilities: when a specific value of a variable is observed, the probability of the connected variables is updated according to the rules of Probability. FRBSs are the main subject of this book and belong to the category of transparent models as well. They act as the common classification rules, with the important difference that conditions apply with a certain degree and the rules contribute altogether to determine the final decision.

Of course, interpreting the transparent models enumerated above will become a simpler or harder task depending on the background of the user. In addition, it is worth noting that the main drawback of transparent models is the accuracy they can achieve when faced to highly complex problems. From a theoretical viewpoint, some of these models could be made complex enough to represent the solution of highly complex problems, but in this case simulatability—which is a key ingredient of transparency—is lost. The famous principle of incompatibility, postulated by

Zadeh in 1973, still applies to these models [73]: "[...] as the complexity of a system increases our ability to make precise and yet significant statements about its behavior diminishes until a threshold is reached beyond which precision and significance (or relevance) become almost mutually exclusive characteristics."

Alternatively, more sophisticated models can be used, such as random forests, neural networks (possibly with a deep architecture), support vector machines and others. In this case, however, transparency is completely lost and interpretability can be only partially recovered through *opening the black box* [32]. Techniques to introduce intepretability in black-box models can be categorized as: (i) model explanation, where a transparent model is designed with the aim of imitating/reproducing the output of the black-box model which is taken as an oracle while the transparent model acts as a surrogate model (the result is a kind of *reverse engineering* of the entire black-box model); (ii) outcome explanation, where a transparent model is designed from a trained black box by using a test sample and a collection of generated data, often in the neighborhood of the test sample (the result is a local reverse engineering of the black-box model) (iii) model inspection, where visualization techniques (e.g., saliency maps) highlight some key properties of the working mechanism that the black-box model puts into action in order to produce its outputs.

Notwithstanding the huge amount of research devoted to introducing and improving post-hoc interpretability in black-box models, interest in transparent models is very high. Indeed, some authors [59] claim for carefully designing interpretable models with enough accuracy to substitute to black-box models. Black-box models have, on their side, the possibility of tackling highly complex problems, especially involving images and large text corpora. On the other hand, transparent models offer the possibility of taking the control of each single part of the model, being easily interpretable in the problem domain. Thus, transparent models may be the favorite solution in those application domains where interpretability is a critical requirement, provided that the quality of such models (e.g., in terms of predictive accuracy) is satisfactory. Fuzzy rule-based models offer an interesting balance between numerical inference, which is typical of black-box models, and symbolic representation of knowledge, which is more akin to classical transparent models. In the following we are going to provide a suitable setting to discuss interpretability in the context of Fuzzy Modeling (FM).

1.5 Granular Computing and Fuzzy Logic as Paradigms for XAI

Granular Computing (GrC) is a computing paradigm where the object of processing is the *information granule*, i.e., a clump of objects kept together by some relations of indistinguishability, similarity, functionality or alike [76]. GrC is motivated by the need to approach AI through *human-centric information processing* [17], thence its central role in XAI.

Following Zadeh, information granules are the results of granulation which, along with organization and causation, composes the three basic concepts of human cognition [76]. Specifically, granulation is the act of decomposing a whole into meaningful parts – like the decomposition of a satellite image into terrain, rivers, lakes, and so on.

Regardless of the specific formal theories that can be developed under the paradigm of GrC, two common principles are generally preserved: the *multilevel* and the *multiview* principles [72]. On the one hand, according to the multilevel principle, granulation yields a hierarchical granular structure, with levels in the hierarchy corresponding to different degrees of abstraction. On the other hand, each granular structure offers just a partial view of a phenomenon, therefore different granular structures yield multiple views which may be used to provide a more complete understanding of the reality that is modeled. A handy example is the scientific publishing model: title-abstract-content is a multilevel granular structure that is represented in a paper, and several papers are usually published on a subject to highlight the methodology, the application, the implementation, etc.

Information granules at one level are treated as primitives for the higher level of a granular structure. Therefore, each information granule is informally defined as a collection of objects (i.e., information granules of the lower level) related together by some relation that makes objects indistinguishable at the higher level. Similarity, spatial proximity, functionality are examples of such relations. Common sense suggests that this kind of concepts are embedded into the human mind and, similarly to other concepts, they are formed through an act of *perception*, i.e., the organization, identification and interpretation of a sensation in order to form a mental representation [61, Chap. 4]. Perceptions pertain to a continuous reality and concepts shaped through perceptions must reflect this intrinsic continuity. Since information granules represent and process concepts as conceived by human minds, they should be defined to preserve the continuity of perception-based concepts. Fuzzy Set Theory (FST) offers a suitable mathematical framework to define this kind of information granules [79]. In other words, "fuzziness of information granules is a direct consequence of fuzziness of the concepts of indistinguishability, similarity, proximity and functionality" [75].

Very often, perception-based concepts are designated by labels borrowed from Natural Language (NL). Therefore, FST can be used for *computing with words* [77]: propositions in NL are translated into fuzzy constraints on the involved variables; inference is carried out through the machinery offered by FST; the results of inference are eventually expressed in NL. Indeed, the Computational Theory of Perceptions [78] is grounded on FST which is a promising approach for defining the theoretical background to represent perception-based information granules designated by linguistic terms. NL is the preferred modality to provide explanations to human beings[9] and NL explanations are expected to be presented as a narrative/story, because this aids human comprehension [71]. Thus, FST is a natural candidate for designing models in XAI.

FM is a methodology oriented toward the design of explanatory and predictive models using FST. FM is long-standing, with pioneering works dated in the 1970s.

[9] Another way to provide explanation is through visual imaging; however, we will not consider this topic in the book. As a reference for the interested reader we propose [36].

The original intent of FM was to develop knowledge-based systems capable of representing highly non-linear relations between inputs and outputs, while offering an intelligible view of such relations through the use of a simplified NL [46]. This was accomplished by *fuzzy rules*; nowadays, FRBSs are common practice in FM. In FM, *interpretability* is the term coined to denote the property of a system to represent knowledge in a way that is easy to be read and understood by humans [9]. Fuzzy rules offer an excellent way to achieve interpretability, that is why in this book we focus on FRBSs. In Chap. [4], FRBSs are introduced with their key aspects and main variants. Let us go deeper in the following section with the definition of interpretability in Fuzzy Logic.

1.6 A Definition of Interpretability in Fuzzy Logic

The quest for interpretability calls for the identification of several features. Among them, resorting to an appropriate framework for knowledge representation is a crucial element and the adoption of a fuzzy inference engine based on fuzzy rules is straightforward to approach the linguistic-based formulation of concepts which is typical of human abstract thought.

A distinguishing feature of an FRBS is the double level of knowledge representation (see Fig. 1.2). The lower level of representation is constituted by the formal definition of the fuzzy sets in terms of their membership functions, as well as the aggregation functions used for inference. This level of representation defines the *semantics* of an FRBS as it determines the behavior of the model, i.e., the input/output mapping for which it is responsible.

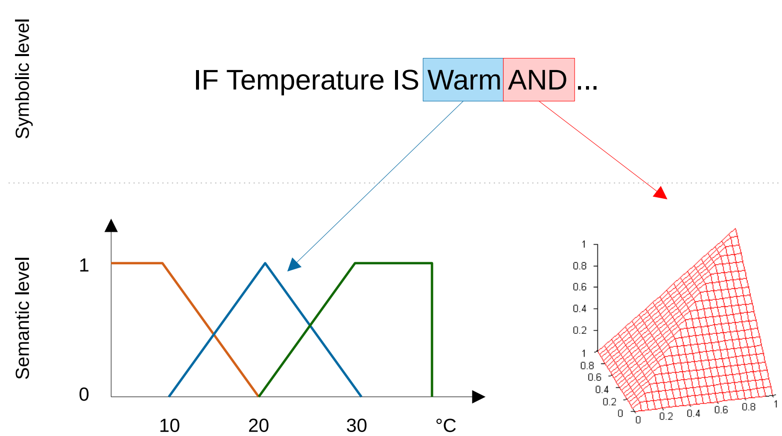

Fig. 1.2: The two representation levels in a fuzzy system.

On the higher level of representation, knowledge is represented in form of rules. They define a symbolic structure where linguistic variables are involved and reciprocally connected by some formal operators, such as AND, THEN, and so on. Linguistic variables correspond to the inputs and outputs of the model. The (symbolic) values they assume are related to linguistic terms which, in turn, are mapped to the fuzzy sets defined in the lower level of representation. The formal operators are likewise mapped to the aggregation functions. This mapping provides the interpretative transition that is quite common in the mathematical context: a formal structure is assigned semantics by mapping symbols (linguistic terms and operators) to objects (fuzzy sets and aggregation functions).

In principle, the mapping of linguistic terms to fuzzy sets is arbitrary. It just suffices that identical linguistic terms are mapped to identical fuzzy sets[10]. Nevertheless, the mere use of symbols in the high level of knowledge representation implies the establishment of a number of semiotic relations that are fundamental for the quest of interpretability of a fuzzy model. In particular, linguistic terms—as usually picked from NL—must be fully meaningful for the expected reader since they denote concepts, i.e., mental representations that allow people to draw appropriate inferences about the entities they encounter.

Mental concepts and fuzzy sets, therefore, are both denoted by linguistic terms. Additionally, concepts and fuzzy sets play a similar role: the former (being part of the human knowledge) contribute to determine the behavior of a person; the latter (being the basic elements of an FRBS) contribute to determine the behavior of a system to be modeled. As a consequence, concepts and fuzzy sets are implicitly connected by means of the common linguistic terms they are related to, which refer to object classes in the real world. The key essence of interpretability is therefore the property of *cointension* [79] between fuzzy sets and concepts, consisting in the possibility of referring to similar classes of objects: such a possibility is assured by the use of common linguistic terms.

Semantic cointension is a key issue when dealing with interpretability of fuzzy systems. It has been introduced and centered on the role of fuzzy sets, but it can be easily extended to refer to some more complex structures, such as the fuzzy rules or the whole fuzzy models. In this regard, a crisp assertion about the importance of cointension pronounced at the level of the whole model is given by the Michalski's *comprehensibility postulate* [49]:

> The results of computer induction should be symbolic descriptions of given entities, semantically and structurally similar to those a human expert might produce observing the same entities. Components of these descriptions should be comprehensible as single "chunks" of information, directly interpretable in natural language, and should relate quantitative and qualitative concepts in an integrated fashion.

It should be observed that the above postulate was formulated in the general area of Machine Learning. Nevertheless, the assertion made by Michalski has important consequences in the specific area of FM too. According to the comprehensibility postulate, results of computer induction should be described symbolically. Symbols

[10] In the case of formal operators, the same terms could map to different functions, though.

are necessary to communicate information and knowledge, hence pure numerical methods, such as neural networks, are not suited for meeting interpretability unless an interpretability-oriented post-hoc processing is performed.

The key point of the postulate is the human centrality of the results of a computer induction process and the importance of the human component implicitly suggests a peculiar aspect to be taken into account in the quest for interpretability. Thus, on one hand, interpretability in FM is related to Lipton's transparency (with special emphasis on simulatability) to face the cognitive limitations of people in understanding a model and replicating its function; on the other hand, interpretability concerns the semantic interpretation performed by a user determined to comprehend such model, therefore it is also related to decomposability and comprehensibility.

This twofold nature of interpretability led some researchers in FM to distinguish between [12]:

Readability concerns the structural aspects of a fuzzy model, which can be quantified in terms of number of involved linguistic terms, structural complexity of the knowledge base, etc.;

Comprehensibility concerns all the semantic aspects of a fuzzy model and relates to the adequacy of the model representation in mapping its components to users' known concepts.

Readability and comprehensibility represent two facets of a common quality and both of them are to be considered while assessing interpretability. In particular, this distinction should be acknowledged when criteria are specifically designed to provide a quantitative definition of interpretability.

1.7 Interpretability Issues in Fuzzy Systems

The definition of interpretability cannot be formulated in strict mathematical sense because it involves the human factor which is hard, if not impossible, to formalize. As described in the previous section, the basic principle underlying interpretability can be found in Michalski's comprehensibility postulate, which parallels the results of a learning algorithm with the description that a human expert may produce by observing the same entities [49]. Roughly speaking, the perception-based concepts acquired by a human should be *cointensive* with the information granules that are automatically generated by a learning algorithm, provided that the same objects are observed [47]. In particular, since we use symbolic terms drawn from NL to communicate knowledge, then the implicit semantics that a term conveys when used in a context should overlap with the explicit semantics a term is given by interpreting it by a fuzzy set.

Interpretability calls for both semantic and structural requirements: the semantic facet is related to the cointension of information granules with perception-based concepts, whereas the structural facet plays a role in coping with the limited capabilities of the human brain to process information [9]. In order to achieve an effective definition of interpretability, a collection of interpretability constraints and criteria can be adopted. Chap. [7] is devoted to formally define interpretability constraints and criteria that are most commonly found in literature. This collection is not standardized, because different constraints can be selected according to the needs of the designer. As a consequence, there is not a unique computational definition of interpretability. A common taxonomy of interpretability constraints relies on the level of modeling. Therefore, there are interpretability constraints for fuzzy sets, for linguistic variables, for multi-dimensional information granules, for fuzzy rules and for entire fuzzy models [48].

Also evaluation of interpretability can be hardly formalized because humans are involved in the process of knowledge communication. Roughly speaking, three main approaches can be identified to assess interpretability [26]: (i) in functionally grounded evaluation, interpretability is assessed through formalized measures that quantify the degree of fulfillment of interpretability constraints by any model component (structural and semantic measures are used and aggregated to define a global evaluation of interpretability [30]); (ii) in human grounded evaluation, the objective is to conduct simple experiments involving human subjects to test some general notions of explanation quality (well-designed experiments are required, preferably in the form of randomized controlled trials); (iii) the most complex form of evaluation is application grounded, which is aimed at conducting human experiments in a real application; the objective is to compare human-produced vs. machine-produced explanations with respect to some end-tasks (such as committing less error, performing less discrimination, etc.). The most widely used approach for interpretability evaluation is functionally grounded. Chap. [8] is devoted to describe the topic of interpretability assessment in FRBSs.

In the 1980s, FST met Machine Learning [66, 67]. Since then, several methods for automatically deriving FRBSs from data arose. As a result, fuzzy models were mainly designed for accuracy, while the original intent of FST to represent perception-based knowledge got secondary relevance. However, FRBSs that are not interpretable are akin to black-box models, like neural networks, for which an equipment of powerful learning techniques already exists. Therefore, interpretability is not given for granted by the mere use of FST but it requires a methodology that is still in development. Moreover, designing interpretable fuzzy models requires some additional steps with respect to the usual modeling stages. In particular, the source of knowledge may be twofold: the available data and the expert's knowledge. The way of considering these sources is critical for building an effective model. An iterative approach is recommended to integrate induced knowledge with expert rules in order to drive the design toward a model that is balanced in terms of predictive and explanatory capabilities [15]. Also, several modeling approaches are available, which may favor interpretability over accuracy or vice versa; other approaches recognize that interpretability and accuracy are conflicting objectives and adopt multi-objective techniques to achieve a Pareto front of solutions [16]. Alternatively, ad-hoc algorithms may be used to incorporate interpretability constraints within the algorithms that induce fuzzy rules from data [23]. Chap. [6] is devoted to introduce design guidelines to embody interpretability in FRBSs acquired from data, possibly with the integration of expert's knowledge.

There are not many software tools to support designers in developing interpretable FRBSs [3]. FisPro[11] is an open-source software that facilitates interpretability in all FM steps [33]. JFML[12] is an open source Java library that offers a complete implementation of a standard of the Institute of Electrical and Electronics Engineers (IEEE) that is designed to provide a unified and well-defined representation of fuzzy systems for problems of classification, regression, and control [65]. GUAJE[13] is another open-source software with the aim of supporting the design of interpretable FRBSs by means of combining several preexisting software tools [11]. GUAJE implements the Highly Interpretable Linguistic Knowledge (HILK) methodology for interpretable FM [12]. These tools offer the possibility of deriving interpretable FRBSs from data and expert knowledge. These systems can be employed to give some forms of explanations by adopting NL Generation (NLG) techniques. Chap. [5] describes step by step how to design, implement and validate an explainable FRBS which is endowed with explanation capability through NLG.

1.8 An Historical Background on Interpretable Fuzzy Systems

Notwithstanding the difficulties in defining interpretability, this quality has been often recalled in literature since the early years of AI. If we confine our discussion within the narrower field of Fuzzy Logic, we recognize four main periods where the use of the term *interpretability* bore distinct connotations, as depicted in Figs. 1.3 and 1.4, which show the distribution of publications and citations, respectively, in the research field of interpretable fuzzy systems [13][14]. It is worth noting that the drop in 2020 is due to the fact that only 9 months are considered.

In essence, the following main phases can be identified regarding FM:

1965–1990 This period coincides with the early development of Fuzzy Logic following the pioneering works of Zadeh. The FST was conceived as a novel approach to computerize perception-based concepts, which can be eventually

[11] The Fuzzy Inference System Professional (FisPro) is an FM tool freely available online at `https://www7.inra.fr/mia/M/fispro/`

[12] The Java Library for the Fuzzy Markup Language (JFML) complies with the IEEE Standard 1855-2016 and is freely available online at `http://www.uco.es/JFML/`

[13] The Java Environment for Generating Understandable and Accurate fuzzy systems (GUAJE) is a FM tool freely available at `https://gitlab.citius.usc.es/jose.alonso/guaje/`

[14] With the aim of covering most key terms (interpretable, understandable, etc.) usually considered in publications related to interpretability issues, we introduced the following query in the Clarivate Analytics Web of Science (`https://clarivate.com/products/web-of-science/`):

TI=(fuzzy AND ((“interpretab*”) OR (“understandab*”) OR (“comprehensib*”) OR (“intelligib*”) OR (“transpar*”) OR (“readab*”) OR (“complexity”)))

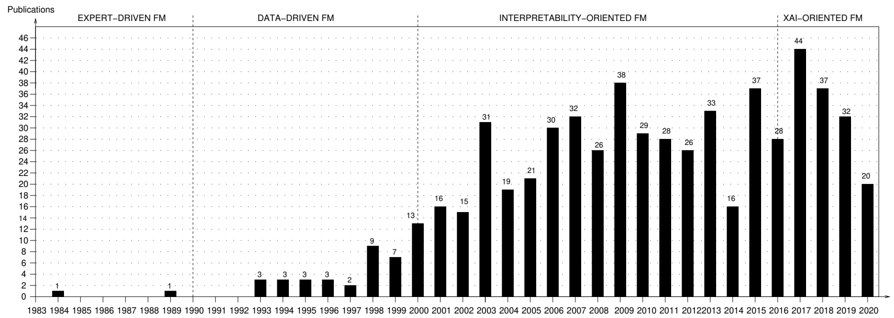

Fig. 1.3: Publications per year related to interpretability issues. (Update: September 2020)

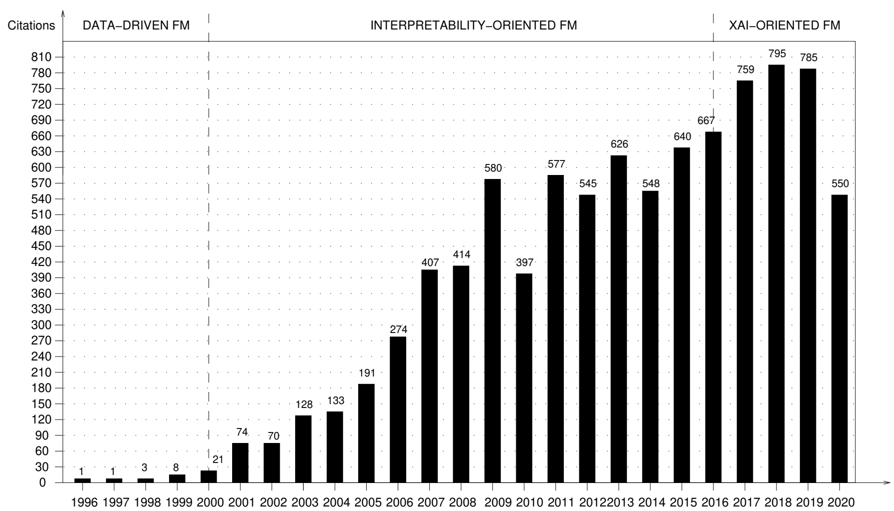

Fig. 1.4: Citations per year related to interpretability issues. (Update: September 2020)

expressed in NL. As a consequence, during this initial period interpretability emerged naturally as the main advantage of fuzzy models. Researchers concentrated on building fuzzy models mainly working with expert knowledge and a few simple linguistic variables [74] and linguistic rules usually referred to Mamdani rules [45]. As a result, those hand-designed fuzzy models were characterized by their high interpretability. Moreover, interpretability was assumed as an intrinsic property of fuzzy models. Therefore, there were a few publications explicitly dealing with interpretability issues.

The first proposal of an FRBS was made by Mamdani who was able to augment Zadeh's initial formulation allowing the application of fuzzy systems to a control problem [44]. These kinds of fuzzy systems are also referred to as *fuzzy* *logic controllers*, as proposed by the author in his pioneering paper [46]. Very soon, Mamdani FRBSs became the main tool to develop linguistic fuzzy models, even though alternative rule formats were arising and gaining importance to overcome some limitations of these systems, especially in terms of computational efficiency. In addition to Mamdani FRBSs, probably the most famous FRBSs are those proposed by Takagi, Sugeno and Kang [67], the so-called TSK fuzzy systems, which required less processing efforts at the price of reduced interpretability.

1990–2000 Fuzzy systems were increasingly appealing and attracted researchers to deal with more and more complex problems. This trend required the development of methods for acquiring and fine-tuning fuzzy systems from data: in other words, Fuzzy Logic met Machine Learning [40]. During the 1990s, the focus was therefore set on developing methods to acquire FRBSs that optimized some accuracy criteria. Accordingly, those data-driven fuzzy models became composed of extremely complicated fuzzy rules with high accuracy but at the cost of disregarding interpretability as a side effect. Obviously, automatically generated rules were seldom readable, with the consequence that the original idea of the FST as instrumental for perception-based knowledge representation got obfuscated.

Along this period some researchers started claiming that fuzzy models are not interpretable *per se*. Interpretability is a matter of careful design [68]. Thus, interpretability issues must be deeply analyzed and seriously discussed. Although the amount of publications related to interpretability issues is still small, they begin to grow exponentially at the end of this period.

2000–2016 Researchers realized that both expert-driven (from 1965 to 1990) and data-driven (from 1990 to 2000) design approaches have their own advantages and drawbacks, but they are somehow complementary. For example, expert knowledge is general and easy to interpret but hard to formalize, especially for complex problems. On the other hand, knowledge derived from data can be extracted automatically but it becomes quite specific and its interpretation is usually hard. Moreover, researchers were aware of the need of taking into account simultaneously interpretability and accuracy during the design of fuzzy models. As a result, during this third period the main challenge was how to combine expert knowledge and knowledge extracted from data, with the aim of designing compact and robust systems with a good interpretability-accuracy trade-off.

When considering both interpretability and accuracy in FM, two main strategies establish themselves in a natural way [2]: *Linguistic Fuzzy Modeling* (LFM) and *Precise Fuzzy Modeling* (PFM). On the one hand, in LFM system designers first focus on the interpretability of the model, and then they try to improve its accuracy [21]. On the other hand, in PFM designers first build a fuzzy model maximizing its accuracy, and then they try to improve its interpretability [22]. As an alternative, since accuracy and interpretability represent conflicting goals by nature, multi-objective FM strategies (considering accuracy and interpretability as objectives) have become very popular [25, 37].

At the same time, there has been a great effort for formalizing interpretability issues. As a result, the number of publications has grown a lot. Researchers have actively looked for the right definition of interpretability. In addition, several interpretability constraints have been identified [48, 80]. Moreover, interpretability assessment has become a hot research topic [14]. In fact, several interpretability indexes (able to guide the FM design process) have been defined. Nevertheless, a universally accepted index is still missing, thus demanding further research on the topic.

In 2003, two pioneering books were published on the topic [21, 22], which contributed to make the fuzzy community aware of the need to take into account again interpretability as a main design concern. In 2005, an axiomatic treatment of interpretability at the level of linguistic variables was provided by Bodenhofer and Bauer [19]. These and other main works attracted attention on the topic, resulting on an exponential growth of publications until 2009, when the trend settled or even decreased (2010, 2012, 2014), perhaps because of the advent of attracting research subjects such as deep learning. Nevertheless, the theme of interpretability in fuzzy systems is by no means exhausted because of the challenging research questions it poses.

2016–now With the XAI challenge launched by DARPA in 2016, the interest on research topics such as fairness, accountability, or interpretability, is blooming in the entire community of AI researchers [1, 32, 50]. Moreover, as described in [10], about 30% of publications in Scopus (20 October 2017) regarding XAI came from authors well recognized in the field of Fuzzy Logic, what highlights the relevance of the FST in the context of XAI. Thus, we can state that the XAI challenge has revitalized the interest toward interpretable fuzzy systems as promising tools to embody explainability in intelligent systems. This fact is reflected in the continuous growing in the number of citations (see Fig. 1.4). In addition, there is a significant increment in the number of publications from 2016 to 2017 (see Fig. 1.3) even if there had been an important decrement from 2015 to 2016, which may be due to the boom of deep learning that was attracting the attention of most researchers at that time. In our opinion, the reduction in the number of publications that we observe in 2018 and 2019 (see Fig. 1.3) may be due to a modified attitude took by researchers: they are considering research on interpretability of fuzzy systems as a mature area, thus focusing their interest on approaching other emerging related topics (such as fairness, accountability and explainability). Such topics go beyond interpretability issues and involve other interdisciplinary research fields, such as NL Processing, Argumentation Technology or Human-machine Interaction. It is worth noting that one of the hottest open problems nowadays in the XAI research field is how to generate and how to evaluate the effectiveness of interactive explanations associated to decisions made by AI-based systems [60, 64].

1.9 Final Remarks

Interpretability is a necessary quality to develop XAI systems, and Fuzzy Logic offers a suitable background to design interpretable intelligent systems. In particular, FRBSs belong to the class of transparent systems according to Lipton's categorization. With respect to other transparent systems, FRBSs offer a double layer of knowledge representation: a symbolic layer and a semantic layer, whereas the former is constituted by linguistic terms and the latter by the mathematical constructs pertaining to FST. Linguistic terms convey an implicit semantics to the user through their linguistic meaning; as a consequence, interpretability in FRBSs is obtained when there is semantic cointension between the implicit semantics and the explicit semantics defined by interpreting the terms in the semantic layer. Therefore, interpretability of fuzzy systems cannot be taken for granted for the mere use of FST (an assertion that is sometimes found in literature) but it requires a design methodology that is the core of the present book.

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Chapter 2
An Overview of Fuzzy Systems

Abstract Fuzzy systems have found widespread application in several contexts and proved their suitability in tackling a number of diverse real-world problems. However, the realization of such systems must be well grounded on some solid theoretical bases that scientists and developers should properly master. In this chapter we discuss the key elements of fuzzy systems, as well as the different instances of their implementation. To this aim, an overview of Fuzzy Set Theory is preliminarily proposed, as an indispensable foundation which is useful to fix the basic concepts, the mathematical underpinning, and the formal notation. In this way, we are able to suitably introduce the concept of fuzzy rule-based systems, illustrating the different components involved in their structure. We review also various types of fuzzy rule-based systems, focusing a special attention on details related to the design of interpretable models.

2.1 Introduction

Rule-based systems have been successfully used to model human problem-solving activity and adaptive behavior, providing a suitable way to represent human knowledge through the use of the classical IF-THEN rules. The fulfillment of the rule antecedent gives rise to the execution of the consequent, i.e., one action is performed. Conventional approaches to knowledge representation are based on bivalent logic, which is associated with a serious shortcoming: the inability to deal with the issues of uncertainty and imprecision. As a consequence, conventional approaches do not provide an adequate model for a kind of familiar thinking embracing most of the commonsense reasoning performed by humans on a daily basis.

The application of Fuzzy Logic (FL) to rule-based systems leads to Fuzzy Rule-Based Systems (FRBSs). These systems consider IF-THEN rules whose antecedents and consequents are composed of fuzzy statements (i.e., these systems work with IF-THEN fuzzy rules), thus presenting two essential advantages over classical rule-based systems:

- the key features of knowledge captured by fuzzy sets involve handling of uncertainty;
- inference becomes more robust and flexible with the employment of approximate reasoning methods.

Knowledge representation is enhanced with the use of linguistic variables (LVs) and their linguistic values, that are defined by context-dependent fuzzy sets whose meanings are specified by gradual membership functions [35, 36, 37]. On the other hand, FL inference methods (such as generalized *modus ponens*, *modus tollens*, etc.) form the bases of approximate reasoning [34]. Hence, FL provides a unique computational base for inference in rule-based systems.

This chapter presents an outline of the general concepts underlying the theory of FRBSs. After a brief introduction to the Fuzzy Set Theory (FST), FRBSs are introduced in their structure and operation. We focus on Mamdani FRBSs because they are of main interest for interpretability; however, other variants of FRBSs can be used for designing interpretable fuzzy systems: they are outlined in the final section.

2.2 A Brief Overview of Fuzzy Set Theory

This section presents an overview of the FST; for deeper details, the reader is referred to specialized textbooks [14, 16, 42].

2.2.1 Introducing Fuzzy Sets

FRBSs rely on the FST [33] where the concept of fuzzy set is identified by a *membership function*, i.e. a mapping from a Universe of Discourse (UoD) U to a completely distributive lattice (usually, the interval $[0,1]$). Formally:

$$\mu_A : U \mapsto [0,1]$$

is the membership function of a fuzzy set labeled by the symbol A.

In our discussion, we are going to distinguish between the symbolic representation of a fuzzy set (denoted by its associated label) and its semantic representation (denoted by its membership function). However, for the sake of conciseness, we will generally refer to a fuzzy set by simply mentioning its label. Thus, the short form "fuzzy set A" will be adopted in place of the more accurate form "fuzzy set labeled with A and identified by the membership function μ_A". Without loss of generality, therefore, given $x \in U$, its *membership degree* to the fuzzy set A will be expressed as $\mu_A(x)$. Moreover, the set of all possible fuzzy sets definable on U is denoted with $\mathscr{F}(U)$.

For a fuzzy set A, the following concepts can be defined:

Height The height of a fuzzy set corresponds to the maximum membership degree of its element, i.e.

$$\mathrm{hgt}(A) = \sup_{x \in U} \mu_A(x) \tag{2.1}$$

α-cut A fuzzy set can be "cut" horizontally at a given membership level $\alpha \in [0,1]$; the result is a (crisp) set defined as

$$[A]_\alpha = \{x \in U : \mu_A(x) \geq \alpha\} \tag{2.2}$$

Strict α-cut The strict α-cut of a fuzzy set is the crisp set containing all the elements in the UoD exhibiting membership strictly greater than α:

$$[A]_{\underline{\alpha}} = \{x \in U : \mu_A(x) > \alpha\} \tag{2.3}$$

Support The support of a fuzzy set is the crisp set containing all the elements in the UoD exhibiting non-zero membership:

$$[A]_{\underline{0}} = \{x \in U : \mu_A(x) > 0\} \tag{2.4}$$

Core The core of a fuzzy set is the crisp set containing all the elements in the UoD exhibiting full membership:

$$[A]_1 = \{x \in U : \mu_A(x) = 1\} \tag{2.5}$$

Prototype Whenever $[A]_1 = \{x_0\}$, the element x_0 is called *prototype* of A.

2.2.2 Universe of Discourse

It is assumed that the UoD U is a closed interval of the real line[1]. This assumption has some important consequences:

1. U is a subset of the real line. When physical systems have to be represented by fuzzy models, resorting to such a numerical UoD is a due choice, although categorical UoDs may be preferred when more abstract data types (e.g., conceptual categories) must be represented. For fuzzy sets defined on categorical UoDs, however, many interpretability constraints (see Chap. [1]) are not applicable while other constraints —though formally applicable— lose significance. More structured domains, such as complex numbers, are more application-specific and hence have not been considered.
2. U is one-dimensional. As a consequence, fuzzy sets model single attributes of a complex system and composite attributes are represented by more complex information granules. Unidimensionality is often required in interpretable fuzzy

[1] That is reasonable in what concerns the subject of this book: in some other contexts, categorical UoDs may be of interest.

modeling and is often implicitly assumed in literature. Complex concepts are then represented as composition of attributes for which one-dimensional fuzzy sets have been defined. However, some complex concepts cannot be defined in terms of composition of attributes (e.g., the niceness of a person). We may argue that interpretable representations of such complex concepts are not possible in principle, this is the reason why we do not consider them when dealing with interpretability constraints.

3. U has a lower bound
$$m_U = \inf U$$
and an upper bound
$$M_U = \sup U$$
The adoption of closed intervals simplifies modeling and is not particularly limiting when representing physical attributes, which are almost always expressible as values in a range. In quantitative modeling, either m_U or M_U may be infinite. In this case, interpretability constraints that rely on these quantities can be easily extended to deal with infinite limits.

In consequence of the previous specifications on U, the definition of a membership function can be expressed in the more precise form

$$\mu_A : U = [m_U, M_U] \subset \mathbb{R} \mapsto [0,1]$$

2.2.3 *Operations on Fuzzy Sets*

Fuzzy set operators rely on functions that generalize the classical definitions of intersection, union and complement; these are extended through t-norms, t-conorms (or s-norms) and negations.

A t-norm is a function $\otimes : [0,1] \times [0,1] \mapsto [0,1]$ which satisfies the following properties:

- Monotonicity: $a \otimes b \leq c \otimes d$ if $a \leq c$ and $b \leq d$;
- Commutativity: $a \otimes b = b \otimes a$;
- Associativity: $a \otimes (b \otimes c) = (a \otimes b) \otimes c$;
- Neutral element[2]: $a \otimes 1 = a$.

A t-conorm $\oplus$ (or s-norm) has the same properties of t-norms, with the difference that the identity element is 0. Usually, 1-complement is used as negation, i.e. $n(x) = 1 - x$ (although more general definitions can be applied). A triplet $(\otimes, \oplus, n)$ is a De Morgan triplet if the relation $x \oplus y = n(n(x) \otimes n(y))$ applies. Table 2.1 reports a short list of very common choices of t-norms and t-conorms. An in-depth discussion on t-norms, t-conorms and other aggregation operators can be found in specialized literature [13].

[2] Some authors also refer to this property as *Identity element* or as *Boundary condition*.

Table 2.1: Common choices of t-norms and t-conorms forming a De Morgan's triplet under 1-complement as negation.

name	$\otimes$	$\oplus$
Gödel	$\min(x,y)$	$\max(x,y)$
Product	$x \cdot y$	$x+y-xy$
Łukasiewicz	$\max(x+y-1,0)$	$\min(x+y,1)$

A favorite choice (because of several theoretical properties) adopts the min function as t-norm, the max function as t-conorm and the 1-complement function as negation. This leads to the following definition of fuzzy set operators:

$$\mu_{A_1 \cap A_2}(x) = \min\{\mu_{A_1}(x), \mu_{A_2}(x)\}$$

$$\mu_{A_1 \cup A_2}(x) = \max\{\mu_{A_1}(x), \mu_{A_2}(x)\}$$

$$\mu_{\bar{A}}(x) = 1 - \mu_A(x)$$

The subsethood relation directly descends from the natural ordering of membership degrees, i.e.

$$A_1 \subseteq A_2 \Leftrightarrow \forall x \in U : \mu_{A_1}(x) \leq \mu_{A_2}(x)$$

although several generalizations are possible to include fuzzy subsethood.

2.2.4 Fuzzy Relations and Composition

Fuzzy relations are fuzzy subsets of the Cartesian product of two UoDs U_X and U_Y, i.e.

$$\mu_R : U_X \times U_Y \mapsto [0,1]$$

As fuzzy sets, they can be processed with the usual fuzzy set operators; however, another operator can be specifically defined for fuzzy relations: composition.

Composition is particularly important because it constitutes the mathematical underpinning onto which most inference operators are defined. Given two fuzzy relations R_1 and R_2 defined on $U_X \times U_Y$ and $U_Y \times U_Z$ respectively, fuzzy composition is defined as:

$$\mu_{R_2 \circ R_1}(x,z) = \max_{y \in U_Y} \min\{\mu_{R_1}(x,y), \mu_{R_2}(y,z)\}$$

Fuzzy composition can be easily adapted to compose a fuzzy set with a fuzzy relation, i.e.

$$\mu_{R \circ A}(y) = \max_{x \in U_X} \min\{\mu_A(x), \mu_R(x,y)\} \tag{2.6}$$

which is known as *compositional rule of inference* [34]. In essence, the usefulness of this rule consists in providing a matching between a fact (expressed as a fuzzy set A_X in the UoD U_X) and a relation R, in order to infer a new fact (which can be expressed by a fuzzy set A_Y in the UoD U_Y). In this way, R can be reviewed as a fuzzy rule in the form $A_X \to A_Y$, and fuzzy composition gives rise to the Generalized *Modus Ponens* (GMP):

$$\frac{A'_X, \quad A_X \to A_Y}{A'_Y}$$

to be applied to any fuzzy set $A'_X \in U_X$ to infer by means of R a new fuzzy set $A'_Y \in U_Y$. The GMP stands as the basis of approximate reasoning.

2.2.5 Fuzzy Sets as Information Granules

Information granulation refers to the concept of decomposing a whole into significant parts [38] (for example: decomposing a satellite image into parts like rivers, forest, roads, etc.). Computing with information granules leads to Granular Computing, an emerging discipline for computing in complex environments [3]. FST offers a nice mathematical apparatus for Granular Computing since fuzzy sets provide a granular decomposition of an UoD into semantically significant parts, which are usually labeled by linguistic terms.

With respect to other mathematical theories for Granular Computing (such as Rough Sets Theory or Interval Analysis), FST allows to define information granules that are inherently imprecise (due to the lack of boundaries) and provide a mathematical machinery for making inference with information granules. In particular, multi-dimensional fuzzy relations enable to decompose a complex, multi-dimensional UoD into fuzzy "patches" (which are fuzzy information granules) wherein simple inference models can be identified. As a consequence, the whole UoD is decomposed into possibly overlapping fuzzy information granules, whose local models can be taken together to define a complex model which, however, can be described by simple components. This is the very essence of an FRBS, which will be described in greater detail in the following sections.

2.3 Fuzzy Rule-based Systems

The previous sections suggest the contribution of two different concepts in FRBSs: knowledge and reasoning. This clear separation between knowledge and processing structures (as shown in Fig. 2.1) is the key aspect of knowledge-based systems. An FRBS, therefore, can be considered as a type of knowledge-based system.

The first type of FRBS dealing with real inputs and outputs was proposed by Mamdani [22], who was able to augment Zadeh's initial formulation allowing the application of FST to a control problem. These kinds of systems are also referred to as *FL Controllers* (FLCs), as proposed by the author in his pioneering paper [23].

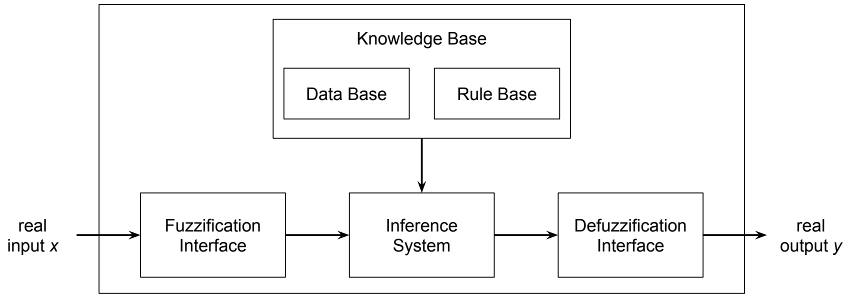


Fig. 2.1: General structure of a Mamdani fuzzy rule-based system.

The term FLC became popular and, since the beginning, control system design constituted the main application of Mamdani FRBSs.

The generic structure of a Mamdani FRBS is shown in Fig. 2.1. The *knowledge base* (KB) stores the available knowledge about the problem in the form of fuzzy IF-THEN rules. The processing structure exploits those rules to put into effect the inference process on the system inputs. The fuzzification interface establishes a mapping between crisp values in the input domain U and fuzzy sets defined on the same UoD. On the other hand, the defuzzification interface performs the opposite operation by defining a mapping between fuzzy sets defined in the output domain and crisp values defined in the same UoD.

The next two subsections analyze in depth the two main components of an FRBS, namely the KB and the processing structure, considering the case of a Mamdani FRBS.

2.3.1 Knowledge Base

The KB serves as the repository of the problem-specific knowledge —which models the relationship between input and output of the underlying system— upon which the inference process is triggered. A KB is based on two different information levels, i.e., the *semantic* level defined by a Data Base (DB) of LVs and the related fuzzy partitions, and the *symbolic* level made up of fuzzy conditional statements expressed in form of linguistic IF-THEN rules which constitute the Rule Base (RB). These levels are reported in Fig. 2.1 and are going to be discussed in the following sections.

2.3.1.1 Data Base: Linguistic Variables and Fuzzy Partitions

The basic elements of a DB in an FRBS are the LVs. An LV requires the specification of a symbol denoting the name of the LV. Usually the name of an LV coincides with the attribute name of a complex object (e.g., the attribute "Age" of an object "Person"). The LV can take values from a set of linguistic terms (also known as linguistic values), which are symbols that usually coincide with natural language terms. Often, adjectives or other linguistic qualifications are used as linguistic terms (e.g., OLD, ABOUT 70, NOT VERY YOUNG, etc.) and apply to a domain (e.g., the natural numbers from 0 to 120). In some cases, the set of available linguistic terms is complex; hence a generative grammar is specified, so as to provide for a computational machinery that is capable of enumerating all the linguistic terms, as well as to verify whether a term belongs to the admissible set or not. An LV maps its linguistic terms into corresponding fuzzy sets, thus endowing them with a semantics. This mapping is crucial for translating linguistic structures (like rules) into fuzzy information granules, which can be involved in approximate inference.

On the overall, an LV can be formalized as follows [35]:

$$\langle X, T, U, G, \mu \rangle \tag{2.7}$$

where

- X is the name of the variable (a symbol) modeled by the LV;
- T is a (usually finite) set of linguistic terms (symbols);
- U is the domain of applicability of the LV (i.e., a UoD in the way it has been defined in the previous section);
- G is a grammar that generates the symbols in T;
- $\mu : T \mapsto \mathscr{F}(U)$ is the semantic interpretation of linguistic terms. Given a term $A \in T$ its interpretation as a membership function will be denoted as μ_A.

With reference to the fuzzy set characterization introduced in Sect. 2.2.1, it can be argued how the formalization of an LV embeds both the symbolic representation of a fuzzy set (that is a symbol in T) and the semantic representation of a fuzzy set (that is the function μ which is conveniently adopted here to provide a mapping between terms and fuzzy sets). Without loss of generality, therefore, whenever ambiguity is avoided we will refer to fuzzy sets by using the associated term A or the membership function μ_A interchangeably.

The set of all semantic interpretations of linguistic terms gives rise to a *fuzzy partition* (a.k.a. frame of cognition [26]), which provides a granular view of the domain U:

$$\mathscr{P}(U) = \{\mu_A \in \mathscr{F}(U) \, | A \in T\} \tag{2.8}$$

Figure 2.2 shows an example of a fuzzy partition comprising seven triangular-shaped fuzzy membership functions. A special class of fuzzy partitions are Ruspini partitions, in which the membership degrees always sum up to one for each element of the UoD [29]. Formally:

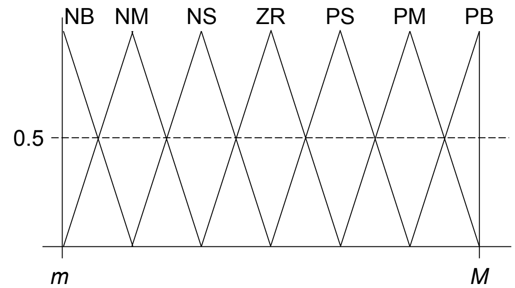


Fig. 2.2: Example of a (strong) fuzzy partition.

$$\forall x \in U : \sum_{A \in T} \mu_A(x) = 1 \tag{2.9}$$

This concept initially defined in the framework of clustering problems, and somehow related to the additivity property of probabilities, has been latter adapted by explicitly imposing some additional conditions (that could be considered as implicit in Ruspini's definition), being now usually referred to as Strong Fuzzy Partitions (SFPs) (see Sect. 3.3.5 in Chap. [1] for a more complete description and discussion about SFPs).

The adoption of SFPs is quite common in fuzzy modeling (especially in fuzzy clustering) [27] because of a number of convenient properties, including interpretability. An SFP assures that the membership of an element is distributed among all fuzzy sets of an LV. In Fig. 2.2, a fuzzy partition is represented in form of SFP.

It is important to observe that the variable X actually takes values in U. An LV on X specifies imprecise information about the actual value of X, whereas such imprecise (i.e., granular) information is represented by the fuzzy sets labeled with the linguistic terms of T. In order words, an LV enables the representation of information as a *restriction* on the values a variable can assume [41].

Restrictions operated by LVs are formalized through *soft constraints* [39]. Soft constraints define a general formalization that includes constraints of different semantics; for the purposes of interpretability analysis, however, *possibilistic* constraints will suffice. Soft constraints of possibilistic type are represented as formal expressions that resemble logical propositions. Given an LV $\langle X, T, U, G, \mu \rangle$, an atomic soft constraint is represented as:

$$X \text{ is } A \tag{2.10}$$

where $A \in T$. The soft constraint described by expression (2.10) is intended as a possibilistic constraint in the sense that it defines a possibility distribution for the value of the variable X on U that corresponds to the fuzzy set μ_A. In essence, (2.10) is a way to express that X can assume only one value in A, i.e., for each $x \in U$ the value $\mu_A(x)$ is interpreted as a *degree of possibility* that x can be the actual value of X. For example, if $\mu_A(x) = 0$, then it is impossible (or inadmissible) that $X = x$; on the other hand, if $\mu_A(x) = 1$, then it is fully possible (or admissible) that $X = x$. Possibility (or admissibility) is a matter of degree[3], which is specified by $\mu_A(x)$. (More detailed information on Possibility Theory can be found in specialized literature [9, 40].)

Given a soft constraint as in (2.10) and a fuzzy set B, we can calculate the degree of satisfaction of the constraint "X is A" if the variable X assumes a linguistic value B. This degree of satisfaction can be evaluated through the possibility measure defined as:

$$\mathrm{Poss}(A,B) = \sup_{x\in U} \min\{\mu_A(x), \mu_B(x)\} \tag{2.11}$$

When B is a singleton fuzzy set, i.e. $B = \{x\}$, then (2.11) reduces to the well-known formula:

$$\mathrm{Poss}(A,B) = \mathrm{Poss}(A,\{x\}) = \mu_A(x) \tag{2.12}$$

More complex soft constraints can be composed by conjunction, disjunction or negation of several atomic soft constraints, possibly defined in different UoDs. Complex soft constraints are also called *linguistic propositions*.In the most common case, conjunctive soft constraints are used, i.e. forms of the type:

$$X_1 \text{ is } A_{X_1} \text{ and } X_2 \text{ is } A_{X_2} \text{ and } \dots \text{ and } X_n \text{ is } A_{X_n} \tag{2.13}$$

The interpretation of such type of complex soft constraint depends on the interpretation of the atomic soft constraints and on the interpretation of the conjunction (and). Usually, a t-norm $\otimes$ is associated to the conjunction (such as the minimum or the product function). Being $\mathbf{X} = (X_1, \dots, X_n)$, we can consider $\mathbf{B} = B_1 \times, \dots, \times B_n$: the evaluation of the possibility that each LV $X_1, \dots, X_n$ may assume the value identified by the terms $B_1, \dots, B_n$ is given by:[4]

$$\mathrm{Poss}(\mathbf{A},\mathbf{B}) = \mathrm{Poss}(A_{X_1}, B_1) \otimes \cdots \otimes \mathrm{Poss}(A_{X_n}, B_n) \tag{2.14}$$

where $\mathbf{A} = A_{X_1} \times \cdots \times A_{X_n}$. In case the terms are identified by fuzzy singletons, i.e. $\mathbf{B} = \{\mathbf{x}\} = \{(x_1, \dots, x_n)\}$, (2.14) becomes:

$$\mathrm{Poss}(\mathbf{A},\{\mathbf{x}\}) = \mu_{A_{X_1}}(x_1) \otimes \cdots \otimes \mu_{A_{X_n}}(x_n) \tag{2.15}$$

Equation (2.15) coincides with the membership function of an information granule defined as Cartesian product of one-dimensional fuzzy sets. Information granules of this type can be employed in fuzzy rules to define pieces of knowledge that linguistically describe some relationship among variables.

[3] For example, the possibility degree that $X = x$ can be related to the similarity of x with respect to a known prototype, e.g. the similarity of a face impressed on a picture w.r.t. a known person.

[4] This evaluation holds as long as the involved LVs are *non-interactive*. For a more general discussion of this issue the reader is addressed to [35, 36, 37].

2.3.1.2 Rule Base

The usual way to organize knowledge in natural language is to express IF-THEN rules of the type:

IF antecedent THEN consequent

which allow to infer facts matching the consequent when some facts matching the antecedent are provided. The biggest power of FRBSs is to provide inference even in the case of partial matching between facts and rules.

Mamdani FRBSs employ a particular kind of IF-THEN rules, whose structure involves the use of LVs, with the antecedent and consequent being expressed by atomic and/or complex soft constraints (linguistic propositions) [35, 36, 37]. Hence, when dealing with Multiple Inputs Single Output (MISO) systems, linguistic rules expressed in the following form:

$$\text{IF } X_1 \text{ is } A_{X_1} \text{ and } \dots \text{ and } X_n \text{ is } A_{X_n} \text{ THEN } Y \text{ is } A_Y \tag{2.16}$$

are considered, with X_i and Y being the names of input and output LVs, respectively. A_{X_i}, A_Y are the linguistic terms involved in the atomic soft constraints associated to those variables. These linguistic terms, therefore, stand as the names of the fuzzy sets identified by the membership functions $\mu_{A_{X_i}}, \mu_{A_Y}$.

Since $\mathbf{U_X} = U_{X_1} \times \cdots \times U_{X_n}$, any given rule R of the form (2.16) is interpreted as $\mu_R : \mathbf{U_X} \times U_Y \to [0,1]$ with

$$\mu_R(\mathbf{x}, y) = \mu_{A_{X_1}}(x_1) \otimes \cdots \otimes \mu_{A_{X_n}}(x_n) \to \mu_{A_Y}(y) \tag{2.17}$$

As concerning the implementation of the "$\to$" operator, it depends on the specific approach followed to intend the fuzzy implication expressed by the IF-THEN statement. Two main interpretations are considered in literature: *implicative* and *conjunctive* [10]. In implicative systems, antecedents are combined with consequents by genuine fuzzy implications which can be different in nature, such as s-implications $A_X \to A_Y \equiv \neg A_X \vee A_Y$. In conjunctive systems, antecedents are combined with consequents by a fuzzy conjunction since implication $A_X \to A_Y$ is interpreted as a "case" where inputs take values in A_X and the output takes values in A_Y. In this case, (2.17) can be rewritten as:

$$\mu_R(\mathbf{x}, y) = \mu_{A_{X_1}}(x_1) \otimes \cdots \otimes \mu_{A_{X_n}}(x_n) \otimes \mu_{A_Y}(y) \tag{2.18}$$

It is worth noting that the RB can be represented in different ways. The usual one is the list of rules, although a decision table (also called rule matrix) becomes an equivalent and more compact representation for the same set of linguistic rules when only a few input variables (usually one or two) are considered by the FRBS. As an example, we may consider an FRBS where two input variables (X_1 and X_2) and a single output variable (Y) are involved. Being $\{\text{SMALL}, \text{MEDIUM}, \text{LARGE}\}$, $\{\text{SHORT}, \text{MEDIUM}, \text{LONG}\}$ and $\{\text{BAD}, \text{MEDIUM}, \text{GOOD}\}$ the corresponding associated term sets, the following RB:

Table 2.2: Example of a decision table.

		X_1 SMALL	MEDIUM	LARGE
X_2	SHORT	BAD	MEDIUM	
	MEDIUM	BAD		MEDIUM
	LONG			GOOD

R_1 : IF X_1 is SMALL and X_2 is SHORT THEN Y is BAD
R_2 : IF X_1 is SMALL and X_2 is MEDIUM THEN Y is BAD
R_3 : IF X_1 is MEDIUM and X_2 is SHORT THEN Y is MEDIUM
R_4 : IF X_1 is LARGE and X_2 is MEDIUM THEN Y is MEDIUM
R_5 : IF X_1 is LARGE and X_2 is LONG THEN Y is GOOD

is also represented by the decision table shown in Table 2.2. This kind of rules is the most commonly adopted in fuzzy modeling literature; however other alternatives are possible [10], as we shall see in Sect. 2.4.

2.3.2 Processing Structure

The processing structure of a Mamdani FRBS is composed of the following three elements:

- a *fuzzification interface* that transforms the crisp input data into fuzzy values (representing the input of the fuzzy reasoning process);
- an *inference system* that infers from the fuzzy input to several resulting output fuzzy sets according to the information stored in the KB;
- a *defuzzification interface* that converts the resulting fuzzy sets into a crisp value (representing the output of the FRBS).

They are briefly introduced in the following sections.

2.3.2.1 The Fuzzification Interface

The LVs involved in a Mamdani FRBS may take the values specified by linguistic terms which can be either fuzzy sets or scalar values. In both cases, the input of the system must be intended as a collection of membership degrees: one for each involved linguistic term. Fuzzification takes place as described in Sect. 2.3.1.1: being B a possible input value for an LV X defined on U, the soft constraint "X is A" can be evaluated according to (2.11) for each term $A \in T$ occurring in at least one rule.

Whenever B is a singleton (i.e., the input is a scalar value x_0, which is the most common case), the soft constraint is simply reduced to $\mu_A(x_0)$.

2.3.2.2 The Inference System

The inference system is the component that derives the fuzzy outputs from the inputs according to the relations defined through fuzzy rules. In Sect. 2.2.4 we introduced the GMP as a convenient way to match a fact and a relation to infer a new fact. Such a machinery is specially suitable to model human reasoning since it is quite common in our everyday practice to apply *modus ponens* in an approximate manner. In this sense, the inference process triggered by an FRBS aims at replicating the human reasoning process, thus managing fuzzy rules through a kind of approximate reasoning. Particularly, the fuzzy inference scheme employs the GMP:

$$\frac{X \text{ is } A'_X, \quad \text{IF } X \text{ is } A_X \text{ THEN } Y \text{ is } A_Y}{Y \text{ is } A'_Y} \tag{2.19}$$

The expression " IF X is A_X THEN Y is A_Y", which is a rule described in terms of a fuzzy conditional statement, corresponds to a fuzzy relation between the LVs $X$and Y involved in the rule. This fuzzy relation R corresponds to a fuzzy set μ_R on $U_X \times U_Y$, and the inferred result can be expressed through the compositional rule of inference (2.6):

$$\mu_{A'_Y}(y) = \mu_{A'_X \circ R}(y) = \sup_{x \in U_X} \min \left\{ \mu_{A'_X}(x), \mu_R(x,y) \right\} \tag{2.20}$$

By recalling the definition of interpretation of a rule μ_R as in (2.18), whenever the t-norm involved in the interpretation is the min function, (2.20) can be translated through an algebraic manipulation into the following form:

$$\mu_{A'_Y}(y) = \min \left\{ \sup_{x \in U_X} \min \left\{ \mu_{A'_X}(x), \mu_{A_X}(x) \right\}, \mu_{A_Y}(y) \right\} \tag{2.21}$$

When A'_X is a singleton, i.e. $A'_X = \{x_0\}$, Eq. (2.21) can be simplified as:

$$\mu_{A'_Y}(y) = \min \{ \mu_{A_X}(x_0), \mu_{A_Y}(y) \}$$

When a rule is composed by several soft constraints as in (2.13), we resort to (2.14) to compute the inference. In the case of singleton inputs, (2.21) can be written as:

$$\mu_{A'_Y}(y) = \min \{ h, \mu_{A_Y}(y) \}$$

being

$$h = \mu_{A_{X_1}}(x_1) \otimes \cdots \otimes \mu_{A_{X_n}}(x_n)$$

the *activation strength* of the rule.

When we turn to consider the case of several rules composing the KB of an FRBS, there is a need to define a method for producing a single inference result. This task can be accomplished with two different approaches: First Aggregate Then Infer (FATI) and First Infer Then Aggregate (FITA). Mamdani originally suggested the FATI approach in his first conception of FLCs [22]. However, later, the FITA approach has become more popular [7, 8, 31], in particular in real-time applications which demand a fast time response. These two inference mechanisms are individually sketched in the following, assuming that *m* rules are involved in the KB.

First Aggregate, Then Infer (FATI)

In this case, the inference process operates the following sequence of steps:

- aggregate the individual rules $\mu_{R^{(r)}}(x,y)$ into an overall fuzzy relation $\mu_R(x,y)$ by means of a fuzzy aggregation operator G:

$$\mu_R(x,y) = G\left\{\mu_{R^{(1)}}(x,y), \mu_{R^{(2)}}(x,y), \ldots, \mu_{R^{(m)}}(x,y)\right\}$$

- apply the compositional rule of inference on the resulting aggregated to generate the output fuzzy set A'_Y.

First Infer, Then Aggregate (FITA)

In this case, the inference process operates the following sequence of steps:

- apply the compositional rule of inference on each rule to generate m output fuzzy sets $A'^{(r)}_Y$, $r = 1,\ldots,m$;
- aggregate the individual fuzzy sets $A'^{(r)}_Y$ into an overall fuzzy set A'_Y by means of a fuzzy aggregation operator G (known as the *also* operator):

$$\mu_{A'_Y}(y) = G\left\{\mu_{A'^{(1)}_Y}(y), \mu_{A'^{(2)}_Y}(y), \ldots, \mu_{A'^{(m)}_Y}(y)\right\} \tag{2.22}$$

Usually, the aggregation operator G is implemented by a t-conorm (e.g., the max function).

2.3.2.3 The Defuzzification Interface

The defuzzification interface is useful to provide the final output of an FRBS as a crisp value. To this aim, a defuzzification method D is employed which transforms the output fuzzy set A'_Y into a crisp output value y_0:

$$y_0 = D(\mu_{A'_Y}(y)) \tag{2.23}$$

In order to implement (2.23), different operators may be adopted, such as the “Center of Gravity” (CG) or the “Mean of Maxima” (MOM), whose expressions are reported below:

- CG:

$$y_0 = \frac{\int_Y y \cdot \mu_{A'_Y}(y)dy}{\int_Y \mu_{A'_Y}(y)dy} \tag{2.24}$$

- MOM:

$$\begin{aligned} y_{inf} &= \inf\left\{z|\mu_{A'_Y}(z) = \sup \mu_{A'_Y}(y)\right\} \\ y_{sup} &= \sup\left\{z|\mu_{A'_Y}(z) = \sup \mu_{A'_Y}(y)\right\} \\ y_0 &= \frac{y_{inf} + y_{sup}}{2} \end{aligned} \tag{2.25}$$

2.4 Variants of Fuzzy Rule-based Systems

A Mamdani FRBS provides a natural framework to include expert knowledge in form of linguistic rules. This knowledge can be easily combined with rules which are automatically generated from datasets describing the relation between system input and output. In addition, this knowledge is highly interpretable. The fuzzy rules are composed of input and output LVs which take values from their term sets having a meaning (a semantics) associated to each linguistic label. Therefore, each rule is a description expressed as a premise-conclusion statement that exhibits a clear interpretation to humans —for this reason, these systems are usually called *linguistic* or *descriptive* Mamdani FRBSs. This property makes Mamdani FRBSs appropriate for applications where the emphasis lies on model interpretability, such as fuzzy control [8, 18, 19] and linguistic modeling [25, 31].

However, although Mamdani FRBSs possess several advantages, they also come with some drawbacks. One of the problems, especially in linguistic modeling applications, is their lack of accuracy in some complex problems, which is due to the structure of the linguistic rules. Such limitations are analyzed in [4] and [5] highlighting how the structure of the fuzzy linguistic IF-THEN rule is subject to certain restrictions because of the use of LVs:

- there is a lack of flexibility in the FRBS due to the rigid partitioning of the input and output spaces;
- when the input variables are mutually dependent, it becomes difficult to find a proper fuzzy partition of the input space;
- the homogeneous partition of the input and output space becomes inefficient and does not scale well as the dimensionality and the complexity of the input-output mapping increases;
- the size of the KB increases rapidly with the number of variables and linguistic terms in the system. In order to obtain an accurate FRBS, a fine level of granularity is needed, which requires additional linguistic terms. This increase in granularity causes the number of rules to grow, which complicates the interpretability of the system by a human. Moreover, in the vast majority of cases, it is possible to obtain an equivalent FRBS that achieves the same accuracy with a

fewer number of rules whose fuzzy sets are not restricted to a fixed input space partition.

Both variants of linguistic Mamdani FRBSs described in this section address the previous issues by making the linguistic rule structure more flexible.

2.4.1 Mamdani Fuzzy Rule-based Systems with Input/Output Scaling

A first attempt to introduce more flexibility in an FRBS is to transform inputs before the fuzzification process and outputs after defuzzification. The initial idea for this transformation was the use of scaling factors with a tuning purpose [28]. The simplest approach requires the use of a linear scaling function of the form:

$$f(z) = \lambda \cdot z + \nu \tag{2.26}$$

where the scaling factor λ enlarges or reduces the operating range, which in turn decreases or increases the sensitivity of the system in respect of that input variable, or the corresponding gain[5] in case of an output variable. The parameter ν shifts the operating range and plays the role of an offset to the corresponding variable.

Finally, it is possible to use more complex mappings generating *non-linear scaling functions*. A common non-linear scaling function is:

$$f(z) = \mathrm{sign}(z) \cdot |z|^{\alpha} \tag{2.27}$$

This non-linear scaling increases ($\alpha > 1$) or decreases ($\alpha < 1$) the relative sensitivity in the region closer to the origin and has the opposite effect when moving far from the origin.

2.4.2 DNF Mamdani Fuzzy Rule-based Systems

Disjunctive Normal Form (DNF) Mamdani FRBSs embed a different rule structure which has the following form [11, 12, 20, 21]:

$$\text{IF } X_1 \text{ is } \widetilde{A_{X_1}} \text{ and } \ldots \text{ and } X_n \text{ is } \widetilde{A_{X_n}} \text{ THEN } Y \text{ is } A_Y, \tag{2.28}$$

where each input variable X_i takes as its value a set of linguistic terms $\widetilde{A_{X_i}}$ whose members are joined by a disjunctive operator, whilst the output variable is still an LV associated to a single term. Thus, each soft constraint in (2.28) should be interpreted as

[5] The gain is a proportional value that shows the relationship between the magnitude of the input and the magnitude of the output signal in a system.

$$X_i \text{ is } \widetilde{A_{X_i}} \equiv X_i \text{ is } A_{i,1} \text{ or } \dots \text{ or } X_i \text{ is } A_{i,l_i}$$

being l_i the total number of terms included in each set $\widetilde{A_{X_i}}$ $(i = 1, \dots, n)$ including the terms connected by disjunction. For convenience, each set $\widetilde{A_{X_i}}$ is represented as

$$\widetilde{A_{X_i}} = \{A_{i,1} \text{ or } \dots \text{ or } A_{i,l_i}\}$$

An example of this kind of rule is shown as follows. Let X_1, X_2, X_3, and Y be LVs and let the following sets be their associated linguistic terms:

$$T_1 = \{\mathrm{A}_{11}, \mathrm{A}_{12}, \mathrm{A}_{13}\} \qquad T_2 = \{\mathrm{A}_{21}, \mathrm{A}_{22}, \mathrm{A}_{23}, \mathrm{A}_{24}, \mathrm{A}_{25}\}$$
$$T_3 = \{\mathrm{A}_{31}, \mathrm{A}_{32}\} \qquad T_4 = \{\mathrm{A}_{41}, \mathrm{A}_{42}, \mathrm{A}_{43}\}$$

then it is possible to write a DNF rule such as

$$\text{IF } \mathrm{X}_1 \text{ is } \{\mathrm{A}_{11} \text{ or } \mathrm{A}_{12}\} \text{ and } \mathrm{X}_2 \text{ is } \{\mathrm{A}_{24} \text{ or } \mathrm{A}_{25}\} \text{ and } \mathrm{X}_3 \text{ is } \{\mathrm{A}_{31} \text{ or } \mathrm{A}_{32}\}$$
$$\text{THEN Y is } \mathrm{A}_{42}$$

The main advantage of this rule structure comes from the possibility to integrate in a single expression (a single DNF rule) the information corresponding to several elemental rules (i.e., the rules commonly used in Mamdani FRBSs). This enables a certain level of compression of the rule base, which is quite helpful when the number of input variables increases.

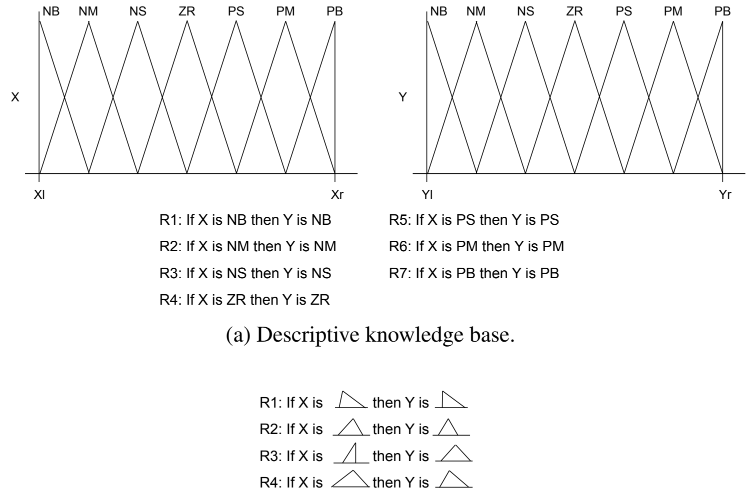
2.4.3 Approximate Mamdani Fuzzy Rule-based Systems

The DNF fuzzy rule structure does not implicate some relevant loss in the interpretability of the resulting Mamdani FRBS. The systems known as approximate Mamdani FRBSs [2, 5, 6, 15], instead, are able to achieve a better accuracy which is paid off in terms of reduced interpretability.

The structure of an approximate FRBS is similar to the one adopted in a descriptive system (shown in Fig. 2.1). The main difference is related to the rules which do not refer in their definition to predefined fuzzy partitions of the LVs. Each rule in an approximate FRBS defines its own fuzzy sets instead of using a linguistic label pointing to a particular fuzzy set of the partition of the underlying LV. Thus, an approximate fuzzy rule has the following form:

$$\text{Rule } r: \text{ IF } X_1 \text{ is } A_{X_1}^{(r)} \text{ and } \dots \text{ and } X_n \text{ is } A_{X_n}^{(r)} \text{ THEN } Y \text{ is } A_Y^{(r)}$$

where the input variables X_i and the output variable Y are special degenerate LVs where linguistic terms are the mathematical expressions of their corresponding membership functions. A simple example of rule for such type of system is:



(a) Descriptive knowledge base.

(b) Approximate fuzzy rule base.

Fig. 2.3: Comparison between a descriptive knowledge base (a) and an approximate fuzzy rule base (b).

$$
\begin{aligned}
&\text{Rule } r: \text{ IF X}_1 \text{ is } \mathtt{exp}((X_1 - c_1^{(r)})/\sigma_1^{2(r)}) \text{ and } \ldots \\
&\ldots \text{ and X}_n \text{ is } \mathtt{exp}((X_n - c_n^{(r)})/\sigma_n^{2(r)}) \text{ THEN Y is } \mathtt{exp}((Y - c_Y^{(r)})/\sigma_Y^{2(r)})
\end{aligned}
\tag{2.29}
$$

In other words, rules of approximate nature are *semantic free* whereas descriptive rules operate in the context formulated by means of the linguistic semantics. Therefore, approximate FRBSs contain no DB that defines a semantic context in the form of LVs and linguistic terms. The former separate entities DB and RB merge into a fuzzy rule base in which each rule subsumes the definition of its underlying input and output fuzzy sets as shown in Fig. 2.3.

Approximate FRBSs provide some specific advantages over linguistic FRBSs and are particularly suitable for certain types of applications [5]:

- each rule employs its own distinct fuzzy sets resulting in additional degrees of freedom and increase in expressiveness;
- the number of rules can be adapted to the complexity of the problem: simple input-output relationships are modeled with a few rules, but still more rules can be added as the complexity of the problem increases. Therefore, approximate FRBSs constitute a potential remedy to the curse of dimensionality that emerges when scaling to multi-dimensional systems.

These properties allow approximate FRBSs to achieve a better degree of accuracy than linguistic FRBSs in complex problem domains. However, their main drawback compared to the descriptive version is the degradation in terms of interpretability of the fuzzy rule base as the fuzzy variables no longer share a unique linguistic interpretation. However, unlike other kinds of approximate models storing knowledge implicitly (such as neural networks), approximate FRBSs still express knowledge in an explicit way, since the system behavior is described by local rules. As an additional drawback, the capability to approximate a set of training data accurately can lead to overfitting and therefore to a poor generalization capability to cope with previously unseen input data.

2.4.4 Takagi–Sugeno–Kang Fuzzy Rule-based Systems

Takagi, Sugeno and Kang proposed a different model based on rules whose antecedent is composed of LVs and the consequent is represented by a function of the input variables [30, 32]. This model is commonly known as Takagi–Sugeno–Kang (TSK) FRBS and is widely used in many engineering applications, especially in fuzzy control [24, 32].

In the most usual form, TSK FRBSs use a polynomial function to define the consequent, where the order of the polynomial provides the order of the TSK FRBS. As an example, 1^{st} order TSK FRBSs express rules as follows:

$$\text{IF } X_1 \text{ is } A_{X_1} \text{ and } \dots \text{ and } X_n \text{ is } A_{X_n} \text{ THEN } Y = p_0 + p_1 \cdot X_1 + \cdots + p_n \cdot X_n$$

where X_i are the input variables, Y is the output variable, and $\mathbf{p} = (p_0, p_1, \dots, p_n)$ is a vector of real parameters. Also, 0^{th} order TSK FRBSs are common, with rules expressed in the form:

$$\text{IF } X_1 \text{ is } A_{X_1} \text{ and } \dots \text{ and } X_n \text{ is } A_{X_n} \text{ THEN } Y = p_0$$

As concerning the antecedents, they can be represented as those considered in either approximate or linguistic Mamdani FRBSs, depending on the relative importance of accuracy with respect to interpretability.

TSK FRBSs are preferred when the efficiency of inference is of utmost importance (e.g., in embedded systems). In fact, when dealing with a KB composed by m rules, the output of a TSK FRBS can be obtained as the weighted average involving the individual outputs provided by each rule, $Y^{(r)}$, $r = 1, \dots, m$, and the single values of rule activation strength $h^{(r)}$:

$$\frac{\sum_{r=1}^{m} h^{(r)} \cdot Y^{(r)}}{\sum_{r=1}^{m} h^{(r)}} \tag{2.30}$$

As a consequence of their design, TSK systems do not need defuzzification, being their outputs real numbers.

This type of FRBS splits the input space into several fuzzy sub-spaces and defines a polynomial[6] input-output relationship in each one of these sub-spaces [32]. In the inference process, these partial relationships are combined as described in (2.30) to obtain the global input-output relationship, taking into account the dominance of the partial relationships in their respective areas of application and the conflicts emerging in the overlapping zones.

These systems have been successfully applied to a large variety of practical problems. Their main advantage consists in the presence of compact system equations allowing the estimation of the parameters p_i by means of classical methods, thus facilitating the design process. However, the main drawback associated to TSK FRBSs is the form of the rule consequents, which does not provide a natural framework for representing expert linguistic knowledge. Nevertheless, in the case of 0^{th}-order TSK FRBSs, the parameter p_0 can be interpreted as the modal point of a linguistic term of the output LV. This stands as a compromise between full linguistic (yet not efficient) FRBSs and efficient (but not interpretable) FRBSs.

2.4.5 MIMO Fuzzy Rule-based Systems

In the previous sections, we have assumed the presence of a single output variable in the definition of an FRBS. However, this is not the case in some problems, where two or more output variables are involved. In this case, the systems are called Multiple Inputs Multiple Outputs (MIMO) FRBSs and the rules take the form:

$$\text{IF } X_1 \text{ is } A_{X_1} \text{ and } \ldots \text{ and } X_n \text{ is } A_{X_n} \text{ THEN } Y_1 \text{ is } A_{Y_1} \text{ and } \ldots \text{ and } Y_m \text{ is } A_{Y_m}$$

where output linguistic terms are replaced by polynomials in the case of TSK FRBSs.

Inference in MIMO FRBSs acts by determining the value for each output variable independently. Thus, an MIMO FRBS is a compact representation of several MISO FRBSs, with the additional advantage that the antecedents of the rules are shared among these systems. However, the representation of the consequents in MIMO FRBSs does not offer an immediate understanding, because the meaning of the corresponding soft constraints does not follow the same semantics of the antecedents as defined in (2.13).

2.4.6 Fuzzy Rule-based Classifiers

A Fuzzy Rule-Based Classifier (FRBC) is a classification system that uses fuzzy rules as knowledge representation tools. From a functional point of view, while a traditional FRBS can be seen as a mapping from a (usually multi-dimensional) nu-

[6] In many cases, 1^{st} order TSK FRBSs are used, therefore the input-output relation is linear.

merical input domain to a (usually one-dimensional) numerical output domain, an FRBC maps a numerical domain into a symbolic domain of class labels. Therefore, the fuzzy classification rule structure is as follows:

$$\text{IF } X_1 \text{ is } A_{X_1} \text{ and } \dots \text{ and } X_n \text{ is } A_{X_n} \text{ THEN } Y \text{ is } c$$

with Y being a categorical variable taking values in a set of class labels C.

In its simplest approach the *winner takes all* mechanism is used to perform inference: given an input $\mathbf{x}$, inference takes place by considering the rule with maximum activation strength; then, the corresponding class label is returned. If two or more fuzzy rules are fired with the same activation strength, either one of the corresponding class labels is randomly selected or the classification is not performed [17].

A more sophisticated processing scheme for an FRBC returns the possibility degree of each class label for a given input. Namely, given an input $\mathbf{x}$ and a class label $c \in C$, the maximum firing strength $h(c)$ is computed from all the rules with c as class label in the consequent part. Then, the set of tuples

$$\{(c, h(c)) \,|\, c \in C\}$$

is returned. This type of FRBC enables a more sophisticated reasoning, for example by allowing the classifier to refuse to classify an input when there is not enough evidence of the most appropriate class label (e.g., when the maximum value of $h(c)$ is below a threshold, or if there is not enough difference between the top two values of possibility).

2.5 Final Remarks

Since their first inception by Mamdani in 1975 [23], FRBSs have been used in almost every field where numerical data can be exploited to represent the perceptual nature of linguistic terms in a computational way. That can be properly accomplished through the employment of fuzzy sets. Also, Mamdani FRBSs constituted the subject of several theoretical studies, eventually leading to a number of variant versions that enhance the original proposal in terms of efficiency (such as TSK FRBSs), applicability (such as FRBCs), and so on. Furthermore, many machine learning schema have been developed, in order to automatically design FRBSs from data.

The possibility of designing FRBSs from data opens the door to their application in domains that were unconceivable when such systems had to be manually designed from scratch. Nevertheless, with automatic design procedures, the original intention of developing systems pursuing good accuracy and acceptable interpretability is hampered. This is the reason why the research on interpretability of FRBSs flourished, with the objective of developing criteria, metrics and design procedures for acquiring FRBSs from data that offer an acceptable balance between accuracy and interpretability. They are conflicting qualities which altogether provide FRBSs with an added value over the plethora of models that can be used for knowledge extraction from data. The following chapters are devoted to give a systematic description and a critical discussion of all these dimensions of interpretability in FRBSs.

Current research on FRBSs goes in several directions. Apart from the multitude of applications where FRBSs are increasingly applied, theoretical research explores more performing learning schemes and more flexible representations for FRBSs, including those exploiting type-2 fuzzy sets and variants thereof, or hierarchical representations of rules with the intended objective of endowing FRBSs with a structure that is capable of organizing complex rules in a hierarchy of simpler representations. These are new developments that, while being promising at this stage of research, are currently neither yet mature nor stabilized from theoretical and methodological viewpoints. Therefore, we choose to provide a more in-depth analysis of well-established methods and models with an outlook of these new proposals.

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Chapter 3
Interpretability Constraints and Criteria for Fuzzy Systems

Abstract Fuzzy systems are commonly considered suitable tools to express knowledge in a human comprehensible fashion. This kind of characterization makes them eligible for being applied in several contexts where interpretability is a major issue and humans may profit from a self-explanatory form of automatic computation. However, fuzzy systems are not interpretable *per se* and the simple adoption of a natural language representation does not guarantee full acceptance and plain confidence among human experts. A number of constraints and criteria have been proposed in literature to drive the design and the construction of fuzzy systems so that they can be deemed *interpretable*. The aim of this chapter is to provide a thorough exposition of constraints and criteria which have been variously adopted in the research community in this context of investigation. Due to their heterogeneity and multiplicity, we resorted to a particular arrangement of their presentation. By following a hierarchical organization, we start from the basic constituents of a fuzzy system (namely, the fuzzy sets) and we go through the other design levels where compound elements are involved: for each level, an exhaustive review of interpretability constraints and criteria is expounded supplying formal definitions, illustrative examples, and bibliographical references.

3.1 Introduction

Interpretability must be characterized in some computational way in order to be effectively used for designing interpretable Fuzzy Rule-Based Systems (FRBSs). A common approach is to define a number of interpretability constraints and criteria that must be imposed on the design process. Furthermore, in many cases these constraints and criteria can be quantified in order to assess the interpretability of an FRBS.

In this chapter we focus on the presentation of common interpretability constraints and criteria. By interpretability *constraints* we denote mathematically defined properties, which can be either crisp or fuzzy in their nature. Fuzzy interpretability constraints are common as they reflect the blur nuance of the interpretability concept. On the other hand, interpretability *criteria* are more generic rules that should be followed in order to achieve interpretability.[1] Also, when criteria are discussed some assessment metrics can be defined in order to quantify how much fuzzy systems are able to satisfy such criteria. Interpretability assessment is the main subject of Chap. [3].

It must be remarked that there is not a general consensus on a set of interpretability constraints and criteria that are *required* to achieve interpretability. More properly, many interpretability constraints and criteria are just design options driven by experience or specific requirements. Furthermore, some of them are applicable to specific types of FRBSs (e.g., only to Takagi-Sugeno-Kang systems or TSK systems in short).

There are several interpretability constraints and criteria for fuzzy systems. Here, they are organized according to the following hierarchy of components defining an FRBS:

- (one-dimensional) fuzzy sets;
- fuzzy partitions;
- multi-dimensional fuzzy information granules;
- fuzzy rules;
- fuzzy models;
- fuzzy model adaptation.

3.2 Constraints and Criteria for Fuzzy Sets

The first level of interpretability analysis concerns each fuzzy set involved in the knowledge base of a fuzzy system. These fuzzy sets define the semantic interpretation of the linguistic terms of each linguistic variable (LV) used in an FRBS (see Chap. [1]). Such LVs designate perception-based concepts, which must be highly cointensive with some elementary concepts conceived by a human mind. In order to guarantee a high semantic cointension, each fuzzy set must be shaped so as to satisfy some requirements, mostly justified by common sense.

3.2.1 Normality

Constraint 1 *A fuzzy set A ought to be* normal*, i.e. there exists at least one element with full membership:*

$$[A]_1 \neq \emptyset \tag{3.1}$$

[1] For ease of referencing, both constraints and criteria will be defined as “constraints” when they are introduced.

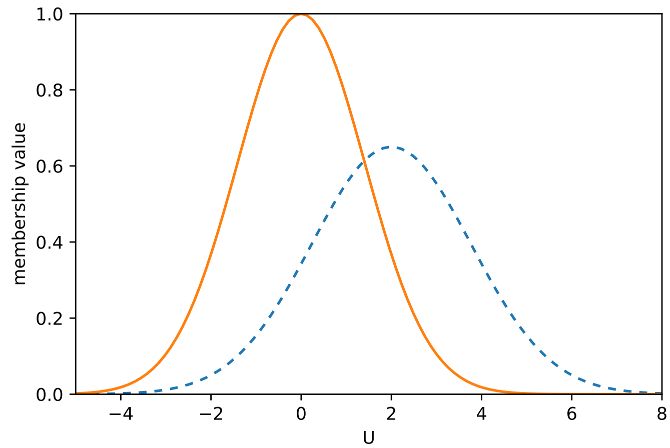

Fig. 3.1: Example of a normal fuzzy set (solid) and a sub-normal fuzzy set (dashed). If the two fuzzy sets represent qualities, then the sub-normal fuzzy set expresses a quality that is never completely met by any element of the domain.

A fuzzy set is sub-normal if it is not normal. In Fig. 3.1, normality and sub-normality are graphically depicted for two fuzzy sets. The normality requirement is very frequent. Indeed, it is implicitly assumed in almost all the literature concerning interpretable fuzzy modeling, while only few authors require it explicitly [13, 35, 62, 66].

Although there are attempts to provide for linguistic interpretations of sub-normal fuzzy sets [40], actually normality is almost always required: it implies that at least one element of the Universe of Discourse (UoD) is going to exhibit full matching with the concept semantically represented by the fuzzy set [16, 59, 72].

Sub-normality is usually avoided also because it is an indication of inconsistency of a fuzzy set [58]. Indeed, if a fuzzy set A is sub-normal, then some inclusion indicators can report a non-zero inclusion degree of A in the empty set $\emptyset$. As an example, a common inclusion indicator is [67]:

$$\mathscr{I}^{*}(A,B) = \inf_{x\in U} \min\left[1, 1-\mu_A(x)+\mu_B(x)\right] \tag{3.2}$$

If $[A]_1 = \emptyset$ and[2] $\mathrm{hgt}(A) < 1$ then

$$\mathscr{I}^{*}(A,\emptyset) = \inf_{x\in U}\left[1-\mu_A(x)\right] = 1-\mathrm{hgt}(A) > 0$$

[2] It is possible that $[A]_1 = \emptyset$ and $\mathrm{hgt}(A) = 1$ if the membership function μ_A is asymptotically approaching 1; nevertheless, this is quite a rare case in fuzzy modeling.

Normality is fundamental because the fuzzy sets involved in the knowledge base of an FRBS are semantic interpretations of linguistic terms that designate elementary perception-based concepts. It is expected that such elementary concepts are well-defined, i.e. they do not carry inconsistencies. As an example, every adult human being has a clear idea of what "young" means in the domain of human ages. In particular, it is certain that at least one age can be clearly identified as a full member of the "young" concept. This requirement is translated into the normality constraint.

On the other hand, a concept for which no object can belong to with maximal membership is not fully consistent. Complex concepts can be inconsistent if they are the result of some inference processes based on imprecise facts and knowledge. Humans naturally deal with partial inconsistency (which may result in some kind of indecision or hesitancy), therefore fuzzy systems should be able to deal with inconsistent concepts too. As a matter of fact, the fuzzy sets resulting from an inference process are usually sub-normal. This is a consequence of reasoning with imprecise knowledge and data; thus partially inconsistent fuzzy sets are perfectly legitimate and can be used to make decisions. The degree of inconsistency (as measured, e.g., by Eq. 3.2) can be used to quantify the degree of hesitancy in making such decisions.

3.2.2 Continuity

Constraint 2 *A fuzzy set A modeling a perception-based concept ought to be* continuous*, i.e. its membership function μ_A should be continuous in the UoD.*

This constraint is motivated by the common assumption that fuzzy sets are models of perception-based concepts, which are conceived through the observation of physical attributes. Since humans perceive the environment as continuously varying[3], it is expected that the membership functions used to interpret linguistic terms vary in continuity within the UoD.

If a fuzzy set A is not defined to represent a perception-based concept, then continuity is not a requirement. This is the case of mathematical or legal concepts, which usually have a crisp nature (e.g., "legal age"). These concepts are effectively modeled by classical sets, which do not have a continuous membership function. The use of these sets along with fuzzy sets does not compromise interpretability.

Actually, continuity is often met when modeling interpretable fuzzy systems, but it is rarely explicated in literature. However, since fuzzy set theory does not guarantee continuity of fuzzy sets *ipso facto*, this constraint should be included in interpretability analysis.

[3] *Natura non facit saltus* (Carl von Linné, Philosophia Botanica, 1751).

3.2.3 Convexity

Constraint 3 *A fuzzy set A ought to be* convex,[4] *i.e. the membership values of elements belonging to any interval are not lower than the membership values at the interval's boundaries:*

$$\forall a,b,x \in U : a \leq x \leq b \rightarrow \mu_A(x) \geq \min\{\mu_A(a), \mu_A(b)\} \tag{3.3}$$

Alternatively, α-cuts *can be used to define convexity; in particular, the* α-cuts *of convex one-dimensional fuzzy sets are always intervals:*

$$\forall \alpha \in [0,1], \exists m_\alpha, M_\alpha \in U : [A]_\alpha = [m_\alpha, M_\alpha] \tag{3.4}$$

Roughly speaking, a convex fuzzy set represents a concept that, if applicable to two elements to a certain degree, then it is applicable to all elements between the two to that degree at least. In this way, the concept represented by the fuzzy set can be conceived as *elementary*, i.e. it is related to a single specific property of a perceived object (see Fig. 3.2).

More precisely, a convex fuzzy set implies the existence of a similarity relation centered on a specific set of elements, called *prototypes* (which constitute the core of the fuzzy set), such that the similarity of elements never increases as their distance from the prototypes increases. Put in other words, the UoD represents an elementary feature: a concept is defined in terms of similarity with respect to a prototype (or a set thereof) so that the smaller the distance between an element and the prototype, the higher its similarity.

Non-convex fuzzy sets can be used for modeling complex concepts, e.g. "MEAL TIME", which may have three peaks around 8:00 am, 1:00 pm and 8:00 pm, and low membership outside these peaks [20]. However, it is easy to observe that such fuzzy sets could be easily redefined as a union of elementary fuzzy sets. Continuing the previous example, we could redefine "MEAL TIME" as union of the elementary (convex) fuzzy sets "BREAKFAST TIME", "LUNCH TIME" and "DINNER TIME". Fig. 3.3 depicts a possible configuration of the fuzzy sets involved in the example.

Convexity is a defining property of fuzzy numbers, fuzzy intervals and fuzzy vectors[5]. A fuzzy number represents an imprecise quantity labeled, e.g., with "ABOUT x" where x is a rational number. Convexity in fuzzy numbers is required to represent the decrease of similarity of a number to the prototype of the fuzzy number as long as their difference increases. In addition, convexity enables to easily extend arithmetic operations on fuzzy numbers without excessive computational burden by using interval arithmetic [39]. The same can be said for fuzzy intervals, although in this case an interval is characterized by a core of prototypes instead of a single element.

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[4] It should be remarked that the notion of convexity of fuzzy sets is different from the notion of convexity of functions. Actually, convex fuzzy sets are often characterized by membership functions that are not convex.

[5] A fuzzy vector is a vector of fuzzy numbers.

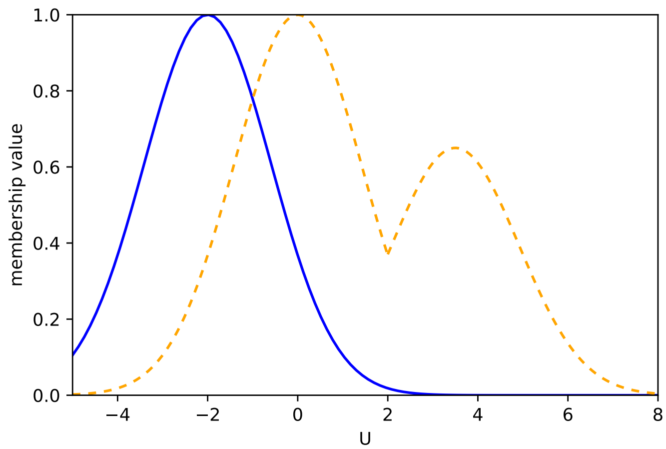

Fig. 3.2: A convex fuzzy set (solid) and a non-convex fuzzy set (dashed). It is difficult to label the non-convex fuzzy set with a linguistic term.

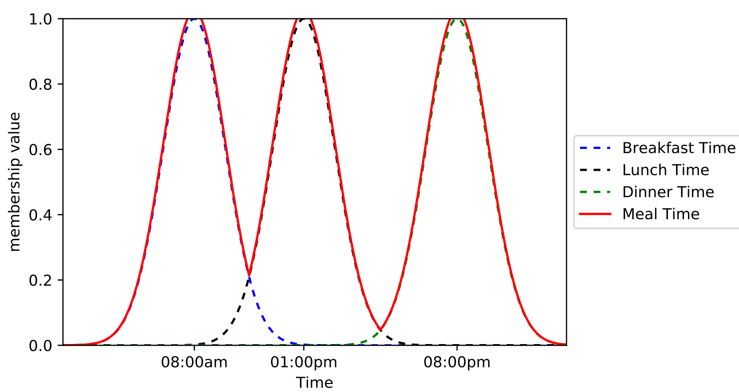

Fig. 3.3: A non-convex fuzzy set (MEAL TIME) defined as the union of three convex fuzzy sets (BREAKFAST TIME, LUNCH TIME, and DINNER TIME).

Notwithstanding its importance in modeling interpretable fuzzy systems, some well-known fuzzy clustering procedures do not yield convex fuzzy sets. As an example, in Fig. 3.4 a typical membership function is shown, deriving from the application of the well-known Fuzzy C-Means (FCM) algorithm [7] to one-dimensional data. To avoid interpretability loss due to non-convex fuzzy sets derived from such clustering algorithms, a technique to approximate the derived clusters with convex fuzzy sets can be applied. As an example, the clustering process can be followed

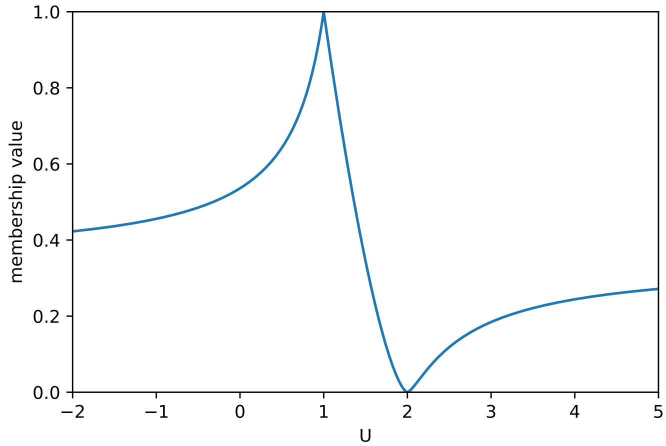

Fig. 3.4: An example of one-dimensional membership function that can be derived by the application of FCM algorithm.

by an optimization procedure that provides convex fuzzy sets by minimizing the approximation error expressed in terms of fuzzy set distance [12]. In this way, the exploratory feature of cluster analysis is exploited, and interpretability of the resulting fuzzy sets is protected.

3.2.4 Unimodality

Constraint 4 *A fuzzy set A ought to be* unimodal[6]*, i.e. there exists only one element with maximum membership function, which is the prototype:*

$$\exists! p \in U : \mu_A(p) = \operatorname{hgt} A \tag{3.5}$$

In quantitative modeling, unimodality is required to represent fuzzy numbers, while it is not necessary to represent fuzzy intervals. In qualitative modeling, unimodality is not a mandatory requirement, as long as it is semantically significant to have a *set* of prototypes (which is always an interval in convex fuzzy sets) instead of just one. However, attention must be paid when unimodality is not considered, because prototypes become indistinguishable and may lead to oversimplified models, as illustrated in the following example.

[6] In this book we adopt the unimodality concept as generally defined in Mathematics. Alternative definitions exist in fuzzy set literature, e.g. [22].

Example 3.1. Consider a system using the fuzzy set labeled TALL, which is characterized by the following membership function defined in the UoD $U = [0, 220]$ (cm):

$$\mu_{\text{TALL}}(x) = \begin{cases} 0 & \text{if } \quad 0 \leq x \leq 150 \\ \frac{x-150}{30} & \text{if } 150 < x < 180 \\ 1 & \text{if } 180 \leq x \leq 220 \end{cases} \tag{3.6}$$

Such a fuzzy set could be used to compare people according to their tallness, so that it is possible to affirm that person x is taller than person y if $\mu_{\text{TALL}}(x) > \mu_{\text{TALL}}(y)$. Nevertheless, with the considered definition, a subject of height 180 cm and a subject of height 200 cm are considered equally tall by the system. Depending on the context, this definition may not conform to the concept of tallness conceived by humans in their experience. A more cointensive definition of the TALL fuzzy set could make use of a strictly monotonic membership function mapping m_U to 0 and M_U to 1. A membership function like this reflects the perception that tallness increases as the actual height increases, and comparing two persons in terms of "taller than" can be done by comparing the respective membership degrees.

3.3 Constraints and Criteria for Linguistic Variables and Fuzzy Partitions

In interpretability analysis, much concern is directed towards linguistic terms in T and their semantic interpretation $\mu(T)$. In particular, linguistic terms that do not correspond to natural language structures should be avoided as they would not convey any meaning to the final user. This requirement is often application oriented; however exotic linguistic constructs should be always avoided (e.g., VERY VERY VERY MEDIUM [46]).

The requirement of using natural language terms in T poses a semantic problem. Natural language terms, indeed, convey an implicit semantics that is roughly shared among all users that use those terms for communicating information. Of course, each user may have a different semantic for the same term; however, the semantics of the same term to different users should highly match: this is necessary for enabling communication among users. As a consequence, to be interpretable, the explicit semantics of a linguistic term (as defined by the corresponding fuzzy set) should highly match with the implicit semantics possessed by a user that reads the term in a knowledge base. In other words, implicit and explicit semantics must be *cointensive* [49]. This problem is non-trivial and hardly formalizable, because the implicit semantics conveyed by a linguistic term cannot be expressed formally. Thus, the objective of interpretability analysis is to define a set of constraints to partially meet the cointension requirement.

3.3.1 Relation Preservation

Suppose that an LV is equipped with a subset of terms

$$A_1, A_2, \ldots, A_n \in T$$

for which a *natural* ordering $\preceq$ is provided, i.e. all users agree upon an ordering of the terms on the basis of their implicit semantics (e.g., LOW $\preceq$ MEDIUM $\preceq$ HIGH). Then, we should expect that the explicit semantics of these terms also satisfy the same ordering relation. However, this requires a formal specification of ordering among fuzzy quantities, which is not trivial [76]. For the sake of interpretability analysis, we observe that when two linguistic terms A', A'' can be ordered according to their implicit semantics, say $A' \prec A''$, then they partition the UoD into two adjacent parts, namely U' and U'', such that A' is more appropriate than A'' on U' and, vice versa, A'' is more appropriate than A' on U''. This requires a threshold $t \in U$ such that $U' = [m_U, t]$ and $U'' = [t, M_U]$ with $U = U' \cup U''$. In these two parts, the explicit semantics of the two linguistic terms A' and A'' should follow the same relation of implicit semantics, that is $\mu_{A'}(x) > \mu_{A''}(x)$ for any $x < t$ and, vice versa, $\mu_{A''}(x) > \mu_{A'}(x)$ for any $x > t$.

The above considerations can be extended to a finite set of linguistic terms, thus leading to the following constraint:

Constraint 5 *Given* LV $= \langle X, T, U, G, \mu \rangle$*, if a strict total order* $\prec$ *is defined for a subset of linguistic terms* $\{A_1, A_2, \ldots, A_n\} \subseteq T$*, then for each couple* $\{A_i, A_j\}$ *with* $A_i \prec A_j$ *there exists a threshold* t *such that*

$$\begin{aligned} &\forall x \in U: \\ &\qquad \Big(x \in [m_U, t) \to \mu_{A_i}(x) > \mu_{A_j}(x)\Big) \wedge \\ &\qquad \Big(x \in [t, M_U] \to \mu_{A_i}(x) \leq \mu_{A_j}(x)\Big) \end{aligned} \tag{3.7}$$

Such a constraint will be referred to as *proper ordering* and poses some limitations to the choice of fuzzy sets used in an LV. Fig. 3.5 illustrates the case of an LV with three corresponding fuzzy sets which appear to be labeled in an unsuitable way (provided that three linguistic terms $A_1 =$ LOW, $A_2 =$ MEDIUM, $A_3 =$ HIGH are involved, with the obvious relationship $A_1 \prec A_2 \prec A_3$).

Additionally, the following example shows that, when Gaussian fuzzy sets are used, proper ordering can be easily violated.

Example 3.2. Let A and B be two fuzzy sets with the following membership functions:

$$\mu_A(x) = \exp\left(-\frac{(x-\omega_A)^2}{2\sigma_A^2}\right) \tag{3.8}$$

and

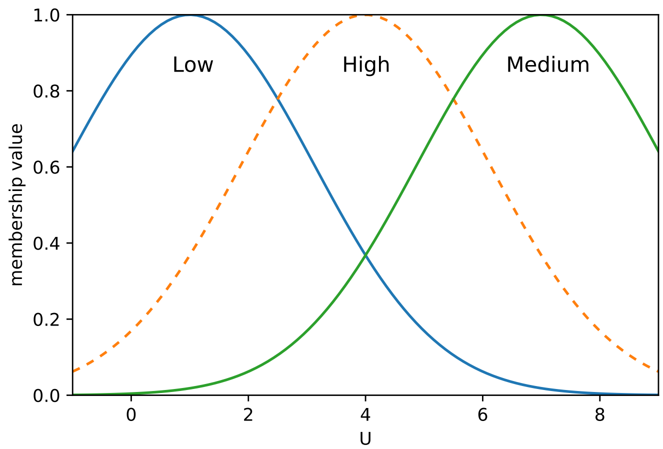

Fig. 3.5: A linguistic variable with three fuzzy sets. While each individual fuzzy sets has a well defined semantics, they are not properly labeled thus hampering their interpretability.

$$\mu_B(x) = \exp\left(-\frac{(x-\omega_B)^2}{2\sigma_B^2}\right) \tag{3.9}$$

The points where $\mu_A(x) = \mu_B(x)$ are:

$$\mu_A(x) = \mu_B(x) \iff x = \begin{cases} \frac{\sigma_B\omega_A \mp \sigma_A\omega_B}{\sigma_B \mp \sigma_A} & \text{if } \sigma_A \neq \sigma_B \\ \frac{\omega_A + \omega_B}{2} & \text{if } \sigma_A = \sigma_B \end{cases} \tag{3.10}$$

Hence there are at most two intersection points, which can be denoted by t_1 and t_2 assuming $t_1 \leq t_2$.

These two fuzzy sets could be used in an LV as interpretations of two linguistic terms $A =$"SMALL" and $B =$"HIGH", where obviously SMALL$\prec$ HIGH. However, the explicit semantics of the two fuzzy sets may differ both in the width and in the center values of the Gaussian functions ($\sigma_A \neq \sigma_B$, $\omega_A \neq \omega_B$): in such a case two thresholds t_1, t_2 exist and, for each $x > t_2$, x is more SMALL than HIGH, thus violating proper ordering (see Fig. 3.6).

The problem depicted in the previous example can be alleviated by choosing more adequate families of fuzzy sets (e.g., trapezoidal) or by enforcing the same width to all fuzzy sets when they are of Gaussian type. Nevertheless, there are cases in which Gaussian fuzzy sets are used with different widths (e.g., in the case of rules tuned by neuro-fuzzy networks). In those cases, the proper ordering condition could be weakened by requiring that the prototypes of the Gaussian fuzzy sets are ordered according to the natural ordering of the related linguistic terms. This is a kind of

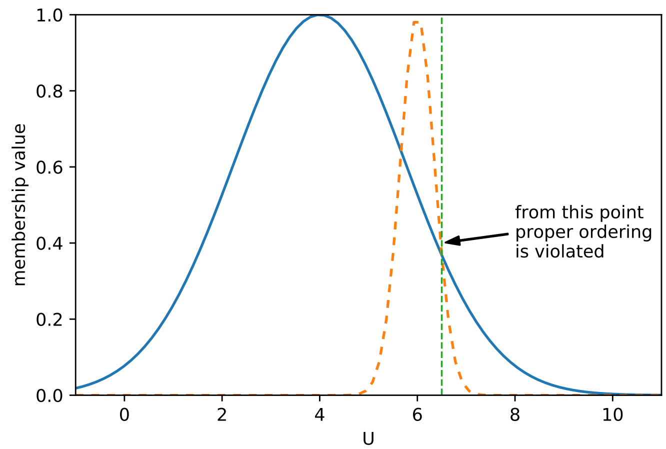

Fig. 3.6: An example of proper ordering violation.

trade-off which is aimed at avoiding too much bias on the learning algorithm that tunes the fuzzy sets without compromising too much the readability of the LVs.

The constraint on term ordering can be extended to any relation that holds on the implicit semantics of the linguistic terms. As an example, if the implicit semantics of a linguistic term is fully contained in the semantics of another linguistic term, then the fuzzy set interpreting the first should be included in the fuzzy set of the second, according to the usual inclusion relationship. On a more general level, if the linguistic terms of an LV are ontologically related, then the corresponding fuzzy sets should be related in an analogous way. This requirement calls for a more general criterion of *relation preservation*, which can be formalized as follows [10]:

Constraint 6 *Given* $\mathrm{LV} = \langle X, T, U, G, \mu \rangle$*, if a relation* R *exists on* T*, i.e.* $R \subseteq T^n$*, then a relation* $Q \subseteq \mathscr{F}(U)^n$ *ought to be defined such that*

$$(A_1, A_2, \ldots, A_n) \in R \implies (\mu_{A_1}, \mu_{A_2}, \ldots, \mu_{A_n}) \in Q \tag{3.11}$$

According to (3.11), whenever a relation holds among linguistic terms, a corresponding relation must hold among the fuzzy sets that are interpretations of these terms. The converse is generally not true, thus justifying the directionality of the constraint.

3.3.2 Justifiable Number of Elements

Constraint 7 *The number of linguistic terms in an LV ought to not be too high, preferably less than* 7 ± 2*.*

This criterion is motivated by a psychological experiment reported in [52] and considered in a large number of works concerning interpretable fuzzy modeling, e.g. in [16, 31, 33, 34, 47, 56, 60, 65, 72]. In his test battery, Miller proved that the span of absolute judgment (assigning scores to stimuli of different perceived magnitudes) and the span of immediate memory (the number of events that are reminded in the short term) impose a strong limitation on the amount of information that a human being is able to perceive, process and remember. Experimentally, the author found that the number of entities that can be clearly remembered for a short time is around 7, plus or minus 2, depending on the subject under examination. Interestingly, and quite surprisingly from a cognitive standpoint, complex information can be mentally organized into unitary chunks of information that are treated as single entities. In this way, the human brain is capable to perceive, process or remember in its short term memory both simple perceptions (e.g., tones) or more complex entities (e.g., faces), provided that the number of entities is less than "*the magical number seven, plus or minus two*".

Miller's findings stimulated further works, which either confirmed or reduced this number [15, 64]. As a general result, human brain is capable of handling few chunks of information in the short memory, and this should be kept in mind when designing interpretable fuzzy systems. In particular, this general criterion requires that the number of linguistic terms in an LV should be kept as small as possible, since these terms denote attribute qualities that can be associated to stimuli of different perceived magnitudes.

This constraint imposes a limit on the complexity of an LV and can be extended to other components of a fuzzy system (number of attributes, information granules and rules). As a consequence, the constraint introduces a bias on the final model which may affect its predictive accuracy. This is a long-stated problem in modeling interpretable fuzzy systems: often, most of the designer effort is to find the right trade-off between accuracy and interpretability. To alleviate this problem, the type of knowledge representation can be considered as a design variable. In particular, multilevel representations can be considered to model complex relationships among data, where higher levels of representation provide for rough yet comprehensible knowledge, whilst lower levels of representation enable accurate prediction without paying too much attention to interpretability [50].

3.3.3 Distinguishability

Constraint 8 *Different terms in an LV ought to be interpreted by well distinguishable fuzzy sets.*

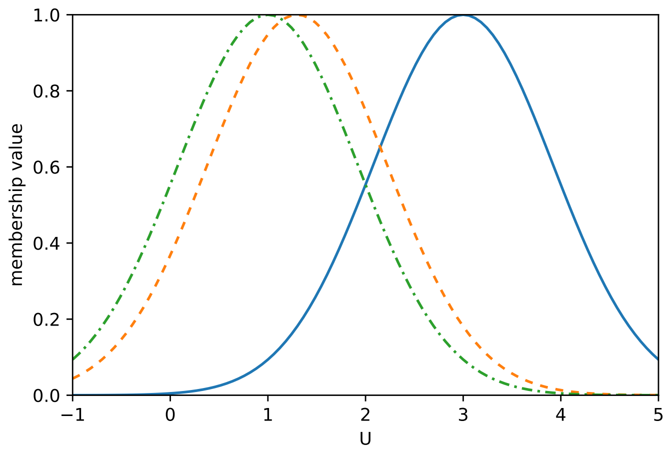

Fig. 3.7: Example of non-distinguishable fuzzy sets (dash vs. dash-dots) and distinguishable fuzzy sets (solid vs. dash or solid vs. dash-dots). Non distinguishable fuzzy sets represent almost the same concept, hence it is difficult to assign different labels.

Distinguishability is one of the most common interpretability constraints adopted in fuzzy modeling literature [5, 13, 16, 24, 31, 33, 36, 47, 55, 56, 60, 62, 65, 73]. Roughly speaking, distinguishable fuzzy sets are well disjoint so they represent distinct concepts and can be assigned to semantically different linguistic labels (see Fig. 3.7). In a slightly more precise sense, distinguishability is a (fuzzy) relation between fuzzy sets defined on the same UoD that is fully satisfied when the fuzzy sets are completely disjoint[7], while it is not satisfied at all when the two fuzzy sets coincide.

Completely disjoint fuzzy sets are maximally distinguishable. However, usually linguistic concepts partially overlap so that the passage from a concept to another is usually smooth (e.g., from SMALL to TALL). This explains the gradual nature of the property of distinguishability. Well distinguishable fuzzy sets represent well separated concepts; this makes their linguistic interpretation easier and less subjective [65, 71]. Distinguishable fuzzy sets are also useful to avoid model inconsistency by reducing redundancy [73]. As a bonus, computational efforts required for inference are alleviated.

There are several ways to quantify the criterion of distinguishability, which depend on how overlapping of fuzzy sets is measured. In Chap. [3] a thorough review of such measures is reported.

[7] i.e. $\forall x \in U : \min\{A(x), B(x)\} = 0$.

3.3.4 Coverage

Constraint 9 *Each element of the UoD ought to be well represented by at least one fuzzy set of an LV. Formally, given* $\mathrm{LV} = \langle X, T, U, G, \mu \rangle$*, then:*

$$\forall x \in U, \exists A \in T \text{ s.t. } \mu_A(x) > 0$$

Completeness is a property of deductive systems that has been used in the context of Artificial Intelligence to indicate that the knowledge representation scheme can represent every entity within the intended domain [73]. When applied to fuzzy models, completeness states that the fuzzy model should be able to infer a proper conclusion for every input [43]. Completeness requires coverage of LV since, for each element of the UoD, at least one linguistic term must be activated to carry on the inference. Hermann [26] justifies coverage (there called "cover full range") by the fact that in human reasoning there will never be a gap of description within the range of the variable. On the contrary, incompleteness may be a consequence of model adaptation from data and can be considered a symptom of overfitting [37].

Usually, a stronger version of the requirement is adopted, which constrains the membership degree of being greater than a fixed threshold, thus avoiding formally complete LVs that cannot be intended as such in practice (see also [32]):

Constraint 10 (α-coverage) *Given* $\mathrm{LV} = \langle X, T, U, G, \mu \rangle$*, and* $\alpha \in [0, 1]$ *then:*

$$\forall x \in U, \exists A \in T \text{ s.t. } \mu_A(x) \geq \alpha$$

Generally, α-coverage is preferable in interpretable fuzzy modeling because it assures that each element of the UoD is *well* represented by the LV, i.e. each element possesses a quality (represented by a fuzzy set) with a minimum degree α (see also Fig. 3.8).

On a semantic level, α-coverage assures that for each element in the domain of the LV there exists at least one linguistic term that is appropriate for the element to a minimum degree α. Stated in an equivalent way, the constraint also means that for each element in the domain of the LV there exists at least one linguistic term whose *negation*[8] is appropriate for the element to a *maximum* degree $1 - \alpha$. When $\alpha < 0.5$ a possibility is given to linguistic terms whose negation is more representative than its positive form. This leads to potential negative impact on the interpretability of the LV, hence it is not advisable to select the threshold α below half the unity. Furthermore, when the distinguishability of fuzzy sets in an LV is evaluated through the possibility measure, then the distinguishability threshold and the coverage threshold coincide: if the threshold is greater than 0.5 more emphasis on coverage is given than on distinguishability; the contrary is true when the common threshold is smaller than 0.5. As an immediate consequence, to balance both distinguishability and cov-

[8] We are assuming 1-complement as negator operator. This is the simplest definition of negation and is widely used in literature.

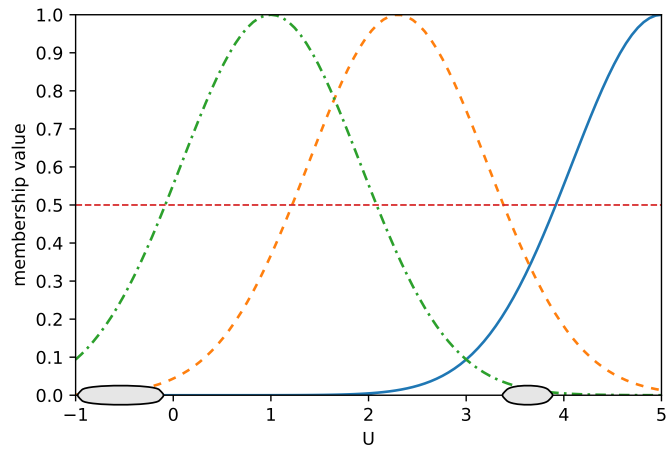

Fig. 3.8: Example of coverage violation. In the highlighted regions of the UoD (ellipses) 0.5-completeness is not verified. Elements in such region are not well represented by any fuzzy set in the LV.

erage, the common threshold 0.5 is the best choice. This is the case for Strong Fuzzy Partitions (SFPs), which will be presented in the next section.

3.3.5 Complementarity (a.k.a. $\Sigma 1$-criterion)

Constraint 11 *Given* LV $= \langle X, T, U, G, \mu \rangle$*, then the corresponding fuzzy partition should form an* SFP*, i.e.:*

$$\forall x \in U : \sum_{A \in T} \mu_A(x) = 1 \tag{3.12}$$

where the involved fuzzy sets are all normal and convex.

Complementarity is mainly motivated by the following properties, which are direct consequences of the adoption of an SFP [26]:

Full acceptance: whenever a membership value decreases from 1, another value increases from zero. From a semantic perspective, full acceptance implies that when a fuzzy set does not fully represent an element, then there exists another fuzzy set that (at least partially) represents the element.

Negations: the negation of a fuzzy set is expressed as the sum of the remaining membership values.

$$\forall x \in U, \forall A \in T : \mu_{\neg A}(x) = \sum_{A' \in T \setminus \{A\}} \mu_{A'}(x) \tag{3.13}$$

Not three a time: normality and convexity of the involved fuzzy sets assure that, for any element of U, no more than two fuzzy sets have non-zero membership degree.

The most significant contribution of complementarity to interpretability appears to be full acceptance, especially when the semantics of fuzzy sets can be interpreted in terms of similarity with respect to a prototype. Full acceptance assures that an element becomes less similar to a prototype, then it becomes more similar to another prototype. As similarity interpretation implies convex and normal fuzzy sets, the third property ("not three a time") implies that each element is similar (with different degrees) to at most two prototypes. This implication is very important in the inference context, because it assures that the number of elements involved in the inference process is limited, thus making inference more efficient. However, such property may appear too stringent in some application contexts, since it precludes that more than two linguistic terms partially represent the same element. In these cases, the *Complementarity* criterion could be replaced with some weaker criteria.

In Fig. 3.9 two examples of fuzzy sets defined for an LV are depicted. Both examples show high interpretability and express almost the same semantics, but only the first one satisfies complementarity. The first configuration of fuzzy sets may enable a faster inference (due to the "not three a time") but the second configuration of fuzzy sets is more expressive since it gives a non-zero membership degree for all elements of the domain and all the fuzzy sets. The use of one of the two configurations is a design choice that depends on how much emphasis must be given to efficiency and expressiveness of the final system.

3.3.6 Uniform Granulation

Constraint 12 *Given* LV $= \langle X, T, U, G, \mu \rangle$*, then the cardinalities*[9] *of all the involved fuzzy sets in* LV *should be almost the same:*

$$\forall A', A'' \in T : |\mu_{A'}| \approx |\mu_{A''}| \tag{3.14}$$

Uniform granulation is a criterion required in [44] as a discriminating property for distinguishing understandable models from those models where linguistic understandability is not of major concern. In Fig. 3.10, a configuration of fuzzy sets with uniform granulation is graphically compared with a frame that violates this constraint.

The criterion can be formalized by fixing a class of membership functions (e.g., Gaussian) characterized by a set of parameters (e.g., center and width). Uniform

[9] The simplest definition of the cardinality of a fuzzy set is $|A| = \int_U \mu_A(x)\,dx$. A deeper discussion on fuzzy set cardinalities can be found in [77].

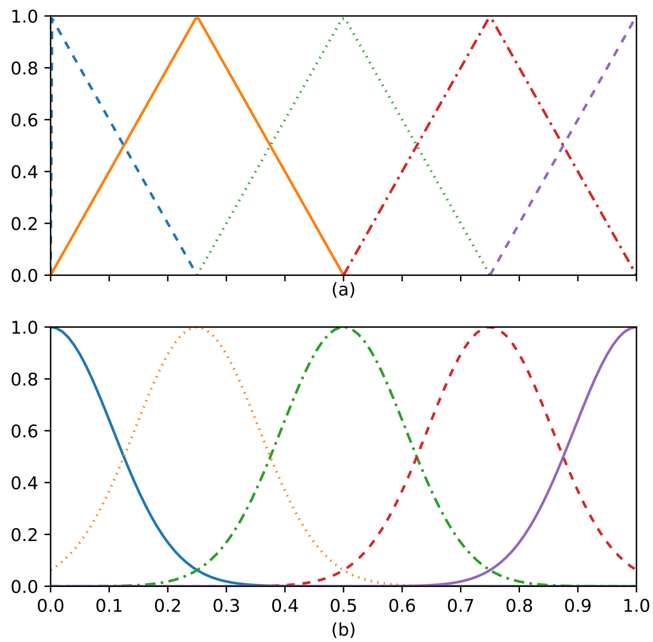

Fig. 3.9: Example of two configurations of fuzzy sets with complementarity constraint verified (a) and violated (b). Both of them appear to be interpretable.

granulation can be imposed either by restricting the range of variability of such parameters (e.g., the width should vary only in a predefined range) or by assigning a penalty value depending on the actual values of each parameter.

While it is recognizable that uniform granulation appears as a desirable property for interpretable fuzzy models, it should be noted that such constraint is not often used in literature. This can be motivated by the necessity of specifying a high number of hyper-parameters (i.e., range of variability and the dispersion index), especially when fuzzy models are acquired from data. An additional problem may arise when data used for model induction have not a uniform distribution: in such a case, the uniform granulation may oppose to an accurate – yet understandable – description of data. Finally, it should be observed that human beings do not perceive stimuli with a uniform scale (e.g., the intensity of perceived sound is non-linearly related to the actual intensity of sound). Such nonlinearities, which translate into non uniformity of scales, should be taken into account in interpretable fuzzy modeling.

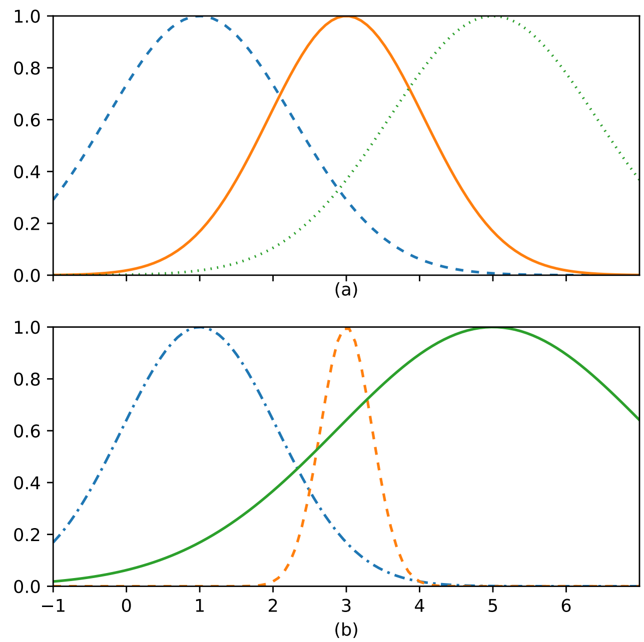

Fig. 3.10: Comparative example of two configurations of fuzzy sets where Uniform Granulation is verified (a) and violated (b).

3.3.7 Leftmost/rightmost Fuzzy Sets

Constraint 13 *Given* LV $= \langle X, T, U, G, \mu \rangle$*, then the lower and upper bounds of* U *ought to be prototypes for some fuzzy sets of the variable:*

$$\exists A', A'' \in T \text{ s.t. } \mu_{A'}(m_U) = \mu_{A''}(M_U) = 1 \tag{3.15}$$

This constraint is often implicitly used in interpretable fuzzy modeling (with some exceptions where it is explicitly mentioned, like in [13]). The constraint states that there exist two *boundary* fuzzy sets that fully represent the limit values of the UoD. Fig. 3.11 depicts two configurations of fuzzy sets, one verifying leftmost/rightmost fuzzy sets and the other violating the constraint.

In interpretability analysis, this constraint is of prominent importance. As a counter-example, we may consider an LV where linguistic terms are normal and

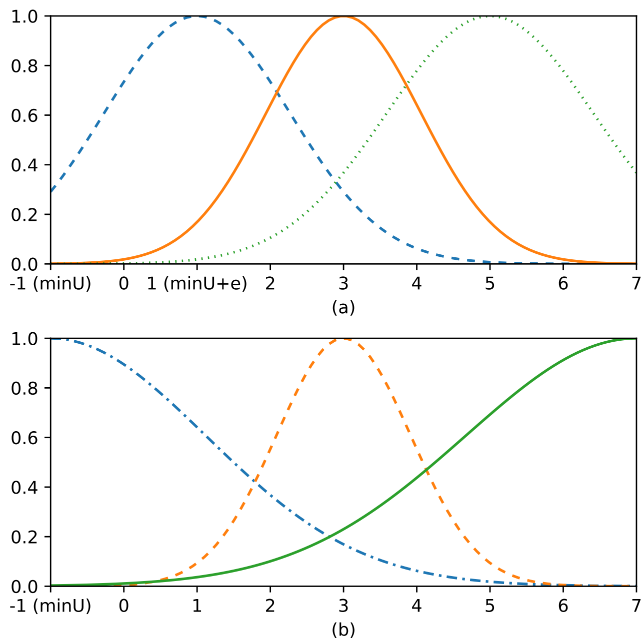

Fig. 3.11: Configurations of fuzzy sets violating leftmost/rightmost fuzzy sets (a) and verifying such a constraint (b).

unimodal fuzzy sets, and where the first fuzzy set[10] is labeled as LOW and has not its prototype in m_U, i.e. $\mu_{\text{LOW}}(m_U) < 1$. Since the fuzzy set is normal, there exists a prototype which could be located at $m_U + \varepsilon$, being $\varepsilon > 0$. Each element of U belonging to $[m_U, m_U + \varepsilon)$ has a degree of membership in LOW that is lower than 1. This means each element in $[m_U, m_U + \varepsilon)$ has an evidence of "being low" that is smaller than element $m_U + \varepsilon$. This is not cointensive with the perception of the quality represented by the linguistic term, hence the LV is not interpretable.

Leftmost/rightmost fuzzy sets are hence important in those LVs that express qualities on the UoD (e.g., LOW, MEDIUM, HIGH), whenever the boundaries of the UoD have a special meaning. However, such a constraint is not necessary for LVs that express fuzzy quantities (e.g., ABOUT 0, PLUS OR MINUS 5, etc.), since these labels do not convey any meaning of extremity.

[10] In a weak sense, i.e. with the smallest prototype.

3.3.8 Natural Positioning

Constraint 14 *Given* LV $= \langle X, T, U, G, \mu \rangle$*, if the elements of a finite subset* $S_U \subset U$ *have some special meaning, then each element in* S_U *ought to be prototype of some fuzzy sets in* LV.

This constraints has been initially proposed as "natural zero positioning" because the null value has a special role in control problems, where FRBSs are widely used [72, 73]. Natural positioning is a generalization of the previous constraint on leftmost/rightmost fuzzy sets because of the special role of extreme values of the UoD for any LV.

Beyond boundary values, other "special" values are only problem specific. Simple examples may include 0 and 100 for an LV describing water temperature (in Celsius degrees), or 37 for an LV describing body temperature (again, in Celsius), etc. These values are natural prototypes for linguistic terms like ICING POINT, BOILING POINT, NORMAL TEMPERATURE when they make sense for the model to be designed.

3.4 Constraints and Criteria for Multi-dimensional Fuzzy Information Granules

Fuzzy sets belonging to fuzzy partitions can be combined together to form multi-dimensional fuzzy information granules, which express fuzzy relations over data and are, as a consequence, the building blocks of a knowledge base. As fuzzy partitions are embodied in LVs, information granules inherit a linguistic representation that depends on the LVs being used, as well as on the way the fuzzy sets are combined. The interpretability of a fuzzy information granule thus depends on the interpretability of the LVs that are combined together. However, the augmented structure of fuzzy information granules through complex soft constraints requires additional interpretability constraints to be satisfied. They are outlined in the following subsections.

3.4.1 Description Length

Constraint 15 *The description length of a fuzzy information granule ought to be as small as possible.*

The description length of the information granule is the number of soft constraints involved in its description. If such number is too high, the information granule description is difficult to read and understand. The maximum number of soft constraints is suggested by psychological studies and should not exceed seven, plus or minus two (see also Sect. 3.3.2). If the number of variables is higher than such limit, it is advisable to apply some simplification procedure, e.g. by selecting a subset of the variables to be included in the information granule.

In modeling contexts, an information granule can be described by a subset of the variables actually used in a model. For implementation purpose, in such situation the remaining variables can be included in the granule representation by special soft constraints of the form X is ANY [11, 53]. The semantics of the linguistic term ANY is defined by a fuzzy set with unitary membership value for all elements in the UoD. In this way, since unity is the neutral element of any t-norm, these special soft constraints do not modify the semantics of the information granule. As an example, the following soft constraints are semantically equivalent:

TEMPERATURE is LOW and PRESSURE is HIGH

TEMPERATURE is LOW and PRESSURE is HIGH and HUMIDITY is ANY

From a computational viewpoint, both expressions are equivalent (the latter being easier to process if all attributes need to be evaluated), but the former is easier to read and understand.

3.4.2 Attribute Correlation

Constraint 16 *Information granules could be represented in terms of correlation between* LVs*.*

In some cases interpretability of a model is improved if some information granules represent a relation among LVs (usually two). In this case a more complex structure on the representation of information granules is imposed, because soft constraints should not restrict the values of a variable, but define a fuzzy relation between the values of two or more variables.

An approach for representing variable correlation is proposed by Gaweda and Zurada [21], which is based on the representation of empirical correlation between LVs. Such correlations are expressed in linguistic terms such as:

TEMPERATURE OUTSIDE is
MORE OR LESS THE SAME AS TEMPERATURE INSIDE

This type of description is quite effective in specific cases but should not be overused because it becomes hard to understand when the number of correlations in a single information granule is high.

3.5 Constraints and Criteria for Fuzzy Rules

A rule is a formal structure that expresses a piece of knowledge. In fuzzy modeling a fuzzy rule defines a relationship between two information granules, one for the *antecedent* and another for the *consequent*. The structure of a fuzzy rule leads to an interpretable form of knowledge, because it is close to the one characterizing the expert rules usually applied in human reasoning and decision making. This enables a tight integration of expert rules and rules acquired from data within the same modeling framework. Notwithstanding the clear structure of a fuzzy rule, some specific constraints can be useful for further improving its interpretability.

3.5.1 Rule Length

Constraint 17 *The length of the rule ought to be as small as possible.*

The length of a rule is the sum of the length of its antecedent and its consequent. When the consequent is defined by an atomic soft constraint, the length of the rule is totally related to the length of the antecedent, i.e. to the description length of the corresponding information granule as discussed in Sect. 3.4.1.

3.5.2 Local Models

Constraint 18 *The consequent of a rule should describe a local model with a satisfactory level of precision.*

A fuzzy rule defines a local model determined by the rule consequent that is confined (in a fuzzy sense) by the rule antecedent. Depending on the application needs, the structure of the consequent may be either simple and immediately understandable, or more complex but more accurate in describing the local model.

When rules are used to identify unknown functions, consequents may provide a description of the shape of the function locally represented by the rule. In [29], consequents are used to describe the values of the function and its derivatives in the local neighborhood covered by each rule. An example of such rules is the following:

IF X is NEAR 0 THEN G(X) HAS A STEEP LOCAL MINIMUM NEAR 0

The rule can be rewritten as:

IF X is NEAR 0 THEN Y is NEAR 0 and Y' is NEAR 0 and Y'' is POSITIVE HIGH

Such rule richly describes the shape of the function $G(x)$ in the neighborhood of zero by providing a fuzzy quantification of the value of the function as well as its first (Y') and second (Y'') derivatives. However, such rules are useful only when the function to be described is defined on a one- or two-dimensional space, because they help the user in imaging a geometrical representation of the identified function, which is impossible for high-dimensional domains.

When TSK fuzzy systems are used, this interpretability constraint can be specialized by expressing the rule consequents as truncated Taylor series of the identified function. This representation is especially useful if the rule antecedent has a single (though multi-dimensional) prototype $\mathbf{v}$, because the Taylor series can be centered on this prototype, leading to the following rule schema [8]:

$$\text{IF } \textit{Antecedent} \text{ THEN } y = b_0 + \mathbf{b}^\top (\mathbf{x} - \mathbf{v}) \tag{3.16}$$

Rules could be extended with second order derivatives by using second-order TSK rules.

3.5.3 Granular Consequents

Constraint 19 *Rule outputs should be represented by linguistic terms.*

In general, interpretability of a model is improved when precise values are replaced with linguistic terms. This is the case of Mamdani fuzzy rules, where only linguistic terms are involved in representation and computation. However, TSK rules are widely used too, because of their high efficiency in computation due to the use of precise output values in the rule consequents.

In order to reconcile interpretability and accuracy of rules in TSK systems, it is advisable to use linguistic terms to describe the output; however to avoid the inefficiencies of Mamdani fuzzy rules, the use of linguistic terms could be used in rule representation only, while the classical TSK representation is used for inferring outputs. This is simple for 0^{th}-order TSK rules, where rule consequents are replaced by soft constraints on an LV that takes value in a set of pre-defined linguistic terms [65]. Unfortunately, there is not a clear way to do the same on TSK rules with order greater than zero.

3.6 Constraints and Criteria for Fuzzy Models

From a purely functional viewpoint, an FRBS can be seen as a a complex function that transforms input numerical data into a response through a set of fuzzy rules and a number of processing operators.

More specifically, an FRBS can be characterized by the following structure:

$$\mathscr{M} = \langle \mathbf{X}, \mathbf{R}, \otimes, \rhd, \oplus, [\mathscr{D}] \rangle \tag{3.17}$$

where:

- $\mathbf{X} = \{X_1, X_2, \ldots, X_{n_I+n_O}\}$ is the set of LVs involved in the model, being n_I and n_O the number of input and output variables respectively (usually $n_O = 1$);
- $\mathbf{R}$ is a finite set of n_R rules defined as in Sect. 3.5. It is the *knowledge base* of the model. Rules can be either in TSK form (of order k) or in Mamdani form. Models are named TSK of Mamdani accordingly.
- $\otimes$ is a t-norm used to combine soft constraints in rules antecedents;
- $\rhd$ is an operator used to interpret the THEN part of each rule;
- $\oplus$ is an operator used to aggregate all fuzzy sets derived by all rules of the knowledge base;
- $\mathscr{D}$ (optional) is a defuzzifier, i.e. a function that collapses a fuzzy set into a single value of the UoD.

A fuzzy model is flexible enough to allow for several interpretations. More specifically, the choice of the implication and aggregation function gives rise to two main interpretations:

Conjunctive Interpretation. The interpretation of THEN in a rule is performed by a conjunction, i.e. the operator $\rhd$ is a t-norm. Rules are consequently disjointed and the aggregation of rules is made by a t-conorm. Rules of this type define a *fuzzy graph* [79];

Implicative Interpretation. The interpretation of THEN in a rule is a real implication, which may be interpreted in several ways [70]. Rules are consequently conjoined and the aggregation is realized by a t-norm.

A number of interpretability issues arise when considering a model as a whole system. In the following, some interpretability constraints are described. For the sake of simplicity, the interpretation of a rule (assumed with a single output) will be denoted as

$$\mu_R = (\mathbf{A}, B) \text{ or } \mu_R = (\mathbf{A}, f)$$

to indicate that the rule R in the rule set $\mathbf{R}$ has all fuzzy sets in the antecedent part whose Cartesian product is $\mathbf{A}$ and in the consequent part has a fuzzy set B or a function f (TSK rule).

3.6.1 Compactness

Constraint 20 *The total number of rules in a model must be small.*

This criterion is strictly related to the "justifiable number of elements" for LVs (see Sect. 3.3.2), as well as the "rule length" criterion for rules (see Sect. 3.5.1), and finds the same psychological motivations. In general, the principle of reducing the descriptive complexity of the model can be considered as an application of the well-known "Occam's razor" principle, which states that among all possible correct hypotheses, the one that reflects the reality is most probably the simplest.

More pragmatically, the Occam's razor principle can be motivated by our need to comprehend phenomena with the simplest possible explanation of it. This principle practically translates in fuzzy modeling by keeping the number of fuzzy sets, LVs and rules as small as possible.

In a learning scenario, the Occam's razor principle is related to the generalization ability of the acquired predictive model, because a model with a small number of rules is a candidate for endowing a knowledge base that is general enough, while complex models are more prone to overfitting. However, if the model is too simple, it may be useless since the space of mappable functions is too poor [69].

Compactness is a widely required constraint for designing interpretable fuzzy models. This requirement is needed because the upper bound of fuzzy rules is exponentially related to the input dimensionality. Indeed, it is easy to show that the total number of combinations of linguistic terms coming from different LVs is exponentially related to the number of LVs used in a model. Thus, if a rule is generated for each combination, then the maximum number of rules is

$$R_{\max} = \prod_{i=1}^{n_I} |T_i| \tag{3.18}$$

being T_i the set of linguistic terms if the LV is X_i. In non-trivial scenarios, this number of rules gives rise to unreadable models even with a modest number of LVs, not to mention the related inefficiency of the inference process. Combinatorial rule explosion can be attenuated by a number of techniques, including sharing of fuzzy sets and removal of redundant and irrelevant rules.

3.6.2 Number of Firing Rules

Constraint 21 *The number of rules simultaneously activated (fired) by an input ought to be small.*

A rule in a model is said to be activated (or fired) if the antecedent of the rule has non-zero membership degree (also called "activation strength"). This constraint concerns the semantic facet of a fuzzy model, since it poses a restriction in the behavior of the model during inference. In particular, the collection of fired rules provides a *local view* of the model for a particular input [60]. By keeping the number of firing rules as small as possible, it is therefore easy for the user to understand how an output has been derived by the model. Moreover, a small number of firing rules assures an efficient inference process since the complexity of the inference procedure is linearly related to the number of activated rules.

When limited support fuzzy sets are used in the model, such as triangular or trapezoidal fuzzy sets, the number of fired rules is smaller than the total number of rules, because the number of fuzzy sets with non-zero membership is small (usually less than two) on each dimension. However it is easy to prove that the number of fired rules is still upper bounded by a number exponentially related to the dimensionality of the UoD. As a corollary, the adoption of limited support fuzzy sets may not be enough for satisfying this interpretability constraint.

The upper bound of firing rules should also be lowered by reducing the total number of rules in the knowledge base. For this reason clustering techniques are widely used to design fuzzy models, since the number of generated rules is usually upper bounded by the number of clusters. When a low number of rules defines the rule base of a fuzzy model, even infinite support fuzzy sets may be used, like Gaussian fuzzy sets; in this case further efficiency optimization can be attained in the inference process by dropping fuzzy rules whose activation strength is less than a small threshold.

3.6.3 Shared Fuzzy Sets

Constraint 22 *Fuzzy sets ought to be shared among rules.*

This constraint is a direct consequence of the distinguishability constraint (see Sect. 3.3.3), which requires that linguistic terms in an LV refer to well distinct concepts in the UoD. When these LVs are used in fuzzy rules, it is very common that the soft constraints in different rules refer to the same concepts, like in the following example:

$$R_1: \text{ IF X is S}\text{HORT and Y is W}\text{IDE THEN Z is S}\text{MALL} \tag{3.19}$$

$$R_2: \text{ IF X is S}\text{HORT and Y is N}\text{ARROW THEN Z is M}\text{EDIUM} \tag{3.20}$$

$$R_3: \text{ IF X is M}\text{IDSIZE and Y is W}\text{IDE THEN Z is M}\text{EDIUM} \tag{3.21}$$

In the example, R_1 and R_2 share the same fuzzy set for the input variable X (SHORT), while R_2 and R_3 share the same fuzzy set for the output variable Z (MEDIUM). Shared fuzzy sets are helpful for representing the model's knowledge with a small number of different symbols.

3.6.4 Rule Locality

Constraint 23 *The mapping realized by a TSK fuzzy system ought to result from the composition of independent local models, being each local model defined by a rule of the rule base.*

TSK fuzzy systems are capable of smoothly approximating functions and are therefore widely used in control applications [17, 38, 44, 65, 78]. However the output of a TSK system, given an input, is defined by an aggregation of contributions coming from different rules that are fired by the input. When too many rules significantly contribute to the generation of the output, it becomes hard to explain the output in terms of the rule base.

According to rule locality, each rule defines a local model that describes the behavior of the model in the context determined by the rule antecedent. Therefore, an input value that satisfies the antecedent of a rule with a high degree, should determine an output that is highly compliant with the local model represented in the rule consequent. On the other hand, if an input value satisfies the antecedent of a rule only partially, it is expected that the resulting output comes from an interpolation of the local models defined by two or more rules. In this way, the interpolative behavior of a fuzzy system is preserved but, at the same time, the global behavior is decomposed into simpler local models that are expected to be more comprehensible.

The adoption of rule locality requires learning techniques that do not take into account only accuracy maximization. Combined global/local learning is a promising approach to reach rule locality while maintaining good global accuracy of the model [78].

3.6.5 Modus Ponens

Constraint 24 (Fuzzy Mapping) *If an input coincides with the antecedent of a rule, the output of the system should be included in the rule consequent:*

$$\forall R \in \mathbf{R} : \mu_R = (\mathbf{A}, B) \implies \mathscr{M}(\mathbf{A}) \subseteq B \tag{3.22}$$

where $\mathscr{M}(\mathbf{A})$ is the output of the FRBS when the fuzzy *input* $\mathbf{A}$ *is provided.*

In most cases, FRBSs are used to compute numerical outputs (e.g., by defuzzifying outputs of Mamdani systems). In such cases, a slightly different definition of the constraint is required:

Constraint 25 (Numerical Mapping) *If a rule in the system has full activation strength for a given input, the output of the system should belong to the core of the rule consequent:*

$$\forall R \in \mathbf{R} : \mu_R = (\mathbf{A}, B) \implies \mu_B(\mathscr{M}(\mathbf{A})) = 1 \tag{3.23}$$

In classical logics, if a rule base contains a rule in the form $A \implies B$ and the antecedent A is verified by the input, then the consequent B must be verified by the output. This inference meta-rule, called *Modus Ponens*, is fundamental for any reasoning scheme. As a consequence, it is desirable that Modus Ponens could be verified also in fuzzy inference systems. Modus Ponens is the logical counterpart of Rule Locality, which offers high comprehensibility as it requires to decompose the global behavior of a system into several local models represented by each rule.

Unfortunately, Modus Ponens is not verified *ipso facto* but must be guaranteed by an appropriate design of the fuzzy model. This is due to possibility of choosing different inference schemes and many different interpretations of the logical connectives. As an example, in [41], a study is made where proper interpretations of logical connectives verify Modus Ponens in fuzzy models. In [61], it is proved that the adoption of constrained triangular fuzzy sets allows for Modus Ponens verification in 0^{th}-order TSK fuzzy inference systems. It should be noted, however, that such conditions are only sufficient, and the existence of other design configurations that enable Modus Ponens verification are possible. Finally, it is worth observing that the implicative interpretation of fuzzy rules ensures Modus Ponens as well as other logical properties.

3.6.6 Consistency

Constraint 26 *If two or more rules have simultaneously high activation strength, (e.g., antecedents are very similar), the consequents should be very similar.*

Consistency is one of the most crucial requirements of any expert system [63]. The knowledge base of an expert system can be viewed as a formal theory in which rules are the axiomatic postulates. A theory is consistent if it is not possible to derive a statement and its negation. If the theory is inconsistent, then any statement and its negation can be derived from the theory. As a consequence, an expert system with inconsistent knowledge base is useless.

In rule-based expert systems, inconsistency comes up when there are two rules in the form $A \to B$ and $A \to C$ where B and C are mutually exclusive concepts, i.e. either $B = \neg C$ or it is possible to derive $B \to \neg C$ from the knowledge base. In fuzzy logic, consistency is a less stringent requirement because the "non-contradiction" law is no more valid, i.e. the statement $A \wedge \neg A$ can be true with a certain degree greater than zero. This is considered a point of strength of fuzzy logic since it is still possible to make inference even in presence of inconsistency, as often occurs in real world. In fuzzy logic (in the broad sense), consistency is a matter of degree; hence in interpretable fuzzy modeling inconsistency could be tolerated as long as its degree is acceptably small.

Jin and Sendhoff extend the concept of inconsistency to cover all situations that may generate confusion in understanding the knowledge base [36]. Such inconsistency conditions are exemplified as follows:

Condition 1 *Antecedents are the same, but consequents are very different. For example:*

$$R_1: \text{ IF X1 is A1 and X2 is A2 THEN Y is } \textsc{Positive Large}$$

$$R_2: \text{ IF X1 is A1 and X2 is A2 THEN Y is } \textsc{Negative Large}$$

This is the classical inconsistency situation in implicative interpretation, where two nearly mutually exclusive consequents are derived by the same antecedent. The two consequents can have very low overlap and consequently the inferred fuzzy set

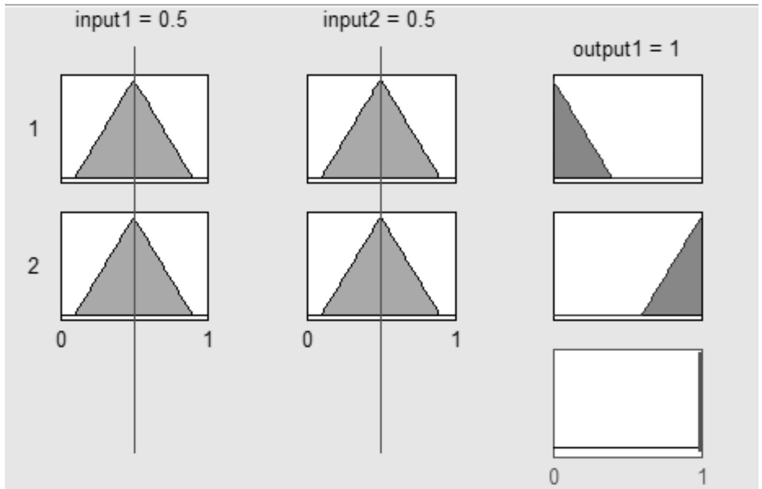

Fig. 3.12: Example of inference with rules in implicative interpretation. The inferred fuzzy set (bottom-right) is empty.

results highly subnormal without any clear indication of a suitable numerical output to be provided (see Fig. 3.12).

Condition 2 *Conditions are apparently different, actually being physically the same:*

$$R_1: \text{ IF CYCLE TIME is LOW THEN Y is POSITIVE LARGE}$$

$$R_2: \text{ IF FREQUENCY is HIGH THEN Y is NEGATIVE LARGE}$$

This is a subtle problem that cannot be checked automatically, unless an ontology on the variables and fuzzy set labels is provided. This kind of inconsistency occurs when there exists a (generally unknown) functional relation between two or more variables (in the example, there exists the relation FREQUENCY $= 1/$CYCLE TIME). Usually, detection of such type of inconsistency is left to the domain experts that analyze the knowledge base.

Condition 3 *Conditions are contradictory:*

$$R_1: \text{ IF SUN is BRIGHT and RAIN is HEAVY THEN } \ldots$$

This situation does not lead to inconsistency in the proper sense because contradicting concepts appear in the antecedents but not in the consequents. However, rules of this type can be considered inconsistent (in a broader extent) since they will never be activated (with high strength) by any input configuration. Rules of this type can be safely removed from the knowledge base because they do not significantly influence the inference process but can generate confusion when reading the knowledge base.

Condition 4 *Consequents involve contradictory actions:*

$$R_1: \text{ IF } \ldots \text{ THEN } \textsc{Acceleration} \text{ is } \textsc{High} \text{ and } \textsc{Brake} \text{ is } \textsc{On}$$

This kind of inconsistency is similar to the first type (same antecedent, different consequents) but it is more subtle because consequents are defined on different UoDs and inconsistency cannot be checked automatically. However, a proper integration of expert knowledge can overcome the problem by inserting rules that exclude contradictory actions. In the example the additional rule

IF ACCELERATION is GREATER THAN 0 THEN BRAKE IS OFF

enables the automatic detection of an inconsistency.

Some observations are noteworthy when analyzing consistency of FRBSs. As previously stated, consistency is a requirement of classical expert systems that has been extended to fuzzy models. However, as previously discussed, two interpretations exists of a fuzzy knowledge base: implicative interpretation and conjunctive interpretation. Implicative interpretation is a direct generalization of classical rule-based expert system within the fuzzy framework. In such interpretation, consistency issues occur also in fuzzy models and can be detected and quantified. But the most common interpretation of the fuzzy knowledge base is conjunctive, which is not equivalent to the implicative interpretation and gives rise to different considerations.

To illustrate the consistency issues in a conjunctive interpretation of fuzzy rules, a simplified setting can be considered. Let be **R** a trivial crisp knowledge base consisting of only two rules. If such rules are interpreted in a conjunctive modality, the entire knowledge base can be formalized in classical logics as

$$(Ant_1 \wedge Cons_1) \vee (Ant_2 \wedge Cons_2) \tag{3.24}$$

suppose that $Ant_1 = Ant_2$ while $Cons_1 = \neg Cons_2$. If the common antecedent is verified by an input, then it is possible to derive $Cons_1 \vee \neg Cons_1$ *which is a tautology, not a contradiction*. Actually, contradiction cannot occur because rules are tied by disjunction, so the "non-contradiction" law cannot apply. In light of such considerations, consistency conditions should be reviewed according to the new interpretation. For example, if two rules have the same antecedent **A**, but different consequents B_1 and B_2 then, for an input that fully verifies the antecedent **A**, the model infers $B_1 \cup B_2$, meaning that both B_1 and B_2 are possible fuzzy sets in which the output can belong. As an illustrative example, in Fig. 3.13 an inference process is graphically depicted, which illustrates the non-existence of inconsistency when rules are interpreted in the conjunctive modality. It is interesting to compare such inference with that depicted in Fig. 3.12, where inconsistency degree is maximal.

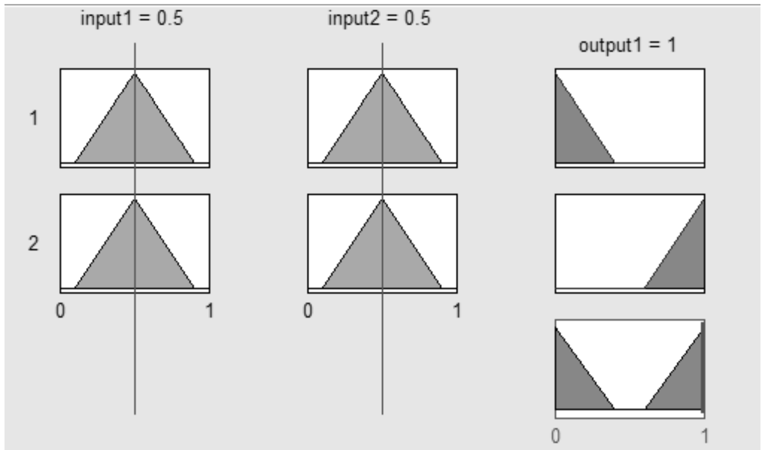

Fig. 3.13: Example of inference with rules in conjunctive interpretation. The inferred fuzzy set (bottom right) is normal.

3.6.7 Completeness

Constraint 27 (Completeness) *For each input, at least one rule ought to have non-zero activation strength:*

$$\forall \mathbf{x} \in \mathbf{U}, \exists R \in \mathbf{R} : \mu_R = (\mathbf{A}, B) \wedge \mu_{\mathbf{A}}(\mathbf{x}) > 0 \tag{3.25}$$

Constraint 28 (α-completeness) *For a given threshold $\alpha \in (0,1]$, and for each input, at least one rule must have activation strength greater or equal to α:*

$$\forall \mathbf{x} \in \mathbf{U}, \exists R \in \mathbf{R} : \mu_R = (\mathbf{A}, B) \wedge \mu_{\mathbf{A}}(\mathbf{x}) \geq \alpha \tag{3.26}$$

Completeness is a property of deductive systems that has been used in the context of Artificial Intelligence to indicate that the knowledge representation scheme can represent every entity within the intended domain. When applied to fuzzy models, this property informally states that the fuzzy model should be able to infer a proper action for every input [28, 42]. On the other hand, incompleteness is symptom of overfitting [37].

As stated in [68], a complete fuzzy model can achieve proper operation avoiding undesirable situations. Such undesirable situations can be easily understood by analyzing the behavior of the fuzzy model when no rules are activated by an actual input. Suppose that there exists an input vector that does not activate any rule in a knowledge base with conjunctive interpretation: the inference procedure yields an empty set that does not offer any possibility of decision. However, many defuzzification procedures always provide a numerical outcome even if the inferred fuzzy set is an empty set. This is necessary when an action is always required from the model (e.g., in control applications). In this case, however, the final outcome is arbitrary and is not a logical consequence of the knowledge base embodied in the model.

In the case of implicative interpretation, a completely opposite situation is likely to happen. Since material implication is always verified when the antecedent is false, the fuzzy set derived for each rule coincides with the entire output UoD U_O. The conjunctive aggregation of all rules coincides to the output UoD, hence the inferred fuzzy set gives full possibility to any element of the UoD and the defuzzification procedure can legitimately choose any random element from this universe. As a consequence, the behavior of the fuzzy model is undefined in input regions where no rules are fired. Undesirable situations may occur even if some rule is fired by the input, but the activation strength is too low. This phenomenon is also called "weak firing". For such reason, stronger α-completeness is preferred over simple completeness to guarantee a proper behavior of the fuzzy model [57].

Coverage (see Sect. 3.3.4) and completeness are tightly related, in the sense that completeness implies[11] coverage for all LVs. However, the contrary is not granted, i.e. if coverage is satisfied for all LVs, completeness may still be violated; this happens when some inputs satisfy a combination of linguistic terms that does not appear in any rule of the knowledge base. Nevertheless, all combinations of fuzzy sets lead to a combinatorial explosion of the number of rules, making the model impractical. Therefore, the selection of the smallest number of rules covering all plausible inputs is a matter of careful design.

If the rules do not cover the entire UoD, then it is assumed that data lie in a subset of it, which is eventually covered by the rules of the knowledge base. However, outliers may still be possible and they should be correctly detected and handled. In such a case, model completeness should be restored but, in order to avoid combinatorial rule explosion, two approaches may be used:

- Infinite support fuzzy sets can be adopted to define fuzzy partitions. In such case, all inputs of the UoD activate all rules. This approach is simple but may decrease the inference efficiency and can originate the weak-firing phenomenon;
- A "default rule" (or "completion meta-rule") can be introduced [45, 60, 75]. A default rule is a special rule that is activated only when no rules of the remaining knowledge base are fired. The default rule may be associated with a special output signaling "out of range operation" or an indecision status. Default rules may be crisp or fuzzy. In the latter case, the activation strength is defined as the unary complement of the maximum activation strength among all ordinary rules. Default fuzzy rules can be thus used to fight the weak-firing phenomenon but its semantics must be carefully defined due to its special role in the fuzzy model.

[11] If completeness is satisfied, then for each input $\mathbf{x}$ there exists a rule R whose antecedent $\mathbf{A}$ is satisfied with $\mu_{\mathbf{A}}(\mathbf{x}) > 0$. Since the antecedent $\mathbf{A}$ is defined by the conjunction of fuzzy sets (one for each feature), i.e. $\mathbf{A} = A_1 \times ... \times A_n$ then $0 < \mu_{\mathbf{A}}(\mathbf{x}) = \mu_{A_1}(x_1) \otimes ... \otimes \mu_{A_n}(x_n)$. But this implies that $\forall i : \mu_{A_i}(x_i) > 0$. Therefore, $\forall x_i, \exists A_i : \mu_{A_i}(x_i) > 0$, i.e. coverage is satisfied.

3.6.8 Number of Variables

Constraint 29 *The number of input and output variables ought to be as small as possible.*

The number of input/output variables are strictly related to the dimensionality of the input/output UoDs. A high number of input/output variables is negative in fuzzy modeling for several reasons, such as:

1. The length of each rule becomes too long, thus hampering interpretability. Unless sophisticated techniques are adopted to reduce the number of variables in each rule, a high number of variables – typically more than seven – reduces the readability of the knowledge base.
2. Fuzzy models are usually acquired from data by means of identification techniques. Like all data-driven models, fuzzy models are subjected to the so-called “curse of dimensionality” [6]. According to Bellman, the number of parameters needed for approximating a function to a prescribed degree of accuracy increases exponentially with the dimensionality of the input space [25]. In fuzzy models, such parameters correspond to the number of fuzzy sets and the number of fuzzy rules. The basic reason for the curse of dimensionality is that a function defined in high dimensional space is likely to be much more complex than a function defined in a lower dimensional space, and those complications are harder to discern [19].

According to the above-mentioned motivations, the number of input variables and that of output variables have a different effect on the model degradation. Specifically, while a high number of output variables hampers the interpretability of the model by lengthening the rules, a high number of input variables damages interpretability and causes the curse of dimensionality. For such reason, input and output variables are treated differently in the attempt of reducing their number.

The reduction of the number of input variables can improve the readability of the knowledge base and avoids the curse of dimensionality. Clearly, a drastic reduction of the number of input variables reduces the number of free parameters in the model and, consequently, the ability of identifying a large set of functions. For such reason, any technique to reduce the number of input variables must be balanced with the desired accuracy of the final model. In literature, there exists an impressive number of techniques to reduce the number of input variables. In Chap. [2] a discussion on variable reduction techniques is outlined.

In many application problems the number of output variables can be greater than one. In this cases, either a single Multiple Inputs Multiple Outputs (MIMO) FRBS or several Multiple Inputs Single Output (MISO) FRBSs can be used. It must be observed, however, that if several MISO systems are used to deal with multiple output variables, and these systems are tuned independently, then each linguistic term may have multiple interpretations (one for each MISO system) thus severely hampering interpretability.

3.6.9 Granular Output

Constraint 30 *The output of the fuzzy model should be an information granule rather than a single numerical value.*

In most cases, fuzzy models are used to identify an unknown input/output mapping defined on numerical domains. Even though information is processed within the model by means of fuzzy sets manipulation, often the required outcome is a single numerical value that is used for decision making, classification, control, etc. In other cases, the outcome of the fuzzy model is just an information for further processing (e.g., in medical diagnosis, the outcome of the fuzzy model can be used by physicians to take a proper decision). In this case, it is favorable that the output of the fuzzy model is an information granule rather than a single numerical value. Indeed, granular information is richer than numerical information because it conveys uncertainty information that is valuable in a decision making process.

A first approach to provide granulated information is to avoid the defuzzification procedure and just return the resulting fuzzy set attained by aggregating the outcomes inferred by each rule. Such fuzzy set provides uncertainty information about the model output but can be hard to interpret because it generally does not satisfy many (if not all) interpretability constraints for fuzzy sets. An alternative approach consists in enriching the numerical outcome provided by the defuzzifier with uncertainty information about the behavior simulated by the fuzzy model. This new information can be well represented by means of prediction intervals. In [48] a method is described for integrating prediction intervals within fuzzy models.

3.7 Constraints and Criteria for Fuzzy Model Adaptation

Fuzzy information granules are often used as building blocks for designing fuzzy rule-based models, which can be effectively used to solve soft-computing problems such as predictions, classifications, system identification, etc. From a functional point of view, fuzzy models are approximate representations of an underlying functional input/output relationship that has to be discovered. To achieve such goal, often a dataset is available, which provides a finite collection of input/output examples from which the model's knowledge has to be built. Information granulation techniques operate on such dataset to properly extract information granules that will be used to define the knowledge base. In such context, information granulation can be accomplished in two main ways:

1. The complete dataset of examples is granulated, involving both input and output information. At the end of the granulation process, the knowledge base of the fuzzy model is complete and can be immediately used for approximating the underlying functional relationship.
2. The examples are split in the input and output components, and the granulation process is executed only in the input part. This choice is typical in classification

problems, where outputs are class labels that belong to a symbolic domain that cannot be granulated. This form of granulation is also chosen when accuracy has a great impact in the assessment of the model quality.

In most cases, the initial granulation of data—if properly designed—leads to a knowledge base that is interpretable but not very accurate. Therefore the knowledge base of the fuzzy model has to be *refined* (or adapted) in order to better approximate the unknown functional relationship by means of supervised learning schemes.

In an extreme synthesis, the adaptation of a fuzzy model can be viewed as a transformation

$$\mathfrak{L}(\mathscr{M}, D) = \mathscr{M}' \tag{3.27}$$

that maps an original model $\mathscr{M}$ into a new model $\mathscr{M}'$ with the support of a finite set of examples D. The transformation is aimed at optimizing an objective function $\wp$ such that

$$\wp\left(\mathscr{M}'\right) \geq \wp(\mathscr{M}) \tag{3.28}$$

The objective function may be a measure of accuracy, generalization ability, interpretability, etc. or a combination of metrics (see [74] for a deep study of supervised learning). Even though it is not the general case, it is assumed hereafter that the adaptation process does not modify the structure of the fuzzy partitions involved in the model, i.e. their number, their cardinality and labeling. What is usually changed by the learning process are the free parameters that characterize input and output fuzzy sets. In some cases, the information granules of the rules are changed by removing useless soft-constraints. The rule structure is admitted to change, e.g. by removing useless rules or adding new ones. With a slight abuse of notation, hereafter each element χ belonging to the model (especially fuzzy sets) that is subject to adaptation by means of $\mathfrak{L}$ will be indicated with $\mathfrak{L}\chi$.

There are several methods to adapt a model to data, such as genetic algorithms, [18, 23, 27, 30, 33, 51] and neuro-fuzzy networks [14, 32, 54]. In interpretable fuzzy modeling, the adaptation process should preserve interpretability constraints, i.e. the adapted model $\mathscr{M}'$ should satisfy at least the same interpretability constraints of the original model $\mathscr{M}$. More constraints can be added if the objective of adaptation is interpretability improvement. If – as often occurs – the adaptation process is iterative, the intermediate models emerging after each iteration may not satisfy the interpretability constraints. This is usually the case when adaptation is carried out by means of regularized learning [9] or when interpretability constraints are applied after several adaptation stages for efficiency pursuits [4]. However, when the adaptation process can be halted at every stage (e.g., to allow user interaction or in certain real-time application contexts) it is more desirable that all intermediate models satisfy interpretability constraints [53].

In addition to the previously discussed interpretability constraints, the adaptation procedure should be designed so as to not twisting the semantics of the fuzzy sets involved in the knowledge base. Indeed, the adaptation of the model has the role of refining without completely modifying the meaning conveyed by the knowledge base. For such reason, additional constraints could be required on the adaptation process for interpretable fuzzy modeling.

3.7.1 No Position Exchange

Constraint 31 *Two fuzzy sets belonging to the same fuzzy partition should not exchange position after learning.*

This criterion, elicited by Nauck and Kruse, assures that the learning process does not switch the meaning of two linguistic terms in an LV [55]. This is especially important when linguistic terms express qualities on the UoD, like LOW, MEDIUM and HIGH. It would be indeed undesirable that, after learning, fuzzy sets associated to these terms exchange their position such that the final order does not reflect the implicit semantics of the labels, e.g. when LOW $\prec$ HIGH $\prec$ MEDIUM. Note that a re-labeling operation is possible so as to restore the semantic ordering of linguistic labels, but this operation is against the aims of model adaptation, according to which the model should be only finely tuned to better meet the objective function. In such a viewpoint, any re-labeling operation would be seen as a model re-design.

When linguistic terms express vague quantities (like ABOUT_2) or precise quantities (as in the consequents of TSK rules), position exchange is admissible provided that the label changes in agreement to the fuzzy sets. In such case, re-labeling is an admissible operation because the fuzzy sets express quantities that are susceptible to changes during the tuning process.

3.7.2 No Sign Change

Constraint 32 *Fuzzy sets with cores in positive (resp. negative) range should not change sign after learning.*

This criterion is mostly related to fuzzy sets used to express imprecise quantities [53, 55]. Informally speaking, the criterion states that if a fuzzy set represents a signed quantity (either positive or negative), then the corresponding fuzzy set resulting from adaptation should represent a quantity with the same sign. In this way, the learning process $\mathfrak{L}$ does not drastically change the semantics of the knowledge base during fine-tuning. Note that if a fuzzy set A is unimodal and satisfies the "natural positioning" constraint on 0, i.e. core$A = \{0\}$, then "no sign change" requires that, after adaptation, the refined fuzzy set still represents the number 0, i.e. core$\mathfrak{L}A = \{0\}$. In general, if "natural positioning" requires that an element s belongs to the core of a fuzzy set, then it should be included in the core of the fuzzy set even after adaptation.

3.7.3 Similarity Preservation

Constraint 33 *Any adapted fuzzy set should be similar to the original.*

This criterion informally states that fuzzy sets subject of adaptation should not change too much in order to preserve their original semantics. The formalization of the criterion is a generalization of the work of Lotfi, who uses a rough-fuzzy representation of Gaussian membership functions to restrict the variability of parameters (namely, center and width) into a predefined range [44]. This constraint captures the very nature of fine tuning, i.e. adaptation of the model without excessive modification of the internal knowledge. On the other hand, it should be observed that if the adaptation process leads to a strong modification of the model, then a bad initial design of the model could be hypothesized. In such a case, a complete re-design of the model should be considered.

3.8 Final Remarks

Along this chapter, a number of interpretability constraints and criteria have been outlined. Some of them are fundamental for interpretable fuzzy modeling, while others can be included or left apart from a design strategy according to the evaluation of the designer. Furthermore, the application of some interpretability constraints depends on the final representation of the information granules, and, in general, of the knowledge base. As an example, some interpretability constraints make sense only for information granules representing qualities on the UoD, while other constraints are especially suited for information granules representing fuzzy quantities.

Additional interpretability constraints can be added for specific application needs. As an example, in [38] a number of special-purpose interpretability constraints have been proposed for system identification applications. In summary, on the engineering standpoint, a number of issues should be taken into account in the selection of interpretability constraints, including the type of representation of information granules (e.g., qualitative vs. quantitative), the type of fuzzy model (e.g., Mamdani vs. TSK), the objectives of modeling (e.g., accuracy, efficiency, interpretability), the semantic assumptions (e.g., implicative vs. conjunctive rules) and, last but not least, the application context.

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Chapter 4
Revisiting Indexes for Assessing Interpretability of Fuzzy Systems

Abstract Interpretability is one of the most valuable properties of fuzzy systems. Despite the effort made by the research community for characterizing interpretability, there is not a consensus about how to measure interpretability yet. It is admitted that the analysis of interpretability is subjective because it depends on the background of the person who makes the assessment. Accordingly, an index flexible enough to fit with preferences and expectations would be appreciated from the designer but also from the user viewpoints. Hence, subjective indexes become essential for customization purposes. However, they are not enough for measuring interpretability in the broad sense. There is also a need of objective indexes to make feasible fair comparisons among different fuzzy systems designed for solving a given problem. Thus, it is necessary to look for two kinds of complementary indexes, subjective and objective ones. Moreover, they should tackle with interpretability from readability and comprehensibility viewpoints. They also must consider both structural and semantic interpretability properties. In this chapter, we give an overview on existing indexes for assessing all aspects related to interpretability of fuzzy systems. In addition, we present a conceptual framework for assessing interpretability with a customizable index which combines subjective and objective ones.

4.1 Introduction

The expressiveness of linguistic rules is acknowledged to be quite close to natural language [37]. This fact favors the interpretability of fuzzy systems because human understanding is grounded by natural language. Accordingly, humans manage naturally concepts that are verbalized with words. Thus, it would be a good thing if fuzzy researchers looked for synergies with the closely related field of computational linguistics. The interested reader is kindly referred to [21] for farther details with respect to natural language processing and generation.

Drawing an analogy between a set of fuzzy rules and a set of sentences in natural language, interpretability should be assessed regarding both *syntax* and *semantics*.

On the one hand, syntax can be defined as the “arrangement of words in sentences, clauses, and phrases, and the study of the formation of sentences and the relationship of their component parts”. On the other hand, semantics refers to the “philosophical and scientific study of meaning in natural and artificial languages”[1]. In the context of fuzzy systems, syntax is related to the system structure, i.e., it involves the analysis of dimension and complexity of the system. Thus, syntax covers single linguistic fuzzy rules but also the whole rule base. Semantics in turn deals with the meaning of fuzzy sets and the related linguistic propositions.

Regarding the assessment of text quality, Paul Grice established the next four conversational maxims which arise from the pragmatics of natural language [22]:

i) Maxim of Quality: do not say either what you believe to be false or anything without adequate evidence.
ii) Maxim of Quantity: make your contribution as informative as required for the current purposes of the exchange.
iii) Maxim of Relation: be relevant.
iv) Maxim of Manner: avoid obscurity of expression and ambiguity. Be brief and orderly.

Keeping the Grice’s maxims in mind when designing fuzzy systems can help to make easier their understanding. The rule base must be coherent avoiding the use of inconsistent rules (Maxim of Quality), redundant rules (Maxim of Quantity), and ambiguous rules (Maxim of Manner). In addition, selecting the most relevant rules (Maxim of Relation) will yield more compact and robust systems.

Section 4.2 briefly introduces all elements to take into account when characterizing and assessing interpretability. These elements are organized in accordance with the structural and semantic constraints and criteria described in Chap. [3]. Sect. 4.3 presents a taxonomy which groups most interpretability indexes in the scientific literature. Then, Sect. 4.4 sketches how to adapt interpretability indexes to the context and preferences of users. Finally, Sect. 4.5 summarizes the main conclusions and provides the reader with final remarks.

4.2 Structural versus Semantic Interpretability

The quality of fuzzy systems is usually measured in terms of accuracy. However, the main advantage of interpretable fuzzy systems is their ability to be understood or accounted for humans. Accordingly, the evaluation of interpretability is not straightforward because it is inherently subjective by nature. Anyway, interpretability indexes are aimed to be as objective as possible because only objective indexes allow fair comparison among different systems.

The comprehensibility of a Fuzzy Rule-Based System (FRBS) depends on the readability of all its components. As thoroughly discussed in Chap. [3], setting in-

[1] The definitions of *syntax* and *semantics* are taken from the Encyclopedia Britannica, which was accessed by April 12, 2020 (`http://www.britannica.com`).

terpretability constraints is required to build interpretable FRBSs. Notice that comprehensibility depends on the knowledge base transparency but also on the inference mechanism understanding. There are also some crucial psychological factors. For example, some people may think the most interpretable systems are those they are used to work with, disregarding their complexity.

Zhou and Gan [63] established a two-level taxonomy regarding interpretability issues. They distinguished between low level (also called fuzzy set level) and high level (or fuzzy rule level). This taxonomy was extended by Alonso et al. [7] who introduced a conceptual framework for characterizing interpretability. They considered both fuzzy partitions and fuzzy rules at several abstraction levels. Moreover, Mencar et al. [40] remarked the need to distinguish between readability (related to structural issues) and comprehensibility (related to semantic issues).

Here, we consider two main viewpoints (description and explanation) when studying interpretable fuzzy systems. Fig. 4.1 shows a schematic diagram with the main factors to keep in mind when analyzing interpretability of FRBSs. The left hand side of the figure is inspired by the two-level taxonomy introduced in [63]. However, we have added more than two structural levels where we include all the interpretability constraints and criteria described in Chap. [3].

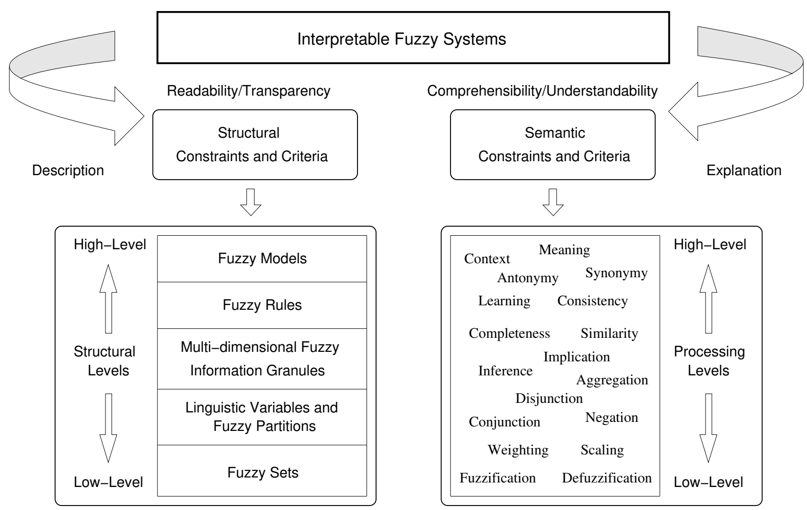

Fig. 4.1: Conceptual framework for characterizing interpretability.

On the one hand, the system description considers the system as a whole, i.e., paying attention to its global behavior and trend. On the other hand, the system explanation analyzes each individual situation with the aim of explaining locally specific behaviors for specific events. For example, if we had a fuzzy controller for driving a car, its description would give an idea on the kind of operations it can do (e.g., go straight forward, turn on the right/left, speed up, brake, etc.) and even on the driver style (e.g., aggressive, sluggish, etc.). In addition, a local explanation would give details about a specific maneuver. It is worth noting that both description (global readability/transparency) and explanation (local comprehensibility/understandability) viewpoints may guide the design of interpretable fuzzy systems while pushing towards contradictory goals. Giving priority to description is likely to yield rules as compact as possible. However, maximizing explanation ability requires minimizing rule overlapping because rule interaction is difficult to explain. In fact, the more general rules are, the more rules are likely to be fired at the same time; thus making harder their interpretation and related explanation.

4.3 Taxonomy of Interpretability Indexes

Gacto et al. [19] proposed a double axis taxonomy regarding semantic and structural properties of fuzzy systems, at both partition and rule base levels. Accordingly, they pointed out four groups of indexes.

Here, we introduce a new taxonomy which extends the one proposed by Gacto et al. As it can be appreciated in Fig. 4.2, we keep the distinction between structural-based and semantic-based interpretability. In addition, we extend the number of structural levels which passes from two (fuzzy partitions and rules) to five (fuzzy sets, partitions, information granules, rules and models) as depicted in Fig. 4.1. As a result, our taxonomy is made up of ten groups of indexes (G1 to G10).

	Fuzzy Sets	Fuzzy Partitions	Fuzzy Granules	Fuzzy Rules	Fuzzy Models
Structural−based Interpretability	G1	G3	G5	G7	G9
Semantic−based Interpretability	G2	G4	G6	G8	G10

Fig. 4.2: Taxonomy of interpretability indexes.

The rest of this section gives an overview of some of the most popular interpretability indexes that are available in the literature. They are categorized and presented in accordance with the taxonomy depicted in Fig. 4.2. Although we are aware the border among some categories is blurry, we did the effort to assign each index to the best fitted group in the taxonomy. Anyway, we recognize that some of the most complex indexes involve several metrics at different levels and they may have been assigned to more than one group.

4.3.1 Structural-based Interpretability at Fuzzy Sets Level

These indexes are aimed at evaluating the structural complexity of a fuzzy set in terms of the underlying membership function (MF) μ_A. In practice, the shape of μ_A is characterized by a set of parameters. The more complex the fuzzy set, the larger the number of parameters to define. For example, a triangular μ_A is defined by three parameters (a, b, c), as depicted in Fig. 4.3. Such parameters may be tuned in order to build customized MFs which will be defined by new parameters (a', b', c'). As described in [23], there are many clustering methods (e.g., fuzzy c-means [11] or support vector clustering [36]) that can be applied to learn MFs from data, i.e., to automatically determine the most suitable parameters of μ_A according to data.

The usual way to measure the structural complexity of a fuzzy set consists of just counting the number of parameters in the related μ_A. As far as we know, there are no other more formal indexes in the literature ready to evaluate this level of interpretability.

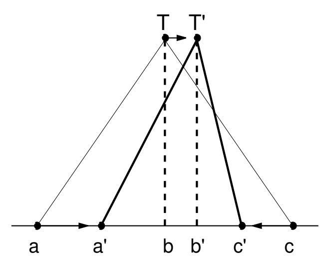

Fig. 4.3: Refinement of the shape of a triangular MF by tuning its parameters.

4.3.2 Semantic-based Interpretability at Fuzzy Sets Level

The meaning of a fuzzy set is directly related to the use of the associated linguistic term in a specific context. Of course, this meaning must be in accordance with the mathematical definition of the underlying μ_A.

Even if fuzzy sets can be automatically extracted from data, elicitation of meaningful fuzzy sets is still a matter of research. In [59], four different methods were analyzed for membership acquisition and fuzzy sets construction. However, human validation was missing. In [52], fuzzy sets representing human perceptions were automatically generated from empirical data that were previously collected through fuzzy rating scale-based questionnaires. Again, there was a lack of semantic grounding when analyzing the meaning associated to each fuzzy set.

To sum up, as far as we know, there are no indexes in the literature ready to evaluate this level of interpretability. Notice that evaluating the meaningfulness of a fuzzy set is not straightforward because it strongly depends on the background of the person who is due to interpret and understand it. In practice, these indexes evaluate the degree of fulfillment of interpretability constraints (e.g., normality, continuity, convexity and unimodality) at fuzzy sets level. See Sect. 3.2 in Chap. [3] for additional details about how these constraints are defined.

4.3.3 Structural-based Interpretability at Fuzzy Partitions Level

These indexes are usually considered to preserve the readability of fuzzy partitions automatically generated from data. They consist of mathematical formulas which evaluate the degree of fulfillment of interpretability constraints (such as justifiable number of elements) at fuzzy partition level. The simplest index just counts the number of MFs defined for each single variable. Nevertheless, it is worth noting that there are other indexes which consider the relationship among fuzzy sets in a given partition. As described in the Sect. 3.3 of Chap. [3], additional constraints (e.g., distinguishability, completeness or complementarity) must be taken into account.

Setnes et al. [58] proposed the use of similarity measures to merge fuzzy sets that were so close that they were likely to produce almost the same rules. Similarity measures can be divided into two groups: (1) geometric similarity measures and (2) set-theoretic similarity measures. Geometric measures are better for disjoint fuzzy sets while set-theoretic measures are better for overlapping fuzzy sets.

If we have a distance measure, *Dm* (e.g., Minkowsky or Hausdorff distance), we can define a geometric similarity measure as follows:

$$Sm(A,B) = \frac{1}{1+Dm(A,B)} \tag{4.1}$$

This definition slightly changes in the case of set-theoretic similarity measures:

$$Sm(A,B) = \frac{area(A \otimes B)}{area(A \oplus B)} \quad \text{with} \quad area(A) = \int_{-\infty}^{\infty} A(x)\,dx \tag{4.2}$$

Guillaume and Charnomordic [24, 25] also proposed merging fuzzy sets with the aim of building interpretable fuzzy partitions with a moderate number of MFs. They focused on classification problems and, given a fuzzy partition with f MFs, defined the following distance function:

$$DD^f = \frac{1}{N(N-1)} \sum_{q,r=1,2,\ldots,N;q\neq r} Dm(x_q, x_r) \tag{4.3}$$

where N is the number of training data and the pairwise distance $Dm(x_q, x_r)$ considers the membership degree of two given data points (x_q and x_r) by combining the respective parts of internal and external distances.

The *internal distance* (Dm_{int}^{A}) regards a single fuzzy set A and it is computed as the difference between the two membership degrees:

$$Dm_{int}^{A}(x_q, x_r) = \left|\mu_A(x_q) - \mu_A(x_r)\right| \tag{4.4}$$

The *external distance* ($Dm_{ext}^{A,B}$) combines internal and prototype distances. It takes into account the point location within the fuzzy set and the relative position of the fuzzy set in the fuzzy partition. The external distance between two points which belong to the A and B fuzzy sets respectively is:

$$Dm_{ext}^{A,B}(x_q, x_r) = \left|\mu_A(x_q) - \mu_B(x_r)\right| + Dm_{prot}^{A,B} + CF \tag{4.5}$$

where CF is a constant correction factor, which ensures that the external distance is always superior to any internal distance, and $Dm_{prot}^{A,B}$ is the prototype (numerical or symbolic) distance between the centers of fuzzy sets A and B. The prototype numerical distance can be computed as follows:

$$Dm_{prot}^{A,B} = \sqrt{(c_A - c_B)^2} \tag{4.6}$$

where we consider the Euclidean distance between the kernel locations (c_A and c_B) of fuzzy sets A and B. In addition, the prototype symbolic distance is computed as follows:

$$Dm_{prot}^{A,B} = \frac{i_A - i_B}{f - 1} \tag{4.7}$$

where f is the partition size, i.e., the number of fuzzy sets in the partition. Moreover, i_A and i_B are the indexes of fuzzy sets A and B sorted in ascending order with respect to their kernels.

It is worth noting that $Dm^{A,B}(x_q, x_r)$ is an internal distance if $A = B$ or an external distance otherwise. Thus, $Dm(x_q, x_r)$ is computed as follows:

$$Dm(x_q, x_r) = \frac{1}{\sum_{i=1}^{f} \mu_{A_i}(x_q)} \cdot \sum_{i=1}^{f} \left| \frac{\mu_{A_i}(x_q)}{\sum_{k=1}^{f} \mu_{B_k}(x_r)} \cdot \sum_{k=1}^{f} \left[\mu_{B_k}(x_r)\, Dm^{A_i,B_k}(q,r)\right] \right| \tag{4.8}$$

4.3.4 Semantic-based Interpretability at Fuzzy Partitions Level

Fazendeiro and Oliveira [16] analyzed the relationship between accuracy and semantics in fuzzy modeling and fuzzy control. They paid special attention to interpretability in terms of expressiveness. They stated that the essence of *semantic integrity* of fuzzy partitions[2] is captured through the following fundamental requirements:

(Req. 1) natural zero positioning;
(Req. 2) distinguishability, i.e., limited overlap between neighboring fuzzy sets;
(Req. 3) coverage of the Universe of Discourse (UoD);
(Req. 4) unimodality of fuzzy sets.

Later, Fazendeiro et al. [17] added a fifth requirement (Req. 5) that is related to the number of fuzzy sets in a given partition. To evaluate all these five requirements, the next index is computed [16]:

$$J = \sum_i K_i \cdot J_i \tag{4.9}$$

where K_i is a constant value to weigh the relevance of J_i:

- J_1 promotes the natural zero positioning (Req. 1);
- J_2 penalizes MFs with a poor distinguishability level (Req. 2);
- J_3 penalizes MFs with a low level of coverage of the UoD (Req. 3).

In addition, they imposed the use of normal unimodal MFs to satisfy (Req. 4) and a justifiable number of fuzzy sets to satisfy (Req. 5). Accordingly, J_i is computed as follows:

$$J_1 = |c_{ZE}|^2 \tag{4.10}$$

$$J_2 = \left[(M_p(\mathscr{L}_x) - 1)^2 \cdot step\,(M_p(\mathscr{L}_x) - 1) \right] \tag{4.11}$$

$$J_3 = \left[(M_p(\mathscr{L}_x) - \varepsilon)^2 \cdot step\,(\varepsilon - M_p(\mathscr{L}_x)) \right] \tag{4.12}$$

where c_{ZE} denotes the core of the fuzzy set associated to the linguistic term ZERO; *step* is the standard unit step function; $\mathscr{L}_x$ is a fuzzy set representing a real-valued sample from the UoD; and $M_p(\mathscr{L}_x)$ is the *sigma-count operator* which gives a measure of the cardinality of a fuzzy set $\mathscr{L}_x$, as follows:

$$M_p(\mathscr{L}_x) = \sqrt[p]{\mu_1^p + \cdots + \mu_f^p} \tag{4.13}$$

being μ_i $(i = 1, \ldots, f)$ the membership degree of x to the i-th linguistic term. In addition, p is a positive integer.

Mencar et al. [39] also remarked the importance of assuring *distinguishability* among fuzzy sets (especially when they are tagged with natural language terms to represent human-oriented concepts) in a given fuzzy partition. They proposed to handle distinguishability between two fuzzy sets A and B, with a *possibility measure* which is defined as the degree of applicability of the soft constraint "X is B" for $X = A$. Possibility is evaluated according to the following definition:

[2] The interested reader is kindly referred to Sect. 3.3 in Chap. [3] for further details on semantic integrity of fuzzy partitions.

$$\mathrm{Poss}(A,B) = \sup_{x \in U} \min\{\mu_A(x), \mu_B(x)\} \tag{4.14}$$

Then, distinguishability can be formally defined as the complement of the equation given above, that must not be less than a threshold δ:

$$J_D(A,B) = 1 - \mathrm{Poss}(A,B) \geq \delta \tag{4.15}$$

Of course, distinguishability can be formalized in different ways. The most acceptable formalization is by means of *similarity* measures (i.e., considering similarity as the opposite to distinguishability). Commonly, similarity is interpreted as the degree to which two fuzzy sets are equal [64].

Similarity measures characterize the notion of distinguishability, but their calculation is often computationally intensive. As a consequence, many strategies that adopt similarity in the context of assessing interpretability are based on evolutionary algorithms such as genetic algorithms [38, 55, 58], evolution strategies [31], symbiotic evolution [29], coevolution [50], multi-objective genetic optimization [30], etc. Alternatively, distinguishability improvement is realized in separate learning stages, often after some data driven procedure like clustering, in which similar fuzzy sets are merged together [45, 58]. Notice that it is common to find the use of (4.14) to evaluate distinguishability in case of considering neural learning approaches [12, 15, 44].

In addition to similarity and distinguishability, the property of α-completeness (and, dually, α-incompleteness) of a linguistic variable can be quantitatively assessed by measuring the ratio of the UoD U that is covered by the variable [38]. More precisely, given a set of linguistic terms T and the following subset of U:

$$U_X^{\alpha} = \left\{ x \in U : \max_{A \in T} \mu_A(x) \geq \alpha \right\} \tag{4.16}$$

Completeness Degree (CD_α) and its complementary Incompleteness Degree (ID_α) are defined as follows:

$$CD_\alpha = \frac{\int_{U_X^{\alpha}} dx}{\int_U dx} \tag{4.17}$$

$$ID_\alpha = 1 - CD_\alpha \tag{4.18}$$

When working with triangular MFs, 0.5-completeness is guaranteed by a proper placement of MF parameters [15, 25, 34]. More precisely, given two fuzzy sets of triangular MFs (with $i \in \{1,2\}$):

$$\mu[a_i, b_i, c_i](x) = \max\left\{0, \min\left\{\frac{x - a_i}{b_i - a_i}, \frac{x - c_i}{b_i - c}\right\}\right\} \tag{4.19}$$

with $b_1 < b_2$, 0.5-completeness in the interval $[b_1, b_2]$ is guaranteed if $a_2 = b_1$ and $b_2 = c_1$. Thus, 0.5-completeness is satisfied by any linguistic variable made of triangular $\mu[a_i, b_i, c_i]$ where, for $i = 1, 2, \ldots, f-1$:

$$a_{i+1} = b_i, \tag{4.20}$$
$$b_{i+1} = c_i, \tag{4.21}$$
$$b_1 = a_1 = m_U, \tag{4.22}$$
$$b_f = c_f = M_U \tag{4.23}$$

This produces a strong fuzzy partition and it can be easily extended to trapezoidal MFs by transforming the punctual core (b_i) into an interval $\left[b_i^l, b_i^u\right]$. The same can be accomplished when considering Gaussian MFs (see Fig. 4.4):

$$\mu_i(x) = \exp\left(-\frac{(x-\omega_i)^2}{2\sigma_i^2}\right), i = 1, 2, \ldots, f \tag{4.24}$$

If there exist $f-1$ values $t_1 < t_2 < \ldots < t_{f-1}$ in U such that

$$\omega_i = \frac{t_{i-1} + t_i}{2} \tag{4.25}$$

and

$$\sigma_i = \frac{t_i - t_{i-1}}{2\sqrt{-2\ln\alpha}} \tag{4.26}$$

with $t_0 = 2m_U - t_1$ and $t_f = 2M_U - t_{f-1}$, then α-completeness is verified. Indeed, it is easy to see that $\mu_i(t_i) = \mu_{i+1}(t_i) = \alpha$ for $i = 1, 2, \ldots, f-1$; therefore, t_i is the intersection point of the i-th and $(i+1)$-th MFs. Since Gaussian MFs are strictly convex, for each x between ω_i and t_i the membership degree of x to the i-th MF is never smaller than α, while between t_i and ω_{i+1} the membership degree of x to the $(i+1)$ MF is never smaller than α. By observing that $\omega_1 = m_U$ and $\omega_n = M_U$, we derive that for each element in U there exists at least one MF for which the membership degree is never smaller than α. An example is depicted in Fig. 4.4 for four MFs and $\alpha = 0.5$.

In addition, Gacto et al. [18] proposed an interpretability index which combined some structural (see Sect. 4.3.3) and semantic (see Sect. 4.3.4) issues at fuzzy partitions level. Authors distinguished between what they called high-level interpretability (i.e., complexity-based interpretability) and low-level interpretability (i.e., semantic-based interpretability). In addition, they defined an interpretability index which was validated in a framework where whole fuzzy models were built (in this sense, this index may also apply to Sect. 4.3.9 and Sect. 4.3.10). Moreover, this index was used to preserve interpretability while tuning fuzzy partitions with the aim of increasing accuracy. Thus, interpretability of tuned partitions is measured in terms of their deviation with respect to the original partitions that are taken as reference and considered as the most interpretable ones. Such original partitions are either defined by an expert or automatically generated as uniform partitions in the UoD of each variable. Namely, the semantic-based interpretability index (G_{M3M}) is defined as the geometric mean of three measures:

$$G_{M3M} = \sqrt[3]{\delta\gamma\rho},\ G_{M3M} \in [0,1] \tag{4.27}$$

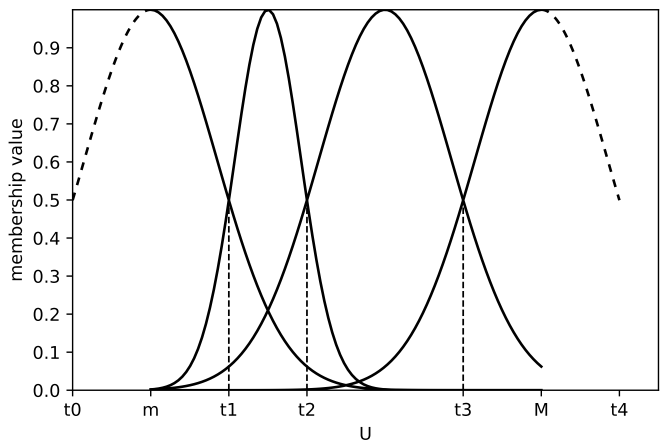

Fig. 4.4: A fuzzy partition with four Gaussian MFs satisfying 0.5-completeness.

The three single measures are defined as follows:

(1) MFs displacement (δ): this metric measures the proximity of the anchor points of the displaced triangular MFs ($\mu[a',b',c']$) to the original ones ($\mu[a,b,c]$). For each MF_j in a given partition, $\delta_j = |b_j - b'_j|/I$, where $I = (I^r_{b_j} - I^l_{b_j})/2$ represents the maximum variation for each parameter. Thus, δ^* is defined as $\max_j \{\delta_j\} \in [0,1]$. Accordingly, values near one show that the MFs present a great displacement. The following transformation is made so that this metric represents proximity (maximization):

$$\delta = 1 - \delta^* \tag{4.28}$$

(2) MFs lateral amplitude rate (γ): this metric measures the left/right rate differences between the displaced and the original MFs. The closer the rates are, the higher the value of γ. Let us define *left* $S_j = |a_j - b_j|$ as the amplitude of the left part of the original MF support and *right* $S_j = |b_j - c_j|$ as the right-part amplitude. Let us define *left* $S'_j = |a'_j - b'_j|$ and *right* $S'_j = |b'_j - c'_j|$ as the corresponding left and right parts in the tuned MFs. The value of γ_j for the j-th MF is computed as follows:

$$\gamma_j = \frac{\min\left\{|a_j - b_j|/|b_j - c_j|,\ |a'_j - b'_j|/|b'_j - c'_j|\right\}}{\max\left\{|a_j - b_j|/|b_j - c_j|,\ |a'_j - b'_j|/|b'_j - c'_j|\right\}} \tag{4.29}$$

$$\gamma_j = \frac{\min\{\mathit{left}\ S_j/\mathit{right}\ S_j,\ \mathit{left}\ S'_j/\mathit{right}\ S'_j\}}{\max\{\mathit{left}\ S_j/\mathit{right}\ S_j,\ \mathit{left}\ S'_j/\mathit{right}\ S'_j\}} \tag{4.30}$$

Values near one mean that the left and right rates are very close in both original and tuned MFs. Then, γ is calculated as the minimum value of all γ_j:

$$\gamma = \min_j \gamma_j \tag{4.31}$$

This metric can be extended to trapezoidal MFs by considering the middle of the core as the anchor point; then computing γ_j with the core and support extremes.

(3) MFs area similarity (ρ): this metric measures the area similarity between the displaced and original MFs. It should be higher if both areas are close. Let us define $area(A_j)$ as the area of the triangle that represents the original j-th MF and $area(A'_j)$ as the area of the tuned MF. The variable ρ_j is computed as follows:

$$\rho_j = \frac{\min\{area(A_j),\, area(A'_j)\}}{\max\{area(A_j),\, area(A'_j)\}} \tag{4.32}$$

The ρ metric is computed as the minimum value of all ρ_j:

$$\rho = \min_j \rho_j \tag{4.33}$$

4.3.5 Structural-based Interpretability at Fuzzy Granules Level

Fuzzy information granules act at the level of linguistic propositions in fuzzy rules. Thus, their analysis lays in the middle between partitions and rules. Therefore, there are not many indexes regarding structural-based interpretability of fuzzy information granules.

Alonso et al. [7] proposed to measure the percentage of input variables used in the premise part of each rule. This is related to the description length of information granules (see Sect. 3.4 in Chap. [3]).

As detailed in Chap. [2], feature selection techniques extract a small subset of the original input variables without degrading too much the accuracy of fuzzy models. The loss of accuracy is due to the fact that some variables are not used by the model anymore. The selection of the most relevant variables can be done by a human expert or by means of statistical methods, based on analyzing the correlation existing between the available variables, or combinatorial methods analyzing the influence of different subsets of variables.

It is worth noting that some of the variables may be relevant for some specific fuzzy information granules but not for all of them. Thus, before making a global feature selection, i.e., for the whole fuzzy model, it makes sense to explore a local feature selection for each single fuzzy information granule. Anyway, this remains an open research problem and more work is required by the community of researchers in Fuzzy Logic.

4.3.6 Semantic-based Interpretability at Fuzzy Granules Level

Again, there are not many indexes regarding semantic-based interpretability of fuzzy information granules. As explained in Chap. [3], semantic of fuzzy information granules should be first analyzed in terms of attribute correlation, i.e., minimizing duplicities or repetitions.

Interpretability analysis at this level regards on how readable and meaningful linguistic propositions associated to fuzzy information granules are. Accordingly, Alonso et al. [7] proposed to distinguish between three types of linguistic terms (i.e., elementary labels[3], OR labels, NOT labels) that may be used to describe fuzzy information granules. In the examples below, let's suppose we have previously designed a fuzzy partition with three elementary labels (LOW, MEDIUM, HIGH).

- Elementary labels are linguistic terms tagged to single fuzzy sets in a given fuzzy partition. For example, if only LOW is considered in one of the fuzzy information granules, then the percentage of elementary labels used in the rule base would be 33.33%. A low percentage of elementary labels means there are many linguistic terms that were defined but not actually used by any fuzzy information granule in the rule base. Hence, expert knowledge is not well aligned with the available data, what denotes something was wrong in the design process. Notice that usually experts provide general knowledge aimed at covering all possible cases while data correspond to specific cases. Thus, some cases considered important by experts may be not well represented by empirical data. It is quite common that experts define more linguistic terms than those really needed in the light of specific data. This may denote that gathering more data is required to properly tune the fuzzy model. Of course, it is also possible that new linguistic terms are required to cover unknown cases, i.e., those initially ignored by experts but becoming relevant in the light of empirical data.
- OR labels represent composed linguistic terms that result of mixing elementary labels (e.g., LOW OR MEDIUM).
- NOT labels can be seen as a special kind of OR composed linguistic terms. For example, NOT(LOW) is equivalent to MEDIUM OR HIGH. In addition, NOT(MEDIUM) is equivalent to SMALLER THAN MEDIUM OR BIGGER THAN MEDIUM.
  The use of OR and NOT labels makes easier to describe compact granules. However, they may be misleading and harder to interpret than elementary labels for some readers.

Guillaume and Magdalena [26] analyzed the properties to be satisfy by OR and NOT labels with the aim of becoming meaningful linguistic level. Namely, they proposed the creation of convex hulls covering all elementary labels involved in the definitions of OR/NOT labels.

In addition, Garcia et al. [20] proposed a genetic method for feature construction. They proved experimentally that a fuzzy information granule built as the combina-

[3] Elementary labels are also known as *primary terms* (see Sect. 5.3.2.3 in Chap. [2]).

tion of several input variables can provide more information than each single isolated variable. However, it is not straightforward to associate a meaning to synthetic variables. In order to guarantee that the synthetic variables are meaningful, the generation procedure must preserve several constraints. Accordingly, only meaningful functions (e.g., sum or product) can be applied over sets of given input variables in order to create new unknown but meaningful composed ones. Thus, two fuzzy information granules are combined into only one fuzzy information granule if and only if they provide together more information (in terms of the Shannon entropy) than separately.

It is worth noting that the definition of meaningful synthetic variables has also been studied in the context of hierarchical fuzzy systems [35]. With the aim of maximizing interpretability, synthetic variables are required to be cointensive with any knowledge in the mind of the reader. Unfortunately, this is not satisfied in most hierarchical fuzzy systems. Moreover, measuring the meaningful/meaningless of synthetic variables remain an open problem.

4.3.7 Structural-based Interpretability at Fuzzy Rules Level

The simplest structural parameter to consider in a fuzzy rule from the point of view of interpretability is the number of features, i.e., attributes or input variables, considered in the premise part of the rule. In other words, the length of the rule measured as the number of conditions (i.e., information granules) in the premise part of the rule. Ishibuchi et al. [27] coined the concept of Total Rule Length (TRL) as the total number of conditions in a rule base.

In addition to TRL, Ishibuchi and Nojima [28] defined other two simple indexes, mainly applied in the context of multi-objective fuzzy genetics-based machine learning, regarding the rule base readability: the *number of rules* (n_R); and the Average Rule Length (ARL=TRL/n_R). It is worth noting that the n_R is the simplest (but also the most widely used) index to measure the structural-based interpretability at fuzzy rules level. In [7], n_R is compared with TRL, ARL and other related indexes such as the number of labels used in the rule base or the percentage of rules which use less than a predefined percentage of fuzzy information granules.

In addition, Alonso and Magdalena [6] defined an index called Rule Base Complexity (RBC) that considered the morphology of fuzzy information granules when assessing the complexity of each single rule in a rule base. The total RBC is computed as follows:

$$RBC = \sum_{j=1}^{n_R} \left[complexity(R^j) \right] \tag{4.34}$$

where $complexity(R^j)$ is the complexity of the $j-th$ rule:

$$complexity(R^j) = \prod_{a=1}^{n_I} \left[complexity(P_a) \right] \tag{4.35}$$

where $complexity(P_a)$ is the complexity of the $a-th$ fuzzy information granule and n_I is the number of input variables. Notice that each fuzzy information granule is associated to just one input in (4.35). *RBC* takes into account all the involved linguistic propositions P_a of form I_a *is* A_a^i. Here, we propose computing the complexity of each P_a as follows (notice that we have redefined the original equation given in [6] with the aim of being more intuitive and natural):

$$complexity(P_a) = \begin{cases} 1 & \text{, if } LT_a^j = NL_a \\ 1 + (\frac{LT_a^j}{MaxNL})^{\frac{1}{NL_a}} & \text{, otherwise} \end{cases} \tag{4.36}$$

where the linguistic term A_a^i which is assigned to the variable I_a can correspond to one of the NL_a elementary labels defined in the fuzzy partition of the input I_a, $NL_a \geq 2$, $MaxNL$ is the maximum number of labels in all inputs, and LT_a^j counts the number of elementary labels actually included in A_a^i. It can take the following values:

- one, for elementary labels;
- number of elementary labels combined with OR (e.g., it equals two for the expression LOW OR MEDIUM);
- $NL_a - 0.5$ for NOT composite labels (this penalizes NOT against OR composite labels involving all the elementary labels minus one);
- NL_a when input I_a is not considered in the rule (it means $complexity(P_a) = 1$).

4.3.8 Semantic-based Interpretability at Fuzzy Rules Level

Semantic of fuzzy rules can be analyzed in terms of rule relevance, rule redundancy, rule overlapping, and rule consistency.

Mikut et al. [42] proposed the next measure of rule relevance:

$$Q = Q_{ac} \cdot Q_{cl}^{\beta} = \left(1 - \frac{E}{E_0}\right) \left[\prod_{r=1}^{n_R} \max_j (\widehat{p}(B_j|P_r))\right]^{\beta} \tag{4.37}$$

where Q measures the compromise between *classification accuracy* ($Q_{ac} \in [0,1]$) and *clearness of the rules* ($Q_{cl} \in [0,1]$); $\beta \geq 0$; E is the minimum quadratic error in terms of membership values of the output classes, and E_0 is the value taken by E in the trivial model (i.e., just one rule with an *always true* premise); n_R is the number of rules; and $\widehat{p}(B_j|P_r)$ is the probability of y *is* B_j when premise P_r is satisfied.

In addition, Dutu et al. [14] defined the next rule Redundancy Index (RI):

$$RI_{R_i} = \frac{1}{N_S^{R_i}} \sum_{j=1}^{N_S^{R_i}} C_{s_j}^{Norm}(R_i, RB) \tag{4.38}$$

where $N_S^{R_i}$ is the number of data samples which fire rule R_i, and $C_{s_j}^{Norm}(R_i,RB)$ indicates the influence of R_i over sample s_j in comparison with all the n_R rules in the rule base RB:

$$C_{s_j}^{Norm}(R_i,RB) = \sum_{k=1}^{n_R} C_{s_j}^{Norm}(R_i,R_k) \tag{4.39}$$

$$C_{s_j}^{Norm}(R_i,R_k) = \frac{C_{s_j}(R_i,R_k)}{D(R_i,R_k)}, \forall s_j \in DS_{R_i} \tag{4.40}$$

where $DS_{R_i} = \left\{ s_j | \mu_{R_i}^{s_j}(\mathbf{x},y) > 0 \right\}$ is the set of data samples firing rule R_i. Notice that $\mu_{R_i}^{s_j}(\mathbf{x},y)$ is the activation strength of rule R_i for a given sample s_j. Given a pair of rules, R_i and R_k, they are projected into a (n_I+1)-dimensional space and then $D(R_i,R_k)$ is the Manhattan distance between them [13]. In addition, $C_{s_j}(R_i,R_k)$ estimates the common influence of rules R_i and R_k over sample s_j:

$$C_{s_j}(R_i,R_k) = \frac{\min(\mu_{R_i}^{s_j}(\mathbf{x},y), \mu_{R_k}^{s_j}(\mathbf{x},y))}{\mu_{R_i}^{s_j}(\mathbf{x},y)}, \forall s_j \in DS_{R_i} \tag{4.41}$$

In addition, Pancho et al. [48] defined the rule CO-Firing index (COF) to measure rule overlapping as follows:

$$COF_{ik} = \begin{cases} \frac{SFR_{ik}}{\sqrt{FR_i \cdot FR_k}} & \text{, if } i \neq k \\ 0 & \text{, if } i = k \end{cases} \tag{4.42}$$

where SFR_{ik} corresponds to the number of data samples for which rules R_i and R_k are fired simultaneously. In addition, FR_i and FR_k account respectively for the total number of data pairs for which rules R_i or R_k are respectively fired, without taking care if they are fired together or not. Notice that (4.42) was later updated in [47], as follows:

$$COF_{ik} = 1 - \frac{\sum_{s_j \in DS_{R_i}} |\mu_{R_i}^{s_j}(\mathbf{x},y) - \mu_{R_k}^{s_j}(\mathbf{x},y)|}{\sum_{s_j \in DS_{R_i}} \mu_{R_i}^{s_j}(\mathbf{x},y)} \tag{4.43}$$

Regarding rule consistency, Jin et al. [32] proposed the following measure:

$$f_{Incons} = \sum_{i=1}^{n_R} \sum_{\substack{k=1 \\ k \neq i}}^{n_R} (1 - Cons(R_i,R_k)) \tag{4.44}$$

where

$$Cons(R_i,R_k) = \exp\left(- \frac{\left(\frac{Sm(\mathbf{A}_i,\mathbf{A}_k)}{Sm(B_i,B_k)} - 1 \right)^2}{\left(\frac{1}{Sm(\mathbf{A}_i,\mathbf{A}_k)} \right)^2} \right) \tag{4.45}$$

being S_m a similarity measure. It is worth noting that inconsistency is evaluated in terms of rule similarity (RS) in [38]:

$$RS = \frac{2}{n_R\,(n_R - 1)} \sum_{i=1}^{n_R} \sum_{\substack{k=1\\k\neq i}}^{n_R} RS\,(R_i, R_k) \tag{4.46}$$

where

$$RS\,(R_i, R_k) = \frac{1}{f+1} \sum_{j=1}^{f} \left(Sm\left(A_i^j, A_k^j\right) + \mathrm{eq}\,(B_i, B_k) \right) \tag{4.47}$$

being $\mathrm{eq}\,(B_i, B_k) = 1$ if $B_i = B_k$, 0 otherwise.

In addition, Pedrycz and Gomide analyzed theoretically the consistency of rules in the specific context of implicative interpretation [49]. Moreover, Yager and Larsen derived a necessary condition for two rules to be inconsistent [61]. Given rules $R_i : A_i \to B_i$ and $R_k : A_k \to B_k$, they are α-inconsistent if there is a fuzzy input $\mathbf{A}$ such that the fuzzy set B inferred by the model has a degree of inconsistency α. If R_i and R_k are α-inconsistent, then the following inequalities hold:

$$\mathrm{Poss}\,(\mathbf{A}_i, \mathbf{A}_k) \geq \alpha \tag{4.48a}$$

$$\mathrm{Poss}\,(B_i, B_k) \leq 1 - \alpha \tag{4.48b}$$

being similar antecedents in (4.48a) and dissimilar consequents in (4.48b), Π is the possibility measure defined by (4.14), $\mathbf{A}_i = A_i^1 \times \cdots \times A_i^f$ and $\mathbf{A}_k = A_k^1 \times \cdots \times A_k^f$.

As an alternative approach, Scalpelli and Gomide applied the following equation to derive potential inconsistencies among couples of rules R_i and R_k [57]:

$$E\,(x) = \mu_{\neg \mathbf{A}_i}\,(x) \otimes \mu_{\neg \mathbf{A}_k}\,(x) \otimes \mathrm{Poss}\,(B_i, B_k) \tag{4.49}$$

4.3.9 Structural-based Interpretability at Fuzzy Models Level

Nauck [43] defined the first structural-based interpretability index regarding a whole fuzzy model. This index focused on fuzzy classifiers and it was computed as the product of three terms:

- *Comp*: this is the complexity of a classifier measured as the number of classes divided by the total number of fuzzy information granules.
- *Part*: this is the average normalized partition index. It averages the inverse of the number of labels minus one regarding all input variables. It is worth noting that two is the minimum number of labels in a fuzzy partition.
- *Cov*: this is the average normalized coverage degree of the fuzzy partition. *Cov* equals zero if the variable is not partitioned or incase it is a crisp partition. In both extreme cases the related variable does not appear in any rule in the fuzzy systems analyzed by Nauck. *Cov* equals one only in case of complete coverage.

Later, Alonso et al. [4, 8] designed a hierarchical fuzzy system for assessing interpretability of linguistic knowledge bases in classification problems. It was inspired by the Nauck's index and considered structural parameters such as the number of fuzzy sets, the number of fuzzy information granules and the number of rules.

It is worth noting that Razak et al. [53, 54] defined later an interpretability index for hierarchical fuzzy systems. In short, for each subsystem in a given hierarchy, they aggregated the values of the indexes previously defined by Nauck [43] and Alonso [4, 8]. In addition, Magdalena [35] has also addressed recently interpretability issues for hierarchical fuzzy systems.

In addition, Antonelli et al. [10] proposed another index to globally evaluate the structural interpretability of Mamdani FRBSs:

$$INT = 1 - \frac{I - 2 \cdot n_{Rmin}}{n_{Rmax} \cdot (n_I + 1) \cdot [1 + \frac{T-2}{2}] - 2 \cdot n_{Rmin}} \tag{4.50}$$

where n_I is the number of inputs; T is the number of fuzzy sets in each partition (considering that all inputs have the same T what is not necessarily true for all Mamdani systems); n_{Rmin} and n_{Rmax} are the admissible minimum and maximum numbers of rules; and I measures the integrity of fuzzy partitions as follows:

$$I = \sum_{r=1}^{n_R} \sum_{i=1}^{n_I+1} (1 + d_i) \cdot u(v_{r,i}) \tag{4.51}$$

with

$$u(v_{r,i}) = \begin{cases} 1 \text{ , if } v_{r,i} > 0 \\ 0 \text{ , if } v_{r,i} = 0 \end{cases} \tag{4.52}$$

being n_R the number of rules and assuming triangular fuzzy sets. The dissimilarity d_i between the current partition (made up of fuzzy sets A_n^i) and a uniform partition (made up of fuzzy sets $\tilde{A_n^i}$) that is taken as reference is computed as follows:

$$d_i = \sum_{n=2}^{T-1} \left| \left[A_n^i\right]_1 - \left[\tilde{A_n^i}\right]_1 \right| \tag{4.53}$$

On the other hand, Jin et al. [33] proposed a global consistency index based on a fuzzy similarity measure. They applied a *soft t-norm* ($\tilde{\otimes}$) and a BAsic Defuzzification Distribution (BADD), which are defined as follows:

$$\tilde{\otimes}(x_1, \ldots, x_n) = (1 - \alpha) \cdot \frac{1}{n} \cdot \sum_{k=1}^{n} x_k + \alpha \cdot \otimes(x_1, \ldots, x_n)$$

$$BADD \equiv y = \frac{\sum_{k=1}^{N} \left\{ \left[T_{j=1}^{n} \cdot A_j^k(x_k) \right]^{\delta} \cdot p_{k0} \right\}}{\sum_{k=1}^{N} \left[T_{j=1}^{n} \cdot A_j^k(x_k) \right]^{\delta}} \tag{4.54}$$

where $\alpha \in [0,1]$, $\delta \in [0,\infty)$, x_k are the input values, A_j^k represents the label associated to the j-th variable in the premise of the k-th rule, and p_{k0} is a constant coefficient.

Given two rules R_i and R_k, their similarity is measured in terms of the Similarity of Rule Premises (SRP) and the Similarity of Rule Consequents (SRC), as follows:

$$SRP(i,k) = \min_{n=1}^{n_I} Sm\left(A_n^i, A_n^k\right), \quad SRC(i,k) = Sm\left(B_i, B_k\right) \tag{4.55}$$

And then the measure of *consistency* between R_i and R_k is as follows:

$$Cons\left(R_i, R_k\right) = \exp\left(-\frac{\left(\frac{SRP(\mathbf{A}_i,\mathbf{A}_k)}{SRC(B_i,B_k)} - 1\right)^2}{\left(\frac{1}{SRP(\mathbf{A}_i,\mathbf{A}_k)}\right)^2}\right) \tag{4.56}$$

In addition, Pota et al. [51] proposed the Readability Global Index (RGI) as a structural-based interpretability index at the level of fuzzy models:

$$RGI = \left(\left(n_I + n_O\right)^v \cdot par^s \cdot M^p \cdot ARL^r \cdot \left(n_R\right)^b \cdot ANFR^i\right)^{\frac{-1}{v+s+p+r+b+i}} \tag{4.57}$$

where $n_I + n_O$ is the total number of variables (I stands for inputs and O refers to outputs); par is the average number of parameters in the related fuzzy sets (e.g., $par = 3$ in the case of fuzzy sets described by triangular MFs); M is the average number of fuzzy sets; ARL is the average rule length; n_R is the number of rules; and $ANFR$ is the average number of fired rules. Notice that this index takes into account elements related to all the structural levels previously discussed. Moreover, the designer can use the exponents (always positive values) for tuning the relevance of all the involved elements: v for variables; s for fuzzy sets; p for fuzzy partitions; r for rules; b for the rule base; and i for the inference process.

4.3.10 Semantic-based Interpretability at Fuzzy Models Level

Alonso and Magdalena [5] defined a quality index to guide the design of fuzzy models with a good balance among accuracy, interpretability and stability. More precisely, interpretability was measured in terms of readability and understandability at fuzzy models level. On the one hand, readability considers structural-based issues at level of fuzzy sets, fuzzy partitions, fuzzy information granules and fuzzy rules. On the other hand, understandability pays attention to semantic-based issues such as similarity of concepts represented by fuzzy sets and fuzzy information granules, simultaneous fired rules, and cointensive inference. This semantic-based interpretability index was implemented by means of a hierarchical decision-making approach where the Analytic Hierarchy Process (AHP) [56] and the Ordered Weighted Averaging (OWA) operators [60] were combined. It is worth noting this index can be customized in accordance to the user's preferences.

The analysis of simultaneous fired rules is directly related to the rule CO-Firing based Comprehensibility Index (COFCI) which was initially defined in [9] and later refined in [48]. COFCI measures the complexity of understanding the fuzzy inference process in terms of information related to co-firing rules (i.e., rules simultaneously fired by a given input vector):

$$COFCI = \begin{cases} 1 - \sqrt{\frac{\sum_{i=1}^{n_R}\sum_{k=1}^{n_R}[(P_i+P_k)\cdot m_{ik}]}{MaxThr}} & \text{, if } \sum_{i=1}^{n_R}\sum_{k=1}^{n_R}[(P_i+P_k)\cdot m_{ik}] \leq MaxThr \\ 0 & \text{, otherwise} \end{cases} \quad (4.58)$$

where n_R is the total number of rules in the fuzzy rule base, P_i and P_k count the number of premises (antecedent conditions) in rules R_i and R_k, while m_{ik} is the measure of co-firing, computed by (4.42) or (4.43), for the rules R_i and R_k, and $MaxThr$ is a maximum value heuristically established to get a normalized measure in the interval [0,1].

It is worth noting that COFCI emerges in relation with a novel approach for fuzzy system comprehensibility analysis that is based on visual representations of the fuzzy rule-based inference process. Such representations are called fuzzy inference-grams (*fingrams* in short) [48]. Given a fuzzy rule base, the related fingram shows graphically the relations among rules in the form of a social network that is made up of nodes representing fuzzy rules and edges connecting nodes in terms of rule interaction at inference level. Edge weights are computed by paying attention to the number of co-firing rules. Thus, the structure and behavior of the represented fuzzy model can be understood indirectly by means of interpreting carefully all the information provided in a fingram. Notice that the software tool FingramsGenerator [46] lets users visualize and analyze fingrams by means of social network analysis techniques in order to go in depth with the interaction of fuzzy rules at inference level. This software provides insights on semantic-based interpretability issues at fuzzy model level.

In addition, the semantic cointension based index [41] exploits the cointension concept coined by Zadeh [62]. Informally, two different concepts referring almost to the same entities are taken as cointensive. Thus, a fuzzy system is deemed as comprehensible only when the explicit semantics (defined by fuzzy sets attached to linguistic terms as well as fuzzy operators) embedded in a fuzzy model is cointensive with the implicit semantics inferred by a user who reads the linguistic representation of the related rules. Assuming that linguistic propositions in a fuzzy rule base resemble logical propositions, then semantic cointension can be evaluated through a logical view approach which evaluates the degree of fulfillment of a number of logical laws (which are expected to hold) by a given fuzzy rule base [40]. This logical view approach has been successfully applied to classification problems.

Pota et al. [51] also proposed a semantic-based interpretability index at the level of fuzzy models. It is called Semantic Interpretability Global Index (SIGI) and it can be seen as an extension of (4.57):

$$SIGI = (SIV^v \cdot SIFS^s \cdot SIFP^p \cdot SIFR^r \cdot SIRB^b \cdot SII^i)^{\frac{1}{v+s+p+r+b+i}} \tag{4.59}$$

where the prefix *SI* stands for Semantic Interpretability and the rest of each acronym makes reference to the related structural level in the fuzzy model: *V* is related to Variables; *FS* corresponds to Fuzzy Sets; *FP* is for Fuzzy Partitions; *FR* refers to Fuzzy Rules; *RB* makes reference to the whole Rule Base; and *I* refers to the Inference process. Notice that the exponential values (i.e., v, s, and so on) are always positive values and express the importance given to each structural level in the model. It is worth noting they follow the same notation used in the case of the *RGI* index.

The *SIV* associated to variables is computed as follows:

$$SIV = \begin{cases} 1, & \text{variables defined by an expert} \\ 0, & \text{variables derived from data} \end{cases} \tag{4.60}$$

Regarding fuzzy sets, *SIFS* is computed as follows[4]:

$$SIFS = \begin{cases} 1, & \text{if all semantic constraints are satisfied} \\ 0, & \text{otherwise} \end{cases} \tag{4.61}$$

The *SIFP* of fuzzy partitions is computed as follows:

$$SIFP = ((1-J_D)^d \cdot (1-J_C)^c \cdot PO^o \cdot LR)^{\frac{1}{d+c+o+1}} \tag{4.62}$$

where J_D is a distinguishability measure and J_C is a coverage measure[5]. *LR* takes the value one if leftmost and rightmost fuzzy sets are used, and zero otherwise. *PO* evaluates the proper ordering of fuzzy sets in the given partition as follows[6]:

$$PO = \begin{cases} 1 & \text{, if strong condition is satisfied} \\ 0.5 & \text{, if weak condition is satisfied} \\ 0 & \text{, otherwise} \end{cases} \tag{4.63}$$

The *SI* of rules is defined as follows:

$$SIFR = (AW \cdot JRC^{rc})^{\frac{1}{1+rc}} \tag{4.64}$$

$$AW = \begin{cases} 1, & \text{if antecedent weights are not used} \\ 0, & \text{otherwise} \end{cases} \tag{4.65}$$

JRC is a rule clearness index and the exponent *rc* establishes its importance in the whole measure. C_r is the consequent of rule r.

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[4] The required semantic constraints are: one-dimensionality; normality; continuity; convexity; unimodality of each fuzzy set; and symmetry for special elements if any. These constraints are defined in Chap. [3].

[5] The interested reader is kindly referred to [51] for the complete definition of J_D and J_C. In addition, other J_i definitions apart from those given in (4.11) and (4.12) are provided in [17].

[6] Weak and strong conditions for proper ordering are defined in Chap. [3].

$$JRC = (\prod_{r=1}^{n_R} [C_r])^{\frac{1}{n_R}} \tag{4.66}$$

The *SIRB* of the rule base is as follows:

$$SIRB = (MCL \cdot CI^{ci} \cdot RW^{rw})^{\frac{1}{1+ci+rw}} \tag{4.67}$$

where *MCL* verifies the use of modus ponens, consistency and locality. *CI* measures the rule base completeness in terms of the ratio of data samples firing at least one rule. In addition, *RW* penalizes the use of rule weights.

$$MCL = \begin{cases} 1, & \text{if modus ponens, consistency and locality are satisfied} \\ 0, & \text{otherwise} \end{cases} \tag{4.68}$$

$$RW = \begin{cases} 1/ANFR & \text{, if rule weights are used} \\ 0 & \text{, otherwise} \end{cases} \tag{4.69}$$

where *ANFR* is the average number of fired rules.

Finally, the *SII* associated to the inference process is evaluated as follows:

$$SII = \begin{cases} 1, & \text{if all norms are of the type min/max, product/sum, or Lukasiewicz} \\ 0, & \text{otherwise} \end{cases} \tag{4.70}$$

4.4 Conceptual Framework for Measuring Interpretability

In the light of the review of interpretability indexes that we presented in the previous section, we observe that there has been a huge effort to define indexes which cover all elements in a fuzzy system, regarding both structural and semantic constraints and criteria. However, there is still the lack of a universal index recognized worldwide.

It is worth noting that the process of measuring something consists in comparing it with a reference (standard unit of measurement) such as a meter for measuring length. However, finding out the suitable reference is not always feasible and this task is especially difficult when measuring non-physical properties like interpretability (which strongly depends on the ability of users to interpret fuzzy sets, partitions, and so on, in accordance with their own background).

In addition, it is widely admitted that interpretability assessment is subjective and context dependent. There is not a universal reference; on the contrary the reference will change depending on the problem and depending on the person who makes the assessment. Therefore, defining a universal index becomes a non-straightforward task.

As alternative, we can either select the most suitable interpretability index (among all those introduced in the previous section) or define a new index for each single specific problem. Moreover, in the case of reusing some of the available indexes, customization is likely to be required in order to properly fit the specific features of the given problem under consideration as well as the user's background and preferences.

Notice that, the total number of rules (n_R) may be qualitatively assessed by means of a linguistic variable with five linguistic terms (VERY SMALL, SMALL, MEDIUM, LARGE, VERY LARGE). Nevertheless, the meaning of each linguistic term needs to be contextualized and defined carefully.

For example, it is arguable which value (e.g., three, ten, one hundred, etc.) should be taken as a prototype or reference value for SMALL regarding n_R. If we were analyzing an FRBS for classification among three kinds of wines, obviously the minimum n_R should be three. Nevertheless, it would be a good thing to ask to the specific people who are going to interact with the system in order to really know how to characterize the related linguistic variables. In fact, the perception of interpretability will change depending on the kind of user. The point of view of a system designer who is used to work with fuzzy systems is likely to be very different from the point of view of a domain expert who may perfectly know the problem and how the system behavior should be, but who probably knows nothing about fuzzy logic. Moreover, the point of view of a final user (who may have only a superficial knowledge of the problem and who probably has never heard anything about fuzzy logic) may be even more different.

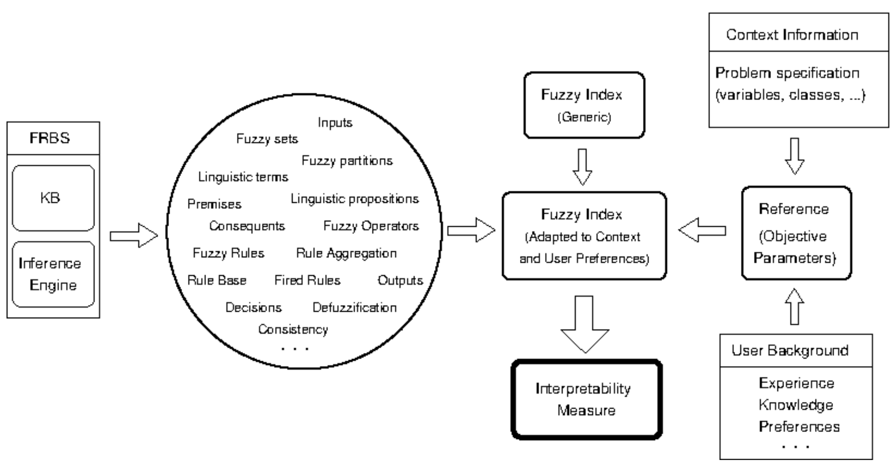

Fig. 4.5: Adapting interpretability indexes to the context of the problem and user preferences.

Figure 4.5 shows a general framework to adapt interpretability indexes to specific problems and specific users. A given (generic) index has to be tuned and adapted for each problem regarding both the problem definition and the user preferences. This framework is flexible enough for making such adaptation. Firstly, we must set a reference with all collected information about the problem and the evaluator (system designer, domain expert, and/or final user). Such reference should be defined in an *objective* manner. It should consider structural parameters to be measured without any subjectivity, e.g., the ranges of the variables, the number of modal points in the fuzzy partitions, n_R or TRL. In addition, it should assess interpretability in a natural way as humans do. Finally, it should provide as output not only a numeric value but also an explanation about how such value was computed.

4.5 Final Remarks

We have provided readers with a complete review of existent interpretability indexes. To make easier the selection of the right index for a specific problem we have also provided a taxonomy which is in accordance with the structural and semantic constraints and criteria thoroughly described in Chap. [3].

We would like to remark that in the last years many different indexes have been proposed. They successfully cover most elements involved in the design of fuzzy systems. However, there is still a lot of work to be done. Let us briefly highlight those issues that in our opinion require future research:

- even if there has been a great effort to characterize and measure interpretability at all structural levels (especially at fuzzy partitions, fuzzy rule, and fuzzy model levels), more work at level of fuzzy sets and fuzzy information granules is required;
- additional objective and subjective indexes for quantification of semantic-based interpretability are needed;
- hierarchical fuzzy systems require deeper attention because there are almost no indexes for measuring their interpretability;
- there is still the lack of a universal index (easy to adapt to specific problems and specific users) recognized worldwide. We think this is due to the subjective nature of the interpretability assessment task.

We envision a new generation of Explainable Artificial Intelligence (XAI) based systems ready not only to measure interpretability naturally as humans do but also to provide users with plausible explanations of the assessments that they carry out [1]. Such explanations should be multimodal, i.e., they should combine textual representations (in natural language) with graphical representations (as well as other modalities) efficiently, but they should be also multilingual and easy to adapt to users with different backgrounds and preferences. Moreover, many of the interpretability indexes defined for fuzzy systems so far are likely to be farther extended to measure the interpretability of other intelligent systems in the context of XAI.

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Chapter 5
Designing Interpretable Fuzzy Systems

Abstract Fuzzy sets and fuzzy logic are powerful tools widely used to represent human knowledge and mimic human reasoning capabilities, being the main constituents of fuzzy systems. Among the different approaches to fuzzy systems, fuzzy rule-based systems represent the one offering a better framework for interpretability considerations. Their applications range from classification to modeling and control. Independently on its purpose, the behavior of a fuzzy rule-based system can be evaluated in terms of its reliability and comprehensibility, two concepts usually represented by accuracy and interpretability in the context of fuzzy systems. Indeed, achieving the highest possible levels of accuracy and interpretability is one of the central aspects of designing fuzzy systems. The present chapter will go through the design process considering its different steps and analyzing the multiple options allowing us to improve interpretability or to achieve a better interpretability-accuracy balance, in the search for interpretable fuzzy systems which represent an interesting approach in the framework of explainable Artificial Intelligence. We will consider questions related to the knowledge extraction and refinement process; some examples are complexity reduction and semantic improvement. We will also analyze other questions linked to the design of the processing structure, such as the effects of applying different aggregation and implication operators.

5.1 Introduction

Rule-based systems have been widely and successfully applied as a tool to represent human knowledge in the form of IF-THEN rules. These rules are capable of reproducing some human problem solving capabilities. Initially, the rules where adapted to the conception and operations of bivalent logic, but new ideas related to fuzzy sets and fuzzy logic gained presence and importance in the field of rule-based systems. As a result of these new ideas, Fuzzy Rule-Based Systems (FRBSs) became a quite important tool to represent human knowledge and implement human reasoning. FRBSs are not the only option when considering fuzzy systems (e.g., most fuzzy clustering techniques do not rely on fuzzy rules), but are the one considered when talking of interpretable fuzzy systems. Consequently this chapter will concentrate on designing interpretable FRBSs, and the terms fuzzy system and FRBS will be used interchangeably.

It is possible to apply FRBSs to quite different problems, particularly to modeling [108], classification [36] and control [48]. But when considering their design from the point of view of interpretability, there are no significant differences among them. Consequently, and without loosing generality, we will concentrate in modeling but assuming that the main ideas presented here can be directly transferred to classification and control problems. In some particular cases, specific comments will remark the potential differences.

A model usually represents a part, or some specific aspects, of a certain reality (i.e., a phenomenon, a situation, a problem) using rules, concepts, mathematical equations, etc. In that sense, modeling is the process of creating a representation of a certain reality (usually referred to as a system). The central idea when creating a model is obtaining a *scheme* that simplifies the real system or the complex reality with the aim of easing its understanding. Thanks to these models, the real system can be explained, controlled, simulated, predicted, and even improved.

As the model represents something, its quality will be strongly related to the purpose of that representation. Some models have as their solely purpose replicating the behavior of the system after a certain stimulus. In this case, the model *simply* needs to replicate the input-output relations of the system, no matter how it does. The modeling tools known as black-box models are well suited for this kind of situation. This is the case when the purpose of the model is simulation or prediction.

But we have previously said that the purpose of the model could also be to explain the system. In such a case the main interest does not concentrate on knowing *what* will be the output, but on knowing *why* will it be such that. In this case it is clear that the pure input-output relation is not enough, black-box models will not be a solution. It is mandatory the existence of (explicit or implicit) pieces of knowledge describing or explaining the input-output relation. Consequently, the quality of a model will be concerned by its *reliability*, but also by its *comprehensibility*.

Previous paragraphs have mentioned the use of black-box models for some specific situations. Those systems are built entirely from data using no additional knowledge in advance, and focusing on accuracy, but at present, its use is not limited to situations where our only concern is accuracy . Under the umbrella of eXplainable Artificial Intelligence (eXplainable AI or XAI in short) there is a growing interest in creating methods to *explain* black-box models, not without some criticism [121].

Explaining black-box models could be an option when needing comprehensibility, but obviously is not the only one. There is always the possibility to work with intrinsically interpretable models (i.e., interpretable by design). One option is to resort to traditional *white-box models*, assuming a sufficient knowledge of system's nature providing a suitable mathematical scheme to represent it. But it is also possible to relay on intermediate approaches as *gray-box modeling* [66], where certain known parts of the system are modeled considering the prior knowledge while the unknown or less certain parts are identified with black-box procedures. This approach provides a better balance between reliability and comprehensibility, being fuzzy modeling one of the most successful tools to develop gray-box models [86].

We have already mentioned that there are at least two potential objectives for fuzzy modeling: knowing *what* will be the output of a system and knowing *why* will it be such that. These ideas are usually considered under two main concepts when evaluating the quality of a fuzzy model:

- *Accuracy* refers to the capability of the fuzzy model to faithfully represent the modeled system. It is evaluated in terms of the proximity or similarity between the responses of the real system and the fuzzy model. This is why the terms reliability, accuracy and approximation have been used as synonyms in this context.
- *Comprehensibility* refers to the capability of the fuzzy model to express the behavior of the system in an understandable way. Being this the main topic of the book, no further description is required.

Determining the interaction between these two objectives is not straightforward. As Zadeh stated in its Principle of Incompatibility [146], "*as the complexity of a system increases, our ability to make precise and yet significant statements about its behavior diminishes until a threshold is reached beyond which precision and significance (or relevance) become almost mutually exclusive characteristics*." In this sense, complexity and interpretability seem to be colliding concepts. In fact, the structural complexity is one of the metrics usually considered to evaluate interpretability of fuzzy systems. But the relation between complexity and accuracy is not so clear. Obviously, the more degrees of freedom a model has, the more flexible it is, and consequently, the higher approximation capability will have. Intuitively that means that more complex implies more accurate, but this is not always true when comparing different modeling paradigms. When considering problems that have structured data and meaningful variables, a modeling tool able to profit from these characteristics will produce simpler models while maintaining performance in terms of accuracy. According to [121], "*it is a myth that there is necessarily a trade-off between accuracy and interpretability*." But it is important to notice that this consideration is made when comparing models based on different paradigms.

If we reduce our scope to a single paradigm, in this case fuzzy modeling, it is commonly accepted that improving interpretability and increasing accuracy are two colliding objectives, so that one should prevail over the other. The result is the consideration of two main approaches to fuzzy modeling:

- Linguistic Fuzzy Modeling (LFM) has as is primary objective producing fuzzy models with a good interpretability.
- Precise Fuzzy Modeling (PFM) considers as first objective obtaining fuzzy models with a good accuracy.

The starting point for fuzzy modeling (from a chronological point of view) was LFM, directed towards exploiting the descriptive power of the concept of a linguistic variable [146, 147, 148, 149]. However, the proliferation of fuzzy models has induced a certain deviation from that seminal purpose producing, mainly during the last decade of the past century, a significant amount of research focused on increasing accuracy as much as possible, assuming interpretability as an intrinsic property. As usually happens, after considering the two extreme positions (only interpretability and only accuracy), the interest for achieving a good balance between interpretability and accuracy [8, 29, 30] gained presence.

After more than forty years of application of FRBSs to real problems, a wide repertory of design approaches is available, and the way they focus on questions as interpretability is quite diverse and multifaced. The present chapter will not try to select the best design approach, or to prioritize or rank the different solutions. We will only describe different approaches to the problem, from different points of view, and apply different arguments to convince the reader. As said, it is impossible to rank the approaches, since the arguments, concepts and metrics they use are in most cases not comparable. As a result, the present chapter is mostly an ordered catalogue of concepts and ideas.

5.2 Designing Fuzzy Rule-based Systems

FRBSs do not have a single common structure (see Sect. 2.4 in Chap. [5]), but represent a family of different approaches: Takagi-Sugeno-Kang (TSK), Mamdani, descriptive, approximate, etc. Consequently, designing an FRBS includes as a very first step the decision on what kind of fuzzy system to use. As clearly stated in previous chapters, any of those approaches offers a different potential for interpretability and accuracy. But from now on, we will consider that a decision was taken to choose a specific type of FRBS. So, the discussion from this point will concentrate on particular design aspects, and not on the high level decision of choosing TSK, Mamdani or any other type of fuzzy systems.

Assuming the type of FRBS was decided, the behavior of the obtained system directly depends on two aspects, the way in which it implements the fuzzy inference process and the content of the Knowledge Base (KB). Therefore, the FRBS design process includes two main tasks:

1. *Conception of the inference engine*, that is, the choice of the different operators that are employed by the inference process: fuzzification, conjunction, implication, aggregation and defuzzification. These questions are considered in Sect. 5.2.1.
2. *Generation of the fuzzy* KB in order to formulate and describe the knowledge that is specific to the problem domain. The KB contains different pieces of information structured in the form of linguistic variables, fuzzy partitions and rules (see Sect. 2.3.1 in Chap. [5]). These questions are considered in Sect. 5.2.2 to 5.2.4.

5.2.1 Design Decisions at Fuzzy Processing Level

The implementation of the inference engine of an FRBS requires the following design choices:

1. Selecting the fuzzification interface enabling Mamdani or TSK FRBSs to handle crisp input values. The most common choice for the fuzzification operator F is the *point wise* or *singleton fuzzification*, where A' is built as a singleton with support x_0, i.e., it presents the following membership function (MF):

$$\mu_{A'}(x) = \begin{cases} 1, \text{ if } x = x_0 \\ 0, \text{ otherwise} \end{cases} \tag{5.1}$$

   Non-singleton options [103] are also possible and have been considered in some cases as a tool to represent the imprecision of measurements.
2. Select the conjunctive operator implementing the *and* appearing in most rule antecedents. This is related to the fact that in systems with multiple inputs, the premises of the rules usually have the form:

$$\text{IF } X_1 \text{ is } A_{X_1} \text{ and } \ldots \text{ and } X_n \text{ is } A_{X_n} \tag{5.2}$$

   There are different operators belonging to the t-norm family that are usually considered to implement it [42, 65].
3. Selecting the implication operator ($\triangleright$) used in the fuzzy implication of IF-THEN rules. Mamdani [98] proposed the use of the *minimum* operator (a *t-norm*) for implication:

$$\triangleright(x,y) = \min\{x,y\} \tag{5.3}$$

   Since then, various other t-norms have been suggested as implication operator [65], such as the *algebraic product*:

$$\triangleright(x,y) = x \cdot y \tag{5.4}$$

   Apart from t-norms, an important family of implication operators are the fuzzy implication functions [136], being one of the most usual the Lukasiewicz's one:

$$\triangleright(x,y) = \min(1, 1 - x + y) \tag{5.5}$$

   Less common implication operators such as force-implications [49], t-conorms and operators not belonging to any of the most known implication operator families [42] have also been considered.
   The choice of a certain implication operator highly influences the system behavior and accuracy. Only a few of those options have been considered when analyzing interpretability.
4. Selecting the aggregation operator used to merge the individual rule's output. In a system where multiple rules are simultaneously *fired* by a certain input value (almost any fuzzy system belongs to this group), it is needed to define a method

to merge the individual outputs (fuzzy outputs in the form of fuzzy sets) from each single rule. This process is usually denoted as the *also* operator to be implemented by using a t-conorm, being the most commonly used, the maximum and the bounded sum:

$$
\begin{aligned}
a \oplus b &= \max\{a, b\} \\
a \oplus b &= \min\{1, a+b\}
\end{aligned}
\tag{5.6}
$$

Different interpretations of the aggregation of rules are also possible, using in those cases t-norms as the minimum, or even some more complex aggregation operators as proposed in [18], where an analysis of their properties, as well as a comparative study of their behavior, is presented.

The previous paragraphs consider the situation where the consequent of rules is fuzzy. In those cases, the individual output of each rule is a fuzzy set (for each output variable). This is the common situation for most Mamdani systems, but TSK models produce crisp values as output. In those cases where individual outputs are not fuzzy, the aggregation takes usually place by means of an average, or most commonly a weighted average of the individual outputs. In the case of weighted average, the weighting factors are related to the importance degree of the corresponding rule, being that usually computed as the *matching degree* between the antecedent of the rule and the input value.

5. Selecting the defuzzification operator, usually being the Center of Gravity (CG) or the Mean Of Maxima (MOM). Obviously, this step is only required for systems using rules with fuzzy output, since in any other case the output is already crisp.

The approach to rule aggregation and output defuzzification described in previous paragraphs corresponds to a system applying the so-called First Aggregate Then Infer (FATI) approach (see Sect. 2.3.2 in Chap. [5]). On the other hand, when considering the First Infer Then Aggregate (FITA) approach, the situation is slightly different. In this case, the aggregation is usually implemented through the average, the weighted average or the selection of a characteristic value of the individual fuzzy sets, being a function of any importance degree of the rule that generated them in the inference process [42]. Usually, the CG and the MOM are taken as characteristic values, while the *area* and *height of the inferred fuzzy set* or the *matching degree* are taken as importance degrees of the rule [42, 67, 131]. The characteristic value plus the importance degree could be considered as the result of the inference, and they are then used for aggregation.

For example, considering a systems with m rules, the MOM *weighted by the area* is computed as:

$$
y_0 = \frac{\sum_{i=1}^{m} s_i \cdot MOM_i}{\sum_{i=1}^{m} s_i}
\tag{5.7}
$$

with s_i and MOM_i being respectively the area and the MOM of the fuzzy set $A_Y'^{(i)}$ (i.e., the fuzzy set obtained by applying the compositional rule of inference on rule r_i); and the CG *of the fuzzy set with the largest height* is computed as:

$$y_0 = CG_k \tag{5.8}$$

where CG_k is the CG of $A_Y^{\prime(k)}$, and

$$A_Y^{\prime(k)} = \{A_Y^{\prime(i)}; s.t.\ \mathrm{hgt}\left(A_Y^{\prime(i)}\right) = \max_{j=1,\ldots,m}\left\{\mathrm{hgt}\left(A_Y^{\prime(j)}\right)\right\}\}. \tag{5.9}$$

Studies analyzing the behavior of the existing fuzzy operators for different purposes are found in the literature [42, 81], but the question has not been considered as a whole from the interpretability point of view. The aim of those papers was to provide some guidelines for the optimal design of the inference mechanism in Mamdani FRBSs. This question will be considered again from the interpretability point of view in Sect. 5.4

5.2.2 Design Tasks for Obtaining the Knowledge Base

Generating the KB of an FRBS requires some design tasks, that can be different depending on the specific type of FRBS under consideration. The most important tasks are:

1. Selecting the input and output variables that are more relevant to the system, among the set of all possible variables.
2. Selecting the Data Base (DB) structure if dealing with a descriptive FRBS, that is, either an LFM or a TSK FRBS using linguistic variables in its antecedents. In such a case the semantics of the terms associated to the linguistic variables (either input variables for TSK FRBSs or input and output variables for linguistic FRBSs) should be defined. This includes some design subtasks:
   - Definition of the main properties of the fuzzy partitions: granularity, coverage, overlapping, use or not of Strong Fuzzy Partitions (SFPs), etc.
   - Choice of the type of MF to be used: most commonly triangular, trapezoidal, Gaussian or exponential-shaped [48]. The latter two having the advantage of achieving a smoother transition and being differentiable (what can be needed in some design approaches), whilst the former two being computationally simpler.

     Different studies have analyzed the influence of the shape of these functions in accuracy of FRBSs [35], although authors of [47] enunciated that trapezoidal-shaped functions might adequately approximate the remaining ones, presenting the advantage of their simplicity as well. Moreover, it has been proven how the choice of trapezoidal functions allows to overcome the limitations deriving from different design strategies [102].
3. Selecting the type of MFs that will be used in the fuzzy rules if considering approximate FRBSs. In this particular case, this is the only task to be performed regarding the DB design, since the rest of the process will be reduced to the

definition of the rules. It is important to notice that in these approaches the fuzzy sets are not independently defined but are an intrinsic part of the corresponding rule.

4. Deriving the DB assuming the previously defined structure.
   - Defining of the scaling functions (if considered).
   - Choosing the possible term sets for each linguistic variable.
   - Defining the MF for the fuzzy set associated to each linguistic label.
5. Deriving the linguistic, approximate or TSK rules that will form the Rule Base (RB) of the system. In some cases a number of rules to be generated has to be set in advance.

5.2.3 Selection of Variables

The design tasks defined in previous section starts with the selection of those input and output variables relevant to the system. This is a central aspect of modeling, regardless the applied modeling technique. As a consequence, we will consider this question as a preliminary step, previous to those others that are directly related to fuzzy modeling. A first concept to consider is the relation between the number of variables of a model, and its interpretability.

It is clear that one of the constraints to be considered when trying to design an interpretable system is keeping the number of variables of the system as small as possible (see Sect. 3.6.8 in Chap. [6]). This question is mostly related to complexity matters since considering more input variables will usually result in models with a larger number of rules. This idea is somehow related to the so-called *curse of dimensionality* since the number of rules usually shows an exponential relation with the number of input variables. At the same time, each individual rule will potentially be longer (and less readable) due to the presence of more variables on it. This latest question applies not only to input but also to output variables: the more (input and/or output) variables the system has, the longer the rules will be.

As the number of input variables and that of output variables have a different effect on interpretability degradation, different considerations will be applied in the attempt of reducing their number. But before considering how to reduce the number of variables of the system, it is important to notice that it is also possible to work on the basis of a large number of variables, then applying techniques to palliate the effects of this situation on interpretability. In that sense, rule selection techniques (Sect. 5.3.3.2) will be also considered.

5.2.3.1 Input Variables

On the one hand, the reduction of the number of input variables improves the readability of the KB and avoids the curse of dimensionality. On the other hand, reducing the number of input variables produces a reduction in the number of free parameters in the model and consequently limits its ability to identify a complex system. Consequently, any attempt to reduce the number of input variables must be balanced with the desired accuracy of the final model. In literature, there are many techniques to reduce the number of input variables, which can be classified in two main categories:

(1) Feature extraction transforms the multi-dimensional Universe of Discourse (UoD) $\mathbf{U}$ into another multi-dimensional UoD $\mathbf{U}'$ of lower dimension than $\mathbf{U}$. Such transformation is usually achieved by exploiting the distribution of the available dataset and can be either linear or non-linear. There is a wide number of feature extraction techniques and a review of them can be found in [74]. While effective for black-box models, feature extraction techniques are dangerous in interpretable fuzzy models since the new set of variables defined in the transformed $\mathbf{U}'$, will usually lack of any physical meaning. That is the main reason for avoiding feature extraction techniques when designing interpretable fuzzy systems.

(2) Feature selection techniques aim to extract a small subset of input variables without degrading too much the accuracy of the fuzzy models. Differently to feature extraction techniques, feature selection does not affect interpretability from the semantic point view, since it does not create new synthetic variables, it simply selects a subset of the original variables. As a consequence, the physical meaning associated to each variable is preserved. Feature selection is less effective than feature extraction, but it is much better suited when interpretability is a major concern.

The selection of the variables relevant to the problem may be done by a human expert, by means of statistical methods (considering the existing correlation between variables), or combinatorial methods (analyzing the influence of different subsets of variables) [18].

Many feature selection techniques have been described in literature. Most of those applied to interpretable fuzzy modeling define a ranking of features before selecting a subset of them. Feature ranking produces an ordered list of input variables, giving top positions to variables with a higher influence in determining the input-output relation represented by the fuzzy model [87]. Having a ranking allows the designer or the domain experts to make a decision on how many variables to be considered, forcing some input variables into the model – perhaps because their physical meaning is essential – or discarding some variables even though their relative position in the ranking is high. Such decisions are better supported if the ranking comes along with a correlation analysis. In this case, the qualitative information of the ranking is enriched with the quantitative information of correlation, discouraging the removal of variables that will significantly affect accuracy of the model.

Most feature selection techniques are better suited for classification than for regression problems. In [17], some feature selection methods originally designed for classification are adapted to fuzzy regression problems and the effect on accuracy and complexity reduction is analyzed in several examples.

5.2.3.2 Output Variables

In many application problems the number of output variables is greater than one (see Sect. 2.4.5 in Chap. [5]). Two approaches can be used to deal with a multiple output system: building a single Multiple Inputs Multiple Outputs (MIMO) FRBS or defining several Multiple Inputs Single Output (MISO) FRBSs:

(1) MIMO FRBSs are defined by rules where the consequent is a conjunction of two or more soft constraints. Usually, the inference procedure of these models operates by inferring the output value of each variable independently. This is possible because of the conjunctive relation of the output fuzzy constraints[1]. When the number of output variables in a single system is too high, rules become illegible and must be replaced by models with simpler knowledge representation.

(2) MISO FRBSs are defined by rules with a single consequent. When multiple outputs are required, several systems with single output should be defined, one for each output variable. The main advantage of this approach is legibility of the RB, which is simpler than for MIMO models. Moreover, since MISO models are identified independently, the resulting KBs are better suited to represent the input-output relationships for each output variable. Consequently this approach can lead to higher accuracy w.r.t. a single MIMO model (this accuracy improvement is also justified by the increased number of free parameters that characterize a family of single output models). On the other hand, independently designing several models (one per output variable) with the same input variables can produce inconsistent partitions. The fuzzy partitions generated for the same linguistic variable in two different models (the models for two different output variables) may include similar linguistic labels with different semantics. This situation of ambiguity clearly avoids a clear understanding of the acquired knowledge, hence MISO models should be avoided unless an integrated design process is considered.

Some approaches considering these concepts will be described latter as tools to improve interpretability (5.3.3.1). They will concentrate on feature selection for input variables.

5.2.4 Knowledge Base Generation

Once taken the different structural decisions related to the KB, that have been previously enumerated, and selected the input and output variables, it is time for knowledge extraction. And this is the key aspect of fuzzy systems design. The problem-specific domain knowledge plays a central role in the KB derivation process and its careful consideration can substantially improve the performance of the FRBS.

[1] Some authors generalize the definition of rules by including also disjunctive consequents [110]. In such case, however, inference cannot be accomplished independently for each output variable. For such reason, disjunctive consequents are very rare in fuzzy modeling.

When designing an FRBS to solve a specific problem, no matter what kind of problem (modeling, control, classification, etc), two types of information are available: numerical and linguistic. The former is usually obtained from observing the system, whilst a human expert provides the latter. Hence, there are three main approaches to knowledge extraction in FRBSs:

1. *Derivation from experts*. In this case knowledge is formulated on the basis of the information provided by an expert or group of experts. They produce information enough to specify the linguistic labels associated to each linguistic variable, the meaning of each label, and the set of rules. This method is the simplest one to be applied in case experts are able to express their knowledge in the form of linguistic rules. Of course, it can only be used directly when working with linguistic FRBSs. In any case, if needed it is always possible to transform those linguistic rules into simplified TSK rules to include them in a TSK fuzzy system.
2. *Derivation from examples using automatic learning methods*. Due to the possible absence of experts, researchers developed numerous inductive learning methods to extract knowledge from data. In fact this was one of the hot topics in fuzzy system design during the 1990s. These design techniques are as diverse as: *ad-hoc* data-driven generation methods [18, 39, 106, 140], variants of the least squares method [18, 134], descent methods [105], methods hybridizing the latter two ones [75], neural networks [133], clustering techniques [33, 144] and evolutionary algorithms [40], among others.
3. *Jointly working with experts and data.* When dealing with linguistic FRBSs, instead of defining the DB and RB in isolation, both parts of the KB can be designed simultaneously in case numerical (data) and linguistic (expert) information about the problem is available. In fact, many authors [13, 18, 138] consider this as a major advantage of FRBSs as these are the only systems able to combine both linguistic and numerical information in a seamless way. This question will be widely considered below.

As already said, both expert knowledge and data can be used to design FRBSs. Now we will offer a short description of those situations where expert knowledge and data are jointly considered. When integrating expert knowledge and induced knowledge (i.e., knowledge extracted from data) there are three main policies:

1. *From expert to data.* One common option is generating an initial FRBS in the form of linguistic rules defined by an expert. Subsequently, the numerical information, usually provided as input-output data pairs of the underlying system, is applied in an automatic learning or tuning phase in order to complete or refine the FRBS.
2. *From data to expert.* In this case the data is considered first so that an induced KB is built from it. Then, an expert evaluates and refines that initial KB.
3. *Iterative approaches.* Finally, the simultaneous use of both sources of knowledge in an overall design process has been considered. In [13], the design process considers both expert knowledge and knowledge extracted from data in combination, trying to yield robust compact systems with a good balance between accuracy and interpretability.

Let us assume the expert is not able to define the entire KB, i.e., he/she lacks information about the linguistic labels associated to each variable, the fuzzy sets defining the semantic of each label and the linguistic rules composing the RB. Accordingly, the expert is only able to provide partial knowledge, but there is additional numerical information available that can be used to improve or complete the expert knowledge. In this case there are different possibilities:

- The expert may specify a subset of rules that form an incomplete KB. In this case, the numerical information may be used to complement the missing rules of the KB [18].
- In some cases the expert is able to identify the possible states of the system in a linguistic fashion. However, he/she is not able to indicate the correct output that is associated to these situations. Inductive learning is then used to complete the fuzzy rule set definition by generating the unknown consequents [18].
- The expert is only able to identify the relevant variables and the linguistic terms associated to them, as well as their meaning, but not the rules. In this case, numerical information is used to identify the rules.
- Inductive methods can be used to tune the semantics of the labels, i.e, to define the DB starting from a preliminary estimate of the fuzzy sets (based on expert knowledge or by uniformly partitioning the spaces) and then applying automatic tuning methods that refine the original DB [68, 75, 80].
- In some cases the only information the expert may afford about the DB is related to the UoDs where the problem variables are identified and their associated linguistic labels, but he/she is unable to define the exact MFs defining the semantics of these labels. This situation is usually considered by defining a primary fuzzy partition for each variable by means of a normalization process [48, 84, 85]. This process involves the discretization of the variable domain, partitioning it into a number of intervals equal to the number of linguistic labels considered and associating a label name and a fuzzy set defining its meaning to each interval. Since there is no knowledge available about the form that these fuzzy sets may have, one usually defines uniform fuzzy partitions in which the fuzzy sets are symmetrical and of identical shape. The problem is that this approach often results in a sub-optimal performance of the FRBS [84, 85]. In order to address this problem, different learning and tuning processes for the automatic generation of the DB *a priori* and *a posteriori*, respectively, have been proposed.

Even considering that expert knowledge is usually better exploited when focusing on linguistic rules it is also useful for other FRBS approaches. Approximate FRBSs operate differently than linguistic ones since the RB is not directly defined based on expert knowledge. However, although an automatic learning method based on numerical information must be used to obtain the RB definition, it is also possible to incorporate expert knowledge in the design process. In this case, there are three different ways to utilize expert knowledge:

1. The knowledge is used in order to define the entire preliminary descriptive KB and afterward the linguistic rules are globally adjusted. The KB is transformed

into an approximate fuzzy RB taking into account the numerical information available [68].

2. The linguistic rules provided by the expert are directly included in the RB being generated by the automatic method based on numerical information. This method requires a simple transformation that substitutes the descriptive fuzzy sets by approximate ones. The automatic design process is capable to adjust the definition of the fuzzy sets of these rules as well [69], therefore resulting in an approximate system that demonstrates the best performance.
3. The third way to integrate expert knowledge in the RB derivation is to work with a constrained learning process. Here, the variation intervals are defined from preliminary fuzzy partitions of the input and output variables obtained from the expert information [2]. Moreover, linguistic rules derived by considering the said fuzzy partitions can be individually considered to specify the definition intervals as well as to initialize the learning process.

Finally, TSK FRBSs make use of the available information in a similar way to the approximate FRBSs introduced previously. Although the fuzzy RB may be directly obtained from expert knowledge by transforming the linguistic rules provided by the expert in simplified TSK rules, this transformation often results in poor system performance. Moreover, choosing a TSK system is not a good idea in case the only information available is the one afforded by an expert. TSK systems are supposed to be designed based on numerical information in case there is no (or only incomplete) expert knowledge [134]. Still, expert information remains useful in order to define the labels of the input variables, to specify their meaning and to identify the possible system states in advance. However, the TSK rule consequent parameters are generated by means of an automated learning process alone.

5.3 Interpretability and the Design of the Knowledge Base

This chapter is focused on the design of interpretable fuzzy systems, but we can not miss the point that what we are designing are models, classifiers, controllers, and so on, for which accuracy is a key aspect. No matter how interpretable it is, if the model does not resemble the behavior of the modeled system, it will become useless. As a consequence, interpretability will be quite important, but should always be considered jointly with other performance metrics as accuracy. In most cases a good performance of the system will imply the achievement of an appropriate trade-off between several requirements (e.g., accuracy and interpretability).

When jointly considering accuracy and interpretability, the different design approaches will adapt more naturally to one or the other requirement. This question, that affects any modeling technique, is particularly clear in the case of fuzzy modeling. In that sense, two different fields have been independently considered, the so-called LFM and PFM. Sect. 5.3.1 introduces these two approaches. But in addition to the *basic* approach, the modeling tool can be improved in terms of either accuracy or interpretability. If we start from an LFM it is possible to improve accuracy later (see Sect. 5.3.2) or even to apply techniques for a further improvement on interpretability (see Sect. 5.3.3). As a counterpart, it is also possible to first consider accuracy, that is, applying a PFM technique, and then trying to achieve a certain level of interpretability improvement of the result (see Sect. 5.3.4). An iterative process where we alternate different steps producing improvements of accuracy and interpretability is also an option (see Sect. 5.3.5). Currently, with the support of multi-objective optimization, the most common approach is the simultaneous consideration of accuracy and interpretability as two objectives of a multi-objective optimization process (see Sect. 5.3.6).

Most of the literature on interpretability of fuzzy systems concentrates on *conventional* fuzzy systems. However, it is worth noting that recently the particular cases of Hierarchical Fuzzy Systems (HFSs) and type-2 fuzzy systems have been considered from the point of view of interpretability. That represents a different design paradigm (see Sects. 5.3.7.1 and 5.3.7.2).

5.3.1 Linguistic and Precise Fuzzy Modeling

Different design strategies have been proposed to handle the balance between interpretability and accuracy, usually assigning greater importance to one or the other, depending on the specific application or users needs. In this respect, two main directions have been explored: the already mentioned LFM (paying special attention to interpretability) and PFM (mostly devoted to improve accuracy).

But, independently on the starting point, it is always possible to workout an adequate trade-off offering a suitable level of both, accuracy and interpretability. Some of the options to do so will be considered in this section. The underlying idea is selecting a design approach focused on our main target (either LFM for interpretability or PFM for accuracy), adding then on top of this initial method some mechanisms to improve the levels of accuracy and/or interpretability.

In summary, we will have two starting points (LFM and PFM), extended with mechanism to: (1) improve accuracy; (2) improve interpretability; (3) iteratively improve accuracy and interpretability in a sequential way; or (4) simultaneously improve accuracy and interpretability by means of multi-objective optimization.

The mixture of these options will lead to six different strategies as illustrated in Fig. 5.1:

1. LFM with improved accuracy (described in Sect. 5.3.2).
2. LFM with improved interpretability (described in Sect. 5.3.3).
3. PFM with improved interpretability (described in Sect. 5.3.4).
4. PFM with improved accuracy (out of the scope of this book).
5. Iterative approaches for improving accuracy and interpretability (described in Sect. 5.3.5).
6. Multi-objective design for optimizing the balance between interpretability and accuracy (described in Sect. 5.3.6).

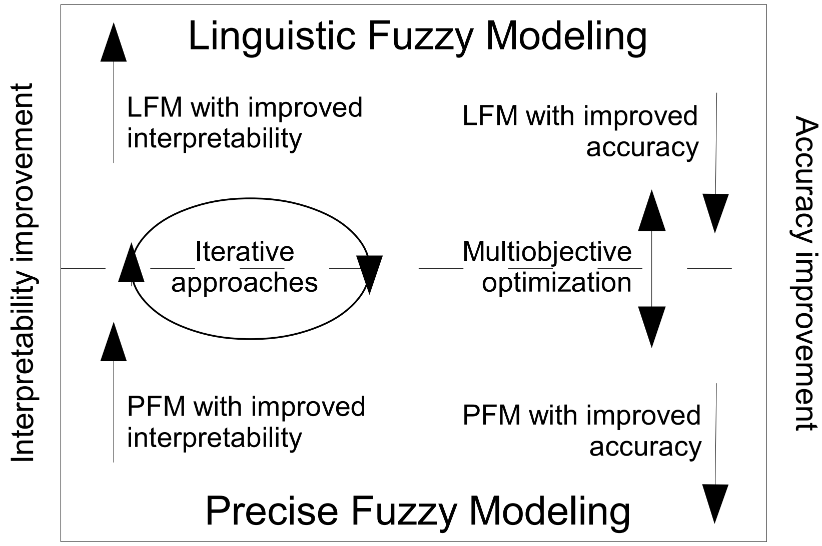

Fig. 5.1: Improvements of interpretability and accuracy in fuzzy modeling.

Both, the iterative and the multi-objective approaches, can start from either LFM or PFM. Although historically accuracy was the main concern of designers, at present the situation is much more balanced. Starting either from a system designed paying initially attention to either interpretability or accuracy, it is possible to introduce improvements of interpretability and accuracy.

The trade-off is particularly considered by those approaches that include both interpretability and accuracy criteria either independently or simultaneously. From the different options shown in Fig. 5.1, we can point out LFM *with improved accuracy* and PFM *with improved interpretability*, where both criteria are considered, one in the initial design and the other in the subsequent refinement. In addition some approaches include both criteria in a single process (iteratively or simultaneously).

It is also interesting the concept of LFM *with improved interpretability*. Although LFM has interpretability as the main concern by considering a model structure with a high descriptive power by itself, some design problems can produce a system not as interpretable as desired. Among those problems, the most common ones are the curse of dimensionality, and the excessive number of input variables or fuzzy rules. In the event of any of those problems occurring, the need of interpretability improvements to restore the desired balance is justified.

We will not pay attention to the approaches that start from a PFM, and try later to improve even more the accuracy of the model. These approaches disregard the comprehensibility of the model. In fact they act almost as black-box techniques and do not follow the original objective of fuzzy modeling, not profiting from the advantages that distinguish fuzzy techniques from other modeling techniques.

5.3.2 From Interpretability to Accuracy

Let us consider first the situation where a fuzzy system is initially designed under the premises of interpretability, and then refined to improve its accuracy. This is mostly the case of LFM with improved interpretability.

The basic structure for LFM is that of a Mamdani FRBS with linguistic variables and rules. Having clear advantages related to interpretability, it can be argued that LFM has certain inflexibility (reducing its approximation potential) due to the use of a global semantics that gives a general meaning to the considered fuzzy sets. Indeed, the use of linguistic variables imposes the following constraints [20]:

1. The use of rigid partitioning of the input and output spaces induces a lack of flexibility in the resulting FRBS, and consequently, a limited approximation potential.
2. By design, the linguistic partitions associated to the different variables are decoupled. As a result, when there is dependence among input variables, it is very hard to make a fuzzy partition of the input space by means of independent partitions.
3. Partitioning of the input and output spaces is usually homogeneous. In most problems the input-output mapping has areas of different complexity. Consequently, an homogeneous partitioning will either produce underrepresented or overrepresented regions. Regions where either the level of description will be too coarse (not being enough) or too fine (being inefficient).
4. The size of the KB directly depends on the number of variables and linguistic terms in the system. The derivation of an accurate linguistic FRBS requires a significant granularity, i.e., it needs the creation of new linguistic terms leading to a course of dimensionality problem. This granularity increase causes the number of rules to rise significantly, which may hamper the capability of being interpretable by human beings. In summary, some LFM approaches do not scale well to high-dimensional spaces or complex problems. In most cases, it would be possible to obtain an equivalent FRBS with a reduced number of rules, by discarding the rigid partitioning.

The final result is that in some cases, as a consequence of these drawbacks, a system of this kind does not achieve the desired accuracy. Having that in mind, it is possible to introduce some changes to alleviate the problem:

- *Modify the modeling process*, through the improvement of the DB design with additional tuning/learning components, but also by improving the RB.
- *Modify the model structure* by slightly changing the rule structure to make it more flexible.

These ideas will be considered in the following subsections, where we introduce some of the improvements described in the literature for designing the DB, learning the RB with sophisticated methods, and extending the model structure.

5.3.2.1 Improving Accuracy through Data Base Design

The simplest LFM methods are solely focused on obtaining the set of fuzzy rules defining the RB of the model [135, 140]. The DB is previously obtained from expert information (if available) or by a normalization process. And once it is defined, the DB remains fixed, not being adapted or refined afterwards.

It has been shown, however, that the automatic design of the DB can increase the approximation capability of a linguistic model [53, 119]. This design offers the following options: (1) defining the optimum number of linguistic terms used in the fuzzy partitions (i.e., the granularity); (2) selecting the most appropriate shapes for the MFs; or (3) distributing the fuzzy sets in the UoD of the corresponding variable.

This process of *learning/tuning* the MFs consist on defining the number, shape and positions of the fuzzy sets giving meaning to the linguistic labels associated to a certain linguistic variable. In summary, creating the fuzzy partitions for the linguistic variables.

In most cases, fuzzy partitions are made up of parameterized fuzzy sets, being those fuzzy sets either piecewise linear (triangular or trapezoidal), or differentiable (Gaussian, sigmoidal, etc.). Consequently, the most typical approach to learn/tune the MFs is that of *learning/tuning the* MF *parameters* . This is an approach broadly considered in literature [38, 68, 77, 80, 88, 111] and it has produced significant results in the field.

The common representation method considers a code with one to four parameters per fuzzy set, where the parameters usually define some characteristic points of the set (support, modal points, etc.). For example, if we consider a triangular MF as follows:

$$\mu(x) = \begin{cases} \frac{x-a}{b-a}, \text{ if } a \le x < b \\ \frac{c-x}{c-b}, \text{ if } b \le x \le c \\ 0, \quad \text{otherwise} \end{cases} \tag{5.10}$$

the most common representation will be that with the three parameters of the function (a, b, c) which represent the support of the set $[a,c]$ and its modal point or prototype (b). Changing any or several of these three parameters will modify the fuzzy set associated to the MF, thus influencing the performance of the FRBS.

It is also possible that the parameters of a fuzzy set in the partition were linked to those of a different fuzzy set in the same partition, in such a way that they cannot be modified independently. For example, it is quite common that for SFPs, the parameter bounding the core of a fuzzy set, defines simultaneously the support of an adjacent set. This approach ensures that the modification of parameters will not break the constraints imposed by an SFP. In addition, by linking some parameters the number of effective parameters to adjust is reduced, simplifying the learning/tuning process. Fig. 5.2 illustrates a situation where a partition with seven terms {VERY LOW (VL), LOW (L), MEDIUM LOW (ML), MEDIUM (M), MEDIUM HIGH (MH), HIGH (H), VERY HIGH (VH)} is codified by means of five anchor points (a, b, c, d, e). In this situation, changing a single parameter (c to c′), will transform three sets (ML′, M′, and MH′). This happens as a consequence of the fact that this single parameter simultaneously represents the upper bound of the support of ML, the core of M and the lower bound of the support of MH. It is quite important to notice that both, the previous partition, and the new one are SFPs. By considering this approach the changes in the parameter do not brake the properties of the SFPs. We can consider this approach either as a tuning of the MFs, or more properly as a joint tuning of the whole fuzzy partition.

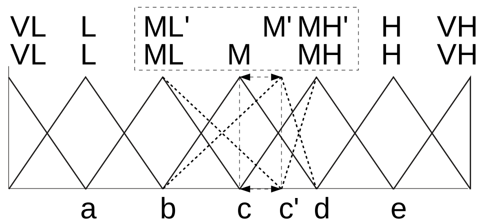

Fig. 5.2: Adapting a fuzzy partition by changing its parameters.

A somehow related approach to this one is that of *establishing the context*. It is usually performed by defining the *scaling functions* that map the input and output variables onto the UoD over which the fuzzy sets (in most cases a normalized partition) are defined. From a linguistic point of view, the scaling function can be interpreted as a sort of context information. We may distinguish between linear and non-linear scaling functions (see Sect. 2.4.1 in Chap. [5]), which are well known in classical control theory. Both *linear scaling* and *non-linear scaling* [62, 93] have been considered, but it is also possible to define methods combining both kinds of context adaptations [41].

It is also possible to directly modify the meaning of a term without effectively changing the fuzzy set representing it. To do so the option is *using linguistic modifiers* [44, 89]. The use of a linguistic modifier is equivalent to the application of a functional transformation to the original MF. The possible transformations are quite different and will be largely considered in Sect. 5.3.2.3. They allow considering more flexible expressions for the MFs.

Some other contributions propose methods to define not only the MF *shapes* but also the MF types (such as triangular, trapezoidal, Gaussian, and sigmoid) [126].

Finally, it can be considered the option of changing the granularity of the partitions. The main difficulty here is that the change of granularity in a system with some predefined rules will either remove a term that was applied by some of those rules, or add a term that was not considered in any of the existing rules. Both options seem to be useless. In [19], the authors obtain the initial set of candidate rules by considering uniform SFPs with $T_{\max}$ terms per variable. The learning of the RBs, and the optimization of the parameters of the fuzzy sets are performed considering such partitions, computed at the very beginning of the process. This process produces a sort of virtual RBs. Then, fuzzy set parameters are tuned using the virtual partitions.

In addition, during the process of learning, the granularity is adapted, and the systems are evaluated according the actual granularity (even considering that they use the virtual rules and partitions). To do so, each time we need to evaluate the fitness, the evaluation is performed on the actual number of fuzzy sets used to partition the single variables (and not on the virtual partition). This process requires the definition of proper mapping strategies, both for the RB and for the fuzzy set parameters, to transform the virtual partition on a partition with the desired granularity.

Interpretability Preservation when Designing the DB

It is possible that the design and adaptation of MFs generate complex semantics disturbing the expert interpretation, and reducing legibility. In an extreme case, the process could lead to a situation as that depicted in Fig. 5.3, where a completely transformed partition significantly reduces its interpretability. In this situation it is almost impossible to assign meaningful terms to the fuzzy sets S_1 to S_7.

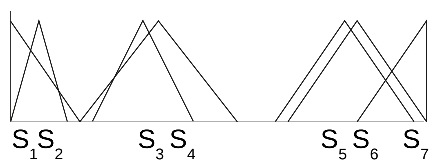

Fig. 5.3: Certain interpretability may be lost when designing the DB.

The third Chap. [6] has established many interpretability constraints both at the level of fuzzy sets (e.g., normality) and fuzzy partitions (e.g., distinguishability, order preservation, coverage, etc.). The fulfillment of those constraints could be seriously compromised by some of the methods and techniques previously described. To avoid that situation, different constraints can be added to the DB design process ensuring a minimum impact on interpretability. Just to offer some key aspects of the problem, some of those interpretability constraints, and its preservation, will be briefly considered:

- *Normality.* Every MF should exhibit full matching with, at least, one element in the UoD. That is, the fuzzy sets should be normal. It is worth noting that the transformations previously described do not affect normality of fuzzy sets, except when a lateral displacement of a set produces its *modal point* to go out of the UoD of the variable. As a consequence, this property will be maintained by forcing the modal points of the extreme MFs to be contained in the UoD.
- *Distinguishability and order preservation.* Each linguistic term should have a clear meaning, the associated fuzzy set should clearly define a range of the UoD, and the semantic ordering of terms should be kept by the corresponding MFs.

This property may be satisfied by constraining the relative variation of MF parameters [91], merging similar MFs [53], or establishing a semantic ordering among the linguistic terms [22].
- *Coverage.* Every value of the UoD should belong, with a certain membership degree, to, at least, a linguistic term. This property may be satisfied by using variation intervals that ensure a certain level of overlapping between the supports of consecutive fuzzy sets [38].

Working with SFPs, we will ensure normality, distinguishability, ordering and coverage (as well as some other properties). Consequently, SFPs are widely used in the field when considering tuning methods as the one described in Fig. 5.2.

Adding constraints to ensure the semantic integrity will obviously make the derivation process less flexible but, at the same time, will reduce the risk of overfitting.

If the design process is based on search/optimization techniques, the preservation of the interpretability constraints for fuzzy sets and fuzzy partitions can be ensured as part of the design process, by simply including interpretability measures as part of the objective function. As an example, measures of completeness, consistency [78], compactness, similarity [76], or even more advanced criteria as conciseness [132], could be considered.

Integrating the DB Design in the whole FRBS Derivation Process

Previous sections have considered DB design, being a component of the whole design process that can interact in different ways with other components of that process. Present section will define various interaction schemes to integrate the DB design into the overall KB design process. Four options will be considered:

- *Previous design.* The DB is induced from available data, usually applying non supervised clustering techniques [90, 109], prior to the definition of the rules and without considering other components of the KB. This schema is also referred to as DB learning approach.
- *Embedded design.* Several DBs are obtained and the corresponding KBs are completed by using a simple learning method that designs other components of each model (e.g., the RB) [41, 119]. Then the DB that generates the overall KB with the best behavior is selected.
- *Integrated design.* The process of designing the DB is jointly developed with the derivation of other components such as the RB in a simultaneous procedure [78, 97, 126]. This schema is also known as RB learning approach.
- *DB tuning.* The initial DB can be obtained by any method, including the use of uniform partitions of the UoD. Later, this predefined DB is refined; once the rest of the KB components have been obtained. This option is widely used and the tuning process changes the MF shapes [38, 68, 80, 89] while considering as the main requirement an accuracy improvement of the linguistic model. Even if this final step is devoted to increase accuracy, interpretability preservation is also

present in most cases. To do so, the tuning process is constrained to avoid losing the main properties of the fuzzy sets or fuzzy partitions. As an example, an MF tuning constrained to keep the SFP structure is proposed in [9]. DB tuning for interpretability improvement is also possible (e.g., merging MFs [53]).

While integrated design of the DB together with other components constitutes a whole derivation process by itself, the other three described schemes should be combined with other methods to perform the overall derivation process in several sequential stages.

Of course, several of these approaches can also be jointly considered. For example, in [88], a two-stage DB design is made by first considering an integrated design of the DB and the RB, and then performing an additional DB tuning. In [77], an initial generation of the RB with a subsequent three-stage DB design (input variable selection, simultaneous DB tuning and RB reduction, and DB fine tuning) is developed. It is quite important to consider that the DB design gives more flexibility to the modeling process while at the same time increases the risk of losing interpretability and overfitting the problem.

As a final and general consideration, the comparison of sequential vs. simultaneous derivation schemes offers pros and cons. The main advantage of sequential approaches is that of a reduction of the search space to be considered for each individual stage. On the other hand, simultaneous derivation offers a better management of the strong dependency of the different components, but producing a process significantly more complex. In this respect, the cooperative coevolutionary paradigm [112] offers its high ability to manage huge search spaces and decomposable problems, allowing the definition of affordable simultaneous derivation methods [31, 111].

5.3.2.2 Improving the RB Learning Method

Once considered DB design, it is also possible to modify the RB learning method to improve accuracy. Maintaining the basic structure of DB and RB, thus resulting in the highest interpretability, the idea is to increase the capability of the rules for *interpolative reasoning* by focusing on the cooperative action of the linguistic fuzzy rules. This is the case for methods like COR (which stands for COoperative Rules) proposed in [28] with the primary objective of inducing a better cooperation among the linguistic rules. To do that, the RB design is made using global criteria that jointly consider the action of the different rules.

In [58], a method is proposed to refine a previously obtained RB in disjunctive normal form. The heuristic process comprises three sequential steps: (1) the precision of the model is firstly improved by removing some linguistic terms from the rule antecedents; (2) the generalization capability is improved afterwards by adding linguistic terms to the antecedents; and (3) the completeness of the RB is finally attained by the addition of new linguistic fuzzy rules.

5.3.2.3 Improving Accuracy by Extending the Mamdani Model

The limited flexibility of the descriptive Mamdani FRBSs is one of the main characteristic reducing the approximation potential of LFM. To cope with this circumstance it is possible to add a certain level of flexibility to these FRBSs by defining extended Mamdani models. Some of those models will be considered below.

Extensions Related to the Linguistic Partitions

Obviously, removing the restriction of using a fixed partition for each variable, and consequently, allowing each rule to consider a different fuzzy set in its definition, will significantly increase the flexibility and approximation potential of the model. But this change will move our model from the field of LFM to that of PFM. An intermediate solution could be allowing a certain relaxation of the fixed partition assumption, assuming some restrictions.

A possible option has, somehow, been mentioned before. It is possible to consider a fixed partition, allowing the use of linguistic modifiers that slightly change the meaning of the linguistic labels transforming the predefined MFs [32, 44, 60]. The idea was already considered in the early years of fuzzy modeling when Zadeh highlighted, in [146], the use linguistic hedges or, in a wider sense, *linguistic modifiers*, as a way to slightly modify the meaning of linguistic labels without losing the description capability. It is important to notice that the inclusion of linguistic modifiers in fuzzy rules differs from their use in the design of the DB [89] (explained in Sect. 5.3.2.1).

A linguistic modifier [23, 24, 145] is an operator that alters the meaning of a linguistic term, by applying a transformation to the MF of the fuzzy set associated to the linguistic term. When working on the basis of descriptive Mamdani rules as the following one

IF TEMP is HIGH THEN REFRIGERATION is HIGH,

the use of linguistic modifiers will allow the generation of a new kind of rule:

IF TEMP is VERYHIGH THEN REFRIGERATION is VERYHIGH.

In this new rule structure, VERY is a linguistic modifier and VERYHIGH is a new linguistic term derived from HIGH. The MF of this generated term has not been defined when designing the fuzzy partition, it has to be computed from the MF of HIGH. Assuming that $lm_{\mathrm{VERY}}(x)$ defines the transformation corresponding to VERY, the MF of VERYHIGH will be $lm_{very}(\mu_{\mathrm{HIGH}})$. It is important to notice that the use of linguistic modifiers, does not change the meaning of the so-called *primary term* (HIGH in this case) and does not require to add new terms in the partition (i.e., it does not require increasing granularity), but allows the generation of new terms with meaning related to the primary terms.

Linguistic modifiers are usually divided into those that reinforce the characterizations (e.g., VERY) and those that weaken them (e.g., SLIGHTLY). But this is merely a linguistic point of view. If we consider the question from a modeling point of view, the usual classification divides linguistic modifiers in three main groups: (1) power, (2) expansion/contraction and (3) displacement.

- *Power modifiers* [145] change the MFs by applying a power operator to the original function (see Fig. 5.4, left). As a result neither membership values equal to zero nor those equal to one are changed (i.e., these modifiers do not change either support or core of the fuzzy sets). Its general expression is:

$$\mu'(x) = (\mu(x))^{\alpha}, \ \ 0 < \alpha \ \ , \tag{5.11}$$

  These modifiers are also called *linguistic hedges* [145], or concentration/dilation modifiers, since values of $\alpha < 1$ dilates the fuzzy set, whilst with $\alpha > 1$ the modifier concentrates the fuzzy set.
  The modifier VERY previously considered is a power modifier obtained with $\alpha = 2$, i.e., $\mu_T^{\text{VERY}}(x) = (\mu_T(x))^2$.
- *Expansion/contraction modifiers* [24] change the support and core of the fuzzy sets enlarging or reducing them, but trying to maintain a center of gravity similar to the original (see Fig. 5.4, center). The effect over the fuzzy sets, though similar to the previous one (in terms of dilation and concentration), is rather more severe and it is implemented with linear variations.
- *Displacement modifiers* [24] are defined by translations of the MFs along their domains (see Fig. 5.4, right). The effect over the linguistic terms is more severe than the aforementioned ones and changes the center of gravity of the sets. This idea is widely considered under the framework of the so-called 2-tuple representation, where the linguistic information is expressed by means of 2-tuples, which are composed of a linguistic term and a numeric value assessed in (-0.5, 0.5) representing the displacement [101]. In this way, the displaced set is related to the originating one.

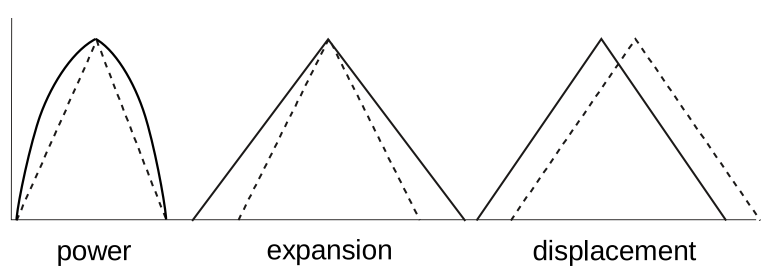

Fig. 5.4: Types of linguistic modifiers.

Extensions Affecting Rules Structure

The usual extension of a modeling tool in the search for a higher accuracy is considering additional degrees of freedom. Obviously, PFM, where the concept of linguistic partition is removed, offers a higher flexibility and consequently a better potential for approximation than LFM. But the extensions we consider in this section will maintain the linguistic approach so that the underlying idea of considering a linguistic partition for each variable, and using rules referring to the fuzzy sets of those partitions, is kept.

The most common transformation of linguistic rules to improve accuracy concerns the modification of the consequent part of the rules. Two main options are possible (either independently or combined): (1) considering more than one consequent per rule, and (2) weighing the consequent (single or multiple) of the rules. Notice that, other options, such as the use of Disjunctive Normal Form (DNF) rules, are mainly devoted to improve interpretability and they will be, consequently, considered in the corresponding section.

The idea of using *rules with double consequent* was first proposed in [106]. The approach involves the use of rules where each combination of antecedents may have two consequents associated when it is necessary to improve the model accuracy [39, 53]. This approach will clearly improve the interpolative capability of the model and its performance. The effect is similar to that of adding new intermediate terms in the fuzzy partitions of the output variables.

The obtained rules will have the following structure

IF TEMP is HIGH THEN REFRIGERATION is {MEDIUM,HIGH},

where the usual fuzzy inference system can be applied simply considering that each double-consequent fuzzy rule can be decomposed into two different rules with a single consequent, both of them being fired with the same degree.

It is important to notice that, even though the interpretation of the action performed by every rule may be confusing to some extent, it does not compromise the linguistic interpretability of the rules. In fact, the double-consequent rule previously presented may be interpreted as follows [39]:

IF TEMP is HIGH THEN REFRIGERATION is *between* MEDIUM *and* HIGH.

Another extension is that of using an additional parameter for each rule, indicating its importance degree in the inference process, breaking the usual consideration of all rules are equally important to compute the output. The use of weighted rules offers an additional approximation capability to improve accuracy [3, 107, 122]. The resultant rule will have the structure

IF TEMP is HIGH THEN REFRIGERATION is HIGH *with* $[w]$,

where w is the real-valued rule weight.

In this approach, some changes to the classical inference system must be made to consider the weighted action of each rule. Consequently this question can also be considered from the point of view of the processing structure (see Sect. 5.4.3).

The operator *with*, which attaches a weight to a rule, may be defined in different ways. One of the most usual options is as a multiplicative factor for the matching degree of the antecedent before applying the implication operator.

Some proposals make use of these weights to improve the model accuracy with an automatic learning of weights using different techniques such as heuristic methods [122], gradient descent processes [107], or evolutionary algorithms [3].

It is even possible to combine the two previous approaches by using rules with multiple consequents where each consequent has a different weight.

Regarding the use of weights, its effect on interpretability is somehow controversial. On the one hand, some authors consider rule weights as a negative aspect from a semantic interpretability point of view [104], and recommend its replacement with modifications in MFs. On the other hand, it is also possible to find authors considering that the comprehensibility of the model viewing weights as certainty grades is not deteriorated significantly [73].

5.3.3 Improving Interpretability in Linguistic Models

Interpretability is not an intrinsic property of fuzzy systems, obtaining an interpretable fuzzy model is mostly a matter of design. Some approaches to fuzzy modeling have a better chance for interpretability than others, being that the case for LFM. But even when considering an LFM approach, interpretability is still a matter of design [10]. Questions such as the number of considered variables, the granularity of the fuzzy partitions, the dimension of the rule set or the interpretability of the fuzzy sets, will lead to a more or less interpretable model, even if we consider a similar rule structure and design process. As a result, most approaches considered in the previous section focused interpretability by, first, considering a Mamdani model with fuzzy partitions and interpretable linguistic rules, and second, trying to control the growth of the model with a reduced number of variables, linguistic terms and rules.

A different option to achieve a good balance between interpretability and accuracy is considering the first premise of the previous description, i.e., using fuzzy partitions and linguistic rules (the main characteristics of LFM), but trying to obtain accurate initial models without a significant control of the dimensionality questions or the interpretability of the fuzzy sets, and subsequently applying a process to improve the interpretability of the obtained model even at the expense of losing certain accuracy.

These accurate initial models (in LFM), usually contain a large number of input variables, a great variety of linguistic terms, oversized RBs, or illegible fuzzy sets. This section will analyze different mechanisms to improve the interpretability in these models. Moreover, there is always the chance of indirectly improving the accuracy in terms of generalization capability, with a final refinement of the RB to remove the existing redundancies and inconsistencies. In most cases, this process will increase both interpretability and accuracy.

5.3.3.1 Global and Local Feature Selection

Working with high-dimensional problems, with a large number of input variables, will usually produce large KBs. The course of dimensionality, which is related to the exponential growth in the number of rules due to the homogeneous partitioning of input and output spaces associated to linguistic variables [20], will significantly reduce the interpretability of the system. In addition, rules considering a large number of input variables will have a limited description ability, since the condition to activate the rule will have so many components that will be hardly understood. A potential solution to these problems is the use of feature selection techniques to reduce the number of variables used either by the overall model or by each individual rule. The impact of feature selection and feature extraction techniques on interpretability has been previously considered from a conceptual point of view (Sect. 5.2.3). Feature selection will be considered now from the implementation point of view.

Basically, we may distinguish between two selection processes:

1. *Feature selection at model level* involves the selection of a subset of the input variables to be used in the model or, what is equivalent, the removal of those input variables that do not significantly contribute to model performance.
   This selection process has been widely applied to classification problems [82], where it is quite common that models have to manage a large number of features. Nevertheless, in [77] an interesting contribution is proposed for linguistic FRBSs including feature selection within a more complex deriving process (i.e., with rule generation, DB tuning, and rule selection).
2. *Feature selection at rule level* implies that, even if the whole set of input variables is considered by the model, the selection process could take place at the level of rules. With this approach, the RB will still contain a large number of rules, but those rules will at least be simpler and will allow an easier understanding. That is because the antecedent length of each rule varies avoiding the need of using all the variables involved in the system. It does not mean that a specific input variable ignored in a rule could not be used in another one. In [82], this input variable selection at RB level is considered together with a global selection of the variables and a merging of the rules. On the other hand, the methods proposed in [34, 93] obtain the most significant input variable for each rule during the learning process, instead of performing a previous selection.

The selection could take place either previously, so reducing the variables to be considered by the rules, or afterward, by removing variables from preexisting rules. In the latter case, the removal of some input variables may cause the existence of several rules with identical antecedents that could imply an inconsistency. The most usual solution for that situation is rule merging that will be considered bellow.

5.3.3.2 Selecting/Merging Linguistic Rules

Working with a higher granularity is in many cases the way to increase the accuracy of a fuzzy model, leading to RBs with an excessive size. However, in some cases the excessive size is caused by a deficient learning process tending to generate too many rules. Some of the rules in the RB can even be *redundant rules*, not containing relevant information and considering situations already covered by other rules; *erroneous rules*, which are wrongly defined and distort the FRBS performance; and *conflicting rules*, which perturb the FRBS performance when coexist with others. Besides worsening the accuracy, an excessive number of rules makes difficult to understand the model behavior.

To face this problem, an RB *reduction process* can be developed by merging rules and/or selecting a subset of rules to achieve the goal of minimizing the number of considered rules while maintaining (or even improving) the performance. Indeed, depending on the criteria considered to reduce the RB, this process can be considered as a mechanism to improve not only the interpretability but also the accuracy. This reduction is generally applied as a postprocessing stage, once an initial set of rules has been derived.

It is possible to distinguish two main approaches for this reduction:

1. *Selecting linguistic rules* involves obtaining an optimized subset of rules from a previous RB by selecting some of them. There are several methods to do so with different search algorithms in the literature [38, 39, 69, 72]. This question is particularly important when working on applications with big data, where most of the considered methods will lead to large RBs. Several works, in the Big Data research field, include a rule selection step in the process of knowledge generation [4, 51].
2. *Merging linguistic rules* is an alternative approach that reduces the RB by merging the existing linguistic rules. In [82], the authors propose to merge neighboring rules, i.e., linguistic rules where the linguistic terms used by the same variable in each rule are adjacent. The merger is carried out in three different ways: (1) using a new fuzzy set that groups the adjacent linguistic terms; (2) merging the adjacent fuzzy sets if they are very similar; and (3) giving the set of rules in DNF. Another proposal is presented in [70], where a special consideration to the merging order is made.

A more general simplification process is described in [11, 13] where in addition to the reduction of rules, they obtain a partition reduction. It starts looking for redundant elements (labels, inputs, rules, etc.) that can be removed without altering the system accuracy. Then, it tries to merge elements always used together. Finally, it forces removing elements apparently needed but not contributing too much to accuracy. In fact, it is an iterative process first acting on the rules and then on the partitions at each iteration. This cycle is repeated until no more interpretability improvement is feasible without penalizing accuracy more than a predefined threshold.

From a different point of view, the RB may be reduced by using a disjunctive form for the fuzzy rules that groups several rules within a more general expression, thus easing the interpretability. This approach is explained in next section.

5.3.3.3 Alternative Linguistic Rule Expressions

Another possibility to improve the interpretability of linguistic FRBSs is to use an alternative model structure to give a higher descriptive power to each rule. With this extended description we can represent linguistic models in a more compact structure with minor accuracy loss. To do that, the linguistic fuzzy rule expression is extended to make it more flexible. Some examples are shown in the following:

- DNF *rules* represent a particular case of fuzzy rules with the following structure [59]:
$$\text{IF } X_1 \text{ is } \widetilde{A_{X_1}} \text{ and } \dots \text{ and } X_n \text{ is } \widetilde{A_{X_n}} \text{ THEN } Y \text{ is } A_Y ,$$
where each input variable X_i takes as a value a set of linguistic terms $\widetilde{A_{X_i}} \equiv X_i$ is $A_{i,1}$ or $\dots$ or X_i is A_{i,l_i}, whose members are joined by a disjunctive operator, whilst the output variable remains a usual linguistic variable with a single label associated.
This structure uses a more compact description that improves the interpretability of the model. Moreover, the structure is a natural support to allow the absence of some input variables in each rule (simply making $\widetilde{A_{X_i}}$ be the whole set of linguistic terms). Several learning methods have been proposed considering this structure [34, 59, 82, 93, 97].
- *Exception rules* constitute an interesting possibility to represent a more interpretable and compact description [83]. In [27], the authors make a fine-tuning of the meaning of each linguistic rule by excluding a local region of its firing region. This consideration is especially useful when a rule structure with multiple fuzzy input subspaces (e.g., the DNF structure) is considered.
- *Generic and specific rules* represent a concept that is somehow related to that of exception rules. In this approach a generic rule is applied only when no applicable specific rule is available. The idea is to define a prioritization of rules in such a way that rules with a different level of specificity receive a different priority, being higher for more specific rules, and lower for more generic rules [141]. The main conceptual difference with exception rules is that exceptions are assumed to be quite limited in number, while the concept of generic and specific rules does not imply that assumption.
The structure of generic and specific rules, with levels of priority, has clear effects from the point of view of output explanation, where interpreting the output involves understanding the concept of *level of specificity* of the rule. This approach could be considered as a particular case of hierarchy (defined by the priorities) and then included in Sect. 5.3.7.1. In this case, the hierarchy is the effect of a particular implication mechanism that applies the rules by taking into account its

priority. As a consequence, this kind of approach could also be considered in the context of designing the fuzzy processing structure (see Sect. 5.4).

5.3.3.4 Hierarchical Partitions

A completely different idea, but somehow connected to the previous approach of generic and specific rules, is to modify the overall structure of the KB by establishing for each variable a hierarchy of partitions with different levels of granularity, i.e., a hierarchical DB. In fact, it would be possible to describe this approach as based on *generic and specific partitions*.

With this concept, the KB is composed of a set of layers where each one contains linguistic partitions with different granularity levels (i.e., different layers in the hierarchical DB) and linguistic rules whose linguistic variables take values in these partitions (a layer of the hierarchical RB) [43]. The considered granularity at each layer is related to that of the previous layer by adding a new term in between each two terms of the previous partition, that is, the sequence is two, three, five, nine, seventeen, etc. This structure allows maintaining the preexisting terms (defined by fuzzy sets obtained by contracting the previous ones), and simply adds some new terms. Fig. 5.5 shows this process for the first three layers of a hierarchy.

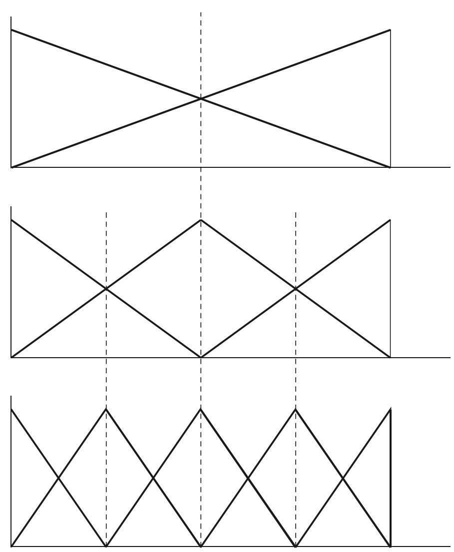

Fig. 5.5: A hierarchy of partitions with 2, 3 and 5 terms.

In this way, each rule proceeding from a certain layer can be directly used at any higher layer since terms are not removed, simply refined (from the semantic point of view). In the figure we can see how each fuzzy set at a certain layer is compressed when moving through the hierarchy.

The idea is clearly related to that of generic/specific rules, but applying a systematic design process. Designing this kind of systems considers either reduction or expansion methods. In [72], a hierarchical KB is built by creating several hierarchical DBs, and then generating the complete set of linguistic rules in each of these partitions. The union of all of these sets of rules is taken and reduced by finally performing a genetic rule selection process (as those previously mentioned in Sect. 5.3.3.2). On the other hand, in [43], an inductive linguistic rule generation method progressively refines the controversial regions (i.e., those regions covered by linguistic fuzzy rules with a bad performance) by defining new rules in a deeper layer with an expansion method. The obtained hierarchical RB is then compacted by a subsequent selection process. A reduction approach by merging fuzzy sets to progressively reducing the granularity is proposed in [63].

As said for the approach based on generic and specific rules, this technique could be considered as a particular case of hierarchy and then included in Sect. 5.3.7.1.

5.3.3.5 Linguistic Approximation

The linguistic approximation [52] is the process that tries to find a linguistic description that represents a given fuzzy set. Given a linguistic FRBS where the DB has been automatically obtained or optimized, it is possible that some fuzzy sets are not directly linked to a certain term. In this case a linguistic approximation procedure may be used to find an interpretation of those fuzzy sets to improve the comprehensibility of the model. During this linguistic approximation, a certain accuracy loss may happen.

The linguistic approximation is usually performed with preexisting linguistic terms and include linguistic modifiers. Some examples of methods that improve the interpretability of the model with linguistic approximation are [50, 131].

5.3.4 From Accuracy to Interpretability

After the initial period of success of fuzzy controllers built on the basis of Mamdani systems with linguistic structure, a second age of fuzzy systems started with the new TSK modeling approach presented by Takagi, Sugeno and Kang [130, 134], and the different design methods introduced under the umbrella of neuro-fuzzy systems [75, 133] and genetic fuzzy systems [80, 135]. While some of those methods based on neural and genetic techniques maintained the idea of fuzzy partitions with a sort of semantic integrity, many others considered fuzzy sets and fuzzy partitions as parametric functions useful for optimization. As a result, tuning these functions (MFs or fuzzy partitions) was included as part of the design process, and the associated semantic (linguistic meaning) disappeared. This situation led to the definition of approximate Mamdani systems, that jointly with TSK systems were included under the umbrella of PFM to achieve more accurate fuzzy models. This fact caused a shift from the seminal intent of fuzzy modeling, and the new approaches increasingly produced black-box models.

But once achieved those higher levels of accuracy, the need for interpretability was again there. As the accuracy improvement is usually obtained by adding degrees of freedom to the fuzzy systems, the effect is in most cases a more complex model. As a consequence, one of the options for recovering some interpretability is reducing that complexity. This complexity reduction can be achieved either at the level of fuzzy partitions/sets, or at the level of rules. Both approaches are considered below.

5.3.4.1 Simplifying the Fuzzy Partitions/Sets

Once broken the semantic approach to linguistic partitions, the design of a fuzzy partition without semantic restrictions will in most cases lead to the coexistence, in that fuzzy partition, of quite similar fuzzy sets, that can even be redundant or irrelevant, making the model unnecessarily more complex [123]:

- *Redundancy* refers to the coexistence of similar fuzzy sets representing compatible concepts. With this kind of fuzzy sets, the model becomes more complex and difficult to be understood (the distinguishability property [137] gets lost).
- *Irrelevancy* is given when fuzzy sets with a constant membership degree equal to one, or close to it, are used. This kind of fuzzy set does not furnish relevant information.

An adequate *filtering* of those sets (that increase complexity without significantly improving accuracy) will surely help in the search for a better balance between interpretability and accuracy. In fact, the effect of this filtering process will not only help in a better interpretation of the partitions, but also in a reduction of the RB, as a result of the disappearing of fuzzy sets. This idea can be applied both to Mamdani and TSK systems.

As an example, to automatically detect these undesired fuzzy sets, the use of similarity measures between fuzzy sets has been proposed [120, 123, 124]. These measures should fulfill different properties as being zero for non-overlapping fuzzy sets, greater than zero for overlapping fuzzy sets, and one for equal fuzzy sets. In addition, the chosen measure should be independent of the scaling domain [123]. Different similarity measures can be considered (e.g., the ratio between the cardinality of the intersection and the union of the compared sets).

The process to simplify the model usually consists of two steps:

1. *Merging/removing fuzzy sets* with a high degree of similarity, to create a unique fuzzy set that represents the whole collection of similar fuzzy sets. In addition, the irrelevant fuzzy sets (i.e., those with a high similarity degree to the universal set) are removed. This step is called RB simplification [124].
2. *Merging fuzzy rules* that after reducing the number of fuzzy sets result in rules with equal antecedents. This step is called RB reduction [124].

This reduction process usually makes the model less complex (i.e., more compact) and more easily interpretable (i.e., more transparent). It is even possible that the complexity reduction produces as a side effect a certain accuracy improvement. This situation is somehow related to that happening in different learning processes where adding new degrees of freedom drives to an overfitting problem (i.e., the model adapts too much to the training examples and loses its generalization ability), so increasing the error when working on unknown situations. In summary, this complexity reduction can, in some cases, palliate a certain overfitting due to the extremely flexible structure allowed to the system.

5.3.4.2 Simplifying TSK Systems by Reducing the Number of Rules

As already mentioned (Sect. 5.3.3), an efficient way to improve the interpretability in fuzzy modeling consists in selecting a subset of significant fuzzy rules that represents (in a more compact way) the system to be modeled. Moreover, this selection of important rules has the interesting advantage of reducing the possible redundancy existing in the RB, thus improving the generalization capability of the system, i.e., its accuracy. This effect was the previously described as an overfitting palliation.

There are different methods to determine the level of contribution of each rule in a system, to the overall output. Due to the particular structure of TSK rules, having a polynomial function as consequent, orthogonal transformations have been widely considered [125, 127, 139, 142]. Once computed their contribution, rules can be sorted accordingly, considering only the best ranked rules as part of the simplified system. To define the number of selected rules it is possible to use statistical information [127], thus reducing the human intervention.

Orthogonal least-squares methods can determine the output contribution and individual error reduction ratio for each rule [139]. A different option is the use of *singular value decomposition* as proposed in [142].

With an approach that can be considered as somehow closer to the rule weights in linguistic FRBSs (see Sect. 5.3.2.3), in [55], the model structure is extended incorporating an intensity parameter to each rule. This parameter allows the method to assign rule importance degrees during the inference process. As a result, this approach produces a more flexible model. In addition, an oversized RB is initially built, and the most redundant fuzzy rules are progressively removed.

An indirect approach to RB reduction is the simplification of the DB by either merging or removing fuzzy sets, as described in previous section.

5.3.4.3 Simplifying Outputs of TSK Rules

In system modeling, a TSK FRBS is usually considered as the combination of several simple models (i.e., the rules) that describe local behavior of the system to be modeled. In other words, a TSK system works on the basis of a sort of model interpolation approach, where each rule is an independent model. Previous section considered the reduction of the number of rules (interpolating a reduced number of models). This section will consider the simplification of each individual rule (interpolating simpler models). To do so, each TSK fuzzy rule could be forced to have as consequent a simpler polynomial function, being the simplest one a zero order polynomial function, that is, a constant value.

As a way to *smooth the consequent polynomial function*, constraints can be imposed to the weights involved in the polynomial function of each rule consequent [54]. A different option is using an objective function for the regression algorithm that combines the classical mean square error (as global error measure) with a local error measure used to induce competition among rules [143]. This competition makes rules describe each region better, thus improving interpretability.

On the other hand, Riid and Rüstern [117] conclude that the description of each TSK fuzzy rule is improved when the overlapping between adjacent input fuzzy sets is reduced, i.e., *isolating the fuzzy rule actions*. This is because the performance region of a particular rule is more clearly defined by avoiding other rules having a high activation strength in such an area. This approach is an alternative proposal to improve the local description of TSK rules by designing the DB instead of the RB.

The proposal in [21] integrates the two said philosophies to improve the local interpretability of TSK FRBSs. On the one hand, the use of a special type of MF based on splines improves the local behavior of the fuzzy system by only firing the most immediate rules to the given input vector. On the other hand, the consequent polynomial structure is modified to interpret the coefficients as a Taylor series expansion around the center of the corresponding rule (i.e., the vector containing the vertex of each MF considered for each input variable in the fuzzy rule).

5.3.5 Iterative Approaches

Many of the previously described works represent sequential processes where accuracy and interpretability are focused on in two steps that are mostly independent. Limiting this process to two steps (one for accuracy and one for interpretability, in any order) is an option, but it is also possible to consider more elaborated processes with additional steps.

As an example, in [120], an iterative fuzzy identification technique starts the process with data-driven fuzzy clustering of an overestimated number of local models. A Genetic Algorithm (GA) is then applied to find redundancy among the local models with a criterion based on maximal accuracy and maximal set similarity. After the reduction steps, the GA is applied again with another criterion searching for minimal set similarity and maximal accuracy. The final result is an automatic identification scheme with fuzzy clustering, RB simplification and constrained genetic optimization. In summary, an iterative process focused on accuracy, then interpretability, and finally accuracy again.

A different option is considering the sequence interpretability, accuracy, interpretability. In [129], the process starts with an LFM (the first step searches for interpretability). In a second step, accuracy is improved by means of an iterative algorithm generating fuzzy partitions with a number of fuzzy sets large enough to achieve the desired accuracy. A linguistic RB, that usually has an excessive size, is then obtained. In case the RB is too complex, then a rule merging process reduces complexity (without losing accuracy) by combining rules that have adjacent fuzzy sets, thus obtaining a set of approximate fuzzy rules where each one has its own semantic. This reduction process is somehow related to that described in Sect. 5.3.4.1, but in this case the starting point is an LFM while in that case it was a PFM.

At the end, those repeated processes could be extended as much as desired, trying to achieve additional refinement in some aspects while reducing negative effects in others. But the advances in techniques designed to simultaneously optimize several criteria (to be considered in following section) has reduced the importance of iterative processes.

5.3.6 Multi-objective Optimization

It is clear that interpretability and accuracy are interesting properties for fuzzy systems. At the same time, it is also recognized that improving one of these two criteria usually produces a negative effect over the other. The concept of interpretability-accuracy trade-off appeared as the way to make compatible a simultaneous improvement of both properties. But the increasing presence of multi-objective optimization techniques offers a new approach allowing to work on a simultaneous improvement, without needing a sequential or iterative process with independent steps for accuracy and interpretability.

Multi-Objective Evolutionary Algorithms (MOEA) [56] are frequently applied in the design of fuzzy systems to deal with the interpretability-accuracy trade-off. The list of techniques and algorithms considered includes Non-dominated Sorting Genetic Algorithm II (NSGA-II) [46], Strength Pareto Evolutionary Algorithm 2 (SPEA2) [151], (2+2) Pareto Archived Evolutionary Strategy ((2+2)PAES) and other evolutionary techniques. But this is not the only option, multi-objective Ant-Colony Optimization (ACO), has also been considered [26].

In a multi-objective search or optimization problem, a solution dominates other when it is better than it according to every objective. But due to the existence of several objectives, it is possible to find a set of (several) solutions that are non-dominated by any other. This set of solutions is called the Pareto front. When considering multi-objective techniques, the implicit trade-off of other approaches is somehow made explicit through the concept of Pareto front, that could be considered as a sort of *mapping* describing it. In fact, moving along the Pareto front provides the whole picture of the trade-off.

The possible uses of MOEA in fuzzy systems design is an almost endless list. The use of a first objective centered on accuracy (usually in terms of error) and a second one related to any interpretability aspect, offers a wide panoply of options. The accuracy component focuses in many cases on a tuning process affecting MFs. But the list of applications is not restricted to the use of two objective functions, several works present three objectives including more than one aspect of interpretability in the process.

Among the interpretability questions considered it is possible to find different options that in most cases are related to complexity:

- The number of rules is probably the most widely considered aspect of interpretability. Reduced complexity in terms of rules is either achieved by directly generating a reduced number of rules, or by means of a selection process on an oversized set of rules as previously considered in Sect. 5.3.3.2 (the usefulness of MOEAs for this particular purpose is reviewed in [1]).
- Complexity in terms of the number of conditions which compose the antecedents of rules is also considered. In [37], low values of complexity correspond to Mamdani systems characterized by a small number of rules and a small number of input variables actually used in each rule. A quite close concept is defined in [71] where a three-objectives function is considered to simultaneously maximize accuracy (i.e., the number of correctly classified training patterns in this case), to minimize the number of fuzzy rules, and to minimize the total number of antecedent conditions.
- Partition granularity affects interpretability directly (in what concerns the semantics of the linguistic variables) and indirectly through the number of rules (because having more terms in the partitions usually induces larger RBs). Granularity is one of the components considered in [15]. In addition, the number of MFs jointly with the number of rules are used to evaluate complexity in [61]. The same concepts are considered in [152], but in this occasion considering them independently, by means of a three-objectives process.
- The number of input variables (either in overall RB or per rule) significantly affects complexity. Feature selection as part of the multi-objective process is applied in [12].

The semantic component of interpretability is a completely different option that is also applied in some cases:

- MF interpretability is one of the options. As an example, interpretability preservation of two-tuples is considered while tuning of MF parameters is performed in [57].
- Partition integrity also affects semantic interpretability. Partition integrity is one of the three criteria to be optimized in [16]. It is measured through an index based on the similarity between the partitions learned during the evolutionary process and the initial interpretable partitions defined by an expert. The total number of conditions in the antecedent is taken as complexity measure.
- Semantic cointension is considered in [25], where a logical view index is defined on its basis. In this case three objectives are considered: accuracy, readability, and comprehensibility.

The use of three-objectives functions opens additional options as the use of a third criteria to join interpretability and accuracy. This is the case for [116] where relevance is the third objective, so that interpretability, accuracy and relevance are simultaneously considered in the design of an FRBS.

The interested reader is kindly referred to [128] which includes a review of MOEA applications regarding interpretability of fuzzy systems.

5.3.7 Interpretability in Non-conventional FRBSs

The FRBSs considered to this point are mostly Mamdani and TSK systems with a flat structure and using conventional fuzzy sets. The use of other structures is also possible and has demonstrated good results in many application domains. The most widely considered options among these other structures are HFSs and type-2 fuzzy systems. Although these systems have proved good results in application, unfortunately quite little attention has been paid to their interpretability. Only in recent years some works start to consider the special characteristics of these systems in what concerns interpretability. How the multilayer structure and intermediate variables of hierarchical systems, or the type reduction process in type-2 fuzzy systems affects interpretability, are still open questions. In any case, the question of interpretability has recently been considered in the framework of HFSs [114, 115, 150] and type-2 FRBSs [79, 92].

5.3.7.1 Hierarchical Fuzzy Systems

The definition of an HFS as a method to solve problems with a higher level of complexity than those usually focused on with non-hierarchical FRBSs, has shown some good results. The main concept underlying the application of HFSs was the mitigation of the problems related to high dimensionality of the KB.

Several methods to establish hierarchies in fuzzy systems have been proposed, considering quite different approaches where the hierarchical decomposition affects different components of the overall system. Three main options can be described, ranging from those that decompose at the level of variables, to those that decompose at the level of fuzzy partitions, with an intermediate option decomposing at the level of rules.

This section will concentrate on the decomposition at the level of variables, being this the option that is usually described as HFSs in literature. Other approaches that somehow define hierarchical structures have been already considered in previous sections. One of those approaches is that considering a prioritization of rules (a hierarchy at the level of rules) as previously described in Sect. 5.3.3.3. A different option is considering a sort of hierarchy at the level of partitions. In this case the hierarchical KB is composed of a set of layers where each layer contains linguistic partitions with different granularity levels, and linguistic rules referring to linguistic variables that take values in these partitions (Sect. 5.3.3.4).

The central idea of HFSs is to split a large system into a cascade of several smaller systems, by decomposing the input space into several input spaces with a reduced number of variables, where each input variable is only considered at a certain level of the hierarchy (Fig. 5.6). To involve all variables in the generation of the overall output, the output of each level (u_k) is considered as one of the inputs to the following level [113]. This decomposition is usually stated as a way to maintain under control the problems generated by the curse of dimensionality (i.e., the exponential growth of the number of rules related to the number of variables in the system).

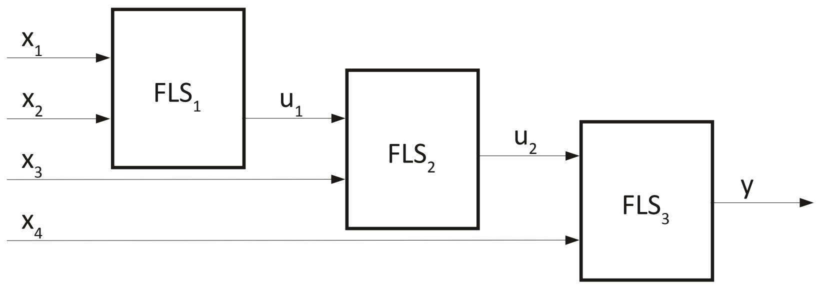

Fig. 5.6: Minimal hierarchical structure.

While the number of rules in an FRBS with n_I input variables and M linguistic terms per variable will be up to M^{n_I}, in an HFS those n_I variables will be grouped into different levels, and the number of rules will be reduced. For an HFS with L levels of rules, n_I system variables, M fuzzy sets per variable, and n_{Ik} variables (including the output variable of the previous level) as inputs to the k^{th} level of the hierarchy, the total number of rules is given by

$$n_R = \sum_{k=1}^{L} M^{n_{Ik}} \tag{5.12}$$

with

$$n_{I1} + \sum_{k=2}^{L} (n_{Ik} - 1) = n_I \tag{5.13}$$

The total number of rules will take on its minimum value when $n_{Ik} = 2$, and this minimum value is

$$n_R = (n_I - 1) * M^2$$

Fig. 5.6 shows this situation, where the subsystems at each layer (FLS_k) have only two inputs (this is the so-called minimal hierarchical structure). The result is that the number of rules in the complete system (involving several subsystems) is a linear function of the number of variables, while in a conventional FRBS it is an exponential function of the number of variables.

In this kind of systems, those variables considered as the most influential ones, are chosen as input variables at the first level, the next most important variables are chosen as input variables at the second level, and so on. The main effect is the reduction on the number of rules of the system. Anyway, the effect of HFSs on interpretability has not been widely studied from a practical point of view. Obviously, the reduction in the number of rules is clear, but the open question is [114]: "a large number of layers with a reduced number of rules is more interpretable than a single layer with a large number of rules?"

As the output variable of each level of the hierarchy is considered as input variable at the following level, the interpretability of those intermediate variables will be crucial for the overall interpretability of the HFS. We have to consider that those intermediate variables are in most cases synthetic variables, and the use of variables without a clear meaning or interpretation will transform our fuzzy model in a black-box model [95].

To cope with this situation, intermediate variables can be considered as a tool to integrate additional expert knowledge. Such an idea is considered in [94], where output variables from a certain level in the hierarchy, are considered to modify scaling functions (in the DB) on a different level producing a sort of *contextualization* of rules on that level. This new approach offers a larger reduction of the number of rules, but, what is more important, gives a meaning to the synthetic intermediate variables, that will usually be linked to properties or characteristic of the system under consideration, so improving interpretability. Then, the main advantage is not the reduction of the number of rules in the system, but the possibility of incorporating the available knowledge to the process of defining the hierarchical structure and generating the rules. This possibility is the result of applying the concept of context during the design process.

It is also possible for some specific problems to create hierarchies where intermediate variables, even being synthetic variables, express or reflect some properties related to the problem under consideration (i.e., the variables have a meaning). This is the approach considered in [96].

5.3.7.2 Type-2 Fuzzy Systems

A different option when designing fuzzy systems is the use of type-2 fuzzy sets for the definition of the terms considered in the fuzzy rules. This option implies the existence of a higher degree of uncertainty in the description of those terms. In the most general case of type-2 fuzzy sets, the membership degree of a certain value to a set is itself a fuzzy set (i.e., a conventional or type-1 fuzzy set). Some simpler options as interval-valued fuzzy sets constitute particular cases of type-2 fuzzy sets.

Two main differences have to be taken into account when considering type-2 fuzzy systems versus type-1 fuzzy systems (i.e., those systems usually deemed as fuzzy systems). The first difference is obvious, the considered fuzzy sets have a different structure, incorporating a higher level of uncertainty. The second difference is not so obvious, but is the one with a higher effect on interpretability. The inference process requires an additional type reduction step prior to the defuzzification of the output. The type reduction process converts a type-2 fuzzy set into its type-1 counterpart, and can be implemented in different ways. Another open question is the interpretability of a type-2 fuzzy set by itself (i.e., how its explicit meaning relates to the implicit meaning of a linguistic term).

Most of works on type-2 interpretability apply the same criteria commonly considered in conventional fuzzy systems (e.g., the number of rules, the total rule length or the distinguishability of the fuzzy sets). However, as far as we know, the effect of the type reduction step on interpretability has not been addressed yet.

As an example, in [92], the use of a small RB, that is made up of simple and comprehensible rules, is considered as a method to achieve interpretability. Simplicity of rules is obtained by considering only some of the conditions in each rule.

Three factors of model interpretability are considered in [79]: transparent fuzzy partitions, simple rules, and a limited number of rules. Fuzzy partition transparency is obtained by considering a moderate number of distinguishable and normal type-2 fuzzy sets, offering a good covering of the UoD. This fuzzy partition transparency is obtained by design applying a parameter learning algorithm that assures the obtained type-2 fuzzy sets being distinguishable, small in number, and properly covering the UoD. To do so, the learning algorithm includes a couple of constraining steps. The first step groups and merges similar fuzzy sets to reduce the number of sets in each input variable. The second step ensures that the merged fuzzy sets are distinguishable and they cover properly the UoD.

Similar ideas are considered in [14]. First, building rules which contain only a subset of the input variables to reduce the total rule length (one of the interpretability metrics defined for type-1 fuzzy systems; see Sect. 3.5.1 in Chap. [7]). Second, considering a limited number of rules ensured by setting an upper bound to the number of rules that the RB can contain. And third, generating partitions that can be easily described by linguistic terms. The main difference with the previous proposal is the use of an MOEA to design the system.

As can be seen, the considered concepts are not quite different from those being present in previous sections, but the question of how the type reduction process affects interpretability is open.

5.4 Interpretability and the Design of the Processing Structure

Designing a fuzzy system has been in most cases considered as generating the KB of a fuzzy system. In other words, most practical applications of fuzzy systems concentrate on the knowledge extraction process and put little attention (if any) on inference. Being that the common situation in fuzzy systems design, it is not different when concentrating on interpretable fuzzy systems design. Consequently not many authors have considered the interaction between inference and interpretability. In any case, considering that the main purpose of this chapter is the design of interpretable fuzzy systems, this section will consider the main options appearing when designing the processing structure of an FRBS, and how the taken decisions will influence the interpretability of the obtained system.

5.4.1 The Processing Structure

The processing structure of an FRBS is usually considered as composed of three main blocks: fuzzification interface, inference engine and defuzzification interface (see Sect. 2.3.2 in Chap. [5]).

On the one hand, fuzzification and defuzzification are processes that are independent of the KB, they are only related to some design decisions that are usually made in advance. On the other hand, inference is the stage in the overall process, where the knowledge (fuzzy partitions and fuzzy rules) is jointly applied with the inputs to produce the outputs. In that sense, inference is extremely related to knowledge, but at the same time is quite influenced by several design decisions that will be considered in this section.

The fuzzification–inference–defuzzification process involves the following five steps of information manipulation:

- fuzzifying the inputs;
- computing the degree of firing for each rule;
- infering the fuzzy output for each rule;
- aggregating the inferred outputs;
- defuzzifying the output.

This general picture may slightly change in some cases. Some examples are FITA systems (see Sect. 2.3.2.3 in Chap. [5]) that apply defuzzification previously to aggregation, or TSK systems not having defuzzification. In any case, this can be considered as a quite general view of the process.

The central step of the reasoning process of an FRBS is fuzzy inference, implying the application of the *compositional rule of inference* [146]. Fuzzy inference produces new (fuzzy) information on the basis of the available one, and in the case of FRBSs the available information has two main aspects: the general information on the problem under consideration (in the form of fuzzy rules and fuzzy partitions), and the specific information on the present situation (in the form of fuzzy inputs).

In general, inputs are not fuzzy, therefore a first step of fuzzification is needed. The usual fuzzification approach is that of singleton fuzzification. Even considering that other types of fuzzification can, in some cases, represent better questions such as measurement dispersion, to the authors knowledge, the interaction between fuzzification and interpretability has never been considered in literature.

5.4.2 Interpretability and Inference

Inference, usually involving the use of aggregation and implication operators, is the key aspect in the whole process. No significant efforts have been made in considering the effect on interpretability regarding the use of different aggregation and implication operators. The effect in complexity has been considered by replacing the standard Mamdani inference with a product-sum inference [118], or even by comparing FITA and FATI systems. But in most cases, the considered complexity was computational, and not structural, being the latter the kind of complexity affecting interpretability.

Some of the effects of aggregation operators on semantic interpretability are considered in [64], where the semantic properties (e.g., convexity) of the fuzzy sets describing the aggregated terms are considered, and an aggregation operator preserving convexity is defined.

In [99], the authors present an evolutionary multi-objective learning model achieving cooperation between the RB and the adaptive fuzzy operators of the inference system in order to obtain simpler, more compact and still accurate linguistic fuzzy models by learning fuzzy inference adaptive operators together with rules. The proposed MOEA generates a set of FRBSs with different trade-offs between interpretability and accuracy, allowing the designers to select the system that involves the most suitable balance for the desired application. Their adaptive inference system works on the basis of adaptive conjunction operators (to be considered for aggregation and implication) where parameters are employed in their expressions to modify their behavior, customizing the way each rule infers.

The use of parameterized aggregation and implication operators (as that of previous paragraph) and the way they affect interpretability is revisited in [45]. The idea is to increase the accuracy of a system without expanding its RB by increasing the precision of aggregation, inference, and defuzzification operators. The implementation leads to the so-called flexible-type systems. The main problem with this approach is the appearance of many parameters (those providing the additional degrees of freedom that improve accuracy) and their (hidden) effect on interpretability. As a matter of fact, the author introduces a list of additional interpretability criteria to reduce the flexibility of the parameters trying to ensure that the parameter values of the flexible operators *provide the best possible match for their non-flexible counterparts* (in words of the author [45]). In the same sense, and considering the inference process, the author states that *a readable inference scheme is the one which the best corresponds to the Mamdani or logical-type inference*. In summary, the only way to avoid a significant interpretability reduction is maintaining some restrictions on the flexibility of the *flexible-type systems*.

5.4.3 Interpretability and Defuzzification

In addition to the previously described options related to the use of parameterized aggregation and parameterized inference, it is also possible to consider parameterized defuzzification functions leading to an adaptive defuzzification process. The general approach to adaptive defuzzification is the use of a parameter per rule that somehow weighs the influence of each rule in the overall output. This is at the end equivalent to the use of weighted rules, an idea that has been present in literature for a long time.

On the one hand, some authors [104] consider rule weights as a negative aspect from a semantic interpretability point of view, recommending their replacement by modifications in MFs (offering similar results with a much more interpretable approach). On the other hand, it is also possible to view these rule weights as certainty grades, using them to create compact FRBSs with good accuracy. Different authors consider that the comprehensibility of the model viewing weights as certainty grades is not deteriorated significantly [73]. In this way, the rule weight can be interpreted as the strength of each rule. Following this second view, adaptive defuzzification is applied in [100] considering three goals: reducing the number of total rules considering that rule weight close to zero can be removed, reducing the rules with weights coupled because rules with weights close to one do not need the weight, and reducing rules triggered jointly.

Parameterized defuzzification is also considered in [45] with a completely different point of view. In this case, discrete defuzzification operators are considered and the flexibility affects the discretization points.

5.5 Final Remarks

An FRBS is a structure with several components involving elements related to knowledge representation as well as others concerned with information processing. As a result, its design process implies many different aspects to be considered and decisions to be made, ranging from the selection of suitable shapes for the MFs, or the definition of rules, to the use of specific aggregation or defuzzification operators. Additionally, other questions as the use of hierarchical structures, the meaning of some components of the KB, the most appropriate structure of a fuzzy rule, the interpretability of the obtained system or its stability, have been considered.

After more than forty years of application of FRBSs to real problems, a wide repertory of design approaches is available, and the way they focus on questions as interpretability is quite diverse and multifaced. The present chapter has only tried to describe different approaches to the problem, that start from different points of view, and apply different arguments to convince the reader. It is impossible to prioritize or rank those approaches, since the arguments, concepts and metrics they use are in most cases not comparable. As a result, the present chapter is mostly an ordered catalogue of concepts and ideas.

Given a specific problem or application, with its properties and characteristics, some of the shown concepts will surely adapt better than others to the specific conditions. But in no way it is possible to say there is an overall best option.

In addition, many questions remain unsolved. Historically, most of the work on interpretability of fuzzy systems concentrated on structural and complexity questions, while the semantic aspects received much less attention. Now, semantic questions are gaining presence, but still, aspects as the influence of the inference process on interpretability should be explored. In summary, a large amount of work has been done, but there is still work to be done in the future.

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Chapter 6
Design and Validation of an Explainable Fuzzy Beer Style Classifier

Abstract We describe step by step how to design, implement and validate an interpretable fuzzy rule-based beer style classifier endowed with explanation capability. First, we revise some preliminary work regarding both interpretable fuzzy modeling methodologies and related software. Second, we introduce the use case on beer style classification. Third, we build and validate a fuzzy rule-based classifier with a good interpretability-accuracy trade-off for this use case. Fourth, we endow this classifier with explanation capability through a general linguistic interface that is tuned *ad-hoc* for the use case under consideration. Fifth, we show how the designed explainable classifier can be refined and exploited with several interoperable software tools. Finally, we compare two kinds of multi-modal (i.e., textual and graphical) explanations: (1) explanations which are inherently natural and fully-meaningful to users, because they are supported by an interpretable fuzzy rule-based classifier which is carefully designed (i.e., it is grounded in common-sense expert knowledge and global semantics); and (2) explanations which are supported by the linguistic approximation of a fuzzy rule-based classifier extracted from data with the focus only on accuracy (thus lacking of linguistic interpretability). The use case is implemented with open source software and all related datasets, tools and scripts are available online for the sake of reproducibility.

6.1 Introduction

The economic impact of Artificial Intelligence (AI), through real-world successful applications, is estimated between 6.5 and 12 € trillion annually by 2025 [68]. In practice, the cohabitation of humans and smart gadgets (e.g., watches, phones, TVs, cars, etc.) makes society demand the development of a new generation of explainable AI systems, i.e., AI systems ready to explain naturally (as humans do) their automatic decisions. This is especially important in the context of Industry4.0 where human-robot teams co-work every day [65] but it is also important in the context of smart cities [15], e-Learning [7], e-Health [41], etc. Accordingly, the European Commission has identified AI as the "most strategic technology of the 21^{st} century"[1]. In addition, the United Nations Global Pulse initiative[2] refers to the need of applying AI techniques (they explicitly refers to Big Data and Data Science) to support sustainable development around, among others, health, education, and industry. Moreover, all these new challenges must be carefully addressed to preserve human rights regarding both legal and ethical issues. Thus, *explanation* is recognized as a basic principle in the search for the required *algorithmic transparency and accountability* to support a new generation of trustworthy and eXplainable AI (XAI for short) models[3].

It is worth noting that about 30% of publications related to XAI in Scopus (until October 2017) came from authors well recognized in the Fuzzy Logic research field [14]. Actually, interpretability is recognized as one of the main strengths of fuzzy systems since Zadeh's pioneer work on fuzzy sets [76]. Moreover, Zadeh made many additional valuable contributions (e.g., linguistic variables [77], computing with words [78], the computational theory of perceptions [79], etc.) that pave the way towards XAI [4].

Of course, several application areas have already taken advantage from the use of interpretable fuzzy systems, as a first step towards XAI. In the following, some of them are briefly outlined, along with a few notes on specific applications and potentialities.

Environment. Environmental issues are often challenging because of the complex dynamics, the high number of variables and the consequent uncertainty characterizing the behavior of subjects under study. Computational intelligence techniques, especially fuzzy logic techniques, come into play when tolerance for imprecision can be exploited to design convenient models that are suitable to understand phenomena and take decisions. Interpretable fuzzy systems show a clear advantage over black-box systems in providing knowledge that is useful to understand complex and non-linear relationships by using linguistic models. Real-world environmental applications of interpretable fuzzy systems include: harmful bioaerosol detection [63]; modeling habitat suitability in river management [73]; modeling pesticide loss caused by meteorological factors in agriculture [49]; and so on.

Finance. This is a sector where human-computer cooperation is very tight. Cooperation is carried out in different ways, including the use of computers to provide business intelligence for decision support in financial operations. In many cases, financial decisions are ultimately made by experts, who can benefit from

[1] European Commission, Artificial Intelligence for Europe, Brussels, Belgium, "Communication from the Commission to the European Parliament, the European Council, the Council, the European Economic and Social Committee and the Committee of the Regions", Tech. Rep., 2018, (SWD(2018) 137 final) `https://ec.europa.eu/digital-single-market/en/news/communication-artificial-intelligence-europe`

[2] `https://www.unglobalpulse.org/`

[3] ACM US Public Policy Council: Statement on Algorithmic Transparency and Accountability, 2017, `https://www.acm.org/binaries/content/assets/public-policy/2017_usacm_statement_algorithms.pdf`

automated analyses of big masses of data flowing daily in markets. To this pursuit, computational intelligence approaches are spreading among the tools used by financial experts in their decisions, including: interpretable fuzzy systems for stock return predictions [56]; exchange rate forecasting [37]; portfolio risk monitoring [46]; etc.

Industry. Industrial applications can take advantage from interpretable fuzzy systems when there is the need of explaining the behavior of complex systems and phenomena, like in fault detection [27]. Also, control plans for systems and processes can be designed with the aid of fuzzy rule-based systems. In such cases, a common practice is to start with an initial expert knowledge (used to design rules which are grounded on common-sense knowledge and thus they are usually highly interpretable) that is then tuned to increase the accuracy of the controller. However, any unconstrained tuning could destroy the original interpretability of the knowledge base (KB). On the contrary, by taking into account interpretability, the possibility of revising and modifying the controller (or the process manager) can be enhanced [67].

Medicine and Health-care. As a matter of fact, in almost all medical contexts intelligent systems can act as appreciated decision support tools. It is worth noting that people are the ultimate actors in any decision process and people need to rely on intelligent systems, whose reliability would be enhanced if their outcomes may be explained in terms that are comprehensible by humans. Interpretable fuzzy systems can play a key role in this area because of the concrete possibility of acquiring knowledge from data and communicating it naturally to users. In literature, several approaches have been proposed to apply interpretable fuzzy systems in different medical problems, like assisted diagnosis [43], prognosis prediction [8, 41], patient subgroup discovery [31], etc.

Robotics. The complexity of robot behavior modeling can be tackled by an integrated approach where a first modeling stage is carried out by combining expert knowledge (provided by humans) and empirical knowledge (acquired from experimental trials). This integrated approach requires that the final KB is provided to experts for further maintenance: this task could be done effectively only if the acquired knowledge is formalized and represented in a way that it is interpretable by experts. Some concrete applications of this approach can be found in robot localization systems [24] and motion analysis [23, 60].

Society. The focus of intelligent systems for social issues has noticeably increased in recent years. For reasons that are common to all the previous application areas, interpretable fuzzy systems have been applied in a wide variety of scopes, including quality of service improvement [28], data mining with privacy preservation [72], social network analysis [25], e-commerce [35], and so on.

Even if interpretable fuzzy systems have been already applied successfully to many varied applications, we would like to remark that interpretability cannot be taken for guaranteed just because of applying fuzzy logic (see Chap. [10]). Conversely, careful design is required in order to get interpretable fuzzy systems (see Chap. [11]). Moreover, several constraints (see Chap. [12]) must be imposed to get good results regarding interpretability assessment (see Chap. [13]).

The rest of this chapter is organized as follows. Sects. 6.2 and 6.3 present some preliminary concepts that are needed to understand the illustrative use case thoroughly described in Sect. 6.4. Finally, Sect. 6.5 concludes the chapter with some final remarks.

6.2 Designing an Interpretable Fuzzy Model

This section briefly introduces the selected fuzzy modeling methodology (HILK) along with the related software (GUAJE). HILK stands for Highly Interpretable Linguistic Knowledge. GUAJE is the acronym of Generating Understandable and Accurate fuzzy systems in a Java Environment.

6.2.1 The HILK Fuzzy Modeling Methodology

The HILK fuzzy modeling methodology was first introduced in [22] and later refined in [19]. We have selected HILK among the fuzzy modeling methodologies described in Chap. [11] because this modeling methodology takes interpretability as the main concern. Moreover, it is aimed at designing interpretable fuzzy models by combining expert knowledge and knowledge automatically extracted from data. Both kinds of knowledge are represented under the same mathematical framework, and integrated into the same KB. On the one hand, expert knowledge comes from human knowledge-elicitation tasks (e.g., interviews, surveys, etc.). On the other hand, data-mining tasks are considered to extract knowledge from data.

Regarding the construction of the KB, the entire modeling process comprises three steps as depicted in Fig. 6.1.

- **Linguistic partition design**. HILK works with Strong Fuzzy Partitions (SFPs) which satisfy all interpretability constraints (i.e., distinguishability, normalization, coverage, overlapping, etc.) that are described in Chap. [12]. Accordingly, HILK fuzzy partitions always satisfy the next equation:

$$\sum_{i=1}^{M} \mu_{A_i}(x) = 1, \ \forall x \in U \tag{6.1}$$

  where U is the Universe of Discourse (UoD), M is the number of linguistic terms, and $\mu_{A_i}(x)$ is the membership degree of x to the A_i fuzzy set.
  Firstly, experts provide information (range, granularity and semantics) about those attributes they think are the most influential ones in the problem under consideration. Secondly, this information is contrasted and completed with information automatically extracted from data. Thirdly, experts refine their knowledge, i.e., they confirm the selection of attributes and give a linguistic term for each membership function. As a result, the selected attributes are characterized

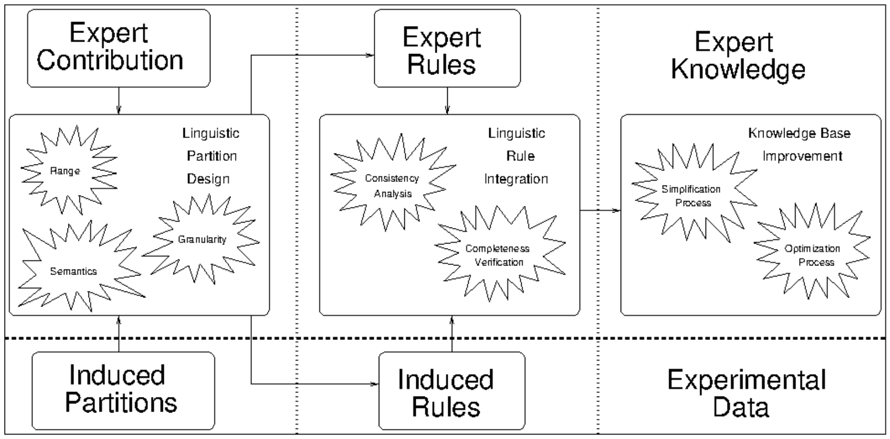


Fig. 6.1: The pipeline of the HILK fuzzy modeling methodology.

by SFPs with meaningful linguistic terms in accordance with both expert knowledge and experimental data.

- **Linguistic rule integration**. Experts describe the system behavior through linguistic rules which combine fuzzy information granules in the form of a conjunction of premises:

$$\text{IF } X_1 \text{ is } A_{X_1} \text{ and } \ldots \text{ and } X_n \text{ is } A_{X_n} \text{ THEN } Y \text{ is } A_Y \tag{6.2}$$

Rule antecedent and consequent include *soft constraints* in the form of linguistic propositions (e.g., X_i is A_{X_i} or Y is A_Y) with X_i and Y being the names of input and output linguistic attributes, respectively, and $i \in [1,n]$. A_{X_i}, A_Y are the linguistic terms involved in the atomic soft constraints associated to those attributes. These linguistic terms, therefore, stand as the names of the fuzzy sets identified by the membership functions $\mu_{A_{X_i}}$, μ_{A_Y}. It is worth noting that all rules share the same linguistic terms previously defined. This fact favors interpretability because there is a global semantics for the whole KB. In consequence, rule consistency and completeness can be checked and verified at linguistic level. In addition to expert rules, it is also possible to induce rules from data. Rule learning algorithms can identify the right combination of premises and conclusions in accordance with experimental data. It is worth noting that rule weights are not considered because even if they can help to increase the accuracy of the whole system they also make harder the rule interpretation.

- **KB improvement**. Once a set of linguistic partitions and rules is defined, then it is time to look for the best interpretability-accuracy trade-off. To do so, HILK provides users with a simplification procedure [21] which increases even more interpretability without penalizing either consistency or accuracy, but also with an optimization procedure [16] which gets better accuracy while keeping interpretability. In addition, HILK comprises several quality indexes regarding both accuracy (e.g., classification rate or mean square error) and interpretability (e.g., number of linguistic terms or number of co-firing rules).

The interested reader is referred to [19, 22] for more details about the HILK methodology.

6.2.2 The GUAJE Open Source Software

GUAJE[4] is an open source Java Environment for Generating Understandable and Accurate fuzzy models [18, 62]. We have selected GUAJE among all fuzzy systems software reviewed in [2] because it implements the HILK fuzzy modeling methodology. Moreover, it is worth noting that GUAJE is interoperable with the IEEE Std 1855-2016 [1] (i.e., the standard for fuzzy systems modeling that is sponsored by the IEEE Computational Intelligence Society). This standard is implemented in the JFML library [70] and GUAJE lets users import and export files in the format established by the standard. GUAJE is also interoperable with other fuzzy modeling software such as the Matlab Fuzzy Logic Toolbox [58] and FisPro [50].

GUAJE implements the pipeline of the HILK fuzzy modeling methodology (see Fig. 6.1) as follows:

- **Linguistic partition design**. Given an input attribute, GUAJE automatically builds a uniform partition (i.e., a standardized uniform SFP in the UoD U) with the selected number of fuzzy sets. Then, the user can edit this partition manually or compare it with partitions extracted from data by the k-Means clustering algorithm [51] or by the Hierarchical Fuzzy Partitioning algorithm (HFP) [48]. In addition, GUAJE assists the user in the selection of the best partition by means of automatic comparison among expert and induced partitions in terms of three quality indexes: the Partition Coefficient [29], the Partition Entropy [29], and the Chen Index [36].
- **Linguistic rule integration**. The next three rule learning methods are available in GUAJE: the Fast Prototyping Algorithm (FPA) [47], the Wang and Mendel algorithm (WM) [74], and the Fuzzy Decision Tree algorithm (FDT) [53].
  Expert rules and rules induced from data are stored in the same rule base. GUAJE assists the user in the rule base consistency analysis. It first automatically looks for consistency errors (e.g., rules with indefinite conclusion or contradictory rules), redundancy warnings (e.g., rule overlapping with the same or different conclusion), unused variables and unused linguistic terms. Then, it suggests[5] actions to fix potential problems, if any. Moreover, GUAJE also assists the user with a completeness verification procedure. It looks for input space regions that

[4] https://gitlab.citius.usc.es/jose.alonso/guaje/

[5] GUAJE makes suggestions but the user always makes the final decision. Anyway, unsolved consistency issues can be automatically fixed during the “KB improvement” stage, if the user selects this option.

are not covered by any rule. It is worth noting that thanks to the global semantics imposed as part of the linguistic partition design, GUAJE makes both consistency and completeness rule checking at linguistic level.

- **KB improvement**. The aim of this stage is looking for a good interpretability-accuracy trade-off. GUAJE offers a linguistic simplification procedure which first ranks rules in terms of relevance. This rule ranking can be made regarding accuracy or interpretability at both global (the whole KB) and local (each single rule) level. Then, simplification is done iteratively (regarding rules and partitions sequentially) until no more changes are allowed. It is worth noting that the simplification procedure automatically stops when a predefined tolerance threshold (that is set by the user) for accuracy degradation is achieved. Rule simplification (RS) comprises four steps[6]: (RS.1) to remove redundant rules; (RS.2) to merge partially overlapping rules; (RS.3) to force rule removing; and (RS.4) to force premise removing. Partition simplification (PS) also comprises four steps: (PS.1) to remove unused variables; (PS.2) to remove unused linguistic terms; (PS.3) to merge pairs of linguistic terms; and (PS.4) to force variable removing.
GUAJE takes profit from the Logical View Index (LVI) [59] as a tool for controlled minimization of the rule base. LVI measures the difference in terms of accuracy between the original rule base and the rule base minimized according to truth-preserving principles. The rationale behind LVI is as follows: if LVI is small, then the rule base is assumed to satisfy logical laws and the minimized version can be retained; otherwise the rule base does not express rules following logical principles and the minimized version should be discarded. The latter case must be considered as a severe warning about the semantic interpretability of the rule base, but only in case the user interpreted the rules in terms of logical propositions.
Accuracy can also be improved (while preserving interpretability) by means of optimization procedures at both partition and rule levels [20]. On the one hand, partition optimization consists of tuning the membership function parameters. There are two optimization strategies implemented in GUAJE: (1) local optimization with the Solis-Wets algorithm [69]; and (2) genetic tuning [40]. On the other hand, rule optimization consists of a genetic rule selection procedure.

In addition, the last release of GUAJE includes new functionality:

- **KB explanation**. GUAJE provides users with graphical tools to facilitate the understanding of the behavior of designed fuzzy systems. These tools assist the fuzzy designer all along the process of building a fuzzy rule-based system with a good interpretability-accuracy trade-off. Obviously, early detection of potential failures is likely to produce better systems with less effort.

[6] There is some overlapping between the two stages named as "Linguistic rule integration" and "KB improvement". They may run independently, but if they ran in sequence then tasks (RS.1), (RS.2), (PS.1) and (PS.2) in the "KB improvement" stage would make no sense because the KB should be free of redundant and/or unused elements after fixing consistency issues previously identified during the "Linguistic rule integration" stage.

In addition to the usual graphical representation of fuzzy partitions, fuzzy rules and fuzzy inference, which is a functionality available in other modeling tools such as the Matlab Fuzzy Logic Toolbox [58] and FisPro [50], the main novelty in GUAJE turns up to be its multi-modal (graphical + textual) explanation capability. On the one hand, the fuzzy inference-grams (*fingrams* in short) module [61, 62] shows graphically the relation among fuzzy rules, at inference level, as if they formed a social network with rules depicted as nodes which are linked through edges which represent the interaction degree between pairs of fired rules. Fingrams convey a lot of information regarding the structure of the fuzzy rule base but also regarding the behavior of the entire fuzzy rule-based system at both local (i.e., paying attention to the inference with a single data instance) and global (i.e., paying attention to the summary of inferences for a dataset) levels[7]. As described in [61], the interpretation of fingrams is simple and intuitive. On the other hand, GUAJE provides users also with meaningful explanations in Natural Language (NL) with the aim of making even easier the KB understanding. Such explanations are built up with the use of the two software tools that are briefly introduced in the next section: rLDCP (see Sect. 6.3.2) and simpleNLG (see Sect. 6.3.3).

- **Interoperability with tools that are complying with the IEEE Std 1855-2016**. Fuzzy rule-based systems designed with GUAJE can be exported/imported into/from the XML format that is defined by the JFML library [70]. This library complies with the IEEE Std 1855-2016 [1]. Thus, GUAJE fuzzy systems can be edited with any software tool that is complying with this standard. For example, py4JFML [3] enables the use of all the functionality of JFML through a Python 3.x module. Accordingly, the goodness of fuzzy systems designed by GUAJE can be compared versus machine learning models which may be designed by using Python packages (e.g., scikit-learn[8]).

The interested reader is referred to [5, 18, 62] for more details about GUAJE.

6.3 Generating Explanations in Natural Language

NL technologies regard both understanding (i.e., NL Processing or NLP in short) and synthesis (i.e., NL Generation or NLG in short). Here, we are mainly interested in the use of NLG technologies for automatic generation of effective and natural explanations related to fuzzy rule-based systems [26].

[7] As far as we know, this kind of explanations are only available in GUAJE and ExpliClas [6]. The latter is a web service for automatic generation of multi-modal explanations associated to Weka classifiers [75].

[8] `https://scikit-learn.org`

6.3.1 The LDCP Linguistic Modeling Methodology

LDCP stands for Linguistic Description of Complex Phenomena. The LDCP modeling methodology is a human-centric design methodology for linguistic modeling which was introduced by Trivino and Sugeno [71]. It has been successfully applied to automatically generate human-like textual descriptions associated to data in some specific contexts (e.g., household energy consumption monitoring [39] or public transport management [15]).

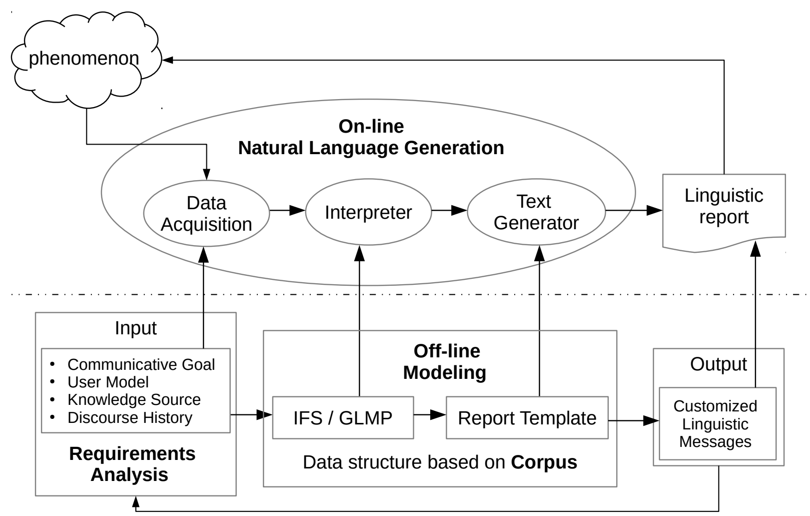


Fig. 6.2: The human-centric LDCP architecture. On-line stage on top and off-line stage at the bottom. The requirement stage is previous to the modeling one and both are carried out off-line.

The LDCP human-centric design methodology comprises the three main stages which are sketched in Fig. 6.2 and described below:

- **Requirements analysis.** First of all, designers have to identify the topics of interest to communicate (i.e., the Communicative Goal). Then, they have to identify both background and expectations of the target audience (i.e., the User Model) in order to select later only the relevant pieces of information to convey. Then, they have to identify the available raw data and their origin (i.e., the Knowledge Source) in order to have a global view of the phenomenon under consideration. Finally, they have to track the temporal series of messages already exchanged with the audience (i.e., the Discourse History) in order to avoid repetitions and provide users with effective and natural messages.

- **Off-line modeling.** The first step consists of looking for a corpus of natural expressions which may be taken as reference and inspiration for the text generation task. If this corpus doesn't exist, then it has to be created *ad-hoc* for the given application. Unfortunately, building a rich corpus to be used by fully end-to-end approaches such as deep learning architectures is highly costly and not feasible in many applications. LDCP opts for a human-centric approach (i.e., for talking to domain experts in order to deeply understand their problems and requirements) instead of just collecting blindly as many data as possible (as it is done in many fully data-driven approaches). Then, an Interpretable Fuzzy System (IFS) is initialized with expert knowledge and later tuned with knowledge extracted from data. This IFS is in charge of automatically interpreting the data related to the specific phenomenon under study. This IFS can be taken as basis of a Granular Linguistic Model of the Phenomenon (GLMP). This model is called IFS/GLMP. It is a general-purpose hierarchical model with different levels of granularity and aware of interpretability issues. As a result, it is expected to be understandable to humans. Building the IFS/GLMP model is a matter of careful design which can be addressed with the assistance of the HILK modeling methodology (i.e., with the assistance of the GUAJE software) that was introduced in Sect. 6.2. In practice, IFS/GLMP is implemented as a hierarchy of Computational Perceptions (CPs) and Perception Mappings (PMs)[9].
  In practice, a CP is a tuple (Z, W_Z, R_Z) which is defined as follows:

  Z $=(z_1, z_2, \cdots, z_n)$ is a vector of Z-numbers [80] which describes the user's perception. Formally, z_i=(a_j,b_k) with
  - $a_j \in A = (a_1, a_2, \cdots, a_n)$, being A a vector of n linguistic expressions, with a_j describing the CP for a specific situation and granularity degree;
  - $b_k \in B = (b_1, b_2, .., b_m)$, being B a vector of m linguistic expressions which rate the reliability of this CP.

  W_Z= (W_A,W_B) with
  W_A= $(w_{a_1}, w_{a_2}, \cdots, w_{a_n})$ is a vector of validity degrees $w_{a_j} \in [0,1]$, with w_{a_j} associated to a_j and $\sum w_{a_j} = 1$;
  W_B= $(w_{b_1}, w_{b_2}, .., w_{b_m})$ is a vector of validity degrees $w_{b_k} \in [0,1]$, with w_{b_k} associated to b_k and $\sum w_{b_k} = 1$.

  R_Z = $(r_{z_1}, r_{z_2}, \cdots, r_{z_n})$ is a vector of relevance degrees $r_{z_i} \in [0,1]$, with r_{z_i} associated to z_i.

  It is worth noting that CPs describe units of information or information granules in the sense of Zadeh, i.e., clumps of elements which are drawn together by indistinguishability, similarity, proximity or functionality. Obviously, a_j and b_k should be meaningful linguistic terms to the target audience for the sake of interpretability.

[9] An illustrative example is provided in the next section (see Fig. 6.14).

In addition, a PM is a tuple (U, y, g, T) which is defined as follows[10]:

U = $(u_1, u_2, \ldots, u_p)$ is a vector of p input CPs where $u_i = (Z_{u_i}, W_{Z_{u_i}}, R_{Z_{u_i}})$. In the case of PMs at the bottom of the hierarchy, Z_{u_i} is just a numerical value usually taken from a sensor or database.

y = (Z_y, W_{Z_y}, R_{Z_y}) is the output CP.

g = (g_{W_Z}, g_{R_Z}) is the processing function for creating or aggregating CPs:
$W_{Z_y} = g_{W_Z}(W_{Z_{u_1}}, W_{Z_{u_2}}, \ldots, W_{Z_{u_p}})$
$R_{Z_y} = g_{R_Z}(R_{Z_{u_1}}, R_{Z_{u_2}}, \ldots, R_{Z_{u_p}})$
In the case of PMs at the bottom of the hierarchy, A and B are defined by fuzzy partitions. Accordingly, g_{W_Z} and g_{R_Z} compute the membership degrees μ_A and μ_B. Otherwise, the aggregation of CPs can be made by means of fuzzy rules, fuzzy quantifiers, etc.

T is a text generation algorithm ready to produce sentences in NL derived from the linguistic expressions in A_y. It is worth noting that the Report Template in Fig. 6.2 combines dynamically all T in the IFS/GLMP with programming code.

- **On-line NLG.** As depicted in the Fig. 6.2, the Data Acquisition module first sets the application parameters in accordance with the Requirements Analysis which is done off-line. Then, Input data are pre-processed with the aim of fitting the format admitted by the Interpreter module. Then, data interpretation is carried out regarding the knowledge embedded into the IFS/GLMP model (designed off-line). Then, the Text Generator module builds up a linguistic report as result of combining the Report Template (also designed off-line) with the most relevant pieces of information provided by the Interpreter (on-line).

6.3.2 The rLDCP Open Source Software

rLDCP[11] is an open source R package for automatic text generation from data [38]. It is aimed for automatic report generation in a general sense and not just for the specific task of generating explanations associated to fuzzy systems. This software implements the LDCP methodology that we briefly introduced in the previous section. Accordingly, the package structure corresponds directly with the LDCP architecture (see Fig. 6.2) and its main functionality is summarized as follows:

- The `ldcp.R` script connects the three main components in the architecture:
  - `data-structure.R` corresponds to the Data Structure handled by the Data Acquisition module.

[10] We adopt the notation given in [15], even if this notation is slightly different from the rest of the book. Namely, U is an input vector while it refers to the universe of discourse in the rest of the book. In addition, T is a text generation algorithm while the same symbol refers to the set of linguistic terms in the rest of the book.

[11] `https://cran.r-project.org/web/packages/rLDCP/index.html`

  - `glmp.R` represents the IFS/GLMP model to be used by the Interpreter module. It makes use of the scripts related to PMs and CPs:
    · `pm.R` builds and handles PMs.
    · `cp.R` builds and handles CPs.
  - `report-template.R` implements the Report Template to be used by the Text Generator module.

- This R package comes with auxiliary tools for users without strong coding skills. Namely, users can directly write LDCP systems into XML files which are later validated and loaded into R. All related R scripts are automatically generated by rLDCP.

It is worth noting that the rLDCP software is already fully integrated with the last release of GUAJE. Given a fuzzy rule-based system, the KB explanation module automatically generates all R scripts that are needed to generate linguistic descriptions with rLDCP. Those linguistic descriptions are taken as pieces of information to build explanations in GUAJE. More precisely, GUAJE runs the rLDCP R scripts and collect all the generated textual descriptions. In addition, GUAJE compares and combines such descriptions through dynamic templates which are enhanced by the simpleNLG library as described in the next section.

6.3.3 The simpleNLG Library

The NLG architectural framework proposed by Reiter and Dale [66] turns up in [44] as one of the most used frameworks for NLG worldwide. It includes three main components:

- **Document Planner** which is in charge of the next two tasks: (1) content determination; and (2) document structuring.
- **Micro Planner** which takes care of the next tasks: (1) lexicalization; (2) denoting and referring; and (3) aggregation.
- **Surface Realizer** which carries out the next two tasks: (1) linguistic realization; and (2) structure realization.

The KB explanation module in GUAJE implements and combines some of the tasks in the framework briefly introduced above (namely content determination, aggregation and linguistic realization) with rLDCP. More precisely, GUAJE is endowed with multilingual linguistic realization (currently in English and Spanish) as result of combining rLDCP templates and natural text dynamically generated with the Java library SimpleNLG-ES[12] [64] which is a bilingual English/Spanish adaptation of the SimpleNLG[13] [45] v4.4.8 library.

---

[12] SimpleNLG in Spanish: `https://github.com/citiususc/SimpleNLG-ES`
[13] SimpleNLG in English: `https://github.com/simplenlg/`

SimpleNLG builds sentences in a very intuitive way. As an illustrative example, the Java code to create a sentence such as "bitterness and strength are high" is sketched below.

```
// Load Lexicon
Lexicon lex= Lexicon.getEnglishLexicon();
// Build linguistic realiser
Realiser real= new Realiser(lex);
// Initialise factory
NLGFactory nlgFactory= new NLGFactory(lex);
...
// Create a hashtable ht with selected pieces of information
Hashtable<String,Vector> ht= new Hashtable();
...
// Select elements from ht
Enumeration k= ht.keys();
Object key= k.nextElement();
Vector v= ht.get(key);
// Create an empty sentence
SPhraseSpec p= nlgFactory.createClause();
// Define the first subject (e.g., s1=bitterness)
NPPhraseSpec s1= nlgFactory.createNounPhrase(v.elementAt(0));
// Define the second subject (e.g., s1=strength)
NPPhraseSpec s2= nlgFactory.createNounPhrase(v.elementAt(1));
// Combine both s1 and s2 into a single subject
CoordinatedPhraseElement subj= nlgFactory.createCoordinatedPhrase(s1, s2);
p.setSubject(subj);
// Set verb
p.setVerb(LocaleKBCT.GetString("toBe"));
// Set object (e.g., key=high)
p.setObject(key.toString());
// Build textual sentence
MainKBCT.getNLGrealiserEN().getRealiser().realiseSentence(p);
```

6.4 An Illustrative Use Case on Beer Style Classification

This section presents an illustrative use case regarding beer style classification (see Sect. 6.4.1) in which we describe step by step how to use the GUAJE[14] software for: (1) building up a fuzzy rule-based classifier with a good interpretability-accuracy trade-off (see Sect. 6.4.2); (2) generating multi-modal (graphical + textual) effective and natural explanations (see Sect. 6.4.3); (3) exporting the fuzzy classifier to be exploited by other software tools (see Sect. 6.4.4); and (4) comparing explanations generated by GUAJE with explanations generated by ExpliClas [6] (see Sect. 6.4.5).

We would like to remark once again that the design of fuzzy systems is a matter of careful design and refinement. Therefore, all the tasks enumerated above are closely related. Moreover, it makes sense to think about an iterative design process in which tasks (1) and (2) are repeated in several iterations. Obviously, explanations should help the designer to have a deeper look into the fuzzy rule-based classifier behavior. Then, the classifier may be refined and as a result a better interpretability-accuracy

[14] We recommend to install and run GUAJE [5] in order to appreciate all details. GUAJE lets you zoom in and zoom out of the figures provided as screenshots in the rest of this chapter. Thus, you are not to miss any detail when analyzing the given pictures even if they were printed into black and white.

trade-off may be achieved. Sometimes, the refinement only can be done with a different software tool. Fortunately, this is feasible thanks to task (3). In addition, we show the greater naturalness and semantic grounding of explanations provided by GUAJE (thanks to the fact that they come from a carefully designed interpretable fuzzy system) versus those provided by ExpliClas (thanks to a linguistic layer designed on top of fuzzy systems automatically learned from data while disregarding interpretability issues and lacking of global semantics).

It is worth noting that the interested reader can find in [9] all software, data and scripts needed to reproduce the experiments described in this section.

6.4.1 The Beer Dataset

This dataset[15] was built up by Castellano et al. from a collection of beer recipes provided by expert brewers [32]. It is made up of 400 instances which correspond to eight beer styles (BLANCHE, LAGER, PILSNER, IPA, STOUT, BARLEYWINE, PORTER, and BELGIAN STRONG ALE). The classification task consists of identifying one out of the eight classes (i.e., beer styles) in terms of three attributes (COLOR, BITTERNESS and STRENGTH). It is worth noting that the dataset is perfectly balanced with 50 instances corresponding to each class.

The Beer dataset was used by Castellano et al. to validate an expert fuzzy classifier which achieved 81.25% of classification rate over 10-fold cross-validation (10-fold CV) [32]. Linguistic terms were defined by expert brewers as follows:

- COLOR $\in$ [PALE, STRAW, AMBER, BROWN, BLACK]
- BITTERNESS $\in$ [LOW, LOW-MEDIUM, MEDIUM-HIGH, HIGH]
- STRENGTH $\in$ [SESSION, STANDARD, HIGH, VERY-HIGH]

In addition, the experts defined the following eight rules (one rule for each output class):

R1. IF COLOR is PALE AND BITTERNESS is LOW AND STRENGTH is SESSION THEN BEER STYLE is BLANCHE
R2. IF COLOR is AMBER AND BITTERNESS is LOW-MEDIUM AND STRENGTH is SESSION THEN BEER STYLE is LAGER
R3. IF COLOR is STRAW AND BITTERNESS is MEDIUM-HIGH AND STRENGTH is SESSION THEN BEER STYLE is PILSNER
R4. IF COLOR is AMBER AND BITTERNESS is HIGH AND STRENGTH is HIGH THEN BEER STYLE is IPA
R5. IF COLOR is BLACK AND BITTERNESS is LOW-MEDIUM AND STRENGTH is STANDARD THEN BEER STYLE is STOUT
R6. IF COLOR is AMBER AND BITTERNESS is HIGH AND STRENGTH is VERY-HIGH THEN BEER STYLE is BARLEYWINE
R7. IF COLOR is BROWN AND BITTERNESS is LOW-MEDIUM AND STRENGTH is SESSION THEN BEER STYLE is PORTER
R8. IF COLOR is BROWN AND BITTERNESS is MEDIUM-HIGH AND STRENGTH is VERY-HIGH THEN BEER STYLE is BELGIAN STRONG ALE

In addition, Castellano et al. used the FISDeT software [33] to automatically generate a fuzzy classifier from the Beer dataset. The generated fuzzy classifier

[15] The Beer dataset in arff format is available online at `https://gitlab.citius.usc.es/jose.alonso/xai/-/blob/master/BEER3.txt.aux.arff`

included nine rules and achieved 87.50% of classification rate (10-fold CV). It was generated by the Double Clustering with A* algorithm (DC*) [34]. As implied by its name, DC* automatically generates interpretable fuzzy information granules from data by means of a DC process: (1) a data compression procedure is first performed through the application of the Learning Vector Quantization algorithm (LVQ1) [54, 55]; (2) then a subsequent search procedure based on A* is carried out to produce the structure of the fuzzy partition which is finally exploited to generate compact fuzzy rules.

Just for comparison purpose, we built a baseline classifier with the Weka software [75]. It is worth noting that Weka is a well-known open source software developed as part of a data mining project leaded by researchers affiliated to the University of Waikato (New Zealand), and with a huge community of users and developers worldwide. We built a Random Forest [30] classifier for the Beer dataset and it achieved 96% of classification rate (10-fold CV). Random Forest is recognized as a very accurate classifier which disregards interpretability issues. In [42], Random Forest turned up as the most accurate classifier in an experimental study where 178 classifiers were compared all over 121 datasets. Accordingly, it is usually taken as a state-of-the-art baseline classifier from the accuracy point of view.

6.4.2 The Interpretable Fuzzy Rule-based Beer Style Classifier

In this section, we go in depth with how to use GUAJE for building a fuzzy rule-based classifier with a good interpretability-accuracy trade-off.

First of all, we loaded the Beer dataset with GUAJE. Then, we selected the 10-fold CV option and 10 folds with the related train and test data pairs were automatically generated. In the rest of this section, just for illustrative purpose, we refer only to the first fold but the same design steps are repeated for all folds and at the end we report the averaged accuracy over the 10 folds[16].

We used GUAJE to fully derive a fuzzy rule-based classifier from the available data. Among the design options provided by GUAJE (see Sect. 6.2) we opted for:

- **Linguistic partition design**. We first generated uniform SFPs with 6 linguistic terms for all the three attributes (COLOR, BITTERNESS, STRENGTH). We set 6 as initial number of linguistic terms because we wanted to be sure to cover all terms defined previously by experts. Please, remind that experts had previously defined 5 terms for COLOR and 4 terms for BITTERNESS and STRENGTH, respectively. Then, we generated all alternative uniform partitions from 2 to 8 linguistic terms but also with the k-Means and HFP algorithms. For example, the right side of Fig. 6.3 shows all generated partitions for BITTERNESS. Moreover, we selected for each input attribute the best partition that is suggested by GUAJE after careful comparison among all generated partitions. Notice that, the data distribution and

[16] It is worth noting that GUAJE can run batch scripts from command line with the aim of building models for all the folds at once.

the suggested best partition are shown in the left side of Fig. 6.3. In this case, GUAJE suggested to work with the partition generated by the k-Means algorithm because it got the best quality indicators as depicted in Fig. 6.4. Actually, the selected partition is the one with the highest values of the Partition Coefficient and Chen Index (but also with the lowest value of the Partition Entropy) among all the three partitions with 6 fuzzy sets. It is worth noting that we can only compare quality indexes among partitions with the same number of fuzzy sets. Otherwise, comparison is not fair.

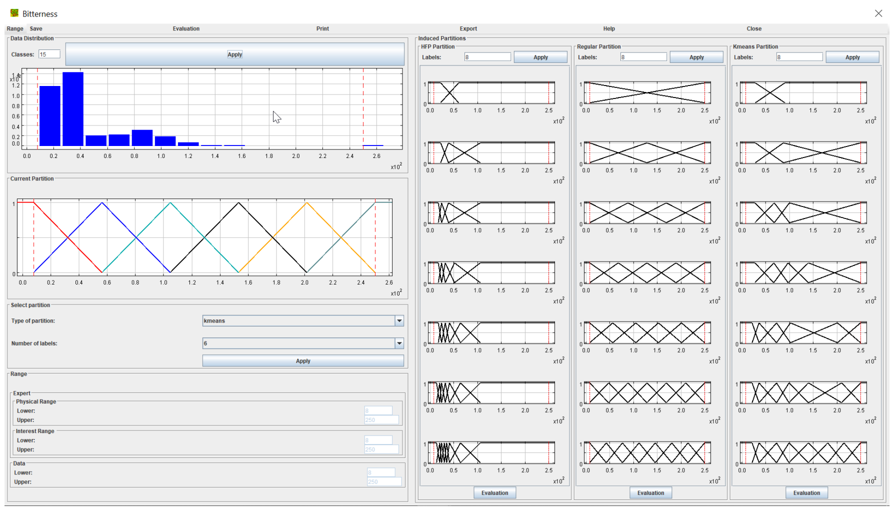


Fig. 6.3: GUAJE screenshot with partition generation details (Attribute = BITTERNESS).

- **Linguistic rule integration**. We generated 29 rules with the FDT algorithm provided by GUAJE. The FDT algorithm was run with the pruning option and 3 (i.e., the number of attributes in the dataset) as maximum tree depth.
  In this case, we ignored the expert rules. At this stage, the fuzzy rule-based classifier contains a KB with one output (i.e., the predicted beer style) and three inputs (i.e., one input associated to each attribute in the dataset). Each input is characterized by an SFP with 6 fuzzy sets. The relation between inputs and output is characterized through 29 fuzzy rules, with a total of 63 premises and 2.65 as average rule length. The classification ratio is 95.8% (Precision=96.1; Recall=95.8; F-measure=95.8) in training and 97.5% (Precision=97.9; Recall=97.5; F-measure=97.5) in test.
- **KB improvement**. With the aim of increasing even more interpretability while preserving accuracy, we applied the linguistic simplification procedure provided by GUAJE. As a result, the number of fuzzy sets for each partition dropped from six to five, the number of rules dropped from 29 to 15, with a total of 33 premises

Evaluation of Fuzzy Partitions

Print Export Help Close

Current Partition

Labels	Partition Coefficient (max)	Partition Entropy (min)	Chen Index (max)
6	0.60080	0.58051	0.66217

HFP Partition

Labels	Partition Coefficient (max)	Partition Entropy (min)	Chen Index (max)
2	0.81603	0.28023	0.73198
3	0.74181	0.38678	0.73988
4	0.71971	0.42047	0.75084
5	0.71562	0.42640	0.76571
6	0.72379	0.41624	0.77985
7	0.70745	0.44263	0.77371
8	0.68901	0.46473	0.75564

Regular Partition

Labels	Partition Coefficient (max)	Partition Entropy (min)	Chen Index (max)
2	0.78332	0.35898	0.71846
3	0.69636	0.47169	0.72363
4	0.67029	0.49999	0.72882
5	0.61454	0.56453	0.67111
6	0.60080	0.58051	0.66217
7	0.61092	0.56796	0.67790
8	0.63037	0.54443	0.70151

Kmeans Partition

Labels	Partition Coefficient (max)	Partition Entropy (min)	Chen Index (max)
2	0.88711	0.18222	0.84841
3	0.87074	0.21150	0.88675
4	0.83140	0.26287	0.86304
5	0.82272	0.27447	0.86252
6	0.73839	0.39541	0.79585
7	0.73465	0.39986	0.79623
8	0.73060	0.40643	0.79535

Fig. 6.4: GUAJE screenshot with evaluation of induced partitions (Attribute = BITTERNESS).

Rules

Rule	Type	Active	If Color	AND Bitterness	AND Strength	THEN Beer_Style
1	I	yes	Pale	very low		Blanche
2	I	yes	Pale	low		Pilsner
3	I	yes	Straw		(Standard) OR (High)	IPA
4	I	yes	Amber		Standard	IPA
5	I	yes	Straw		Session	Lager
6	I	yes	Straw		(Very high) OR (Extremley high)	Barleywine
7	I	yes	Amber		High	Barleywine
8	I	yes	Amber	(low) OR (average low) OR (average high)	Very high	Barleywine
9	I	yes	Amber		Extremley high	Barleywine
10	I	yes	Brown	average low	Extremley high	Barleywine
11	I	yes	(Amber) OR (Brown)		Session	Porter
12	I	yes	Amber	very low	Very high	Belgian_Strong_Ale
13	I	yes	Brown		(Standard) OR (High) OR (Very high)	Belgian_Strong_Ale
14	I	yes	Brown	(very low) OR (low)	Extremley high	Belgian_Strong_Ale
15	I	yes	Black			Stout

Fig. 6.5: GUAJE screenshot with the rule base at the end of the simplification stage.

(almost the half of the initial 63 premises), while accuracy remained the same. The list of simplified rules is depicted in Fig. 6.5.

Then, with the aim of increasing a bit more accuracy while preserving interpretability, we tuned the generated partitions with the Solis-Wets algorithm.

Namely, we tuned locally each fuzzy set with a maximum of 300 iterations (optimization was constrained to preserve SFPs). As a result, accuracy slightly goes up from 95.8% to 97.5% in training.

Unfortunately, as it can be appreciated in Figs. 6.6 and 6.7, the automatically generated partitions are not so meaningful as the ones defined by experts (pay special attention to the difference between partitions derived from data and the rest of partitions). Each fuzzy set is defined by four characteristic parameters $A_h = \{a, b, c, d\}$. In case of fuzzy sets with semi-trapezoidal shape, $a = b$ if decreasing slope, and $c = d$ otherwise. In case of fuzzy sets with triangular shape $b = c$. With the aim of facilitating the reproducibility of pictures in Figs. 6.6 and 6.7, we have included the values of the parameters $\{a, b, c, d\}$ for all fuzzy sets in the Table 6.1.

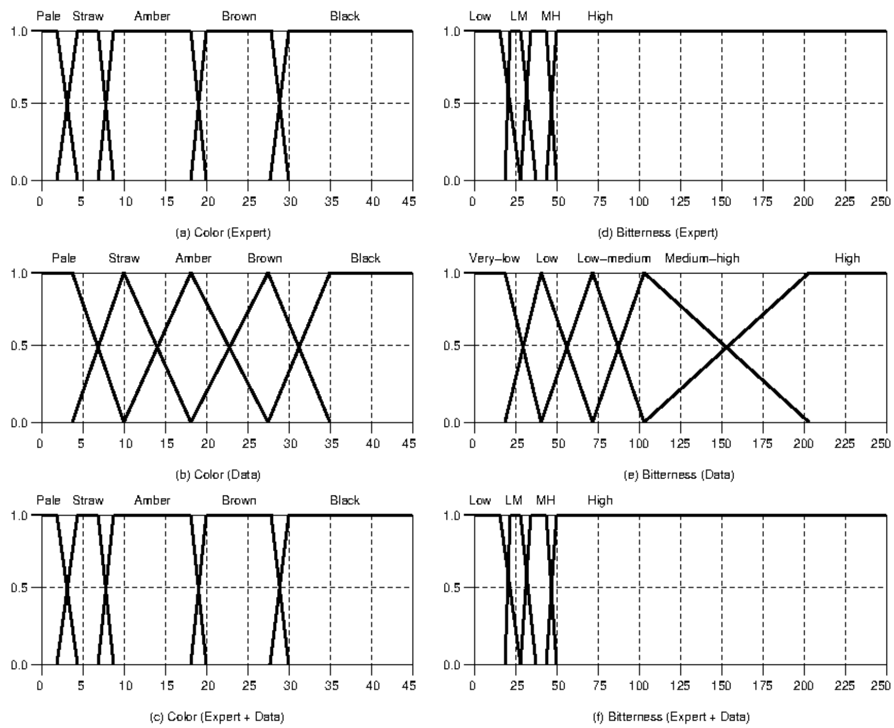


Fig. 6.6: Comparison between expert and induced partitions regarding attributes COLOR (see left hand side of the figure) and BITTERNESS (see right hand side of the figure). LM stands for LOW-MEDIUM and MH stands for MEDIUM-HIGH.

Fortunately, GUAJE helps us to combine expert and induced knowledge, what is likely to produce even better interpretability-accuracy trade-off than only considering knowledge extracted from data. Then, we built another fuzzy rule-based classifier as follows.

Table 6.1: Definition of designed fuzzy sets.

Attribute	Linguistic term	Expert				Data				Expert + Data			
		a	b	c	d	a	b	c	d	a	b	c	d
COLOR	PALE	0.000	0.000	2.000	4.000	0.000	0.000	3.258	9.905	0.000	0.000	2.000	4.000
	STRAW	2.000	4.000	7.000	8.000	3.258	9.905	9.905	17.096	2.000	4.000	7.000	8.000
	AMBER	7.000	8.000	18.000	20.000	9.905	17.096	17.096	27.571	7.000	8.000	18.000	20.148
	BROWN	18.000	20.000	28.000	30.000	17.096	27.571	27.571	34.769	18.000	20.148	28.180	30.000
	BLACK	28.000	30.000	45.000	45.000	27.571	34.769	45.000	45.000	28.180	30.000	45.000	45.000
BITTERNESS	VERY-LOW	-	-	-	-	8.000	8.000	22.007	36.664	-	-	-	-
	LOW	8.000	8.000	20.000	22.000	22.007	36.664	36.664	73.512	8.000	8.000	20.000	23.303
	LOW-MEDIUM	20.000	22.000	30.000	35.000	36.664	73.512	73.512	101.976	20.000	23.303	31.354	35.000
	MEDIUM-HIGH	30.000	35.000	45.000	50.000	73.512	101.976	101.976	201.600	31.354	35.000	45.000	50.000
	HIGH	45.000	50.000	250.000	250.000	101.976	201.600	250.000	250.000	45.000	50.000	250.000	250.000
STRENGTH	SESSION	0.039	0.039	0.050	0.055	0.039	0.039	0.054	0.079	0.039	0.039	0.050	0.055
	STANDARD	0.050	0.055	0.065	0.07	0.054	0.079	0.079	0.092	0.050	0.055	0.065	0.070
	HIGH	0.065	0.070	0.085	0.095	0.079	0.092	0.092	0.104	0.065	0.070	0.085	0.095
	VERY-HIGH	0.085	0.095	0.136	0.136	0.092	0.104	0.104	0.122	0.085	0.095	0.136	0.136
	EXTREMELY-HIGH	-	-	-	-	0.104	0.122	0.136	0.136	-	-	-	-

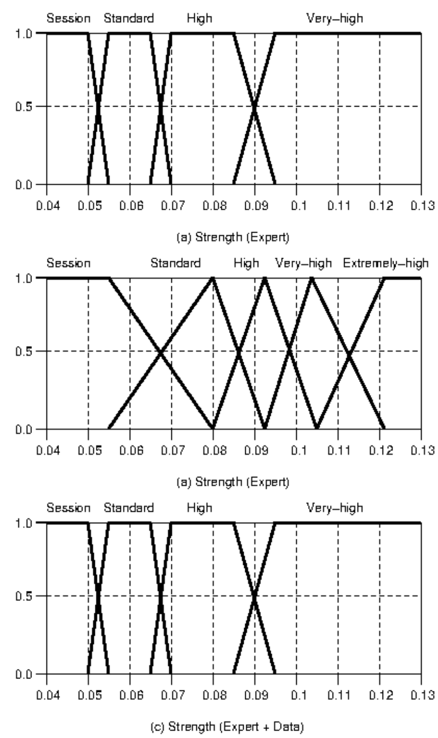
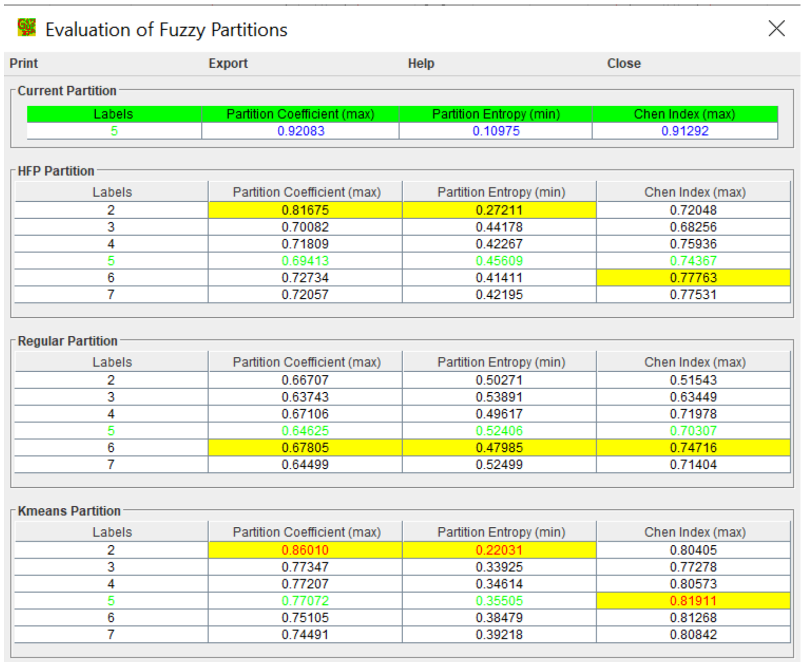


Fig. 6.7: Comparison between expert and induced partitions regarding the attribute STRENGTH.

- **Linguistic partition design**. We took as starting point the expert partitions described in the previous section. Then we generated alternative partitions (i.e., uniform SFPs, and partitions automatically extracted from data with the k-Means and HFP algorithms). As shown in Fig. 6.8, expert partitions turned out as the best partitions for attribute COLOR in terms of the GUAJE quality criteria (i.e., Partition Coefficient, Partition Entropy and Chen Index). Moreover, they are fully meaningful because they were designed by experts. Notice that, the same situation happens for BITTERNESS and STRENGTH.



Evaluation of Fuzzy Partitions

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Current Partition

Labels	Partition Coefficient (max)	Partition Entropy (min)	Chen Index (max)
5	0.92083	0.10975	0.91292

HFP Partition

Labels	Partition Coefficient (max)	Partition Entropy (min)	Chen Index (max)
2	0.81675	0.27211	0.72048
3	0.70082	0.44178	0.68256
4	0.71809	0.42267	0.75936
5	0.69413	0.45609	0.74367
6	0.72734	0.41411	0.77763
7	0.72057	0.42195	0.77531

Regular Partition

Labels	Partition Coefficient (max)	Partition Entropy (min)	Chen Index (max)
2	0.66707	0.50271	0.51543
3	0.63743	0.53891	0.63449
4	0.67106	0.49617	0.71978
5	0.64625	0.52406	0.70307
6	0.67805	0.47985	0.74716
7	0.64499	0.52499	0.71404

Kmeans Partition

Labels	Partition Coefficient (max)	Partition Entropy (min)	Chen Index (max)
2	0.86010	0.22031	0.80405
3	0.77347	0.33925	0.77278
4	0.77207	0.34614	0.80573
5	0.77072	0.35505	0.81911
6	0.75105	0.38479	0.81268
7	0.74491	0.39218	0.80842

Fig. 6.8: GUAJE screenshot with the evaluation of expert and induced partitions (Attribute = COLOR).

- **Linguistic rule integration**. We took as starting point the eight expert rules described in the previous section. Then, we added rules generated with the FDT algorithm (with pruning option and maximum tree depth equals 3). As a result, the new rule base was made up of 32 rules (i.e., the 8 expert rules plus 24 new rules derived from data with FDT).
- **KB improvement**. We compared expert and induced rules in terms of consistency through the linguistic simplification procedure provided by GUAJE. In case of conflict, expert rules are the priority (i.e., they are preserved during the simplification stage). At the end of this stage, the rule base was made up of 21 rules (i.e., the 8 expert rules plus 13 rules coming out of the simplification process applied over the induced rules). Then, we run the Solis-Wets optimization procedure and accuracy went slightly up to 97.5%. It is worth noting that partition tuning preserves SFPs which remain meaningful. As it can be appreciated in Figs. 6.6 and 6.7, with details in Table 6.1 (Expert + Data versus Expert), COLOR

and BITTERNESS are slightly modified with respect to the expert partitions while STRENGTH remains the same. For example, the BROWN value for COLOR is defined by the tuned fuzzy set B′={18.000, 20.148, 28.180, 30.000} while it was originally B={18.000, 20.000, 28.000, 30.000}. In addition, the LOW-MEDIUM value for BITTERNESS is defined by LM′={20.000, 23.303, 31.354, 35.000} after tuning the original fuzzy set LM={20.000, 22.000, 30.000, 35.000}.

Table 6.2 summarizes the results reported in Sects. 6.4.1 and 6.4.2. It is worth noting that GUAJE automatically extracted from data a fuzzy rule-based classifier (i.e., 6UP-SP-FDTP-S-OSW) which exhibits a very good interpretability-accuracy trade-off (95.25% of classification ratio with 11.4 rules and 22.7 premises) in 10-fold CV. It clearly outperforms the fuzzy classifiers described in the previous section (see the rows related to Expert and DC* in the table). In addition, the fuzzy rule-based classifier obtained by combining expert and induced knowledge (see the last row in the table; EXP-SP-FDTP-S-OSW) exhibits similar (even slightly higher) values of accuracy indexes (i.e., Accuracy, Precision, Recall and F-measure) with a greater number of rules (20.4) and premises (52.2). However, it is characterized by more meaningful fuzzy partitions (see pictures in Figs. 6.6 and 6.7) since they were defined in agreement with the knowledge previously provided by experts. As a final remark, pay attention to the fact that the designed fuzzy classifiers are not only interpretable but also accurate since their accuracy is really close to the accuracy baseline (96%) established by Random Forest.

Table 6.2: Quality indexes of classifiers for the Beer dataset (10-fold CV). In the case of GUAJE, acronyms refer to the involved design steps and should be interpreted as follows: EXP means starting with expert partitions. 6UP means starting with uniform fuzzy partitions of 6 linguistic terms each. SP means selecting the best partitions among the original ones and those induced from data. FDTP means inducing rules from data by means of the pruned FDT algorithm (with 3 set as maximum tree depth). S means applying simplification. OSW means applying Solis-Wets optimization (with 300 set as maximum number of iterations and optimization constrained to preserve SFPs).

Tool	**Algorithm**	**Accuracy (%)**	**Precision**	**Recall**	**F-measure**	**Rules**	**Premises**
-	Expert	81.25	-	-	-	8.0	21.0
FISDeT	DC*	87.50	-	-	-	9.0	18.0
Weka	Random Forest	96.00	96.00	96.00	96.00	-	-
GUAJE	6UP-SP-FDTP	93.00	94.51	93.00	92.74	25.8	51.9
	6UP-SP-FDTP-S	93.50	94.78	93.50	93.22	11.4	22.7
	6UP-SP-FDTP-S-OSW	95.25	95.96	95.25	95.09	11.4	22.7
	EXP-SP-FDTP	93.75	94.95	94.00	93.94	31.3	75.6
	EXP-SP-FDTP-S	93.50	94.95	93.75	93.81	20.4	52.2
	EXP-SP-FDTP-S-OSW	**95.25**	**96.27**	**95.25**	**95.31**	**20.4**	**52.2**

Inference: Rules

Print Export Close

Rule	Type	If Color	AND Bitterness	AND Strength	THEN Beer
1	E	Pale	Low	Session	Blanche
2	E	Amber	Low medium	Session	Lager
3	E	Straw	Medium high	Session	Pilsner
4	E	Amber	High	High	IPA
5	E	Black	Low medium	Standard	Stout
6	E	Amber	High	Very high	Barleywine
7	E	Brown	Low medium	Session	Porter
8	E	Brown	Medium high	Very high	Belgian_Strong_Ale
9	I	(Pale) OR (Straw)	Low		Blanche
10	I	Pale	NOT(Low)		Pilsner
11	I	Straw	Medium high		Pilsner
12	I	Straw	Low medium		Lager
13	I	Amber		(Session) OR (Standard)	Lager
14	I	Straw	High		IPA
15	I	Brown	High	Very high	Barleywine
16	I	Brown		(Session) OR (Standard)	Porter
17	I	Brown		High	Belgian_Strong_Ale
18	I	Brown	Low medium	Very high	Belgian_Strong_Ale
19	I	Black	(Medium high) OR (High)	High	Belgian_Strong_Ale
20	I	Black		(Session) OR (Standard)	Stout
21	I	Black	Low medium	High	Stout

Fig. 6.9: GUAJE screenshot with expert and induced rules at the end of the optimization stage (EXP-SP-FDTP-S-OSW in Table 6.2).

6.4.3 *The Explainable Fuzzy Rule-based Beer Style Classifier*

This section illustrates the use of GUAJE with the aim of understanding the behavior, at inference level, of the fuzzy rule-based classifier that we designed in the previous section (i.e., EXP-SP-FDTP-S-OSW in Table 6.2). Once again, even if we consider 10-fold CV, we illustrate here examples associated to the first fold just for illustrative purpose. The fuzzy classifier distinguishes among 8 classes (BLANCHE, LAGER, PILSNER, IPA, STOUT, BARLEYWINE, PORTER, and BELGIAN STRONG ALE) in terms of 3 attributes (COLOR, BITTERNESS and STRENGTH). The fuzzy partitions are depicted in Figs. 6.6 and 6.7 with detail parameters in Table 6.1. As it can be seen in Fig. 6.9, the rule base is made up of 21 rules. The first 8 rules (R1-R8; Type=E) are expert rules while the remaining 13 rules (R9-R21; Type=I) are induced from data.

Figure 6.10 shows a fingram which illustrates how these 21 rules interact at inference level. The fingram view provides the designer with a global view of the rule base behavior at inference level because it takes into account at once all inferences made for the whole dataset. In Fig. 6.10, we identify a main group of 12 rules which are strongly connected. They include expert rules (R4, R5, R6, R7, R8) and induced rules (R15, R16, R17, R18, R19, R20, R21) which correspond to five classes (IPA, BARLEYWINE, PORTER, STOUT, BELGIAN STRONG ALE). It is worth noting that most of these rules identify beer styles with similar (BROWN to BLACK) COLOR. Only R4 and R6 refer to AMBER COLOR.

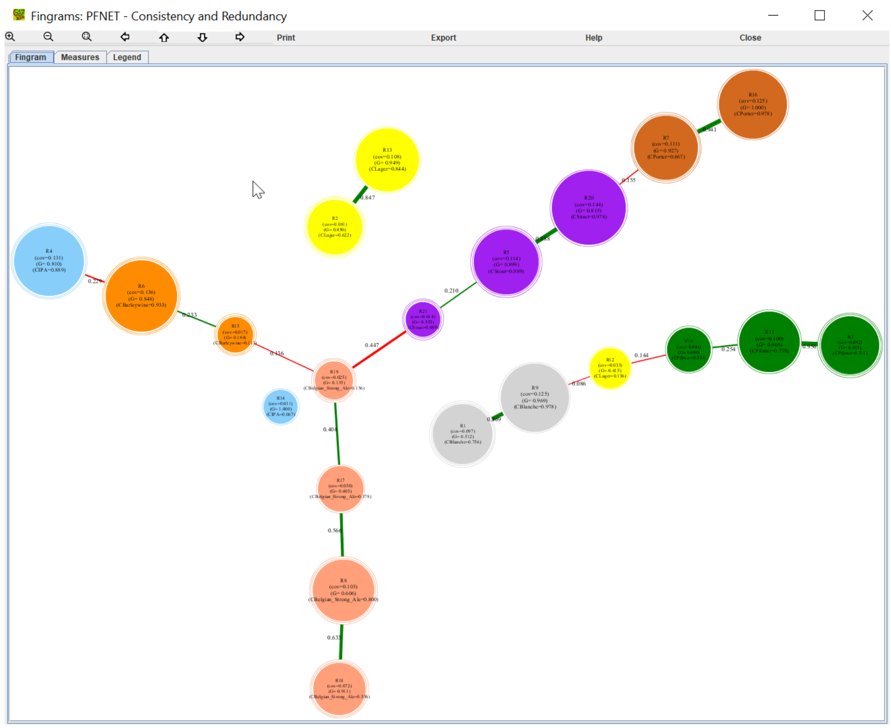

Fig. 6.10: GUAJE screenshot with the fingram view of EXP-SP-FDTP-S-OSW (see Table 6.2).

In addition, there is another group of six rules (listed in the picture from left to right: R1, R9, R12, R10, R11, R3) which are associated to three classes (BLANCHE, LAGER, PILSNER). These three classes identify three beer styles with (PALE to STRAW) COLOR. Indeed, COLOR is the only attribute that is considered in all the 21 rules (see Fig. 6.9) and it turns up as the most discriminant attribute.

The fingram view also facilitates to see how rules R2 ("IF COLOR is AMBER AND BITTERNESS is LOW-MEDIUM AND STRENGTH is SESSION THEN BEER STYLE is LAGER) and R13 ("IF COLOR is AMBER AND STRENGTH is SESSION or STANDARD THEN BEER STYLE is LAGER") are connected but isolated from the others. Actually, both rules are redundant (R2 is covered by R13) because they cover the same input space and point out at the same class (LAGER), so we may opt for removing one of them. For example, it is possible to remove R2 that is more specific than R13 (pay attention to the size of nodes in Fig. 6.10) and the accuracy of the rule base remains the same. However, R2 is an expert rule and that is the reason why this rule was not automatically removed by the simplification procedure that we did previously. It is worth noting that GUAJE always preserves expert knowledge versus knowledge automatically extracted from data. Anyway, in case of detecting this kind of situation we should ask the expert before removing R2. Finally, we also observe that R14 is alone. This seems a very specific rule that we may remove with the aim of improving the interpretability-accuracy trade-off. Actually, R14 only covers 3 instances in the dataset but they are not covered by any other rule. So, we recommend again to ask the expert before removing this rule because unless the related instances were outliers they should be properly handled by the rule base.

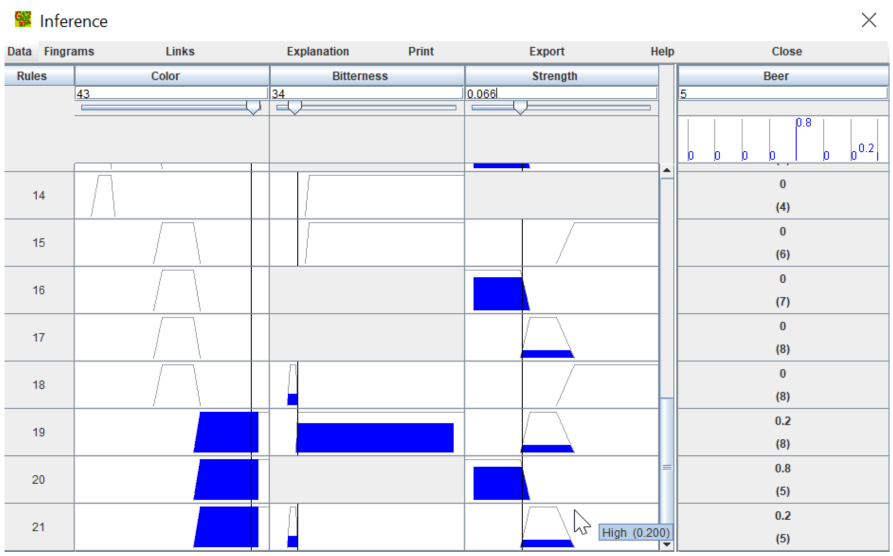

Fig. 6.11: GUAJE screenshot with an illustrative inference view.

Figure 6.11 shows a screenshot with the GUAJE inference window. For example, given a specific data instance {COLOR=43, BITTERNESS=34, STRENGTH=0.066}, the numerical values assigned to the three attributes are associated to the scrolling controls in the upper part of the picture. In addition, rule numbers are given on the left side of the figure. For each rule premise, the activation strength is highlighted with the membership function filled in up to the corresponding height. In addition, the actual membership values are shown as tooltip texts when hovering the mouse over the pictures. In addition, rule activation strengths are given on the right side of the figure, and they are aggregated in the upper right bar chart. Such values are computed by considering minimum as t-norm, maximum as t-conorm and the winner takes all rule inference mechanism. Nevertheless, GUAJE lets users opt for other fuzzy operators. The interested reader is kindly referred to Chap. [11] for further details about how to select the right fuzzy operators for ensuring an effective and meaningful inference.

Figure 6.12 shows an instance-based fingram which corresponds to the inference view illustrated in Fig. 6.11. It is easy to appreciate how this kind of fingram depicts

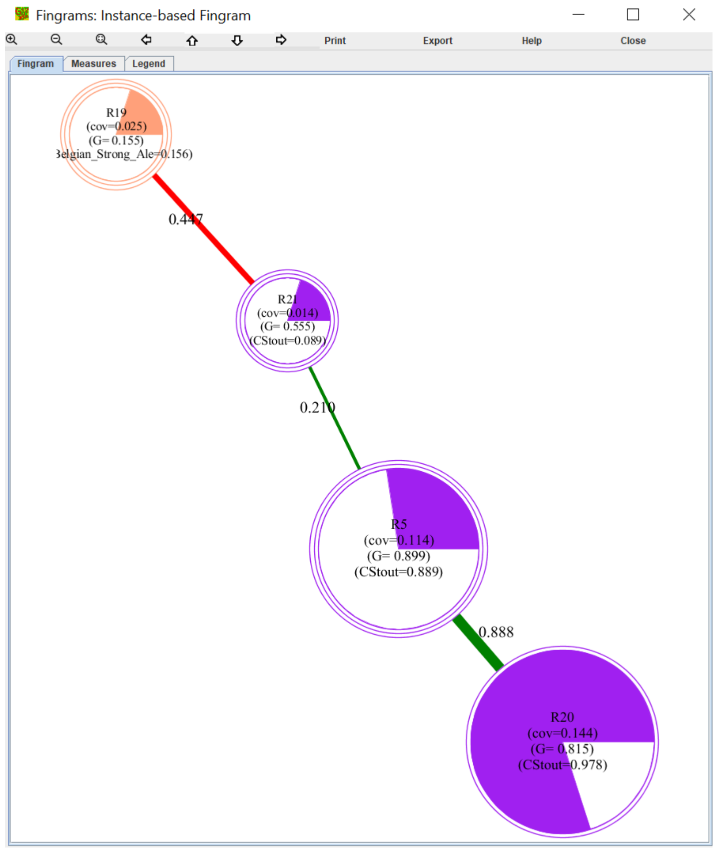

Fig. 6.12: GUAJE screenshot with the instance-based fingram view associated to the inference case illustrated in Fig. 6.11.

graphically the relation among the fired rules (in our example, rules R5, R19, R20 and R21). Rule R20 turns up as the winner rule with an activation strength equals 0.8. The action of this rule is also supported by rules R5 and R21 which point out the same output class. The nodes which represent these three rules are printed in the same color (i.e., purple in this example) and they are connected through green links what means they are partially redundant rules. In addition, rule R19 is represented by a node of different color because it refers to a different class. Moreover, R19 is connected to R21 through a red link what remarks the fact that there may be a potential inconsistency in case both rules were fired with similar degree.

These visualization tools (i.e., both inference and fingram view tools) provide the designer with a lot of valuable information to carefully debug and verify the fuzzy rule-based classifier under design.

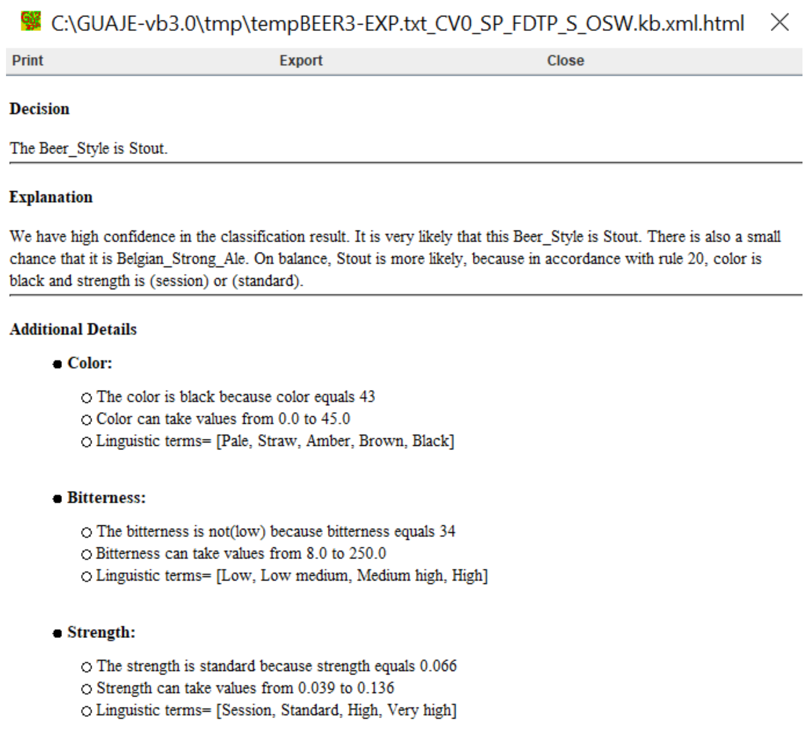

Fig. 6.13: GUAJE screenshot with the explanation in natural language of a non-ambiguous case.

In addition, GUAJE is endowed with self-explanatory capabilities. Fig. 6.13 shows the explanation in NL that is automatically generated for the inference that was illustrated in Figs. 6.11 and 6.12. The beer style (STOUT) is identified by the winner rule (R20) which is activated with an activation strength of 0.8. There is a smaller chance that beer style is BELGIAN STRONG ALE because R19 is activated with an activation strength equal to 0.2. Anyway, there is a big difference (0.6) between the activation strengths of these two rules. Accordingly, the explanation starts with remarking that there is a high confidence in the classification result (because the winner rule activation strength is high and the closer alternative is far enough). Then, it is commented that there is a minor chance for the second most plausible beer style. Then, additional details (i.e., the range of each input attribute along with the list of linguistic terms) are provided to facilitate the understanding of the given explanation.

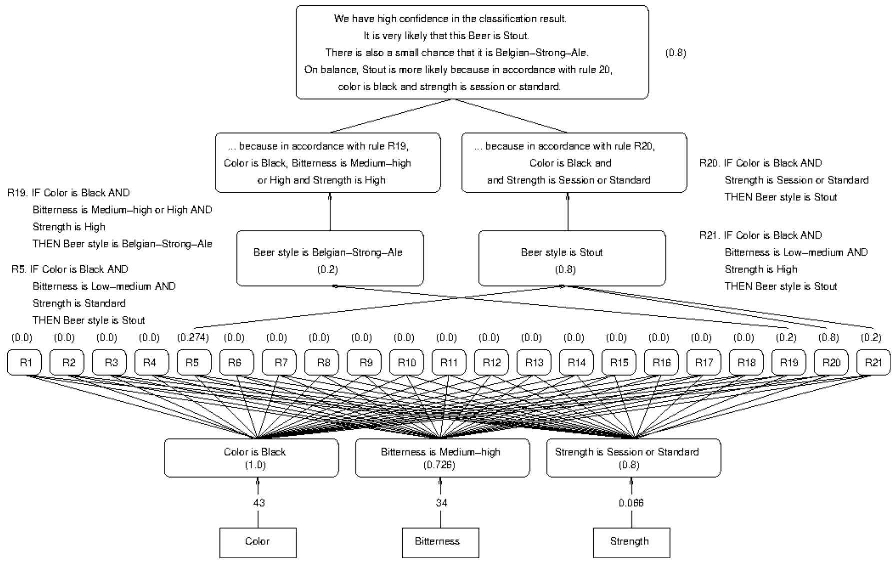

Fig. 6.14: Sketch of the IFS/GLMP for interpreting the inference case illustrated in Figs. 6.11 and 6.12.

This explanation is generated as follows. Firstly, the fuzzy rule-based classifier is translated into the IFS/GLMP[17] that is depicted in Fig. 6.14. This IFS/GLMP can be seen as a hierarchy of nodes with inputs at the bottom, rules in the core, and output at the upper level. Each node in the hierarchy has associated a textual description along with a relevance degree (pay attention to the values in brackets in the picture). Then, the system decision (associated to the node at the top of the hierarchy) is explained in terms of the fired rules and their relations at inference level, what turns up as result of the combination of all the identified pieces of information. Then, additional details are provided in order to facilitate understanding how the given numerical values for each input attribute are translated into the linguistic terms that can match with the linguistic descriptions of the rules (see the rule list in Fig. 6.9 and the associated explanation in Fig. 6.13).

The linguistic realization is done in the following two steps:

1. GUAJE asks rLDCP to interpret the IFS/GLMP (see Fig. 6.14). The text generation is based on pre-defined templates (see Sect. 6.3.2).
2. rLDCP provides GUAJE with a textual explanation that is further refined by GUAJE thanks to simpleNLG (see Sect. 6.3.3). The final explanation is given on top of Fig. 6.14.

---

[17] The interested reader is kindly invited to revisit Sect. 6.3 and the references there in for further details about how an IFS/GLMP is defined. IFS stands for Interpretable Fuzzy System and GLMP is the acronym of Granular Linguistic Model of a Phenomenon.

In case of ambiguity cases in inference, i.e., data instances for which at least two rules (which point at different output classes) are fired with similar degrees (i.e., the difference between activation strengths is smaller or equal than a threshold δ that we have set equals 0.1 in the experiments), then understanding the inference result is harder. Therefore, in such cases, providing users with explanations of the inference process in NL is expected to be highly appreciated.

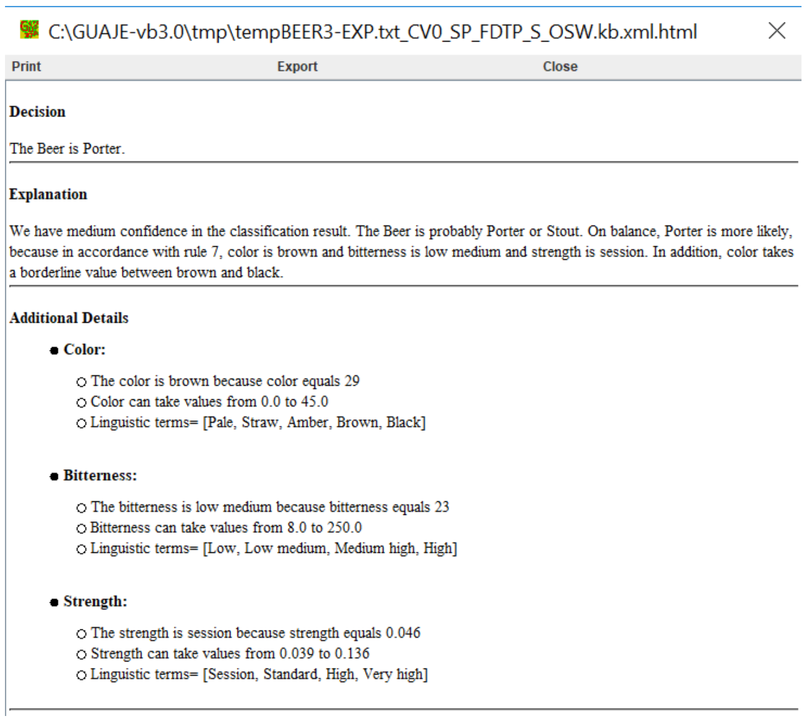

Fig. 6.15: GUAJE screenshot with an example of how to provide users with an explanation in natural language associated to a case of ambiguous inference.

Figures 6.15 and 6.16 show an illustrative example of ambiguity case. Given a data instance {Color=29, Bitterness=23, Strength=0.046}, there are two winner rules that are fired with the same activation strength (0.549): R7 ("IF Color is Brown AND Bitterness is Low-medium AND Strength is Session THEN Beer Style is Porter") and R16 ("IF Color is Brown AND Strength is Session or Standard THEN Beer Style is Porter"). R7 is more specific than R16. Since both rules point out the same class, there is neither inference ambiguity nor inconsistency between these rules. Then, the explanation is based on R7 because this is an expert rule and GUAJE always gives priority to expert knowledge. In addition, R20 ("IF Color is Black AND Strength is

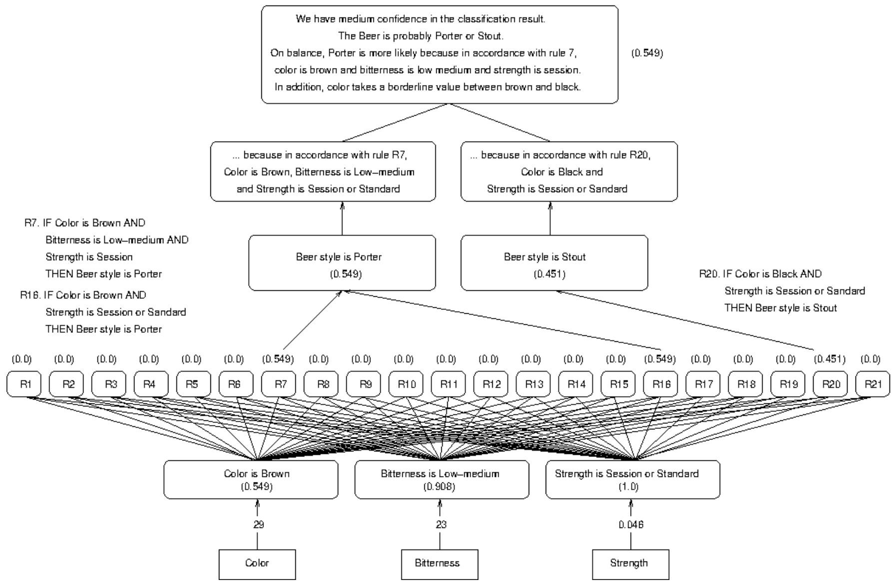

Fig. 6.16: Sketch of how GUAJE generates the explanation given in Fig. 6.15.

SESSION or STANDARD THEN BEER STYLE is STOUT") is fired with activation strength equals 0.451. Comparing R7 with R20, $FD_{R7} - FD_{R20} = 0.098 \leq \delta$. It is easy to see how the difference between rule activation strengths is due to the activation strengths of the premises "COLOR is BROWN" (R7) and "COLOR is BLACK" (R20). In other words, the actual value of COLOR (29) is in the borderline between BROWN and BLACK. As shown in Fig. 6.15 (and detailed in Fig. 6.16), this fact is highlighted in the given explanation. Moreover, we have illustrated with this example how given a specific data instance, GUAJE analyzes those borderline situations that are likely to yield ambiguity in the result of the inference process, and if any of these situations is identified then they are highlighted in the generated explanations.

6.4.4 The Portable Fuzzy Rule-based Beer Style Classifier

This section shows how to use the fuzzy classifier previously designed with other software tools apart from GUAJE. This is feasible thanks to the import/export options (e.g., GUAJE from/to FisPro, GUAJE from/to Matlab, GUAJE from/to JFML, etc.) that are provided by this open source modeling tool.

As an illustrative example, we have exported the explainable classifier described in the previous section (see Fig 6.9) into the XML format that is defined by the IEEE Std 1855-2016. The exported file can be loaded by the JFML library. In order to make a fuzzy inference with JFML and the same data values given in Fig 6.11, it is only needed to type the following instruction in a command window.

```
> java -jar ./lib/JFML-v1.2.jar
  ./XMLFiles/BEER3-EXP.txt_CV0_SP_FDTP_S_OSW.kb.xml.jfml.xml
  1 Beer_Style Color 43 Bitterness 34 Strength 0.066
```

We first set the JFML jar file (./lib/JFML-v1.2.jar). It is recommended to use always the most recent JFML version (e.g., here the considered version is 1.2). Then, we set the file where the fuzzy rule-based classifier is stored. It is worth noting that in this case the file name (BEER3-EXP.txt_CV0_SP_FDTP_S_OSW.kb.xml.jfml.xml) is related to the experiments that we described in the previous sections. We selected the first fold (CV0) and the classifier initialized with expert knowledge (EXP) and then refined with the GUAJE options (SP-FDTP-S-OSW) that yielded the best results (see Table 6.2). Then, we set the number of outputs (1), the name of the given output (BEER STYLE) and a specific value for each input attribute ("COLOR 43", "BITTERNESS 34", and "STRENGTH 0.066").

The result of running this command is printed below. First, the fuzzy classifier is loaded from the XML file. Second, data values are assigned to each input. Third, inference is performed and output decision is printed. In this case, BEER STYLE = 5.0 corresponds to STOUT as shown in Figs. 6.9, 6.11-6.14. Fourth, the linguistic description of the fuzzy classifier is printed. It comprises both linguistic variables and rules with the related activation strengths. Pay attention to rule R20 which turns up as the winner rule for the case under study.

```
1) Loading Fuzzy System from an XML file according the standard IEEE 1855

2) Setting input variables: Color=43.0, Bitterness=34.0, Strength=0.066

3) Making fuzzy inference
RESULTS
 (INPUT): Color=43.0, Bitterness=34.0, Strength=0.066
 (OUTPUT): Beer_Style=5.0

4) Fuzzy System Description

FUZZY SYSTEM: BEER3 - MAMDANI
KNOWLEDGEBASE:
  *Color - domain[0.0, 45.0] - input
Pale -  trapezoid [a: 0.0, b: 0.0, c: 2.0, d: 4.0]
Straw -  trapezoid [a: 2.0, b: 4.0, c: 7.0, d: 8.0]
Amber -  trapezoid [a: 7.0, b: 8.0, c: 18.0, d: 20.148]
Brown -  trapezoid [a: 18.0, b: 20.148, c: 28.18, d: 30.0]
Black -  trapezoid [a: 28.18, b: 30.0, c: 45.0, d: 45.0]
(Pale) OR (Straw) -  trapezoid [a: 0.0, b: 0.0, c: 7.0, d: 8.0]

  *Bitterness - domain[8.0, 250.0] - input
Low -  trapezoid [a: 8.0, b: 8.0, c: 20.0, d: 23.303]
Low medium -  trapezoid [a: 20.0, b: 23.303, c: 31.354, d: 35.0]
Medium high -  trapezoid [a: 31.354, b: 35.0, c: 45.0, d: 50.0]
High -  trapezoid [a: 45.0, b: 50.0, c: 250.0, d: 250.0]
NOT(Low) -  trapezoid [a: 20.0, b: 23.303, c: 250.0, d: 250.0]
(Medium high) OR (High) -  trapezoid [a: 31.354, b: 35.0, c: 250.0, d: 250.0]

  *Strength - domain[0.039, 0.136] - input
Session -  trapezoid [a: 0.039, b: 0.039, c: 0.05, d: 0.055]
Standard -  trapezoid [a: 0.05, b: 0.055, c: 0.065, d: 0.07]
High -  trapezoid [a: 0.065, b: 0.07, c: 0.085, d: 0.095]
Very high -  trapezoid [a: 0.085, b: 0.095, c: 0.136, d: 0.136]
```

```
(Session) OR (Standard) -  trapezoid [a: 0.039, b: 0.039, c: 0.065, d: 0.07]

  *Beer_Style - domain[1.0, 8.0] - Accumulation:MAX; Defuzzifier:MOM - output
Blanche -  singleton [a: 1.0]
Lager -  singleton [a: 2.0]
Pilsner -  singleton [a: 3.0]
IPA -  singleton [a: 4.0]
Stout -  singleton [a: 5.0]
Barleywine -  singleton [a: 6.0]
Porter -  singleton [a: 7.0]
Belgian_Strong_Ale -  singleton [a: 8.0]

RULEBASE:
  *mamdani - rulebase-BEER3: OR=MAX; AND=MIN; ACTIVATION=MIN

RULE 1: rule1 - (0.0)
        IF Color IS Pale AND Bitterness IS Low
        AND Strength IS Session
        THEN Beer_Style IS Blanche [weight=1.0]

RULE 2: rule2 - (0.0)
        IF Color IS Amber AND Bitterness IS Low medium
        AND Strength IS Session
        THEN Beer_Style IS Lager [weight=1.0]

RULE 3: rule3 - (0.0)
        IF Color IS Straw AND Bitterness IS Medium high
        AND Strength IS Session
        THEN Beer_Style IS Pilsner [weight=1.0]

RULE 4: rule4 - (0.0)
        IF Color IS Amber AND Bitterness IS High
        AND Strength IS High
        THEN Beer_Style IS IPA [weight=1.0]

RULE 5: rule5 - (0.2742732)
        IF Color IS Black AND Bitterness IS Low medium
        AND Strength IS Standard
        THEN Beer_Style IS Stout [weight=1.0]

RULE 6: rule6 - (0.0)
        IF Color IS Amber AND Bitterness IS High
        AND Strength IS Very high
        THEN Beer_Style IS Barleywine [weight=1.0]

RULE 7: rule7 - (0.0)
        IF Color IS Brown AND Bitterness IS Low medium
        AND Strength IS Session
        THEN Beer_Style IS Porter [weight=1.0]

RULE 8: rule8 - (0.0)
        IF Color IS Brown AND Bitterness IS Medium high
        AND Strength IS Very high
        THEN Beer_Style IS Belgian_Strong_Ale [weight=1.0]

RULE 9: rule9 - (0.0)
        IF Color IS (Pale) OR (Straw) AND Bitterness IS Low
        THEN Beer_Style IS Blanche [weight=1.0]

RULE 10: rule10 - (0.0)
        IF Color IS Pale AND Bitterness IS NOT(Low)
        THEN Beer_Style IS Pilsner [weight=1.0]

RULE 11: rule11 - (0.0)
        IF Color IS Straw AND Bitterness IS Medium high
        THEN Beer_Style IS Pilsner [weight=1.0]
```

```
RULE 12: rule12 - (0.0)
        IF Color IS Straw AND Bitterness IS Low medium
        THEN Beer_Style IS Lager [weight=1.0]

RULE 13: rule13 - (0.0)
        IF Color IS Amber AND Strength IS (Session) OR (Standard)
        THEN Beer_Style IS Lager [weight=1.0]

RULE 14: rule14 - (0.0)
        IF Color IS Straw AND Bitterness IS High
        THEN Beer_Style IS IPA [weight=1.0]

RULE 15: rule15 - (0.0)
        IF Color IS Brown AND Bitterness IS High
        AND Strength IS Very high
        THEN Beer_Style IS Barleywine [weight=1.0]

RULE 16: rule16 - (0.0)
        IF Color IS Brown AND Strength IS (Session) OR (Standard)
        THEN Beer_Style IS Porter [weight=1.0]

RULE 17: rule17 - (0.0)
        IF Color IS Brown AND Strength IS High
        THEN Beer_Style IS Belgian_Strong_Ale [weight=1.0]

RULE 18: rule18 - (0.0)
        IF Color IS Brown AND Bitterness IS Low medium
        AND Strength IS Very high
        THEN Beer_Style IS Belgian_Strong_Ale [weight=1.0]

RULE 19: rule19 - (0.2000003)
        IF Color IS Black AND Bitterness IS (Medium high) OR (High)
        AND Strength IS High
        THEN Beer_Style IS Belgian_Strong_Ale [weight=1.0]

RULE 20: rule20 - (0.7999997)
        IF Color IS Black AND Strength IS (Session) OR (Standard)
        THEN Beer_Style IS Stout [weight=1.0]

RULE 21: rule21 - (0.2000003)
        IF Color IS Black AND Bitterness IS Low medium
        AND Strength IS High
        THEN Beer_Style IS Stout [weight=1.0]
```

In addition, this fuzzy classifier can be loaded and managed in Python with Py4JFML. We exported into the IEEE Std 1855-2016 format all the 10 classifiers that were previously built with GUAJE under 10-fold CV. Then, we made fuzzy inferences with Py4JFML by using standard Python commands. Moreover, the `confusion_matrix` function available in scikit-learn was used to compute the related confusion matrix for the designed explainable classifier. Just for comparison purpose, we also computed the confusion matrix of the Random Forest model built with Python's scikit-learn and default parameters for the sake of simplicity. Then, a simple procedure based on `pyplot` was written to draw Figs. 6.17 and 6.18.

These figures show how the designed explainable classifier exhibits an accuracy comparable to the Random Forest classifier, taken as baseline, for all target classes. In the worst cases (i.e., Class4=IPA and Class6=BARLEYWINE), the classification ratio drops from 94% to 92%. This lost in accuracy is admissible because we are comparing a black-box classifier (Random Forest) with an explainable classifier (GUAJE EXP-SP-FDTP-S-OSW). Thus, the advantages derived from explainability are worth the slight lost in accuracy. Moreover, in some cases (e.g.,

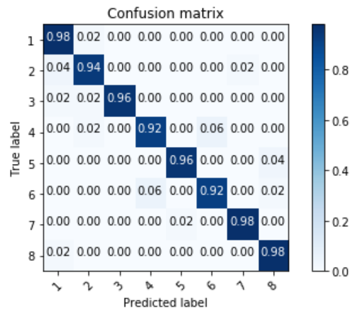

Fig. 6.17: Confusion matrix of the explainable fuzzy classifier exported to Py4JFML.

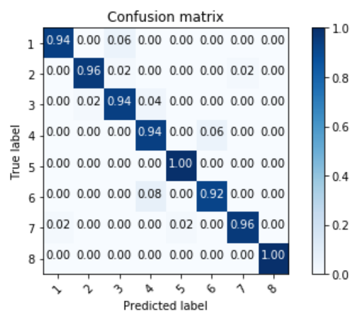

Fig. 6.18: Confusion matrix of the baseline Random Forest classifier (Python's scikit-learn).

Class1=BLANCHE or Class3=PILSNER), the explainable classifier performs even better than the Random Forest classifier. Anyway, let us remind that the aim here is maximizing explainability and we are aware that the accuracy of the Random Forest classifier may be slightly improved if we searched for the optimal parameters instead of just considering the default ones.

Let's sketch below the Python script that we wrote to run the experiments described above[18].

```
from py4jfml.Py4Jfml import Py4jfml
import pandas as pd
from sklearn import tree
from sklearn.ensemble import RandomForestClassifier
from sklearn import svm
from sklearn.metrics import confusion_matrix
import matplotlib.pyplot as plt
import itertools
```

[18] This Python script is available online along with the rest of files needed to reproduce the use cases illustrated in this chapter [9].

```
import numpy as np

features = ["Color", "Bitterness", "Strength"]
target = ["Beer_Style"]
...
for fold_idx in range(n_folds):
...
    fis_filename = ("BEER3-EXP.txt_CV" + str(fold_idx)
    + "_SP_FDTP_S_OSW.kb.xml.jfml.xml")
    fis = Py4jfml.load("./"+fold_path+"/"+fis_filename)
...
    cm_fis = []
    for fis_idx in range(len(fold["fis_results"])):
        actual = fold["fis_results"][fis_idx]
        cm = confusion_matrix(desired, actual,
                              labels=[str(i+1)+".0" for i in range(8)])
        cm_fis.append(cm)
    result["cm_fis"] = cm_fis
...
```

Finally, we can export the fuzzy classifier built with GUAJE into the format recognized by other software tools. For example, FisPro [49, 50] lets us plot the classifier as a 3D surface (see Fig. 6.19). This kind of graph is useful to identify potential relations between pairs of input attributes (associated to axes X and Y) with respect to the output class (associated to axis Z). 3D system surfaces are used worldwide, especially in the context of control and regression problems.

6.4.5 Explanations Supported by Interpretable Fuzzy Classifiers versus Explanations Supported by Fuzzy Classifiers without Linguistic Interpretability

ExpliClas[19] is a web service coded in Java [6, 17]. It provides users with multimodal (textual + graphical) explanations related to Weka[20] classifiers [75]. Among the Weka classifiers currently available in ExpliClas, we have selected the Fuzzy Unordered Rule Induction Algorithm (FURIA) [52].

FURIA rules are generated with the focus on accuracy, while disregarding the interpretability of the whole model. They are fuzzy IF-THEN classification rules with fuzzy sets $A = \{a, b, c, d\}$ of trapezoidal shape in the antecedent of each rule. It is worth noting that only the most relevant features are considered in each rule. For example, a rule R_i with three antecedents may be as follows:

$$\begin{aligned} R_i: \text{ IF } A_1 \in [0.65, 2.7, \infty, \infty] \text{ and } A_2 \in [-\infty, -\infty, 0.27, 20.8] \\ \text{and } A_3 \in [5, 7, 7.6, 12] \text{ THEN } \textsc{class} \text{ is } \mathrm{C}_1 \text{with CF} = 0.75 \end{aligned} \tag{6.3}$$

where CF is the certainty factor of rule R_i. The rule activation strength is computed as the multiplication of the CF by the firing strength which results of applying the

[19] The ExpliClas web service: `https://demos.citius.usc.es/ExpliClas/`

[20] The Waikato Environment for Knowledge Analysis (WEKA): `https://www.cs.waikato.ac.nz/ml/weka/`

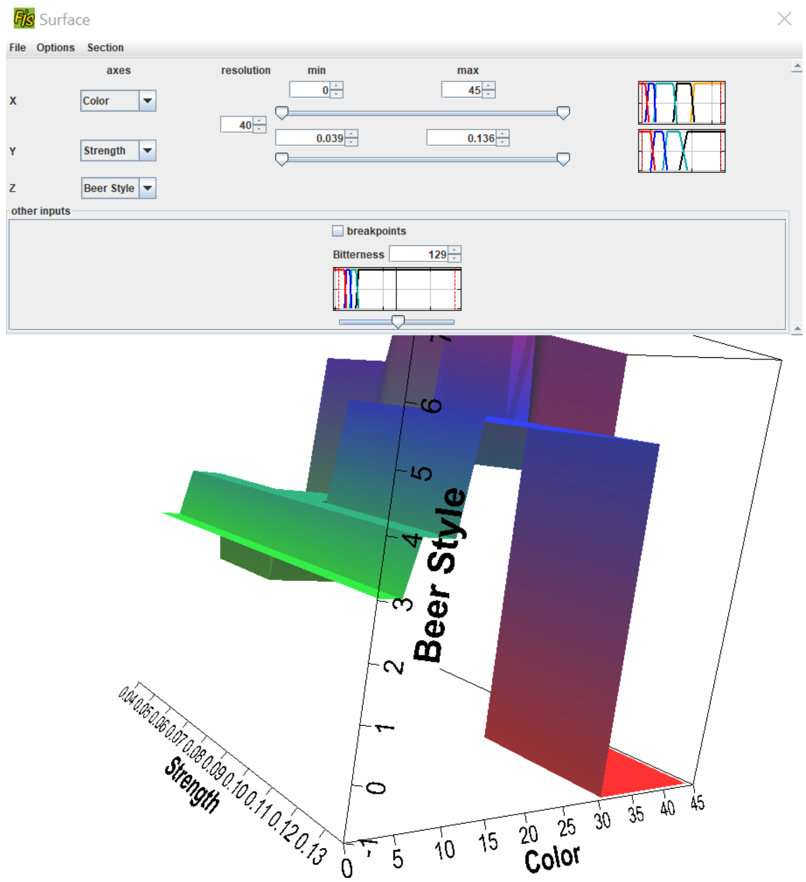

Fig. 6.19: FisPro screenshot with an example of 3D surface.

usual inference mechanism of Mamdani fuzzy systems [57]. Given a data instance, if there is no rule with activation strength greater than zero, then FURIA offers three options to the user: (1) abstain (i.e., no output class is given); (2) voting for the a priori most frequent class in the dataset; and (3) rule stretching (i.e., a new set of rules is dynamically created from the initial rule base by removing antecedents in order one by one, rule by rule, until the instance is covered). FURIA is deemed as a gray-box classifier because it produces a set of rules which can be interpreted (but only at certain extent because it lacks of linguistic interpretability) by users. Nevertheless, FURIA acts as a black-box classifier when applying the stretching mechanism.

ExpliClas provides users with: (1) global and local explanations associated to inferences made by the selected classifier (which is trained by Weka with 10-fold CV); and (2) explanations associated to the classifier confusion matrices (regarding

CV, training and test). Explanations provided by ExpliClas are multi-modal because they are a mixture of graphs along with sentences in NL. Global semantics is enforced by means of defining beforehand SFPs for each decision variable. By default, three linguistic terms (e.g., SMALL, MEDIUM, LARGE) are created but they can be manually edited by the user. Then, non-linguistic fuzzy intervals in FURIA rules are interpreted in terms of the closest linguistic terms previously defined. Those terms are used to verbalize decisions in the form of textual explanations which are realized with simpleNLG [45], in a similar way how explanations are generated by GUAJE.

We used ExpliClas to create FURIA rules for the Beer dataset. FURIA produced 15 rules which achieve 95.5% of classification rate (10-fold CV). Let us print below a sketch of the resultant buffer log file.

```
FURIA rules:
===========
(Bitterness in [-inf, -inf, 19, 30]) and (Color in [-inf, -inf, 4, 7])
 => Beer_Style=1.0 (CF = 0.97)

(Color in [-inf, -inf, 4, 7]) and (Bitterness in [-inf, -inf, 23, 30])
 => Beer_Style=1.0 (CF = 0.97)

(Bitterness in [-inf, -inf, 27, 52]) and (Color in [6, 7, inf, inf])
 and (Color in [-inf, -inf, 14, 16]) => Beer_Style=2.0 (CF = 0.95)

(Color in [-inf, -inf, 6, 8]) and (Bitterness in [29, 30, inf, inf])
 and (Bitterness in [-inf, -inf, 52, 63]) => Beer_Style=3.0 (CF = 0.97)

(Bitterness in [52, 57, inf, inf])
 and (Strength in [-inf, -inf, 0.091, 0.092]) => Beer_Style=4.0 (CF = 0.89)

(Bitterness in [48, 50, inf, inf]) and (Color in [-inf, -inf, 8, 10])
 => Beer_Style=4.0 (CF = 0.84)

(Color in [30, 31, inf, inf]) => Beer_Style=5.0 (CF = 0.96)

(Color in [29, 30, inf, inf]) and (Bitterness in [-inf, -inf, 32, 39])
 and (Strength in [0.046, 0.053, inf, inf]) => Beer_Style=5.0 (CF = 0.96)

(Strength in [0.087, 0.093, inf, inf])
 and (Bitterness in [47, 50, inf, inf]) and (Color in [11, 12, inf, inf])
 => Beer_Style=6.0 (CF = 0.96)

(Bitterness in [93, 95, inf, inf])
 and (Strength in [0.089, 0.092, inf, inf]) > Beer_Style=6.0 (CF = 0.88)

(Bitterness in [41, 77, inf, inf])
 and (Strength in [0.096, 0.097, inf, inf]) => Beer_Style=6.0 (CF = 0.95)

(Bitterness in [62, 79, inf, inf]) and (Color in [13, 14, inf, inf])
 and (Bitterness in [-inf, -inf, 85, 98]) => Beer_Style=6.0 (CF = 0.9)

(Color in [12, 20, inf, inf]) and (Strength in [-inf, -inf, 0.051, 0.053])
 and (Color in [-inf, -inf, 30, 34]) => Beer_Style=7.0 (CF = 0.96)

(Color in [17, 20, inf, inf]) and (Strength in [-inf, -inf, 0.054, 0.082])
 and (Color in [-inf, -inf, 29, 30]) => Beer_Style=7.0 (CF = 0.95)

(Strength in [0.07, 0.087, inf, inf])
 and (Bitterness in [-inf, -inf, 47, 50]) => Beer_Style=8.0 (CF = 0.97)

Number of Rules : 15

Time taken to build model: 0.13 seconds
```

It is easy to appreciate how the rules listed above lack of linguistic interpretability. Fortunately, thanks to the linguistic layer on top of the explaining module, ExpliClas is able to generate explanations in NL which are grounded on approximate linguistic terms associated to the fuzzy intervals given in these FURIA rules. It is worth noting that we consider here the same linguistic terms defined previously (see the expert partitions defined in Figs. 6.6 and 6.7). For example, Fig. 6.20 shows how ExpliClas explains the classification made by FURIA rules regarding the same case explained by GUAJE in Figs. 6.15 and 6.16. Notice that, the ExpliClas inference view is similar to the GUAJE inference view (pay attention to how rules are similarly depicted in Figs. 6.11 and 6.20).

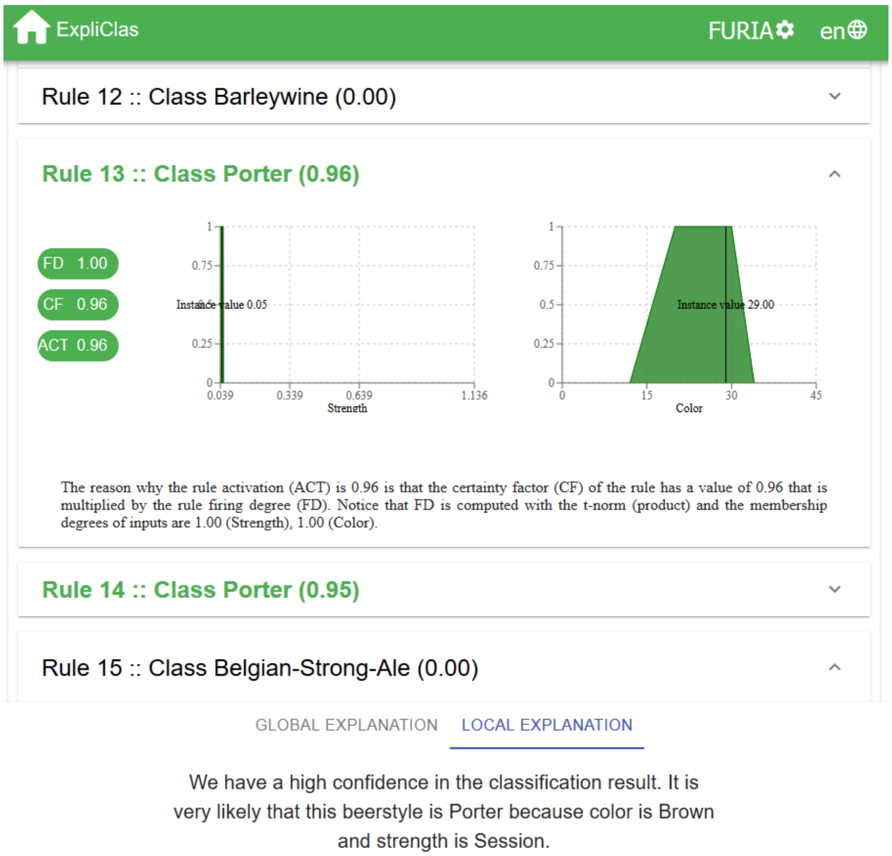

Fig. 6.20: ExpliClas screenshot with an example of multi-modal explanation associated to FURIA rules (the case under study is the same that was considered in Figs. 6.15 and 6.16).

Even if GUAJE dealt with the case under consideration as a doubtful or ambiguous case because two rules were fired almost with the same degree (see Fig. 6.16), ExpliClas considers it as a clear case without any doubt because the two fired rules point out the same beer style. In addition, the activation strength of the winner rule in ExpliClas is higher than it was in GUAJE. This is the reason why the explanation associated to FURIA rules remarks that classification is made with high confidence while GUAJE told that it had a medium confidence in the classification result (see Fig. 6.15). Obviously, different fuzzy rule-based classifiers may differ in their classification and associated explanation similarly to how different people may have different opinions. Anyway, we observe that the two classifiers act consistently since they point out the same beer style regarding the given data instance. Moreover, both explanations are comprehensible and consistent. On the one hand, GUAJE says that the given data instance corresponds to the PORTER or STOUT beer styles, but on balance PORTER is more likely. On the other hand, ExpliClas confirms that the beer style is PORTER. In addition, GUAJE says that STRENGTH is SESSION, BITTERNESS is LOW-MEDIUM, and COLOR takes a borderline value between BROWN and BLACK. ExpliClas confirms that STRENGTH is SESSION and COLOR is BROWN. However, if we compare the fuzzy sets in Fig. 6.20 with those defined by experts in Figs. 6.6 and 6.7, we see they are very different. This fact may make the linguistic approximations on top of FURIA rules being not so meaningful as the linguistic interpretability that is inherent to rules generated with GUAJE, which naturally deals with the given expert partitions.

Indeed, ExpliClas lets us export the linguistic approximation of FURIA rules into the IEEE Std 1855-2016 format which can be then loaded and understood by GUAJE. This classifier covers only 95.75% of instances in the entire Beer dataset and the classification ratio is 64% (Precision=72.8; Recall=64; F-measure=77.7). Obviously, this system is highly interpretable (Rules=15; Premises=31) but less accurate than the original FURIA classifier. It is worth noting that the gain in interpretability (thanks to imposing global semantics to make feasible a linguistic interpretation of FURIA rules) is obtained at the cost of a significant reduction in accuracy. The goodness of the interpretability-accuracy trade-off for this classifier can be judged in comparison with those systems previously generated by GUAJE (see Table 6.2). The linguistic approximated classifier is by far the least accurate and even if it exhibits a good interpretability, GUAJE was able to produce even more interpretable classifiers by considering interpretability issues as a main concern all along a human-centric design process.

Figure 6.21 shows the explanation provided by this classifier in GUAJE, regarding the same data instance that was considered by ExpliClas in Fig. 6.20. Obviously, both linguistic explanations are in agreement but GUAJE provides us with further details. We have a high confident in the classification made by the original FURIA classifier while we have a medium confidence in the classification made by its linguistic approximation.

If we pay attention to the behavior of fuzzy rules at inference level (take a look at Figs. 6.10, 6.22 and 6.23), we observe that overlapping among rules is also different in GUAJE and ExpliClas.

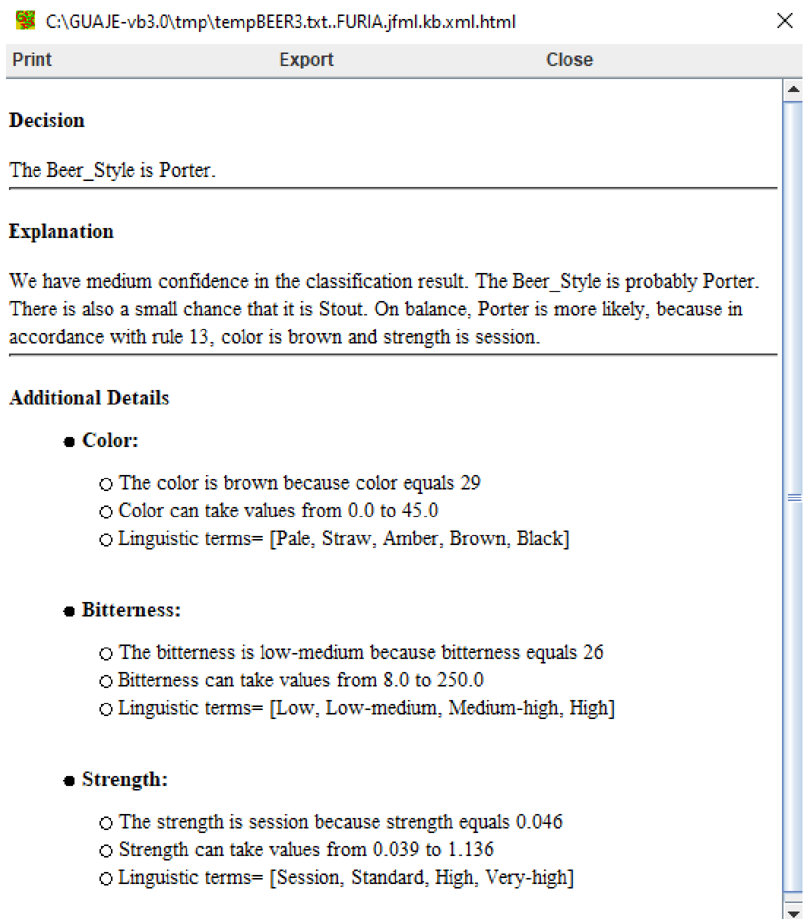

Fig. 6.21: GUAJE screenshot with an explanation associated to the classifier endowed with linguistic interpretation of FURIA rules.

In GUAJE, rules have global semantics, while in ExpliClas, FURIA rules have local semantics. On the one hand, in the case of GUAJE, the SFPs that are created for each attribute (as a mixture of expert knowledge and knowledge extracted from data) define a grid which fully covers the data input space. On the other hand, in the case of ExpliClas, there are no SFPs associated a priori to FURIA rules. Rules are created locally to cover the larger number of data instances that they can but without taking care of covering the whole input space. Thus, sparsity and overlapping among rules differ depending on the way how rules are created. This fact can be appreciated when looking at the fingrams associated to each fuzzy classifier. In the case of ExpliClas (see Fig. 6.22), FURIA rules of the same class tend to be more clearly grouped and separated from the others, forming more isolated clusters. In other words, FURIA

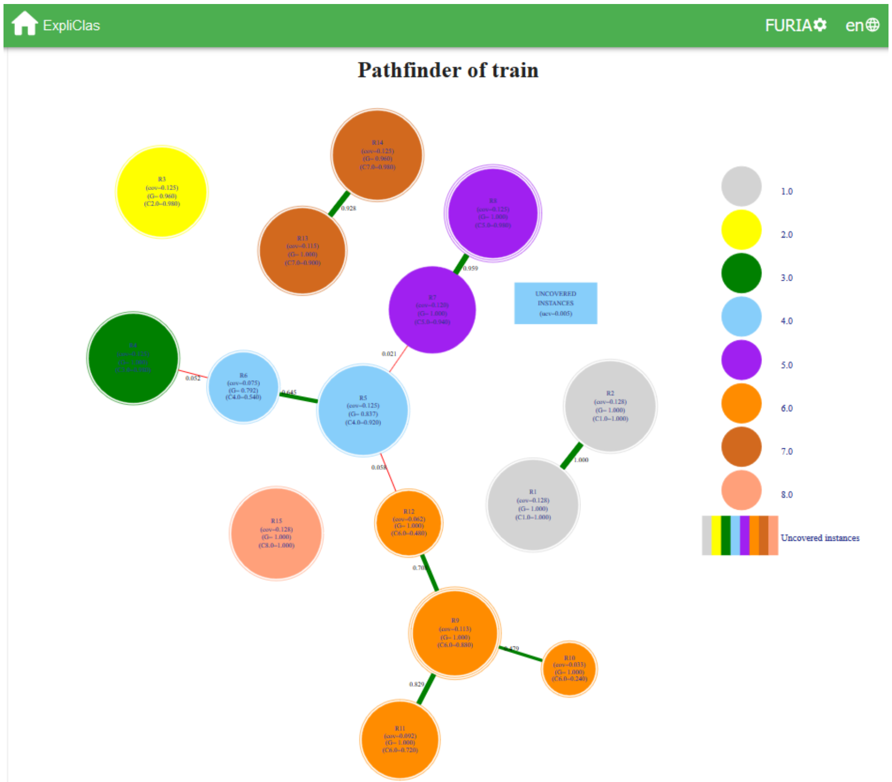

Fig. 6.22: ExpliClas screenshot with the fingrams view of FURIA rules.

rules look like less fuzzy and more crisp than rules handled by GUAJE. Indeed, in the case of GUAJE (see Fig. 6.10), we observe a larger number of connections among rules. It is worth noting that in Fig. 6.22 there is a rectangular node, which is rarely present in fingrams associated to rules in GUAJE, and represents uncovered instances, i.e., data instances that are not firing any rule. As said before, by default FURIA deals with those instances as a black-box classifier by means of the so-called stretching mechanism. Of course, the user may also opt for abstain (i.e., no output class is provided) or for majority voting (i.e., given as output class the most frequent one in the dataset).

Figure 6.23 shows how the linguistic approximated FURIA rules interact once they are endowed with global semantics in GUAJE. In this case, four rules (R8, R9, R10 and R11) cover no data and may be removed. Actually, after running the simplification and optimization procedures provided by GUAJE, the final classifier gets much better balance between interpretability and accuracy (Accuracy=74%; Precision=90.5; Recall=74; F-measure=83.4; Rules=8; Premises=15) but it is still worse than EXP-SP-FDTP-S-OSW (see Table 6.2).

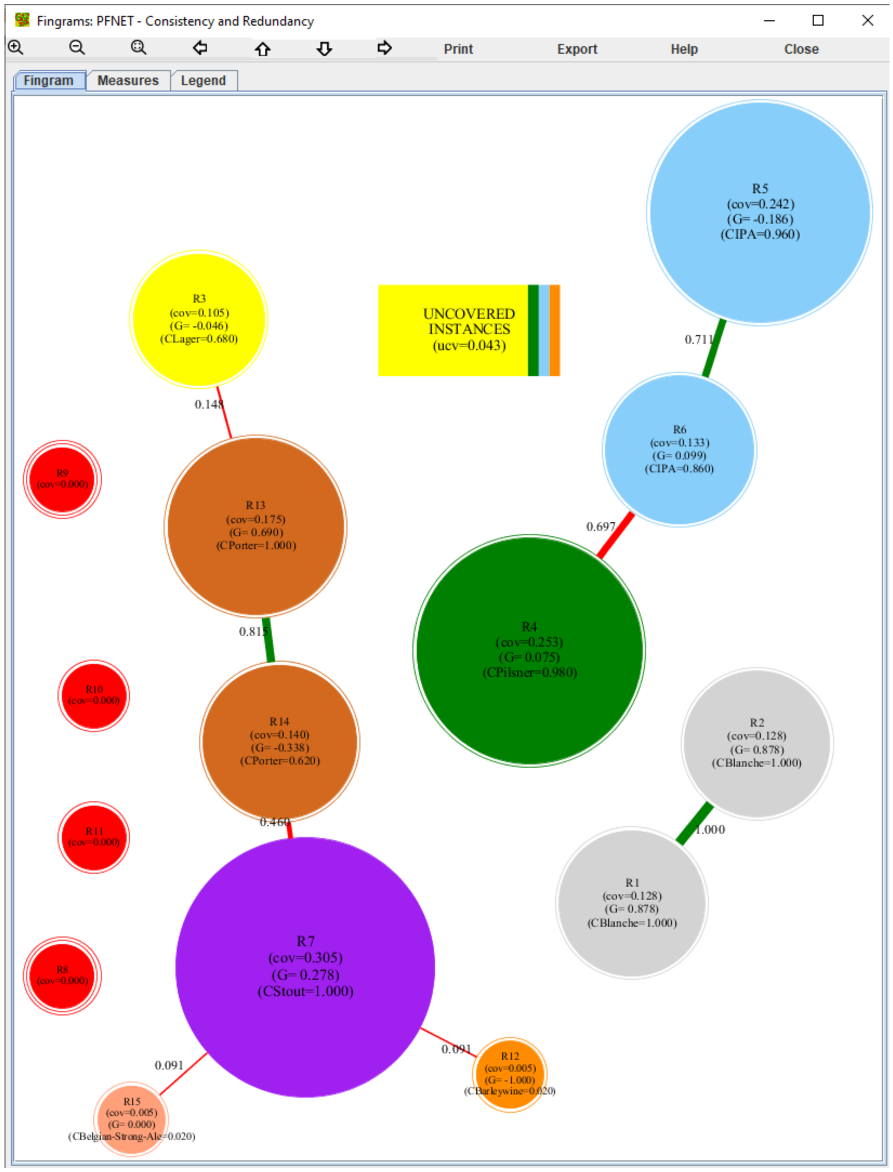

Fig. 6.23: GUAJE screenshot with the fingrams view of the linguistic approximation of FURIA rules.

In addition, in the case of considering the fuzzy inference for just one single data instance (see Figs. 6.12 and 6.24) the so-called instance-based fingram is simpler for FURIA rules. This fact is due to their lack of linguistic interpretability. As a result,

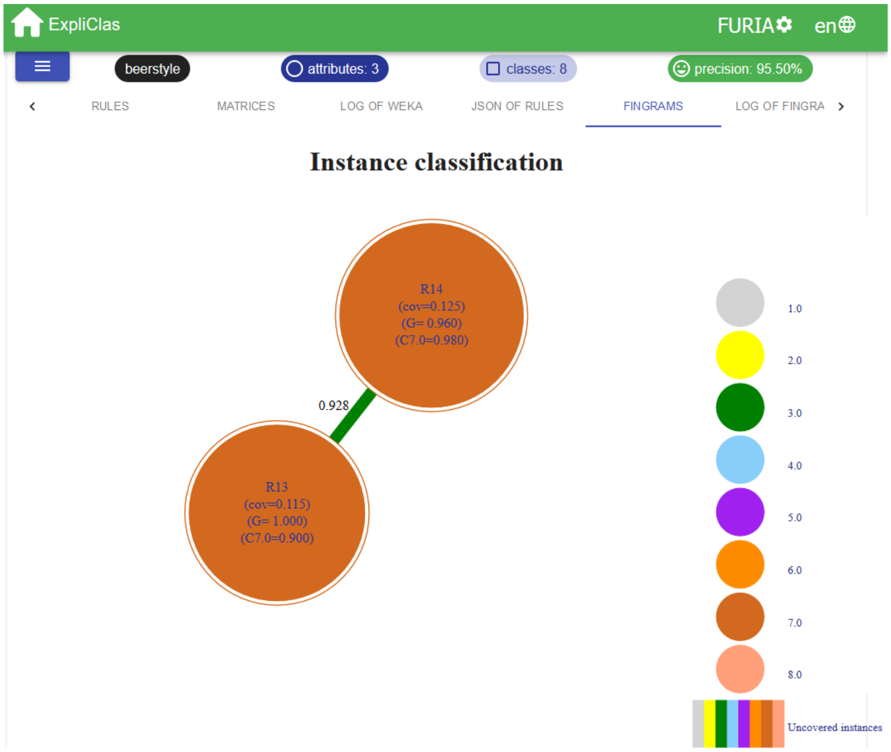

Fig. 6.24: ExpliClas screenshot with an example of instance-based fingrams view.

the explanation provided by ExpliClas (see Fig. 6.20) ignores the effect of having borderline values among linguistic terms what is by the way naturally identified and communicated to users in the case of GUAJE (see Fig. 6.15). Notice that, there is no reference to borderline values in the explanation provided by GUAJE in the case of the linguistic approximation of FURIA rules (see Fig. 6.21).

6.5 Final Remarks

In this chapter, we have provided readers with an illustrative use case regarding how fuzzy systems are ready to pave the way towards XAI: (1) how to carefully design and develop a fuzzy rule-based classifier with a good interpretability-accuracy trade-off for a specific application domain (i.e., beer style classification); and (2) how to endow such “white-box” interpretable fuzzy classifier with explanation capability (in the form of rich and natural textual and graphical explanations). Moreover, we have shown how it is also possible to generate readable and comprehensible linguistic explanations associated to fuzzy rule-based classifiers which are fully extracted from data without taking care of linguistic interpretability. Such classifiers are likely to be more accurate but they are harder to interpret, behaving usually as “gray-box” classifiers and even as “black-box” classifiers sometimes. Obviously, explanations associated to gray and black boxes are less natural and informative than those explanations associated to white boxes.

The use case under consideration was fully conducted with open source software. We used GUAJE for interpretable fuzzy modeling and then one of the generated classifiers was used with other fuzzy software (i.e., JFML, Py4JFML and FisPro), just for illustrating interoperability. In addition, it is worth noting that GUAJE makes use of many open source libraries. For example, Graphviz is used to visualize fingrams. rLDCP and simpleNLG are used to produce explanations in NL. We also used the ExpliClas web service to generate explanations associated to non-interpretable fuzzy classifiers, just for comparison purposes. In addition, we have shown how classifiers built with ExpliClas can be exported into the IEEE Std 1855-2016 format and thus they can be imported and enhanced by GUAJE. Moreover, all data, software and scripts needed to reproduce the examples provided along this chapter are freely available at [9].

As future work, we plan to evaluate the goodness, naturalness and effectiveness of the generated explanations with intrinsic and extrinsic human evaluation. In addition, we will integrate GUAJE and ExpliClas with a chatbot (i.e., with an intelligent dialogue system) ready to provide users with dynamic and interactive factual and counterfactual explanations.

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Chapter 7
Remarks and Prospects on Explainable Fuzzy Systems

Abstract Since 2016, there is an increasing interest on research topics such as fairness, accountability and interpretability, in the entire community of researchers in Artificial Intelligence. However, there were researchers working hard on these research topics much earlier. For example, Michalski published his comprehensibility postulate in the 1980s while Zadeh introduced the notions of fuzzy sets and systems in the 1960s and the paradigm of computing with words in the 1990s. Moreover, Zadeh was talking about precisiated natural language in 2004. All these pioneer contributions to explainable Artificial Intelligence should be recognized properly. Explainable fuzzy systems go a step ahead of interpretable fuzzy systems and are ready to act as a cornerstone in the context of explainable Artificial Intelligence where interdisciplinary research fields (e.g., Computational Linguistics, Argumentation Technology, Human-machine Interaction, Philosophy, Math, Computer Science, Neuroscience, and so on) meet. Explainable fuzzy systems provide users with interactive explanations in natural language, the preferred modality among humans, with visualization as a complementary modality. Fuzzy Logic endows explainable fuzzy systems with a solid mathematical background to model properly the inherent uncertainty, imprecision and vagueness of human language.

7.1 The Role of Explainable Fuzzy Systems in the Context of Explainable Artificial Intelligence

Prof. Zadeh made many highly valuable contributions to the Fuzzy Sets and Systems research field, and beyond. Many of his contributions were pioneering ideas and/or challenging proposals with a lot of potential to be fully developed later by other researchers. Nowadays, eXplainable Artificial Intelligence (eXplainable AI or XAI for short) is a prominent and fruitful research field where many of Zadeh's contributions can become crucial if they are carefully considered and thoroughly developed. Unfortunately, many of these contributions are overlooked out of the community of researchers in Fuzzy Logic.

In this book, we have provided readers with a holistic view of fundamentals and current research trends about how to pave the way from interpretable to explainable fuzzy systems. Moreover, we have pointed out advantages and disadvantages, strengths and weaknesses, threats and opportunities for explainable fuzzy systems to become a cornerstone in the XAI research field. Let us summarize here the main conclusions and remarks chapter by chapter.

In the first Chap. [7], interpretability is introduced as a necessary quality to develop XAI systems. In addition, Fuzzy Logic is highlighted as a tool to offer a suitable background to design interpretable intelligent systems. In particular, Fuzzy Rule-Based Systems (FRBSs) are characterized by transparent models with a double layer of knowledge representation: (1) a symbolic layer and (2) a semantic layer. The former is constituted by linguistic terms and the latter by the mathematical constructs pertaining to Fuzzy Set Theory (FST). Linguistic terms convey an implicit semantics to the user through their linguistic meaning, while fuzzy sets define their explicit semantics. As a consequence, interpretability in FRBSs is obtained when there is semantic cointension between the implicit and explicit semantics defined by interpreting the terms in the semantic layer. Dealing with global semantics or common ground is essential to avoid misleading interpretations. Therefore, interpretability of fuzzy systems cannot be taken for granted by the mere use of FST (an assertion that is sometimes found in literature) but it requires a design methodology that is the core of the present book.

In the second Chap. [2], FST and different types of FRBSs are revisited. Since the pioneer FRBS designed by Mamdani in 1975, FRBSs have been used in almost every field where numerical data can be exploited to represent the perceptual nature of linguistic terms in a computational way, by means of fuzzy sets. The original Mamdani FRBS has been enhanced in terms of efficiency and applicability. Currently, there are many machine learning schemes for automatically extracting FRBSs from data, what opens the door to their application in domains that were inconceivable when such systems were manually designed. Nevertheless, with automatic design procedures, the original intention of developing systems pursuing good accuracy and acceptable interpretability is hampered. This is the reason why the research on interpretability of FRBSs flourished, with the objective of developing criteria, metrics and design procedures for acquiring FRBSs from data that offer an acceptable interpretability-accuracy balance. Interpretability and accuracy are conflicting qualities which altogether provide FRBSs with an added value over the plethora of models that can be used for knowledge extraction from data nowadays. Apart from the multitude of applications where FRBSs are increasingly applied, theoretical research currently explores more performing learning schemes and more flexible representations for FRBSs, including those exploiting type-2 fuzzy sets and variants thereof, or hierarchical representations of rules with the intention of endowing FRBSs with a structure that is capable of organizing complex rules in a hierarchy of simpler representations. It is worth noting that these are new developments that, while being promising at their stage of research, are neither mature yet nor stabilized from theoretical and methodological viewpoints, so they require further research effort in the near future.

In the third Chap. [5], a number of interpretability constraints and criteria have been outlined. Some of them are fundamental for interpretable fuzzy modeling (e.g., distinguishability), while others can be considered as optional depending on the adopted design strategy and the specific application needs. Furthermore, the application of some interpretability constraints depends on the final representation of the information granules, and, in general, of the knowledge base. To sum up with, a number of issues should be taken into account in the selection of interpretability constraints: the type of representation of information granules (e.g., qualitative vs. quantitative), the type of fuzzy model (e.g., Mamdani vs. Takagi-Sugeno-Kang), the objectives of modeling (e.g., accuracy vs. interpretability), the semantic assumptions (e.g., implicative vs. conjunctive rules) and, last but not least, the application context.

In the fourth Chap. [6], readers are provided with a review of interpretability indexes which are categorized in a novel taxonomy in agreement with the structural and semantic constraints and criteria described in the previous chapter. In spite of the hard work already done for defining interpretability indexes, this is still a hot research topic and there is a lot of work to be done in the near future:

- Even if there are many proposals to measure interpretability at all structural levels (especially at fuzzy partitions, fuzzy rule, and fuzzy model levels), more work at fuzzy set and fuzzy granule levels is required.
- Additional objective and subjective indexes for quantification of semantic-based interpretability need to be defined.
- Type-2 and hierarchical fuzzy systems require deeper attention because there are almost no indexes for measuring their interpretability.
- Due to the subjective nature of the interpretability assessment task, there is not a universally accepted quantifiable definition of interpretability. Accordingly, there is not any universal index (easy to adapt to specific problems and specific users) recognized worldwide.
- With XAI-based systems, the focus of researchers should shift from measuring interpretability to assessing explainability. Explanations are expected to be multimodal, i.e., they should combine textual representations (in natural language) with graphical representations (as well as other modalities), but they should be also multi-lingual and easy to adapt to users with different backgrounds and preferences. It is worth noting that most interpretability indexes defined for fuzzy systems so far should be extended (i.e., enriched with other metrics) to measure not only the interpretability but also the explainability of fuzzy systems as well as other intelligent systems.

In the fifth Chap. [4], the design of FRBSs covers many different aspects to be considered and decisions to be taken, ranging from the selection of suitable shapes for the membership functions, or the definition of rules, to the use of specific aggregation or defuzzification operators. Additionally, other questions as the use of hierarchical structures, the meaning of some components of the knowledge base, the most appropriate structure of a fuzzy rule, the interpretability of the obtained system or its stability, have been considered. After decades of application of FRBSs to real problems, a wide repertory of design approaches is available, and the way they focus on questions as interpretability is quite diverse and multifaceted. It is impossible to prioritize or rank those approaches, since the arguments, concepts and metrics they use are in most cases not comparable. Given a specific problem or application, with its properties and characteristics, some approaches adapt better than others. However, there is no way to point out one single specific approach as an overall best option. In addition, historically, most of the work on interpretability of fuzzy systems concentrated on structural and complexity questions, while the semantic aspects received much less attention. Nowadays, with the arrival of XAI, semantic questions gain increasing attention, but still, aspects as the influence of the inference process on interpretability and explainability should be explored.

In the sixth Chap. [3], readers are provided with an illustrative use case regarding how fuzzy systems are ready to pave the way towards XAI: (1) how to carefully design and develop a fuzzy rule-based classifier with a good interpretability-accuracy trade-off for a specific application domain (i.e., beer style classification); and (2) how to endow such "white-box" interpretable fuzzy classifier with explanation capability (in the form of rich and natural textual and graphical explanations). It is worth noting that, with this approach, it is also possible to generate readable and comprehensible linguistic explanations associated to fuzzy rule-based classifiers which are fully extracted from data without taking care of linguistic interpretability. This approach could be extended to non-fuzzy intelligent systems, what opens the door to explaining black-box AI-based systems in terms of explainable fuzzy systems. Classifiers learned from data without considering interpretability constraints are likely to be more accurate but they are harder to interpret, behaving usually as "gray-box" classifiers and even as "black-box" classifiers sometimes. Obviously, explanations associated to gray and black boxes are less natural and informative than those explanations associated to white boxes. Fortunately, explainable fuzzy systems can assist users to understand the internal behavior and the output from gray-box to black-box models. However, there are still many challenging problems that deserve further research.

It is worth noting that all data, software and scripts needed to reproduce the examples provided along the book are freely available at [1].

7.2 Open Challenges

With the aim of paving the way from interpretable fuzzy systems toward XAI systems, it is needed to pay special attention to both theoretical and methodological aspects of explainability. Due to the intrinsic multidisciplinary nature of the XAI research field, researchers are forced to move beyond the usual topics treated by the community of researchers in Fuzzy Logic. They must join forces with researchers from other disciplines, from both academy and industry, working in the fields of Artificial and Computational Intelligence (with special attention to Fuzzy Logic but addressing also XAI challenges on Neural Networks, Evolutionary Computation, Bayesian Networks, Bio-inspired algorithms, etc.). Research on explainable fuzzy systems is open in several directions, which are aligned with the main XAI challenges:

Explainability definition. The notion of explainability goes beyond the notion of interpretability. The interpretability of a fuzzy system, or an AI-based system in a broader sense, relies on the ability of a human being to interpret the knowledge embedded in such system. Then, in the light of such interpretation, the human may elaborate an explanation to answer the question why the system behaves one way or another given a specific situation. In the context of XAI, researchers face the challenge of developing trustworthy self-explanatory AI-based systems, i.e., systems able to interpret themselves and elaborate explanations to answer "why" questions. It is worth noting that finding agreement on the definition of interpretability in the context of fuzzy systems has taken years. Therefore, current definitions should not be overlooked when looking for new definitions of interpretability in the context of machine learning. On the contrary, taking into account lessons learnt in the past should help to consider the multiple facets of explainability. Indeed, the definition of explainability should reflect the fact that there are explanations of different nature, e.g., task-oriented, factual, counterfactual, contrastive, causal, etc. Even if current definitions of interpretability can be generalized from fuzzy systems to other AI-based systems, the blur nature of interpretability requires continuous investigations on possible definitions that enable a computable treatment of this quality in XAI systems. This requirement casts the research on interpretable fuzzy systems towards cross-disciplinary investigations. For example, this research line includes investigations on computable definitions of some conceptual qualities, like vagueness (which has to be distinguished from imprecision and fuzziness).

Explainability assessment. There is a lack of universal evaluation metrics and protocols supported by statistics to measure the effectiveness and naturalness of automatically generated explanations. A prominent challenge is the adoption of a common framework for characterizing and assessing explainability with the aim of avoiding misleading notations. Within such a framework, novel metrics could be devised, especially for assessing different types of explanations. Subjective aspects of interpretability must be also considered, and integrated with objective interpretability measures to define more significant explainability indexes. In addition to data-driven metrics, human users should be involved in the assessment of explainability. Moreover, correlation among data-driven metrics and both intrinsic and extrinsic human evaluation needs to be thoroughly and rigorously studied.

Design of explainable fuzzy systems. The challenge is to design the three components of a self-explaining fuzzy system: (1) an interpretable fuzzy system endowed with fuzzy-grounded knowledge representation; (2) an intelligent module able to generate explanations by automatically interpreting the knowledge embedded in the interpretable fuzzy system in agreement with the background of the intended audience; and (3) an advance human-machine interaction interface (e.g., a conversational agent) ready to provide humans with effective explanations. Such explanations should be semantic grounded, persuasive, trustworthy, interactive and multi-modal (e.g., graphical and textual modalities).

In the case of very complex (high-dimensional) problems, the use of novel forms of representation (different from the classical rule-based) may help in representing complex relationship in comprehensible ways thus yielding a valid aid in designing explainable fuzzy systems. For example, a multi-level hierarchical representation could enhance the interpretability of fuzzy systems by providing different granularity levels for knowledge representation. On the one hand, the highest granulation levels give a coarse (yet immediately comprehensible) description of knowledge, while lower levels provide for more detailed knowledge. The combination of linguistic and graphical approaches could be a promising approach for descriptive and exploratory analysis of explainable fuzzy systems. In addition, a common approach in designing interpretable fuzzy models makes use of multi-objective genetic algorithms in order to deal with the conflicting design objectives of accuracy and interpretability. Producing a pool of fuzzy models with global semantics but different interpretability-accuracy trade-offs may help in the search for alternative plausible explanations personalized to specific users.

7.3 Final Remarks

Quoting to Polanyi [8], "humans know more than they can explain". This is mainly due to all implicit knowledge that we unconsciously acquire through culture, heritage, etc. The same applies to AI-based models learned from data. In order to build good trustworthy models, designers need good cured data, free of bias. Accordingly, they have to pay attention not only to technical but also to ethical and legal issues due to the worldwide social impact of AI-based systems.

In the particular case of fuzzy systems, they integrate both expert knowledge and knowledge automatically extracted from data under the same mathematical formalism. Fuzzy systems are deemed as interpretable when they are built by a human-centered design approach subject to interpretability constraints. Thus, interpretable fuzzy systems are trustworthy systems for designers with background on Fuzzy Logic. In addition, explainable fuzzy systems leverage the use of fuzzy systems even by domain experts (e.g., physicians) who usually don't know anything about Fuzzy Logic. For example, explainable fuzzy systems may provide physicians with trustworthy recommendations or assist other professionals to make better decisions in their daily tasks.

However, as a final remark, it is worth observing that explainability is only one aspect of the multi-faceted problem of *human-centered* design of fuzzy systems. Other facets include acceptability (e.g., according to ethical rules), interestingness of fuzzy rules, applicability (e.g., with respect to law), etc. We are at the dawn of a new era where humans and machines collaborate together as peers. Research in this direction is blooming and the authors of this book believe that Fuzzy Logic will have a central role in the development of XAI systems.

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